

CHAPTER 1: DEFINITIONS, ELIMINATION OF ARBITRARY CONSTANTS

1.1 DEFINITIONS

A **Differential Equation** is an equation in which one or more of the derivatives of one or more unknown functions occur. These functions, generally represent quantities, the derivatives represent their rates of change, and the differential equation defines a relationship between the two.

The following are examples of differential equations:

- (1) $\frac{dy}{dx} = \cos x$
- (2) $\frac{d^2y}{dx^2} + k^2y = 0$
- (3) $(x^2 + y^2)dx - 2xydy = 0$
- (4) $\frac{\partial u}{\partial t} = h^2 \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$
- (5) $\frac{\partial u}{\partial t} = L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = E \omega \cos \omega t$

Independent Variable – when an equation involves one or more derivatives with respect to a particular variable.

Dependent Variable – if a derivative of the variable occurs.

In the equation,

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = E \omega \cos \omega t$$

i is the dependent variable, t is independent variable, and L, R, C, E , and ω are called **parameters**.

Consider the equation,

$$(x^2 + y^2)dx - 2xydy = 0$$

May be written as,

$$(x^2 + y^2) - 2xy \frac{dy}{dx} = 0$$

or

$$(x^2 + y^2) \frac{dx}{dy} - 2xy = 0$$

We may consider either variable x or y to be dependent, the other being the independent

CLASSIFICATION OF DIFFERENTIAL EQUATIONS

1.2 CLASSIFICATION OF DIFFERENTIAL EQUATIONS

Differential equations can be classified according to its order, as ordinary or partial, and as linear or nonlinear equations.

❖ CLASSIFICATION BY ORDER:

This classification of differential equation is the most common of all.

The **order** of a differential equation is the order of the highest-ordered derivative appearing in the equation. For instance,

$$\frac{d^2y}{dx^2} + 2b \left(\frac{dy}{dx} \right)^3 + y = 0$$

is an equation of "order two". It is also referred to as "second-order equation"

The **degree** of a differential equation is the algebraic degree of the highest-ordered derivative involved in the expansion. If the equation is in the fractional form, simplify until a whole numbered exponent is obtained.

❖ CLASSIFICATION AS ORDINARY OR PARTIAL EQUATIONS:

This classification depends on whether you have only ordinary derivatives involved or only partial derivatives.

- a. **ORDINARY DIFFERENTIAL EQUATION** – this equation contains total derivatives and does not contain partial derivatives which means that the unknown function depends on only one independent variable.

Examples:

- (1) $\frac{d^2y}{dx^2} + \frac{dy}{dx} = 3$
- (2) $(x-5)dy = 2xydx$
- (3) $(x^2 + y^2) \frac{dy}{dx} = 5x^2y$

- b. **PARTIAL DIFFERENTIAL EQUATION** – this equation contains partial derivatives which means the unknown function depends on two or more independent variables.

CLASSIFICATION OF DIFFERENTIAL EQUATIONS

Examples:

$$(1) \frac{\partial z}{\partial y} = x$$

$$(2) \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 y}{\partial z^2} = -\frac{\partial^2 V}{\partial z^2}$$

♦ CLASSIFICATION AS LINEAR OR NONLINEAR:

Another way that you can classify differential equations is as *linear* or *nonlinear*.

- a. **LINEAR DIFFERENTIAL EQUATION** - if it exclusively involves linear terms (that is, terms to the power of 1) of $y, y', y'',$ and beyond to $y^{(n)}$. For example, this equation is a linear equation:

$$L \frac{d^3 y}{dx^3} + R \frac{d^2 y}{dx^2} + Q \frac{dy}{dx} = E(x)$$

This kind of differential equation may be written as,

$$Ly''' + Ry'' + Qy' = E(x)$$

- b. **NONLINEAR DIFFERENTIAL EQUATION** - involves nonlinear terms in any of $y, y', y'',$ up to $y^{(n)}$. For example,

$$L \frac{d^3 y}{dx^3} + R \left(\frac{d^2 y}{dx^2} \right)^2 + Q \frac{dy}{dx} = E(x)$$

$$\frac{d^2 y}{dx^2} + 2b \left(\frac{dy}{dx} \right)^3 + y = 0$$

ELIMINATION OF ARBITRARY CONSTANTS

1.3 ELIMINATION OF ARBITRARY CONSTANTS

We shall consider in this section the inverse process of obtaining the differential equation from the general solution. This is done by eliminating the arbitrary constants from the general solution.

From an equation containing arbitrary constants, the constants may be eliminated by the following methods:

1. By differentiation and combination. The general method to eliminate arbitrary constants is to differentiate the equation n times. The n arbitrary constants may then be eliminated from the $n + 1$ equations. The result will be a differential equation of order n .
2. By isolation of the constants. In this method at each step before differentiation, isolate one of the arbitrary constants, free from variable, so that it will disappear after differentiation. Solve continuously until all constants are eliminated.

EXERCISES:

Eliminate the arbitrary constants from the given differential equations.

1. $x \sin y + x^2 y = C$

$$[x \cos y \cdot y' + \sin y (1)] + [x^2 y' + y(2x)] = 0$$

$$xy' \cos y + x^2 y' + \sin y + 2xy = 0$$

$$(x \cos y + x^2) y' + (\sin y + 2xy) = 0$$

$$(x \cos y + x^2) \frac{dy}{dx} + (\sin y + 2xy) = 0$$

$$(x \cos y + x^2) dy + (\sin y + 2xy) dx = 0 \text{ ANS.}$$

2. $3x^2 - xy^2 = C$

$$6x - (2xyy' + y^2) = 0$$

$$6x - [x(2yy') + y^2(1)] = 0$$

$$6x - (2xyy' + y^2) = 0$$

$$6x - 2xyy' - y^2 = 0$$

$$6x - 2xy \frac{dy}{dx} - y^2 = 0$$

$$6xdx - 2xydy - y^2 dx = 0$$

$$6xdx - y^2 dx - 2xydy = 0$$

$$(6x - y^2) dx - 2xydy = 0 \text{ ANS.}$$

3. $xy^2 - 1 = Cy$

$$\frac{xy^2 - 1}{y} = C$$

$$\frac{y[x(2yy') + y^2(1) - 0] - (xy^2 - 1)y'}{y^2} = 0$$

$$\frac{y(2xyy' + y^2) - (xy^2 - 1)y'}{y^2} = 0$$

$$(2xy^2 y' + y^3) - (xy^2 - 1)y' = 0$$

$$y^3 - (xy^2 - 1 - 2xy^2)y' = 0$$

$$y^3 - (xy^2 - 1 - 2xy^2) \frac{dy}{dx} = 0$$

$$y^3 dx - (-xy^2 - 1) dy = 0$$

$$y^3 dx + (xy^2 + 1) dy = 0 \text{ ANS.}$$

4. $Cx^2 + x + y^2 = 0$

$$C = \frac{-x - y^2}{x^2}$$

$$0 = \frac{x^2(-1 - 2yy') - (-x - y^2)(2x)}{x^4}$$

ELIMINATION OF ARBITRARY CONSTANTS

$$\begin{aligned}(-x^2 - 2x^2yy') + (2x^2 + 2xy^2) &= 0 \\(-x^2 - 2x^2yy') + (2x^2 + 2xy^2) &= 0 \\-2x^2yy' - x^2 + 2x^2 + 2xy^2 &= 0 \\-2x^2y \frac{dy}{dx} + x^2 + 2xy^2 &= 0 \\-2xydy + (x + 2y^2)dx &= 0 \\(x + 2y^2)dx - 2xydy &= 0 \text{ ANS.}\end{aligned}$$

5. $x = A \sin(\omega t + \beta)$;
 ω a parameter, not to be eliminated.

$$\begin{aligned}\frac{dx}{dt} &= A\omega \cos(\omega t + \beta) \\ \frac{d^2x}{dt^2} &= -A\omega^2 \sin(\omega t + \beta)\end{aligned}$$

$$\text{But } x = A \sin(\omega t + \beta)$$

$$\begin{aligned}\frac{d^2x}{dt^2} &= -\omega^2 x \\ \frac{d^2x}{dt^2} + \omega^2 x &= 0 \text{ ANS.}\end{aligned}$$

6. $x = C_1 \cos \omega t + C_2 \sin \omega t$;
 ω a parameter

$$\begin{aligned}\frac{dx}{dt} &= -C_1 \omega \sin \omega t + C_2 \omega \cos \omega t \\ \frac{d^2x}{dt^2} &= -C_1 \omega^2 \cos \omega t - C_2 \omega^2 \sin \omega t \\ \frac{d^2x}{dt^2} &= -\omega^2 (C_1 \cos \omega t + C_2 \sin \omega t)\end{aligned}$$

$$\text{But } x = C_1 \cos \omega t + C_2 \sin \omega t$$

$$\begin{aligned}\frac{d^2x}{dt^2} &= -\omega^2 x \\ \frac{d^2x}{dt^2} + \omega^2 x &= 0 \text{ ANS.}\end{aligned}$$

7. $y = Cx + C^2 + 1$

$$\begin{aligned}y' &= C + 0 + 0 \\ C &= y' \\ \text{Thus,}\end{aligned}$$

$$y = xy' + (y')^2 + 1 \text{ ANS.}$$

8. $y = mx + \frac{h}{m}$;
 h a parameter, m to be eliminated.

$$\begin{aligned}y' &= m + 0 \\ m &= y'\end{aligned}$$

Thus,

$$y = xy' + \frac{h}{y'} \text{ ANS.}$$

9. $y^2 = 4ax$

$$\begin{aligned}2yy' &= 4a \\ a &= \frac{2yy'}{4} = \frac{yy'}{2}\end{aligned}$$

Thus,

$$\begin{aligned}y^2 &= 4 \frac{yy'}{2} x \\ y &= 2x \frac{dy}{dx} \\ ydx &= 2xdy \\ 2xdy - ydx &= 0 \text{ ANS.}\end{aligned}$$

10. $y = ax^2 + bx + C$

$$\begin{aligned}y' &= 2ax + b \\ y'' &= 2a \\ y''' &= 0 \text{ ANS.}\end{aligned}$$

11. $y = C_1 + C_2 e^{3x}$

$$\begin{aligned}y' &= 3C_2 e^{3x} \rightarrow (\text{Eq. 1}) \\ y'' &= 9C_2 e^{3x} \rightarrow (\text{Eq. 2})\end{aligned}$$

Multiply (Eq. 1) by 3 then subtract (Eq. 2);

$$\begin{aligned}3y' &= 9C_2 e^{3x} \\ y'' &= 9C_2 e^{3x} \\ 3y' - y'' &= 0\end{aligned}$$

Thus,

$$y'' - 3y' = 0 \text{ ANS.}$$

ELIMINATION OF ARBITRARY CONSTANTS

12. $y = 4 + C_1 e^{3x}$

$$\begin{aligned}y' &= 3C_1 e^{3x} \\ C_1 e^{3x} &= \frac{y'}{3}\end{aligned}$$

Thus,

$$\begin{aligned}y &= 4 + \frac{y'}{3} \\ 3y &= 12 + y' \\ y' - 3y &= -12 \text{ ANS.}\end{aligned}$$

13. $y = C_1 + C_2 e^{-4x}$

$$\begin{aligned}y' &= -4C_2 e^{-4x} \rightarrow (\text{Eq. 1}) \\ y'' &= 16C_2 e^{-4x} \rightarrow (\text{Eq. 2})\end{aligned}$$

Multiply (Eq. 1) by 4 then add (Eq. 2);

$$\begin{aligned}4y' &= -16C_2 e^{-4x} \\ y'' &= 16C_2 e^{-4x} \\ 4y' + y'' &= 0\end{aligned}$$

Thus,

$$y'' + 4y' = 0 \text{ ANS.}$$

14. $y = C_1 e^x + C_2 e^{-x}$

$$\begin{aligned}y' &= C_1 e^x - C_2 e^{-x} \\ y'' &= C_1 e^x + C_2 e^{-x}\end{aligned}$$

$$\text{But } y = C_1 e^x + C_2 e^{-x}$$

Thus,

$$\begin{aligned}y'' &= y \\ y'' - y &= 0 \text{ ANS.}\end{aligned}$$

15. $y = x + C_1 e^x + C_2 e^{-x}$

$$\begin{aligned}y' &= 1 + C_1 e^x - C_2 e^{-x} \\ y'' &= C_1 e^x + C_2 e^{-x}\end{aligned}$$

$$\text{But } y - x = C_1 e^x + C_2 e^{-x}$$

Thus,

$$\begin{aligned}y'' &= y - x \\ y'' - y &= -x \text{ ANS.}\end{aligned}$$

16. $y = C_1 e^{2x} + C_2 e^{3x}$

$$y = C_1 e^{2x} + C_2 e^{3x} \rightarrow (\text{Eq. 1})$$

$$y' = 2C_1 e^{2x} + 3C_2 e^{3x} \rightarrow (\text{Eq. 2})$$

$$y'' = 4C_1 e^{2x} + 9C_2 e^{3x} \rightarrow (\text{Eq. 3})$$

Multiply (Eq. 1) by 2 then subtract (Eq. 2);

$$\begin{aligned}2y &= 2C_1 e^{2x} + 2C_2 e^{3x} \\ y' &= 2C_1 e^{2x} + 3C_2 e^{3x} \\ 2y - y' &= -C_2 e^{3x} \rightarrow (\text{Eq. 4})\end{aligned}$$

Multiply (Eq. 2) by 2 then subtract (Eq. 3);

$$\begin{aligned}2y' &= 4C_1 e^{2x} + 6C_2 e^{3x} \\ y'' &= 4C_1 e^{2x} + 9C_2 e^{3x} \\ 2y' - y'' &= -3C_2 e^{3x} \rightarrow (\text{Eq. 5})\end{aligned}$$

Multiply (Eq. 4) by 3 then subtract (Eq. 5);

$$\begin{aligned}6y - 3y' &= -3C_2 e^{3x} \\ 2y' - y'' &= -3C_2 e^{3x} \\ 6y - 3y' - 2y' + y'' &= 0\end{aligned}$$

Thus,

$$y'' - 5y' + 6y = 0 \text{ ANS.}$$

17. $y = x^2 + C_1 e^{2x} + C_2 e^{3x}$

$$y = x^2 + C_1 e^{2x} + C_2 e^{3x} \rightarrow (\text{Eq. 1})$$

$$y' = 2x + 2C_1 e^{2x} + 3C_2 e^{3x} \rightarrow (\text{Eq. 2})$$

$$y'' = 2 + 4C_1 e^{2x} + 9C_2 e^{3x} \rightarrow (\text{Eq. 3})$$

Multiply (Eq. 1) by 2 then subtract (Eq. 2);

$$\begin{aligned}2y &= 2x^2 + 2C_1 e^{2x} + 2C_2 e^{3x} \\ y' &= 2x + 2C_1 e^{2x} + 3C_2 e^{3x} \\ 2y - y' &= 2x^2 - 2x - C_2 e^{3x} \rightarrow (\text{Eq. 4})\end{aligned}$$

ELIMINATION OF ARBITRARY CONSTANTS

Multiply (Eq. 2) by 2 then subtract (Eq. 3):

$$\begin{aligned} 2y' &= 4x + 4C_1e^{2x} + 6C_2e^{3x} \\ y'' &= 2 + 4C_1e^{2x} + 9C_2e^{3x} \\ 2y' - y'' &= 4x - 2 - 3C_2e^{3x} \rightarrow (\text{Eq. 5}) \end{aligned}$$

Multiply (Eq. 4) by 3 then subtract (Eq. 5):

$$\begin{aligned} 6y - 3y' &= 6x^2 - 6x - 3C_2e^{3x} \\ 2y' - y'' &= 4x - 2 - 3C_2e^{3x} \\ 6y - 3y' - 2y' + y'' &= 6x^2 - 6x - 4x + 2 \end{aligned}$$

Thus,

$$y'' - 5y' + 6y = 6x^2 - 10x + 2 \text{ ANS.}$$

$$18. \quad y = C_1e^x + C_2xe^x$$

$$\begin{aligned} y' &= C_1e^x + C_2(xe^x + e^x) \\ y' &= C_1e^x + C_2xe^x + C_2e^x \\ y' &= y + C_2e^x \rightarrow (\text{Eq. 1}) \end{aligned}$$

$$y'' = y' + C_2e^x \rightarrow (\text{Eq. 2})$$

Subtract (Eq. 1) and (Eq. 2):

$$\begin{aligned} y' &= y + C_2e^x \\ y'' &= y' + C_2e^x \\ y' - y'' &= y - y' \end{aligned}$$

Thus,

$$y'' - 2y' + y = 0 \text{ ANS.}$$

$$19. \quad y = Ae^{2x} + Bxe^{2x}$$

$$\begin{aligned} y' &= 2Ae^{2x} + B(2xe^{2x} + e^{2x}) \\ y' &= 2Ae^{2x} + 2Bxe^{2x} + Be^{2x} \\ y' &= 2y + Be^{2x} \rightarrow (\text{Eq. 1}) \end{aligned}$$

$$y'' = 2y' + 2Be^{2x} \rightarrow (\text{Eq. 2})$$

Multiply (Eq. 1) by 2 then subtract (Eq. 2):

$$\begin{aligned} 2y' &= 4y + 2Be^{2x} \\ y'' &= 2y' + 2Be^{2x} \\ 2y' - y'' &= 4y - 2y' \end{aligned}$$

Thus,

$$y'' - 4y' + 4y = 0 \text{ ANS.}$$

$$20.$$

$$y = C_1e^{2x} \cos 3x + C_2e^{2x} \sin 3x$$

$$y' = C_1(-3e^{2x} \sin 3x + 2e^{2x} \cos 3x) + C_2(3e^{2x} \cos 3x + 2e^{2x} \sin 3x)$$

$$y' = -3C_1e^{2x} \sin 3x + 2C_1e^{2x} \cos 3x + 3C_2e^{2x} \cos 3x + 2C_2e^{2x} \sin 3x$$

$$y' = 2(C_1e^{2x} \cos 3x + C_2e^{2x} \sin 3x) - 3C_1e^{2x} \sin 3x + 3C_2e^{2x} \cos 3x$$

$$y' = 2y - 3C_1e^{2x} \sin 3x + 3C_2e^{2x} \cos 3x \rightarrow (\text{Eq. 1})$$

$$\begin{aligned} y'' &= -3C_1(3e^{2x} \cos 3x + 2e^{2x} \sin 3x) + 2C_1(-3e^{2x} \sin 3x + 2e^{2x} \cos 3x) + 3C_2(-3e^{2x} \sin 3x + 2e^{2x} \cos 3x) + 2C_2(3e^{2x} \cos 3x + 2e^{2x} \sin 3x) \end{aligned}$$

$$\begin{aligned} y'' &= C_1(-9e^{2x} \cos 3x - 6e^{2x} \sin 3x) + C_1(-6e^{2x} \sin 3x + 4e^{2x} \cos 3x) + C_2(-9e^{2x} \sin 3x + 6e^{2x} \cos 3x) + C_2(6e^{2x} \cos 3x + 4e^{2x} \sin 3x) \end{aligned}$$

$$\begin{aligned} y'' &= -9(C_1e^{2x} \cos 3x + C_2e^{2x} \sin 3x) + 4(C_1e^{2x} \cos 3x + C_2e^{2x} \sin 3x) + 12C_2e^{2x} \cos 3x - 12C_1e^{2x} \sin 3x \end{aligned}$$

$$y'' = -9y + 4y + 12C_2e^{2x} \cos 3x - 12C_1e^{2x} \sin 3x$$

$$y'' = -5y + 12C_2e^{2x} \cos 3x - 12C_1e^{2x} \sin 3x \rightarrow (\text{Eq. 2})$$

Multiply (Eq. 1) by 4 then add (Eq. 2):

$$4y' = 8y - 12C_1e^{2x} \sin 3x + 12C_2e^{2x} \cos 3x$$

$$y'' = -5y + 12C_2e^{2x} \cos 3x - 12C_1e^{2x} \sin 3x$$

$$4y' - y'' = 8y + 5y$$

Thus,

$$y'' - 4y' + 13y = 0 \text{ ANS.}$$

ELIMINATION OF ARBITRARY CONSTANTS

$$21.$$

$$y = C_1e^{ax} \cos bx + C_2e^{ax} \sin bx;$$

a and b are parameters.

$$y' = C_1(-be^{ax} \sin bx + ae^{ax} \cos bx) + C_2(be^{ax} \cos bx + ae^{ax} \sin bx)$$

$$y' = a(C_1e^{ax} \cos bx + C_2e^{ax} \sin bx) - bC_1e^{ax} \sin bx + bC_2e^{ax} \cos bx$$

$$y' = ay - b(C_1e^{ax} \sin bx - C_2e^{ax} \cos bx) \rightarrow (\text{Eq. 1})$$

$$y'' = ay' - bC_1(be^{ax} \cos bx + ae^{ax} \sin bx) + bC_2(-be^{ax} \sin bx + ae^{ax} \cos bx)$$

$$y'' = ay' - b^2C_1e^{ax} \cos bx - abC_1e^{ax} \sin bx - b^2C_2e^{ax} \sin bx + abC_2e^{ax} \cos bx$$

$$y'' = ay' - b^2(C_1e^{ax} \cos bx + C_2e^{ax} \sin bx) - ab(C_1e^{ax} \sin bx - C_2e^{ax} \cos bx)$$

$$y'' = ay' - b^2y - ab(C_1e^{ax} \sin bx - C_2e^{ax} \cos bx) \rightarrow (\text{Eq. 2})$$

Multiply (Eq. 1) by a then subtract (Eq. 2):

$$ay' = a^2y - ab(C_1e^{ax} \sin bx - C_2e^{ax} \cos bx)$$

$$y'' = ay' - b^2y - ab(C_1e^{ax} \sin bx - C_2e^{ax} \cos bx)$$

$$ay' - y'' = a^2y - ay' + b^2y$$

Thus,

$$y'' - 2ay' + a^2y + b^2y = 0$$

$$y'' - 2ay' + (a^2 + b^2)y = 0 \text{ ANS.}$$

$$22.$$

$$y = C_1x + C_2e^x$$

$$y' = C_1 + C_2e^x \rightarrow (\text{Eq. 1})$$

$$y'' = C_2e^x \rightarrow (\text{Eq. 2})$$

Subtract (Eq. 1) and (Eq. 2):

$$y' = C_1 + C_2e^x$$

$$y'' = C_2e^x$$

$$y' - y'' = C_1 \rightarrow (\text{Eq. 3})$$

Substitute (Eq. 2) and (Eq. 3) in original equation,

$$y = (y' - y'')x + y'$$

$$y = xy' - xy'' + y'$$

$$xy'' - y'' - xy' + y = 0$$

$$(x-1)y'' - xy' + y = 0 \text{ ANS.}$$

$$23.$$

$$y = C_1x^2 + C_2e^{-x}$$

$$y = C_1x^2 + C_2e^{-x} \rightarrow (\text{Eq. 1})$$

$$y' = 2C_1x - C_2e^{-x} \rightarrow (\text{Eq. 2})$$

$$y'' = 2C_1 + C_2e^{-x} \rightarrow (\text{Eq. 3})$$

Add (Eq. 1) and (Eq. 2):

$$y = C_1x^2 + C_2e^{-x}$$

$$y' = 2C_1x - C_2e^{-x}$$

$$y + y' = C_1x^2 + 2C_1x \rightarrow (\text{Eq. 4})$$

Add (Eq. 2) and (Eq. 3):

$$y' = 2C_1x - C_2e^{-x}$$

$$y'' = 2C_1 + C_2e^{-x}$$

$$y' + y'' = 2C_1 + 2C_1x \rightarrow (\text{Eq. 5})$$

From (Eq. 4):

$$C_1 = \frac{y + y'}{x^2 + 2x}$$

From (Eq. 5):

$$C_1 = \frac{y' + y''}{2 + 2x}$$

$$C_1 = C_1$$

Thus,

$$\frac{y + y'}{x^2 + 2x} = \frac{y' + y''}{2 + 2x}$$

$$(y + y')(2 + 2x) = (y' + y'')(x^2 + 2x)$$

$$2xy + 2y + 2xy' + 2y' = x^2y' + 2xy'' + 2xy'' + 2xy''$$

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$$2xy + 2y + 2y' - x^2y' - x^2y'' - 2xy'' = 0$$

$$2(x+1)y + (2-x^2)y' - x(x+2)y'' = 0$$

$$x(x+2)y'' + (x^2-2)y' - 2(x+1)y = 0 \text{ ANS.}$$

24. $y = x^2 + C_1x + C_2e^{-x}$

$$y = x^2 + C_1x + C_2e^{-x} \rightarrow (\text{Eq. 1})$$

$$y' = 2x + C_1 - C_2e^{-x} \rightarrow (\text{Eq. 2})$$

$$y'' = 2 + C_2e^{-x} \rightarrow (\text{Eq. 3})$$

Add (Eq. 1) and (Eq. 2):

$$y = x^2 + C_1x + C_2e^{-x}$$

$$y' = 2x + C_1 - C_2e^{-x}$$

$$y + y' = x^2 + 2x + C_1x + C_1$$

$$C_1 = \frac{y + y' - x^2 - 2x}{x + 1}$$

From (Eq. 3):

$$C_2e^{-x} = y'' - 2$$

Substitute C_1 and C_2e^{-x} in (Eq. 1),

$$y = x^2 + \frac{y + y' - x^2 - 2x}{x + 1}x + y'' - 2$$

$$(x+1)y = x^2(x+1) + (y+y'-x^2-2x)x + (x+1)(y''-2)$$

$$xy + y = x^3 + x^2 + xy + xy' - x^3 - 2x^2 + xy'' - 2x + y'' - 2$$

$$y = xy - x^2 + xy'' - 2x + y'' - 2$$

$$xy'' + y'' + xy' - y = x^2 + 2x + 2$$

$$(x+1)y'' + xy' - y = x^2 + 2x + 2 \text{ ANS.}$$

25. $y = C_1x^2 + C_2e^{2x}$

$$y = C_1x^2 + C_2e^{2x} \rightarrow (\text{Eq. 1})$$

$$y' = 2C_1x + 2C_2e^{2x} \rightarrow (\text{Eq. 2})$$

$$y'' = 2C_1 + 4C_2e^{2x} \rightarrow (\text{Eq. 3})$$

Multiply (Eq. 1) by 2 then subtract (Eq. 2),

$$2y = 2C_1x^2 + 2C_2e^{2x}$$

$$y' = 2C_1x + 2C_2e^{2x}$$

$$2y - y' = 2C_1x^2 - 2C_1x$$

$$C_1 = \frac{2y - y'}{2x^2 - 2x} \rightarrow (\text{Eq. 4})$$

Multiply (Eq. 2) by 2 then subtract (Eq. 3),

$$2y' = 4C_1x + 4C_2e^{2x}$$

$$y'' = 2C_1 + 4C_2e^{2x}$$

$$2y' - y'' = 4C_1x - 2C_1$$

$$C_1 = \frac{2y' - y''}{4x - 2} \rightarrow (\text{Eq. 5})$$

Equate (Eq. 4) = (Eq. 5)

$$\frac{2y - y'}{2x^2 - 2x} = \frac{2y' - y''}{4x - 2}$$

$$(2y - y')(4x - 2) = (2y' - y'')(2x^2 - 2x)$$

$$8xy - 4y - 4xy' + 2y' = 4x^2y' - 4xy'' - 2x^2y'' + 2xy'$$

$$2y' - 4xy' - 4y + 8xy - 2xy'' + 2x^2y'' - 4x^2y' = 0$$

$$2y' - 4y + 8xy - 2xy'' + 2x^2y'' - 4x^2y' = 0$$

$$2y' - 4x^2y' - 4y + 8xy - 2xy'' + 2x^2y'' = 0$$

$$2(1 - 2x^2)y' - 4(1 - 2x)y - 2x(1 - x)y'' = 0$$

$$(1 - 2x^2)y' - 2(1 - 2x)y - x(1 - x)y'' = 0$$

$$x(1 - x)y'' + (2x^2 - 1)y' - 2(2x - 1)y = 0 \text{ ANS.}$$

FAMILIES OF CURVES

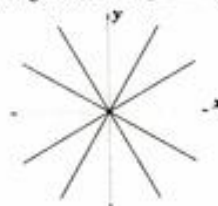
1.4 FAMILIES OF CURVES

A family of curve on a plane is usually defined by an equation containing one or more parameters together with the coordinates of a point on the plane. The differential equation of the family is obtained by eliminating the parameters by the methods discussed in the previous section.

EXERCISES:

In the following problems, obtain the differential equation of the given families of curves whose properties are indicated and sketch its curve.

1. Straight lines through the origin.



General Equation of a Line:

$$y - y_1 = m(x - x_1)$$

Through the origin @ (0, 0):

$$y = mx$$

$$\frac{dy}{dx} = m$$

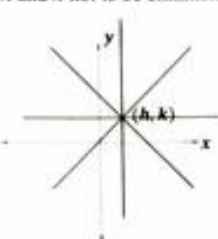
Thus,

$$y = \frac{dy}{dx}x$$

$$ydx = xdy$$

$$ydx - xdy = 0 \text{ ANS.}$$

Straight lines through the fixed point (h, k); h and k not to be eliminated.



General Equation of a Line:

$$y - y_1 = m(x - x_1)$$

Through the fixed point @ (h, k):

$$y - k = m(x - h)$$

$$\frac{dy}{dx} = m$$

Thus,

$$y - k = \frac{dy}{dx}(x - h)$$

$$(y - k)dx = (x - h)dy$$

$$(y - k)dx - (x - h)dy = 0 \text{ ANS.}$$

3. Straight lines with slope and y-intercept equal.

General Equation of a Line:

$$y - y_1 = m(x - x_1)$$

@ (0, b):

$$y - b = m(x - 0)$$

$$y = mx + b$$

$$\frac{dy}{dx} = m = b$$

Thus,

$$y = \frac{dy}{dx}x + \frac{dy}{dx}$$

$$ydx = (x + 1)dy$$

$$ydx - (x + 1)dy = 0 \text{ ANS.}$$

FAMILIES OF CURVES

4. Straight lines with slope and x-intercept equal.

General Equation of a Line:

$$y - y_1 = m(x - x_1)$$

$$@ (a, 0):$$

$$y - 0 = m(x - a)$$

$$y = m(x - a)$$

$$y' = m = a$$

Thus,

$$y = y'(x - y')$$

$$y = xy' - (y')^2$$

$$(y')^2 = xy' - y \text{ ANS.}$$

5. Straight lines with algebraic sum of the intercepts fixed as k.

For x-intercept:

$$y = m(x - a)$$

$$y' = m$$

$$y = y'(x - a)$$

$$a = x - \frac{y}{y'}$$

For y-intercept:

$$y = mx + b$$

$$y' = m$$

$$y = y'x + b$$

$$b = y - xy'$$

Thus,

$$k = a + b$$

$$k = x - \frac{y}{y'} + y - xy'$$

$$ky' = xy' - y + yy' - x(y')^2$$

$$(x(y')^2 - yy' - xy' + y) + ky' = 0$$

$$(xy' - y)(y' - 1) + ky' = 0 \text{ ANS.}$$

7.

Circles with center at the origin.

General Equation of a Circle:

$$(x - h)^2 + (y - k)^2 = r^2$$

Center at the origin @ (0, 0);

$$x^2 + y^2 = r^2$$

$$2x + 2yy' = 0$$

$$x + yy' = 0$$

$$x + y \frac{dy}{dx} = 0$$

$$xdx + ydy = 0 \text{ ANS.}$$

8.

Circles with center on the x-axis.

General Equation of a Circle:

$$(x - h)^2 + (y - k)^2 = r^2$$

With center on the x-axis @ (h, 0);

$$(x - h)^2 + (y - 0)^2 = r^2$$

$$(x - h)^2 + y^2 = r^2$$

First Derivative:

$$2(x - h) + 2yy' = 0$$

$$(x - h) + yy' = 0$$

Second Derivative:

$$1 + yy'' + y'y' = 0$$

$$yy'' + (y')^2 + 1 = 0 \text{ ANS.}$$

9.

Circles with fixed radius r and tangent to x-axis.

General Equation of a Circle:

$$(x - h)^2 + (y - k)^2 = r^2$$

@ (h, k); $k \pm r$

$$(x - h)^2 + (y - k)^2 = r^2$$

$$2(x - h) + 2y'(y - k) = 0$$

$$(x - h) = -y'(y - k)$$

FAMILIES OF CURVES

Squaring both sides:

$$(x - h)^2 = (y')^2(y - k)^2$$

But:

$$k \pm r$$

$$(x - h)^2 = r^2 - (y - k)^2$$

Thus,

$$r^2 - (y \pm r)^2 = (y')^2(y \pm r)^2$$

$$r^2 = (y')^2(y \pm r)^2 + (y \pm r)^2$$

$$r^2 = (y \pm r)^2(y')^2 + (y^2 \pm 2ry + r^2)$$

$$(y \pm r)^2(y')^2 + y^2 \pm 2ry = 0 \text{ ANS.}$$

14.

Parabolas with vertex on the x-axis, with axis parallel to the y-axis, and with distance from focus to vertex fixed as a.

General Equation of a Parabola with axis parallel to the y-axis:

$$(x - h)^2 = 4a(y - k)$$

Vertex on the x-axis @ (h, 0);

$$(x - h)^2 = 4a(y - 0)$$

$$(x - h)^2 = 4ay \rightarrow (1)$$

$$2(x - h) = 4ay'$$

$$(x - h) = 2ay'$$

Square both sides:

$$(x - h)^2 = (2ay')^2$$

Substitute in (1);

$$(2ay')^2 = 4ay$$

$$a^2(y')^2 = y \text{ ANS.}$$

15.

Parabolas with vertex on the y-axis, with axis parallel to the x-axis, and with distance from focus to vertex fixed as a.

General Equation of a Parabola with axis parallel to the x-axis:

$$(y - k)^2 = 4a(x - h)$$

Vertex on the y-axis @ (0, k);

$$(y - k)^2 = 4a(x - 0)$$

$$(y - k)^2 = 4ax \rightarrow (1)$$

$$2(y - k)y' = 4a$$

$$(y - k)y' = 2a$$

Square both sides:

$$(y - k)^2(y')^2 = 4a^2$$

$$(y - k)^2 = \frac{4a^2}{(y')^2}$$

Substitute in (1);

$$\frac{4a^2}{(y')^2} = 4ax$$

$$x(y')^2 = a \text{ ANS.}$$

16.

Parabolas with axis parallel to the y-axis, and with distance from focus to vertex fixed as a.

General Equation of a Parabola with axis parallel to the y-axis:

$$(x - h)^2 = 4a(y - k) \rightarrow (1)$$

First Derivative:

$$2(x - h) = 4ay'$$

$$(x - h) = 2ay'$$

$$x = 2ay' + h$$

Second Derivative:

$$1 = 2ay'' \text{ ANS.}$$

FAMILIES OF CURVES

17.

Parabolas with axis parallel to the x -axis, and with distance from focus to vertex fixed as a .

General Equation of a Parabola with axis parallel to the x -axis:

$$(y-k)^2 = 4a(x-h) \rightarrow (1)$$

$$2(y-k)y' = 4a$$

$$(y-k)y' = 2a$$

Square both sides:

$$(y-k)^2(y')^2 = 4a^2$$

$$(y-k)^2 = \frac{4a^2}{(y')^2}$$

Substitute in (1):

$$\frac{4a^2}{(y')^2} = 4a(x-h)$$

$$\frac{a}{(y')^2} = x-h$$

Differentiate,

$$\frac{(y')^2(0) - 2ay'y''}{(y')^4} = 1$$

$$-2ay'y'' = (y')^4$$

$$2ay'' + (y')^3 = 0 \text{ ANS.}$$

20.

Parabolas with vertex and focus on the x -axis

General Equation of a Parabola with axis parallel to the x -axis:

$$(y-k)^2 = 4a(x-h)$$

with vertex and focus on the x -axis @ $(h, 0)$:

$$(y-0)^2 = 4a(x-h)$$

First Derivative:

$$y^2 = 4a(x-h)$$

$$2yy' = 4a$$

Second Derivative to eliminate a :

$$2yy'' + 2y'y' = 0$$

$$yy'' + (y')^2 = 0 \text{ ANS.}$$

21.

Parabolas with axis parallel to the x -axis

General Equation of a Parabola with axis parallel to the x -axis:

$$(y-k)^2 = 4a(x-h)$$

First Derivative:

$$2(y-k)y' = 4a$$

$$(y-k)y' = 2a$$

Second Derivative:

$$(y-k)y'' + y'y' = 0$$

$$k = \frac{yy'' + (y')^2}{y''}$$

Third Derivative:

$$0 = \frac{y''[yy''' + y''y' + 2y'y''] - [yy'' + (y')^2]y'''}{(y'')^2}$$

$$0 = yy''y''' + y'(y'')^2 + 2y'y''^2 - yy''y''' - (y')^2y'''$$

$$0 = 3y'(y'')^2 - (y')^2y'''$$

$$y'y''' - 3(y'')^2 = 0 \text{ ANS.}$$

24.

The cubics $Cy^2 = x^2(x-a)$ with a held fixed.

$$C = \frac{x^2(x-a)}{y^2}$$

Differentiate,

$$0 = \frac{y^2[x^2 + (x-a)(2x)] - x^2(x-a)(2yy')}{y^4}$$

$$x^2y^2 + 2xy^2(x-a) - 2x^2y'(x-a) = 0$$

$$xy + 2y(x-a) - 2xy'(x-a) = 0$$

$$3xy - 2ay - 2xy'(x-a) = 0$$

$$2x(x-a)y' = y(3x-2a) \text{ ANS.}$$

FAMILIES OF CURVES

25.

The cubics of exercise #24 with C held fixed and a to be eliminated.

$$C = \frac{x^2(x-a)}{y^2}$$

$$(x-a) = \frac{Cy^2}{x^2}$$

Differentiate,

$$1 = \frac{Cx^2(2yy') - Cy^2(2x)}{x^4}$$

$$x^4 = 2Cx^2yy' - 2xCy^2$$

$$x^3 = 2Cxyy' - 2Cy^2$$

$$x^3 = 2Cy(xy' - y) \text{ ANS.}$$

26.

The quartics $C^2y^2 = x(x-a)^3$ with a held fixed.

$$C^2 = \frac{x(x-a)^3}{y^2}$$

Differentiate,

$$0 = \frac{y^2[3x(x-a)^2 + (x-a)^3] - 2yy'x(x-a)^3}{y^4}$$

$$0 = y^2(x-a)^2[3x + (x-a)] - 2yy'x(x-a)^3$$

Divide by $y(x-a)^2$:

$$0 = y(4x-a) - 2xy'(x-a)$$

$$2x(x-a)y' = y(4x-a) \text{ ANS.}$$

29.

The cissoids $y^2 = \frac{x^3}{a-x}$

$$a = \frac{x^3}{y^2} + x$$

Differentiate,

$$0 = \frac{y^2(3x^2) - x^3(2yy')}{y^4} + 1$$

$$3x^2y^2 - 2x^3yy' + y^4 = 0$$

$$3x^2y - 2x^3y' + y^3 = 0$$

$$2x^3y' = y^3 + 3x^2y$$

$$2x^3y' = y(y^2 + 3x^2) \text{ ANS.}$$

33.

The circles $r = 2a(\sin \theta - \cos \theta)$

Differentiate,

$$dr = 2a(\cos \theta d\theta + \sin \theta d\theta)$$

$$2a = \frac{dr}{(\cos \theta + \sin \theta)d\theta}$$

Substitute $2a$,

$$r = \frac{dr(\sin \theta - \cos \theta)}{(\cos \theta + \sin \theta)d\theta}$$

$$r(\cos \theta + \sin \theta)d\theta = dr(\sin \theta - \cos \theta)$$

$$r(\cos \theta + \sin \theta)d\theta = -dr(\cos \theta - \sin \theta)$$

$$(\cos \theta - \sin \theta)dr + r(\cos \theta + \sin \theta)d\theta = 0 \text{ ANS.}$$

34.

The cardioids $r = a(1 - \sin \theta)$

$$a = \frac{r}{1 - \sin \theta}$$

Differentiate,

$$0 = \frac{(1 - \sin \theta)dr - r(-\cos \theta d\theta)}{(1 - \sin \theta)^2}$$

$$(1 - \sin \theta)dr + r \cos \theta d\theta = 0 \text{ ANS.}$$

36.

The strophoids $r = a(\sec \theta + \tan \theta)$

Differentiate,

$$dr = a(\sec \theta \tan \theta d\theta + \sec^2 \theta d\theta)$$

$$dr = a(\sec \theta + \tan \theta d\theta) \sec \theta d\theta$$

But $r = a(\sec \theta + \tan \theta)$

$$dr = r \sec \theta d\theta \text{ ANS.}$$

CHAPTER 2: EQUATIONS OF ORDER ONE

2.1 TYPE 1: SEPARATION OF VARIABLES

In this chapter we study several elementary methods for solving first order differential equations. We begin by studying an equation of the form

$$M dx + N dy = 0$$

Where M and N may be functions of both x and y . Some equations of this type are so simple that they can be put in the form

$$F(x) dx + F(y) dy = 0 \rightarrow (1)$$

That is, the variables can be separated.

Integrating (1)

$$\int F(x) dx + \int F(y) dy = C$$

Note:

1. The solution containing an arbitrary constant is called the **GENERAL SOLUTION**.

2. The solution not containing arbitrary constant is called the **PARTICULAR SOLUTION**.

I. In exercises 1 through 24 obtain the general solution.

1. $(1-x)y' = y^2$

$$(1-x) \frac{dy}{dx} = y^2$$

$$\frac{dx}{1-x} - \frac{dy}{y^2} = 0$$

$$\int \frac{dx}{1-x} - \int \frac{dy}{y^2} = \int 0$$

$$-\ln(1-x) + \frac{1}{y} = \ln C$$

$$y \ln C + y \ln(1-x) = 1$$

$$y \ln[C(1-x)] = 1$$

2. $\sin x \sin y dx + \cos x \cos y dy = 0$

$$\frac{\sin x dx}{\cos x} + \frac{\cos y dy}{\sin y} = 0$$

$$\int \frac{\sin x dx}{\cos x} + \int \frac{\cos y dy}{\sin y} = \int 0$$

$$-\ln \cos x + \ln \sin y = \ln C$$

$$\ln \sin y = \ln C + \ln \cos x$$

$$\sin y = C \cos x$$

3. $xy^3 dx + e^{x^2} dy = 0$

$$\frac{xdx}{e^{x^2}} + \frac{dy}{y^3} = 0$$

$$xe^{-x^2} dx + \frac{dy}{y^3} = 0$$

$$-\frac{1}{2} \int -2xe^{-x^2} dx + \int y^{-3} dy = \int 0$$

$$-\frac{1}{2} e^{-x^2} + \frac{y^{-2}}{-2} = C$$

$$e^{-x^2} + y^{-2} = C$$

TYPE 1: SEPARATION OF VARIABLES

4. $2y dx = 3x dy$

$$\frac{2dx}{x} - \frac{3dy}{y} = 0$$

$$\int \frac{2dx}{x} - \int \frac{3dy}{y} = \int 0$$

$$2 \ln x - 3 \ln y = \ln C$$

$$\ln x^2 - \ln y^3 = \ln C$$

$$\ln \left(\frac{x^2}{y^3} \right) = \ln C$$

$$x^2 = Cy^3$$

5. $my dx = nxdy$

$$\frac{mdx}{x} - \frac{ndy}{y} = 0$$

$$\int \frac{mdx}{x} - \int \frac{ndy}{y} = \int 0$$

$$m \ln x - n \ln y = \ln C$$

$$\ln x^m - \ln y^n = \ln C$$

$$\ln \left(\frac{x^m}{y^n} \right) = \ln C$$

$$x^m = Cy^n$$

7. $dV/dP = -V/P$

$$\frac{dV}{V} + \frac{dP}{P} = 0$$

$$\int \frac{dV}{V} + \int \frac{dP}{P} = \int 0$$

$$\ln V + \ln P = \ln C$$

$$VP = C$$

8. $ye^{2x} dx = (4 + e^{2x}) dy$

$$\frac{e^{2x} dx}{(4 + e^{2x})} - \frac{dy}{y} = 0$$

$$\frac{1}{2} \int \frac{2e^{2x} dx}{(4 + e^{2x})} - \int \frac{dy}{y} = \int 0$$

$$\frac{1}{2} \ln(4 + e^{2x}) - \ln y = \ln C$$

$$\ln(4 + e^{2x}) - 2 \ln y = 2 \ln C$$

$$\ln(4 + e^{2x}) - \ln y^2 = \ln C^2$$

$$\ln \left(\frac{4 + e^{2x}}{y^2} \right) = \ln C^2$$

$$\frac{4 + e^{2x}}{y^2} = C^2$$

$$C^2 y^2 = 4 + e^{2x}$$

6. $y' = xy^2$

$$\frac{dy}{dx} - xy^2 = 0$$

$$xdx - \frac{dy}{y^2} = 0$$

$$\int x dx - \int y^{-2} dy = \int 0$$

$$\frac{x^2}{2} - \frac{y^{-1}}{-1} = C$$

$$\frac{x^2}{2} + \frac{1}{y} = C$$

$$x^2 y + 2 = yC$$

$$y(x^2 - C) + 2 = 0$$

9. $dr = b(\cos \theta dr + r \sin \theta d\theta)$

$$dr - b \cos \theta dr - br \sin \theta d\theta = 0$$

$$(1 - b \cos \theta) dr - br \sin \theta d\theta = 0$$

$$\frac{dr}{r} - \frac{b \sin \theta d\theta}{(1 - b \cos \theta)} = 0$$

$$\int \frac{dr}{r} - \int \frac{b \sin \theta d\theta}{(1 - b \cos \theta)} = \int 0$$

$$\ln r - \ln(1 - b \cos \theta) = \ln C$$

$$\ln \left(\frac{r}{1 - b \cos \theta} \right) = \ln C$$

$$\frac{r}{1 - b \cos \theta} = C$$

$$r = C(1 - b \cos \theta)$$

TYPE 1: SEPARATION OF VARIABLES

10. $xydx - (x+2)dy = 0$

$$\begin{aligned}\frac{xdx}{(x+2)} - \frac{dy}{y} &= 0 \\ \frac{(x+2-2)dx}{(x+2)} - \frac{dy}{y} &= 0 \\ \int \left(1 - \frac{2}{x+2}\right) dx - \int \frac{dy}{y} &= \int 0 \\ x - 2 \ln(x+2) - \ln y &= \ln C \\ x - \ln(x+2)^2 - \ln y &= \ln C \\ x &= \ln C + \ln y + \ln(x+2)^2 \\ x &= \ln[Cy(x+2)^2] \\ e^x &= Cy(x+2)^2\end{aligned}$$

11. $x^2dx + y(x-1)dy = 0$

$$\begin{aligned}\frac{x^2dx}{(x-1)} + ydy &= 0 \\ \frac{(x+1)(x-1)+1}{(x-1)} dx + ydy &= 0 \\ \int \left((x+1) + \frac{1}{x-1}\right) dx + \int ydy &= \int 0 \\ \frac{(x+1)^2}{2} + \ln(x-1) + \frac{y^2}{2} &= -\ln C \\ (x+1)^2 + y^2 + 2 \ln(x-1) + 2 \ln C &= 0 \\ (x+1)^2 + y^2 + 2 \ln[C(x-1)] &= 0\end{aligned}$$

12. $(xy+x)dx = (x^2y^2+x^2+y^2+1)dy$

$$\begin{aligned}x(y+1)dx &= [x^2(y^2+1) + (y^2+1)]dy \\ x(y+1)dx &= (x^2+1)(y^2+1)dy \\ \frac{xdx}{(x^2+1)} - \frac{(y^2+1)dy}{(y+1)} &= 0 \\ \frac{1}{2} \int \frac{2xdx}{(x^2+1)} - \int \frac{(y+1)(y-1)+2}{(y+1)} dy &= \int 0 \\ \frac{1}{2} \ln(x^2+1) - \int \left[y - 1 + \frac{2}{(y+1)}\right] dy &= \ln C \\ \frac{1}{2} \ln(x^2+1) - \frac{y^2}{2} + y - 2 \ln(y+1) &= \ln C \\ &= 2 \ln C\end{aligned}$$

$$\ln(x^2+1) = y^2 - 2y + 4 \ln(y+1) + 4 \ln C$$

$$\ln(x^2+1) = y^2 - 2y + 4 \ln[C(y+1)]$$

13. $x \cos^2 y dx + \tan y dy = 0$

$$\begin{aligned}x dx + \frac{\tan y dy}{\cos^2 y} &= 0 \\ x dx + \tan y \sec^2 y dy &= 0 \\ x dx + \tan y \sec^2 y dy &= 0\end{aligned}$$

$$\begin{aligned}\frac{x^2}{2} + \frac{\tan^2 y}{2} &= \frac{C}{2} \\ x^2 + \tan^2 y &= C\end{aligned}$$

14. $xy^3dx + (y+1)e^{-x}dy = 0$

$$\begin{aligned}\frac{xdx}{e^{-x}} + \frac{(y+1)dy}{y^3} &= 0 \\ xe^x dx + \left(\frac{y}{y^3} + \frac{1}{y^3}\right) dy &= 0 \\ \int xe^x dx + \int (y^{-2} + y^{-3}) dy &= \int 0\end{aligned}$$

Considering:

$$\int xe^x dx$$

Using Integration by Parts:

Let $u = x$ $dv = e^x dx$
 $du = dx$ $v = e^x$

thus,

$$\begin{aligned}\int xe^x dx &= xe^x - \int e^x dx \\ &= xe^x - e^x\end{aligned}$$

Then,

$$\begin{aligned}xe^x - e^x + \frac{y^{-1}}{-1} + \frac{y^{-2}}{-2} &= C \\ xe^x - e^x - \frac{1}{y} - \frac{1}{2y^2} &= C \\ e^x(x-1) - \frac{1}{y} - \frac{1}{2y^2} &= C\end{aligned}$$

TYPE 1: SEPARATION OF VARIABLES

15. $x^2yy' = e^y$

$$\begin{aligned}x^2y \frac{dy}{dx} &= e^y \\ \frac{dx}{x^2} - \frac{ydy}{e^y} &= 0 \\ x^{-2}dx - ye^{-y}dy &= 0 \\ \int x^{-2}dx - \int ye^{-y}dy &= \int 0\end{aligned}$$

Considering $\int ye^{-y}dy$

Using Integration by Parts:

Let $u = y$ $dv = e^{-y}dx$
 $du = dy$ $v = -e^{-y}$

thus,

$$\begin{aligned}\int ye^{-y}dy &= -ye^{-y} + \int e^{-y}dy \\ &= -ye^{-y} - e^{-y} \\ &= -e^{-y}(y+1)\end{aligned}$$

Then,

$$\begin{aligned}\frac{x^{-1}}{-1} + e^{-y}(y+1) &= C \\ e^{-y}(y+1) &= \frac{1}{x} + C\end{aligned}$$

Multiply by xe^y .

$$\begin{aligned}x(y+1) &= e^y + Cxe^y \\ x(y+1) &= e^y(1+Cx)\end{aligned}$$

16. $\tan^2 y dy = \sin^3 x dx$

$$\begin{aligned}\sin^3 x dx - \tan^2 y dy &= 0 \\ \sin x \sin^2 x dx - (\sec^2 y - 1)dy &= 0 \\ \sin x (1 - \cos^2 x) dx - (\sec^2 y - 1)dy &= 0 \\ \int \sin x dx - \int \sin x \cos^2 x dx &= 0 \\ - \int \sec^2 y dy + \int dy &= \int 0\end{aligned}$$

$$\begin{aligned}-\cos x + \frac{\cos^3 x}{3} - \tan y + y &= C \\ -3\cos x + \cos^3 x - 3\tan y + 3y &= 3C \\ \cos^3 x - 3\cos x &= 3(\tan y - y + C)\end{aligned}$$

17. $y' = \cos^2 x \cos y$

$$\begin{aligned}\frac{dy}{dx} &= \cos^2 x \cos y \\ \cos^2 x dx - \frac{dy}{\cos y} &= 0 \\ \cos^2 x dx - \sec y dy &= 0 \\ \int \cos^2 x dx - \int \sec y dy &= \int 0 \\ \int \frac{1 + \cos 2x}{2} dx - \ln(\sec y + \tan y) &= -C \\ \frac{1}{2}x + \frac{\sin 2x}{4} - \ln(\sec y + \tan y) &= -C \\ 2x + \sin 2x - 4 \ln(\sec y + \tan y) &= -C \\ 4 \ln(\sec y + \tan y) &= 2x + \sin 2x + C\end{aligned}$$

18. $y' = y \sec x$

$$\begin{aligned}\frac{dy}{dx} &= y \sec x \\ \frac{dy}{y} &= \sec x dx \\ \sec x dx - \frac{dy}{y} &= 0 \\ \int \sec x dx - \int \frac{dy}{y} &= \int 0 \\ \ln(\sec x + \tan x) - \ln y &= \ln C \\ \ln \frac{(\sec x + \tan x)}{y} &= \ln C \\ (\sec x + \tan x) &= yC\end{aligned}$$

TYPE 1: SEPARATION OF VARIABLES

$$19. \quad dx = t(1+t^2) \sec^2 x dt$$

$$\frac{dx}{\sec^2 x} - t(1+t^2)dt = 0$$

$$\cos^2 x dx - t(1+t^2)dt = 0$$

$$\int \frac{1+\cos 2x}{2} dx - \int t(1+t^2)dt = \int 0$$

$$\frac{1}{2}x + \frac{\sin 2x}{4} - \frac{1}{2} \int 2t(1+t^2)dt = C$$

$$\frac{1}{2}x + \frac{\sin 2x}{4} - \frac{(1+t^2)^2}{4} = C$$

$$2x + \sin 2x = (1+t^2)^2 + C$$

$$20. \quad (e^{2x} + 4)y' = y$$

$$(e^{2x} + 4) \frac{dy}{dx} = y$$

$$\frac{dx}{(e^{2x} + 4)} - \frac{dy}{y} = 0$$

$$\frac{dx}{(e^{2x} + 4)} \left(\frac{e^{-2x}}{e^{-2x}} \right) - \frac{dy}{y} = 0$$

$$\frac{e^{-2x} dx}{1 + 4e^{-2x}} - \frac{dy}{y} = 0$$

$$-\frac{1}{8} \int \frac{-8e^{-2x} dx}{1 + 4e^{-2x}} - \int \frac{dy}{y} = \int 0$$

$$-\frac{1}{8} \ln(1 + 4e^{-2x}) - \ln y = \frac{1}{4} \ln C$$

$$-\ln(1 + 4e^{-2x}) - 8 \ln y = 2 \ln C$$

$$-\ln(1 + 4e^{-2x}) - \ln y^8 = \ln C^2$$

$$\ln((1 + 4e^{-2x})y^8) = -\ln C^2$$

$$(1 + 4e^{-2x})y^8 = C^2$$

$$21. \quad ad\beta + \beta da + a\beta(3da + d\beta) = 0$$

Divide by $a\beta$,

$$\frac{d\beta}{\beta} + \frac{da}{a} + 3da + d\beta = 0$$

$$\int \frac{d\beta}{\beta} + \int \frac{da}{a} + \int 3da + \int d\beta = \int 0$$

$$\ln \beta + \ln a + 3a + \beta = -\ln C$$

$$\ln \beta + \ln a + \ln C = -3a - \beta$$

$$\ln a\beta C = -3a - \beta$$

$$a\beta C = e^{-3a-\beta}$$

$$22. \quad (1 + \ln x)dx + (1 + \ln y)dy = 0$$

$$dx + \ln x dx + dy + \ln y dy = 0$$

$$\int dx + \int \ln x dx + \int dy + \int \ln y dy = \int 0$$

$$x + (x \ln x - x) + y + (y \ln y - y) = C$$

$$x \ln x + y \ln y = C$$

$$23. \quad xdx - \sqrt{a^2 - x^2} dy = 0$$

$$\frac{xdx}{\sqrt{a^2 - x^2}} - dy = 0$$

$$x(a^2 - x^2)^{-1/2} - dy = 0$$

$$-\frac{1}{2} \int -2x(a^2 - x^2)^{-1/2} - \int dy = \int 0$$

$$-\frac{1}{2} \frac{(a^2 - x^2)^{1/2}}{1/2} - y = -C$$

$$y - C = -\sqrt{a^2 - x^2}$$

The lower half of the circle
 $x^2 + (y - C)^2 = a^2$

$$24. \quad xdx + \sqrt{a^2 - x^2} dy = 0$$

$$\frac{xdx}{\sqrt{a^2 - x^2}} + dy = 0$$

$$x(a^2 - x^2)^{-1/2} - dy = 0$$

$$-\frac{1}{2} \int -2x(a^2 - x^2)^{-1/2} + \int dy = \int 0$$

$$-\frac{1}{2} \frac{(a^2 - x^2)^{1/2}}{1/2} + y = C$$

$$y - C = \sqrt{a^2 - x^2}$$

The upper half of the circle
 $x^2 + (y - C)^2 = a^2$

TYPE 1: SEPARATION OF VARIABLES

$$25. \quad a^2 dx = x\sqrt{x^2 - a^2} dy$$

$$\frac{a^2 dx}{x\sqrt{x^2 - a^2}} - dy = 0$$

$$\int \frac{a^2 dx}{x\sqrt{x^2 - a^2}} - \int dy = \int 0$$

From the table of Integration:

$$\int \frac{dx}{x\sqrt{x^2 - a^2}} = \frac{1}{a} \text{Arc cos } \frac{a}{x} + C$$

then,

$$a^2 \left(\frac{1}{a} \text{Arc cos } \frac{a}{x} \right) - y = C$$

$$a \text{Arc cos } \frac{a}{x} = y + C$$

$$\text{Arc cos } \frac{a}{x} = \frac{y + C}{a}$$

$$\text{From Arc cos } \theta = B$$

$$\cos B = \theta$$

Thus,

$$\frac{a}{x} = \cos \left(\frac{y + C}{a} \right)$$

$$x = a \sec \left(\frac{y + C}{a} \right)$$

$$26. \quad y \ln x \ln y dx + dy = 0$$

$$\ln x dx + \frac{dy}{y \ln y} = 0$$

$$\int \ln x dx + \int \frac{dy}{y \ln y} = \int 0$$

Considering

$$\int \frac{dy}{y \ln y}$$

$$\text{Let } u = \ln y \quad du = \frac{dy}{y}$$

$$\int \frac{dy}{y \ln y} = \ln(\ln y)$$

Then,

$$x \ln x - x + \ln(\ln y) = C$$

$$x \ln x + \ln(\ln y) = C + x$$

II. In exercises 27 through 32 obtain the particular solution satisfying the initial condition indicated

$$27. \quad dr/dt = -4rt; \text{ when } t = 0, r = r_0$$

$$\frac{dr}{r} + 4t dt = 0$$

$$\int \frac{dr}{r} + \int 4t dt = \int 0$$

$$\ln r + \frac{4t^2}{2} = \ln C$$

$$\ln r - \ln C = -2t^2$$

$$\ln \left(\frac{r}{C} \right) = -2t^2$$

$$\frac{r}{C} = e^{-2t^2}$$

$$C = \frac{r}{e^{-2t^2}} \rightarrow \text{General Solution}$$

$$\text{when } t = 0, r = r_0;$$

$$C = \frac{r_0}{e^{-2(0)^2}} = r_0$$

Then,

$$r_0 = \frac{r}{e^{-2t^2}}$$

$$r = r_0 e^{-2t^2} \rightarrow \text{Particular Solution}$$

$$28. \quad 2xyy' = 1 + y^2; \text{ when } x = 2, y = 3$$

$$2xy \frac{dy}{dx} = 1 + y^2$$

$$-\frac{dx}{x} + \frac{2y dy}{1 + y^2} = 0$$

$$-\int \frac{dx}{x} + \int \frac{2y dy}{1 + y^2} = \int 0$$

TYPE 1: SEPARATION OF VARIABLES

$$-\ln x + \ln(1+y^2) = \ln C$$

$$\ln\left(\frac{1+y^2}{x}\right) = \ln C$$

$$\frac{1+y^2}{x} = C \rightarrow \text{General Solution}$$

$$\text{when } x = 2, y = 3;$$

$$C = \frac{1+3^2}{2} = 5$$

Then,

$$\frac{1+y^2}{x} = 5$$

$$1+y^2 = 5x$$

$$y = \sqrt{5x-1} \rightarrow \text{Particular Solution}$$

$$29. \quad xyy' = 1+y^2; \text{ when } x = 2, y = 3$$

$$xy \frac{dy}{dx} = 1+y^2$$

$$\frac{dx}{x} - \frac{ydy}{1+y^2} = 0$$

$$\int \frac{dx}{x} - \int \frac{ydy}{1+y^2} = \int 0$$

$$\ln x - \frac{1}{2} \ln(1+y^2) = \ln C$$

$$2 \ln x - \ln(1+y^2) = 2 \ln C$$

$$\ln x^2 - \ln(1+y^2) = \ln C^2$$

$$\ln \frac{x^2}{(1+y^2)} = \ln C^2$$

$$\frac{x^2}{(1+y^2)} = C^2 \rightarrow \text{General Solution}$$

$$\text{when } x = 2, y = 3;$$

$$\frac{2^2}{(1+3^2)} = C^2$$

$$C = \frac{\sqrt{10}}{5}$$

Then,

$$\frac{x^2}{(1+y^2)} = \left(\frac{\sqrt{10}}{5}\right)^2$$

$$\frac{x^2}{(1+y^2)} = \frac{2}{5}$$

$$2(1+y^2) = 5x^2$$

$$y = \sqrt{\frac{5x^2}{2} - 1}$$

$$y = \frac{1}{2} \sqrt{10x^2 - 4} \rightarrow \text{Particular Solution}$$

$$30. \quad 2ydx = 3xdy; \text{ when } x = 2, y = 1$$

$$\frac{2dx}{x} - \frac{3dy}{y} = 0$$

$$\int \frac{2dx}{x} - \int \frac{3dy}{y} = \int 0$$

$$2 \ln x - 3 \ln y = \ln C$$

$$\ln x^2 - \ln y^3 = \ln C$$

$$\frac{x^2}{y^3} = C \rightarrow \text{General Solution}$$

$$\text{when } x = 2, y = 1;$$

$$C = \frac{2^2}{1^3} = 4$$

Then,

$$\frac{x^2}{y^3} = 4$$

$$y = (x/2)^{2/3} \rightarrow \text{Particular Solution}$$

$$31. \quad 2ydx = 3xdy; \text{ when } x = -2, y = 1$$

$$\frac{x^2}{y^3} = C \rightarrow \text{General Solution}$$

$$\text{when } x = -2, y = 1$$

$$C = \frac{(-2)^2}{1^3} = 4$$

TYPE 1: SEPARATION OF VARIABLES

Then,

$$\frac{x^2}{y^3} = 4$$

$$y = (x/2)^{2/3} \rightarrow \text{Particular Solution}$$

$$32. \quad 2ydx = 3xdy; \text{ when } x = 2, y = -1$$

$$\frac{x^2}{y^3} = C \rightarrow \text{General Solution}$$

$$\text{when } x = -2, y = 1;$$

$$C = \frac{2^2}{(-1)^3} = -4$$

Then,

$$\frac{x^2}{y^3} = -4$$

$$y = -(x/2)^{2/3} \rightarrow \text{Particular Solution}$$

III. In exercises 33 through 36 obtain the particular solution satisfying the initial condition indicated

$$33. \quad y' = xe^{(y-x^2)}; \text{ when } x = 0, y = 0$$

$$\frac{dy}{dx} = xe^{(y-x^2)}$$

$$\frac{dy}{dx} = x \left(\frac{e^y}{e^{x^2}} \right)$$

$$\frac{xdx}{e^{x^2}} - \frac{dy}{e^y} = 0$$

$$xe^{-x^2} dx - e^{-y} dy = 0$$

$$-\frac{1}{2} \int -2xe^{-x^2} dx + \int e^{-y} dy = \int 0$$

$$-\frac{1}{2} e^{-x^2} + e^{-y} = C \rightarrow \text{General Solution}$$

$$\text{when } x = 0, y = 0;$$

$$C = -\frac{1}{2} e^{-0^2} + e^{-0} = \frac{1}{2}$$

Then,

$$-\frac{1}{2} e^{-x^2} + e^{-y} = \frac{1}{2}$$

$$-e^{-x^2} + 2e^{-y} = 1$$

$$e^{-y} = \left(\frac{1+e^{-x^2}}{2} \right)$$

$$\ln e^{-y} = \ln \left(\frac{1+e^{-x^2}}{2} \right)$$

$$-y \ln e = \ln \left(\frac{1+e^{-x^2}}{2} \right)$$

$$y = -\ln \left(\frac{1+e^{-x^2}}{2} \right)$$

$$y = \ln \left(\frac{1+e^{-x^2}}{2} \right)^{-1}$$

$$y = \ln \left(\frac{2}{1+e^{-x^2}} \right) \rightarrow \text{Particular Solution}$$

$$34. \quad xy^2 dx + e^x dy = 0; \text{ when } x \rightarrow \infty, y \rightarrow 1/2$$

$$\frac{xdx}{e^x} + \frac{ydy}{y^2} = 0$$

$$xe^{-x} dx + y^{-2} dy = 0$$

$$\int xe^{-x} dx + \int y^{-2} dy = \int 0$$

Considering

$$\int xe^{-x} dx$$

Using Integration by Parts:

$$\text{Let } u = x \quad dv = e^{-x} dx$$

$$du = dx \quad v = -e^{-x}$$

thus,

$$\int xe^{-x} dx = -xe^{-x} + \int e^{-x} dx$$

$$= -xe^{-x} - e^{-x}$$

Then,

$$-xe^{-x} - e^{-x} + \frac{y^{-1}}{-1} = C$$

$$-xe^{-x} - e^{-x} - \frac{1}{y} = C \rightarrow \text{General Solution}$$

when $x \rightarrow \infty, y \rightarrow 1/2$;

$$C = 0 - 0 - \frac{1}{1/2} = -2$$

Then,

$$-xe^{-x} - e^{-x} - \frac{1}{y} = -2$$

Multiply by ye^x

$$-xy - y - e^x = -2ye^x$$

$$-xy - y + 2ye^x = e^x$$

$$y(2e^x - x - 1) = e^x$$

$$y = \frac{e^x}{(2e^x - x - 1)} \rightarrow \text{Particular Solution}$$

36. $v(dv/dx) = g$; when $x = x_0, v = v_0$

$$gdx - vdv = 0$$

$$\int gdx - \int vdv = \int 0$$

$$gx - \frac{v^2}{2} = C$$

$$2gx - v^2 = 2C$$

when $x = x_0, v = v_0$

$$2C = 2gx_0 - v_0^2$$

Then,

$$2gx - v^2 = 2gx_0 - v_0^2$$

$$v^2 - v_0^2 = 2gx - 2gx_0$$

$$v^2 - v_0^2 = 2g(x - x_0)$$

2.2 TYPE 2: HOMOGENEOUS FUNCTIONS

A formal definition of homogeneity is:

The function $f(x, y)$ is said to be homogeneous of degree k in x and y if, and only if,

$$f(\lambda x, \lambda y) = \lambda^k f(x, y)$$

EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

The equation $M(x, y)dx + N(x, y)dy = 0$ is a homogeneous differential equation if M and N are homogeneous function of the same degree. To solve this equation, use substitution;

$$y = vx \quad \text{or} \quad x = vy$$

If N is simpler, use $y = vx$

If M is simpler, use $x = vy$

L Test the homogeneity of the given functions and state its degree.

1. $4x^2 - 3xy + y^2$

$$f(x, y) = 4x^2 - 3xy + y^2$$

$$f(\lambda x, \lambda y) = 4(\lambda x)^2 - 3(\lambda x)(\lambda y) + (\lambda y)^2$$

$$f(\lambda x, \lambda y) = 4\lambda^2 x^2 - 3\lambda^2 xy + \lambda^2 y^2$$

$$f(\lambda x, \lambda y) = \lambda^2(4x^2 - 3xy + y^2)$$

$\therefore$ Homogeneous, 2nd degree!

2. $x^3 - xy + y^3$

$$f(x, y) = x^3 - xy + y^3$$

$$f(\lambda x, \lambda y) = (\lambda x)^3 - (\lambda x)(\lambda y) + (\lambda y)^3$$

$$f(\lambda x, \lambda y) = \lambda^3 x^3 - \lambda^2 xy + \lambda^3 y^3$$

$$f(\lambda x, \lambda y) = \lambda^2(\lambda x^3 - xy + \lambda y^3)$$

$\therefore$ Not Homogeneous!

3. $2y + \sqrt{x^2 + y^2}$

$$f(x, y) = 2y + \sqrt{x^2 + y^2}$$

$$f(\lambda x, \lambda y) = 2\lambda y + \sqrt{(\lambda x)^2 + (\lambda y)^2}$$

$$f(\lambda x, \lambda y) = 2\lambda y + \sqrt{\lambda^2(x^2 + y^2)}$$

$$f(\lambda x, \lambda y) = 2\lambda y + \lambda\sqrt{x^2 + y^2}$$

$$f(\lambda x, \lambda y) = \lambda(2y + \sqrt{x^2 + y^2})$$

$\therefore$ Homogeneous, 1st degree!

4. $\sqrt{x - y}$

$$f(x, y) = \sqrt{x - y}$$

$$f(\lambda x, \lambda y) = \sqrt{\lambda x - \lambda y}$$

$$f(\lambda x, \lambda y) = \sqrt{\lambda(x - y)}$$

$$f(\lambda x, \lambda y) = (\lambda)^{1/2} \sqrt{(x - y)}$$

$\therefore$ Homogeneous, 1/2 degree!

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

5. e^x
 $f(x, y) = e^x$
 $f(\lambda x, \lambda y) = e^{\lambda x}$
 $\therefore$ Not Homogeneous!
6. $\tan x$
 $f(x, y) = \tan x$
 $f(\lambda x, \lambda y) = \tan \lambda x$
 $\therefore$ Not Homogeneous!
7. $e^{x/y}$
 $f(x, y) = e^{x/y}$
 $f(\lambda x, \lambda y) = e^{\lambda x/\lambda y}$
 $f(\lambda x, \lambda y) = e^{x/y}$
 $\therefore$ Homogeneous, 0 degree!
8. $\tan \frac{3y}{x}$
 $f(x, y) = \tan \frac{3y}{x}$
 $f(\lambda x, \lambda y) = \tan \frac{3\lambda y}{\lambda x}$
 $f(\lambda x, \lambda y) = \tan \frac{3y}{x}$
 $\therefore$ Homogeneous, 0 degree!
9. $(x^2 + y^2)e^{2x/y} + 4xy$
 $f(x, y) = (x^2 + y^2)e^{2x/y} + 4xy$
 $f(\lambda x, \lambda y) = ((\lambda x)^2 + (\lambda y)^2)e^{2\lambda x/\lambda y} + 4(\lambda x)(\lambda y)$
 $f(\lambda x, \lambda y) = (\lambda^2 x^2 + \lambda^2 y^2)e^{2x/y} + 4\lambda^2 xy$
 $f(\lambda x, \lambda y) = \lambda^2 (x^2 + y^2)e^{2x/y} + 4\lambda^2 xy$
 $\therefore$ Homogeneous, 2nd degree!

10. $x \sin \frac{x}{y} - y \sin \frac{y}{x}$
 $f(x, y) = x \sin \frac{x}{y} - y \sin \frac{y}{x}$
 $f(\lambda x, \lambda y) = \lambda x \sin \frac{\lambda x}{\lambda y} - \lambda y \sin \frac{\lambda y}{\lambda x}$
 $f(\lambda x, \lambda y) = \lambda^1 \left(x \sin \frac{x}{y} - y \sin \frac{y}{x} \right)$
 $\therefore$ Homogeneous, 1st degree!
11. $\frac{x^2 + 3xy}{x - 2y}$
 $f(x, y) = \frac{x^2 + 3xy}{x - 2y}$
 $f(\lambda x, \lambda y) = \frac{(\lambda x)^2 + 3(\lambda x)(\lambda y)}{\lambda x - 2\lambda y}$
 $f(\lambda x, \lambda y) = \frac{\lambda^2 x^2 + 3\lambda^2 xy}{\lambda x - 2\lambda y}$
 $f(\lambda x, \lambda y) = \frac{\lambda^2 (x^2 + 3xy)}{\lambda (x - 2y)}$
 $f(\lambda x, \lambda y) = \lambda^1 \left(\frac{x^2 + 3xy}{x - 2y} \right)$
 $\therefore$ Homogeneous, 1st degree!
12. $\frac{x^5}{x^2 + 2y^2}$
 $f(x, y) = \frac{x^5}{x^2 + 2y^2}$
 $f(\lambda x, \lambda y) = \frac{(\lambda x)^5}{(\lambda x)^2 + 2(\lambda y)^2}$
 $f(\lambda x, \lambda y) = \frac{\lambda^5 x^5}{\lambda^2 x^2 + 2\lambda^2 y^2}$
 $f(\lambda x, \lambda y) = \frac{\lambda^5 x^5}{\lambda^2 (x^2 + 2y^2)}$
 $f(\lambda x, \lambda y) = \lambda^3 \left(\frac{x^5}{x^2 + 2y^2} \right)$
 $\therefore$ Homogeneous, 3rd degree!

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

13. $(u^2 + v^2)^{3/2}$
 $f(u, v) = (u^2 + v^2)^{3/2}$
 $f(\lambda u, \lambda v) = (\lambda^2 u^2 + \lambda^2 v^2)^{3/2}$
 $f(\lambda u, \lambda v) = (\lambda^2)^{3/2} (u^2 + v^2)^{3/2}$
 $f(\lambda u, \lambda v) = \lambda^3 (u^2 + v^2)^{3/2}$
 $\therefore$ Homogeneous, 3rd degree!
14. $(u^2 - 4v^2)^{-1/2}$
 $f(u, v) = (u^2 - 4v^2)^{-1/2}$
 $f(\lambda u, \lambda v) = ((\lambda u)^2 - 4(\lambda v)^2)^{-1/2}$
 $f(\lambda u, \lambda v) = (\lambda^2)^{-1/2} (u^2 - 4v^2)^{-1/2}$
 $f(\lambda u, \lambda v) = \lambda^{-1} (u^2 - 4v^2)^{-1/2}$
 $\therefore$ Homogeneous, -1 degree!
15. $y^2 \tan \frac{x}{y}$
 $f(x, y) = y^2 \tan \frac{x}{y}$
 $f(\lambda x, \lambda y) = (\lambda y)^2 \tan \frac{\lambda x}{\lambda y}$
 $f(\lambda x, \lambda y) = \lambda^2 y^2 \tan \frac{x}{y}$
 $f(\lambda x, \lambda y) = \lambda^2 \left(y^2 \tan \frac{x}{y} \right)$
 $\therefore$ Homogeneous, 2nd degree!
16. $\frac{(x^2 + y^2)^{1/2}}{(x^2 - y^2)^{1/2}}$
 $f(x, y) = \frac{(x^2 + y^2)^{1/2}}{(x^2 - y^2)^{1/2}}$
 $f(\lambda x, \lambda y) = \frac{((\lambda x)^2 + (\lambda y)^2)^{1/2}}{((\lambda x)^2 - (\lambda y)^2)^{1/2}}$
 $f(\lambda x, \lambda y) = \frac{(\lambda^2 x^2 + \lambda^2 y^2)^{1/2}}{(\lambda^2 x^2 - \lambda^2 y^2)^{1/2}}$
 $f(\lambda x, \lambda y) = \frac{(\lambda^2)^{1/2} (x^2 + y^2)^{1/2}}{(\lambda^2)^{1/2} (x^2 - y^2)^{1/2}}$
 $f(\lambda x, \lambda y) = \lambda^0 \left(\frac{(x^2 + y^2)^{1/2}}{(x^2 - y^2)^{1/2}} \right)$
 $\therefore$ Homogeneous, 0 degree!
17. $\frac{a+4b}{a-4b}$
 $f(a, b) = \frac{a+4b}{a-4b}$
 $f(\lambda a, \lambda b) = \frac{\lambda a + 4\lambda b}{\lambda a - 4\lambda b}$
 $f(\lambda a, \lambda b) = \frac{\lambda(a+4b)}{\lambda(a-4b)}$
 $f(\lambda a, \lambda b) = \frac{a+4b}{a-4b}$
 $\therefore$ Homogeneous, 0 degree!
18. $\ln \frac{x}{y}$
 $f(x, y) = \ln \frac{x}{y}$
 $f(\lambda x, \lambda y) = \ln \frac{\lambda x}{\lambda y}$
 $f(\lambda x, \lambda y) = \ln \frac{x}{y}$
 $\therefore$ Homogeneous, 0 degree!
19. $x \ln x - y \ln y$
 $f(x, y) = x \ln x - y \ln y$
 $f(\lambda x, \lambda y) = \lambda x \ln \lambda x - \lambda y \ln \lambda y$
 $f(\lambda x, \lambda y) = \lambda (x \ln \lambda x - y \ln \lambda y)$
 $\therefore$ Not Homogeneous!
20. $x \ln x - x \ln y$
 $f(x, y) = x \ln x - x \ln y$
 $f(\lambda x, \lambda y) = \lambda x \ln \lambda x - \lambda x \ln \lambda y$
 $f(\lambda x, \lambda y) = \lambda x (\ln \lambda x - \ln \lambda y)$
 $f(\lambda x, \lambda y) = \lambda x \left(\ln \frac{\lambda x}{\lambda y} \right)$
 $f(\lambda x, \lambda y) = \lambda x \left(\ln \frac{x}{y} \right)$
 $f(\lambda x, \lambda y) = \lambda x (\ln x - \ln y)$
 $f(\lambda x, \lambda y) = \lambda^1 (x \ln x - x \ln y)$
 $\therefore$ Homogeneous, 1st degree!

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

II. In exercises 1 through 21, obtain a family of solutions

1. $(x - 2y)dx + (2x + y)dy = 0$

∴ Homogeneous, 1st degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$(x - 2vx)dx + (2x + vx)(vdx + xdv) = 0$$

$$x(1 - 2v)dx + (2vxdx + 2x^2dv + v^2xdx + vx^2dv) = 0$$

$$(1 - 2v)dx + (2vdx + 2x^2dv + v^2dx + vx^2dv) = 0$$

$$(1 - 2v + 2v + v^2)dx + (2x + vx)dv = 0$$

$$(1 + v^2)dx + x(2 + v)dv = 0$$

$$\frac{dx}{x} + \frac{(2 + v)dv}{(1 + v^2)} = 0$$

$$\int \frac{dx}{x} + 2 \int \frac{dv}{(1 + v^2)} + \int \frac{v dv}{(1 + v^2)} = \int 0$$

$$\ln x + 2 \operatorname{Arctan} v + \frac{1}{2} \ln(1 + v^2) = C$$

$$2 \ln x + 4 \operatorname{Arctan} v + \ln(1 + v^2) = C$$

$$\ln[x^2(1 + v^2)] + 4 \operatorname{Arctan} v = C$$

But: $v = y/x$

$$\ln \left[x^2 \left(1 + \left(\frac{y}{x} \right)^2 \right) \right] + 4 \operatorname{Arctan} \frac{y}{x} = C$$

$$\ln \left[x^2 \left(\frac{x^2 + y^2}{x^2} \right) \right] + 4 \operatorname{Arctan} \frac{y}{x} = C$$

$$\ln(x^2 + y^2) + 4 \operatorname{Arctan} \frac{y}{x} = C$$

2. $2(2x^2 + y^2)dx - xydy = 0$

∴ Homogeneous, 2nd degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$2(2x^2 + (vx)^2)dx - xvx(vdx + xdv) = 0$$

$$(4x^2 + 2v^2x^2)dx - vx^2(vdx + xdv) = 0$$

$$(4 + 2v^2)dx - v^2dx - vx^2dv = 0$$

$$(4 + 2v^2 - v^2)dx - vx^2dv = 0$$

$$\frac{dx}{x} - \frac{v dv}{(4 + v^2)} = 0$$

$$\int \frac{dx}{x} - \frac{1}{2} \int \frac{2v dv}{(4 + v^2)} = \int 0$$

$$\ln x - \frac{1}{2} \ln(4 + v^2) = \ln C$$

$$2 \ln x - \ln(4 + v^2) = 2 \ln C$$

$$\ln x^2 - \ln(4 + v^2) = \ln C^2$$

$$\frac{x^2}{(4 + v^2)} = C^2$$

But: $v = y/x$

$$\frac{x^2}{(4 + (\frac{y}{x})^2)} = C^2$$

$$\frac{x^2}{(4x^2 + y^2)} = C^2$$

$$\frac{x^4}{(4x^2 + y^2)} = C^2$$

$$x^4 = C^2(4x^2 + y^2)$$

3. $xydx - (x^2 + 3y^2)dy = 0$

∴ Homogeneous, 2nd degree!

Let: $x = vy$
 $dx = vdy + ydv$

Substitute:

$$vy^2(vdy + ydv) - (v^2y^2 + 3y^2)dy = 0$$

$$v^2dy + vydv - v^2dy - 3dy = 0$$

$$(v^2 - v^2 - 3)dy + vydv = 0$$

$$\frac{-3dy}{y} + vdv = 0$$

$$\int \frac{-3dy}{y} + \int vdv = \int 0$$

$$-3 \ln y + \frac{v^2}{2} = -3 \ln C$$

$$-6 \ln y + v^2 = -6 \ln C$$

$$v^2 = -6 \ln C + 6 \ln y$$

$$v^2 = 6 \ln(y/C)$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

But: $v = x/y$

$$\left(\frac{x}{y} \right)^2 = 6 \ln(y/C)$$

$$x^2 = 6y^2 \ln(y/C)$$

4. $x^2y' = 4x^2 + 7xy + 2y^2$
 $x^2dy - (4x^2 + 7xy + 2y^2)dx = 0$

∴ Homogeneous, 2nd degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$x^2(vdx + xdv) - (4x^2 + 7vx^2 + 2v^2x^2)dx = 0$$

$$(vdx + xdv) - (4 + 7v + 2v^2)dx = 0$$

$$(v - 4 - 7v - 2v^2)dx + xdv = 0$$

$$(-4 - 6v - 2v^2)dx + xdv = 0$$

$$-2(2 + 3v + v^2)dx + xdv = 0$$

$$\frac{-2dx}{x} + \frac{dv}{(2 + 3v + v^2)} = 0$$

$$\int \frac{-2dx}{x} + \int \frac{dv}{(v + 1)(v + 2)} = \int 0$$

Considering:

$$\int \frac{dv}{(v + 1)(v + 2)}$$

Using Integration by Partial Fraction:

$$\frac{1}{(v + 1)(v + 2)} = \frac{A}{(v + 1)} + \frac{B}{(v + 2)}$$

$$1 = A(v + 2) + B(v + 1)$$

If $v = -1$; $A = 1$
 $v = -2$; $B = -1$

Then,

$$\int \frac{dv}{(v + 1)(v + 2)} = \int \frac{dv}{v + 1} - \int \frac{dv}{v + 2}$$

$$= \ln(v + 1) - \ln(v + 2)$$

$$= \ln \frac{v + 1}{v + 2}$$

Thus,

$$-2 \ln x + \ln \frac{v + 1}{v + 2} = -\ln C$$

$$\frac{v + 1}{v + 2} = \frac{x^2}{C}$$

But: $v = y/x$

$$\frac{\frac{y}{x} + 1}{\frac{y}{x} + 2} = \frac{x^2}{C}$$

$$\frac{y + x}{y + 2x} = \frac{x^2}{C}$$

$$x^2(y + 2x) = C(y + x)$$

5. $3xydx + (x^2 + y^2)dy = 0$

∴ Homogeneous, 2nd degree!

Let: $x = vy$
 $dx = vdy + ydv$

Substitute:

$$3vy^2(vdy + ydv) + (v^2y^2 + y^2)dy = 0$$

$$3v(vdy + ydv) + (v^2 + 1)dy = 0$$

$$(3v^2 + v^2 + 1)dy + 3vydv = 0$$

$$\frac{dy}{y} + \frac{3vdv}{4v^2 + 1} = 0$$

$$\int \frac{dy}{y} + \frac{3}{8} \int \frac{8vdv}{4v^2 + 1} = \int 0$$

$$\ln y + \frac{3}{8} \ln(4v^2 + 1) = \ln C$$

$$8 \ln y + 3 \ln(4v^2 + 1) = \ln C$$

$$\ln y^8 + \ln(4v^2 + 1)^3 = \ln C$$

$$y^8(4v^2 + 1)^3 = C$$

But: $v = x/y$

$$y^8 \left(4 \left(\frac{x}{y} \right)^2 + 1 \right)^3 = C$$

$$y^2(4x^2 + y^2)^3 = C$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

6. $(x-y)(4x+y)dx + x(5x-y)dy = 0$

∴ Homogeneous, 1st degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$\begin{aligned} (x-vx)(4x+vx)dx + x(5x-vx)(vdx+xdv) &= 0 \\ x^2(1-v)(4+v)dx + x^2(5-v)(vdx+xdv) &= 0 \\ (4+v-4v-v^2)dx + (5vdv+5xdv-v^2dx-vxdv) &= 0 \\ (4-3v-v^2+5v-v^2)dx + (5x-vx)dv &= 0 \\ (4+2v-2v^2)dx + x(5-v)dv &= 0 \\ 2(2+v-v^2)dx + x(5-v)dv &= 0 \end{aligned}$$

$$\frac{2dx}{x} + \frac{(5-v)dv}{(2+v-v^2)} = 0$$

$$\frac{2dx}{x} - \frac{(5-v)dv}{(v+1)(v-2)} = 0$$

$$\int \frac{2dx}{x} - \int \frac{(5-v)dv}{(v+1)(v-2)} = \int 0$$

Considering:

$$\int \frac{(5-v)dv}{(v+1)(v-2)}$$

Using Integration by Partial Fraction:

$$\frac{(5-v)}{(v+1)(v-2)} = \frac{A}{(v+1)} + \frac{B}{(v-2)}$$

$$5-v = A(v-2) + B(v+1)$$

If $v = -1$: $A = -2$
 $v = 2$: $B = 1$

Then,

$$\begin{aligned} \int \frac{(5-v)dv}{(v+1)(v-2)} &= -2 \int \frac{dv}{(v+1)} + \int \frac{dv}{(v-2)} \\ &= -2 \ln(v+1) + \ln(v-2) \end{aligned}$$

Thus,

$$\begin{aligned} 2 \ln x + 2 \ln(v+1) - \ln(v-2) &= \ln C \\ \frac{x^2(v+1)^2}{(v-2)} &= C \end{aligned}$$

But: $v = y/x$

$$\frac{x^2 \left(\frac{y}{x} + 1\right)^2}{\left(\frac{y}{x} - 2\right)} = C$$

$$\frac{x^2 \left(\frac{y+x}{x}\right)^2}{\left(\frac{y-2x}{x}\right)} = C$$

$$x(y+x)^2 = C(y-2x)$$

7. $(5v-u)du + (3v-7u)dv = 0$

∴ Homogeneous, 1st degree!

Let: $v = uz$
 $dv = udz + zdu$

Substitute:

$$(5uz-u)du + (3uz-7u)(udz+zdu) = 0$$

$$\begin{aligned} (5z-1)du + (3z-7)(udz+zdu) &= 0 \\ (5z-1)du + (3zudz+3z^2du-7udz-7zdu) &= 0 \\ (5z-1+3z^2-7z)du + (3zu-7u)dz &= 0 \end{aligned}$$

$$(3z^2-2z-1)du + u(3z-7)dz = 0$$

$$\frac{du}{u} + \frac{(3z-7)dz}{(3z^2-2z-1)} = 0$$

$$\int \frac{du}{u} + \int \frac{(3z-7)dz}{(3z+1)(z-1)} = \int 0$$

Considering:

$$\int \frac{(3z-7)dz}{(3z+1)(z-1)}$$

Using Integration by Partial Fraction:

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$\frac{(3z-7)}{(3z+1)(z-1)} = \frac{A}{(3z+1)} + \frac{B}{(z-1)}$$

$$3z-7 = A(z-1) + B(3z+1)$$

If $z = 1$: $B = -1$
 $z = -1/3$: $A = 6$

Then,

$$\begin{aligned} \int \frac{(3z-7)dz}{(3z+1)(z-1)} &= 6 \int \frac{du}{(3z+1)} - \int \frac{du}{(z-1)} \\ &= 2 \ln(3z+1) - \ln(z-1) \end{aligned}$$

Thus,

$$\begin{aligned} \ln u + 2 \ln(3z+1) - \ln(z-1) &= \ln C \\ \frac{u(3z+1)^2}{(z-1)} &= C \end{aligned}$$

But: $z = v/u$

$$\frac{u \left(3 \left(\frac{v}{u}\right) + 1\right)^2}{\left(\frac{v}{u} - 1\right)} = C$$

$$\frac{u \left(\frac{3v+u}{u}\right)^2}{\left(\frac{v-u}{u}\right)} = C$$

$$(3v+u)^2 = C(v-u)$$

8. $(x^2+2xy-4y^2)dx - (x^2-8xy-4y^2)dy = 0$

∴ Homogeneous, 2nd degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$\begin{aligned} (x^2+2vx^2-4v^2x^2)dx - (x^2-8vx^2-4v^2x^2)(vdx+xdv) &= 0 \\ -x^2-8vx^2-4v^2x^2(vdx+xdv) &= 0 \end{aligned}$$

$$(1+2v-4v^2)dx - (1-8v-4v^2)(vdx+xdv) = 0$$

$$\begin{aligned} (1+2v-4v^2)dx - (vdx+xdv-8v^2dx-8vxdv-4v^3dx-4v^2xdv) &= 0 \\ (1+2v-4v^2-v+8v^2+4v^3)dx - (x-8vx-4v^2x)dv &= 0 \\ (4v^3+4v^2+v+1)dx - x(1-8v-4v^2)dv &= 0 \end{aligned}$$

$$\frac{dx}{x} - \frac{(1-8v-4v^2)dv}{(4v^3+4v^2+v+1)} = 0$$

$$\int \frac{dx}{x} - \int \frac{(1-8v-4v^2)dv}{(4v^2+1)(v+1)} = \int 0$$

Considering:

$$\int \frac{(1-8v-4v^2)dv}{(4v^2+1)(v+1)}$$

Using Integration by Partial Fraction:

$$\frac{1-8v-4v^2}{(4v^2+1)(v+1)} = \frac{8vA}{(4v^2+1)} + \frac{B}{(v+1)}$$

$$1-8v-4v^2 = 8vA(v+1) + B(4v^2+1)$$

Collecting v terms:

$$-8 = 8A \quad ; \quad A = -1$$

Collecting v^2 terms:

$$\begin{aligned} -4 &= 8A + 4B \\ -4 &= 8(-1) + 4B \\ B &= 1 \end{aligned}$$

Then,

$$\begin{aligned} \int \frac{(1-8v-4v^2)dv}{(4v^2+1)(v+1)} &= - \int \frac{8v dv}{(4v^2+1)} + \int \frac{dv}{(v+1)} \\ &= - \ln(4v^2+1) + \ln(v+1) \end{aligned}$$

Thus,

$$\ln x + \ln(4v^2+1) - \ln(v+1) = \ln C$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$\frac{x(4v^2 + 1)}{(v + 1)} = C$$

But: $v = y/x$

$$\frac{x \left(4 \left(\frac{y}{x} \right)^2 + 1 \right)}{\left(\frac{y}{x} + 1 \right)} = C$$

$$\frac{x \left(\frac{4y^2 + x^2}{x^2} \right)}{\left(\frac{y + x}{x} \right)} = C$$

$$\frac{(4y^2 + x^2)}{(y + x)} = C$$

$$(4y^2 + x^2) = C(y + x)$$

9. $(x^2 + y^2)dx - xydy = 0$

△ Homogeneous, 2nd degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$(x^2 + v^2x^2)dx - vx^2(vdx + xdv) = 0$$

$$(1 + v^2)dx - v^2dx - vxdv = 0$$

$$(1 + v^2 - v^2)dx - vxdv = 0$$

$$\frac{dx}{x} - vdv = 0$$

$$\int \frac{dx}{x} - \int vdv = \int 0$$

$$\ln x - \frac{v^2}{2} = \ln C$$

$$2 \ln x - v^2 = 2 \ln C$$

$$v^2 = 2 \ln \frac{x}{C}$$

But: $v = y/x$

$$v^2 = 2 \ln \frac{x}{C}$$

$$\left(\frac{y}{x} \right)^2 = 2 \ln \frac{x}{C}$$

$$y^2 = 2x^2 \ln \frac{x}{C}$$

10. $x(x^2 + y^2)^2(ydx - xdy) + y^4dy = 0$

Divide by y^2 .

$$\frac{x(x^2 + y^2)^2(ydx - xdy)}{y^2} + y^4dy = 0$$

Let: $v = \frac{x}{y}$; $x = vy$

$$dv = \frac{ydx - xdy}{y^2}$$

Substitute:

$$vy(v^2y^2 + y^2)^2dv + y^4dy = 0$$

$$vy^5(v^2 + 1)^2dv + y^4dy = 0$$

$$v(v^2 + 1)^2dv + \frac{dy}{y} = 0$$

$$\int v(v^2 + 1)^2dv + \int \frac{dy}{y} = \int 0$$

$$\frac{(v^2 + 1)^3}{6} + \ln y = \ln C$$

$$(v^2 + 1)^3 + 6 \ln y = 6 \ln C$$

$$(v^2 + 1)^3 = 6 \ln C - 6 \ln y$$

$$(v^2 + 1)^3 = 6 \ln \frac{C}{y}$$

$$(v^2 + 1)^3 = 6 \ln \frac{C}{y}$$

But: $v = x/y$

$$\left(\frac{x^2}{y^2} + 1 \right)^3 = 6 \ln \frac{C}{y}$$

$$\left(\frac{x^2 + y^2}{y^2} \right)^3 = 6 \ln \frac{C}{y}$$

$$(x^2 + y^2)^3 = 6y^6 \ln \frac{C}{y}$$

11. $(x^2 + y^2)dx + xydy = 0$

△ Homogeneous, 2nd degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$(x^2 + v^2x^2)dx + vx^2(vdx + xdv) = 0$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$(1 + v^2)dx + v^2dx + vxdv = 0$$

$$(1 + v^2 + v^2)dx + vxdv = 0$$

$$(1 + 2v^2)dx + vxdv = 0$$

$$\frac{dx}{x} + \frac{vdv}{(1 + 2v^2)} = 0$$

$$\int \frac{dx}{x} + \frac{1}{4} \int \frac{4vdv}{(1 + 2v^2)} = \int 0$$

$$\ln x + \frac{1}{4} \ln(1 + 2v^2) = \ln C$$

$$4 \ln x + \ln(1 + 2v^2) = 4 \ln C$$

$$x^4(1 + 2v^2) = C^4$$

But: $v = y/x$

$$x^4 \left(1 + 2 \frac{y^2}{x^2} \right) = C^4$$

$$x^2(x^2 + 2y^2) = C^4$$

12. $xydx - (x + 2y)^2dy = 0$

$$xydx - (x^2 + 4xy + 4y^2)dy = 0$$

△ Homogeneous, 2nd degree!

Let: $x = vy$
 $dx = vdy + ydv$

Substitute:

$$vy^2(vdy + ydv) - (v^2y^2 + 4vy^2 + 4y^2)dy = 0$$

$$v^2dy + vydv - (v^2 + 4v + 4)dy = 0$$

$$(v^2 - v^2 - 4v - 4)dy + vydv = 0$$

$$-4(v + 1)dy + vydv = 0$$

$$\frac{-4dy}{y} + \frac{vdv}{(v + 1)} = 0$$

$$\int \frac{-4dy}{y} + \int \frac{(v + 1) - 1}{(v + 1)} dv = \int 0$$

$$\int \frac{-4dy}{y} + \int dv - \int \frac{1}{(v + 1)} dv = \int 0$$

$$-4 \ln y + v - \ln(v + 1) = -\ln C$$

$$v = 4 \ln y + \ln(v + 1) - \ln C$$

$$v = \ln \frac{y^4(v + 1)}{C}$$

$$e^v = \frac{y^4(v + 1)}{C}$$

But: $v = x/y$

$$e^{x/y} = \frac{y^4 \left(\frac{x}{y} + 1 \right)}{C}$$

$$e^{x/y} = \frac{y^4 \left(\frac{x + y}{y} \right)}{C}$$

$$y^3(x + y) = Ce^{x/y}$$

13. $v^2dx + x(x + v)dv = 0$

△ Homogeneous, 2nd degree!

Let: $x = zv$
 $dx = zdv + vdz$

Substitute:

$$v^2(zdv + vdz) + zv(zv + v)dv = 0$$

$$(zdv + vdz) + (z^2 + z)dv = 0$$

$$(z + z^2 + z)dv + vdz = 0$$

$$(z^2 + 2z)dv + vdz = 0$$

$$\frac{dv}{v} + \frac{dz}{(z^2 + 2z)} = 0$$

$$\int \frac{dv}{v} + \int \frac{dz}{z(z + 2)} = \int 0$$

Considering:

$$\int \frac{dz}{z(z + 2)}$$

Using Integration by Partial Fraction:

$$\frac{1}{z(z + 2)} = \frac{A}{z} + \frac{B}{(z + 2)}$$

$$1 = A(z + 2) + B(z)$$

If $z = 0$; $A = 1/2$
 $z = -2$; $B = -1/2$

Then,

$$\int \frac{dz}{z(z + 2)} = \frac{1}{2} \int \frac{dz}{z} - \frac{1}{2} \int \frac{dz}{(z + 2)}$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$= \frac{1}{2} \ln(z) - \frac{1}{2} \ln(z+2)$$

Thus,

$$\ln v + \frac{1}{2} \ln(z) - \frac{1}{2} \ln(z+2) = \frac{1}{2} \ln C$$

$$2 \ln v + \ln(z) - \ln(z+2) = \ln C$$

$$\frac{v^2 z}{(z+2)} = C$$

But: $z = x/v$

$$\frac{v^2 \frac{x}{v}}{(\frac{x}{v}+2)} = C$$

$$\frac{vx}{(\frac{x+2v}{v})} = C$$

$$xv^2 = C(x+2v)$$

14. $[x \csc(y/x) - y]dx + xdy = 0$

∴ Homogeneous, 1st degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$[x \csc(vx/x) - vx]dx + x(vdx + xdv) = 0$$

$$[\csc(v) - v]dx + (vdx + xdv) = 0$$

$$[\csc(v) - v + v]dx + xdv = 0$$

$$\csc(v)dx + xdv = 0$$

$$\frac{dx}{x} + \frac{dv}{\csc(v)} = 0$$

$$\int \frac{dx}{x} + \int \sin v dv = \int 0$$

$$\ln x - \cos v = \ln C$$

$$\ln x - \ln C = \cos v$$

$$\ln \frac{x}{C} = \cos v$$

But: $v = y/x$

$$\ln \frac{x}{C} = \cos \frac{y}{x}$$

15. $x dx + \sin^2(y/x) [y dx - x dy] = 0$

Divide by x^2 ,

$$\frac{dx}{x} + \frac{\sin^2(y/x) [y dx - x dy]}{x^2} = 0$$

Let: $v = \frac{y}{x}$; $y = vx$

$$dv = \frac{xdy - ydx}{x^2}$$

Substitute:

$$\frac{dx}{x} - \sin^2(vx/x) dv = 0$$

$$\frac{dx}{x} - \sin^2(v) dv = 0$$

$$\int \frac{dx}{x} - \int \frac{1 - \cos 2v}{2} dv = \int 0$$

$$\ln x - \frac{1}{2}v + \frac{1}{4}\sin 2v = \ln C$$

$$4 \ln x - 2v + \sin 2v = 4 \ln C$$

$$4 \ln x - 4 \ln C = 2v - \sin 2v$$

$$4 \ln \left(\frac{x}{C}\right) = 2v - \sin 2v$$

But: $v = y/x$

$$4 \ln \left(\frac{x}{C}\right) = 2 \frac{y}{x} - \sin 2 \frac{y}{x}$$

$$4x \ln \left(\frac{x}{C}\right) = 2y - x \sin \frac{2y}{x}$$

$$4x \ln \left(\frac{x}{C}\right) - 2y + x \sin \frac{2y}{x} = 0$$

16. $(x - y \ln y + y \ln x)dx + x(\ln y - \ln x)dy = 0$

$$[x - (y \ln y - y \ln x)]dx + x \ln \left(\frac{y}{x}\right) dy = 0$$

$$(x - y \ln \frac{y}{x})dx + x \ln \frac{y}{x} dy = 0$$

∴ Homogeneous, 1st degree!

Let: $y = vx$
 $dy = vdx + xdv$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

Substitute:

$$(x - vx \ln \frac{vx}{x})dx + x \ln \frac{vx}{x} (vdx + xdv) = 0$$

$$(1 - v \ln v)dx + \ln v (vdx + xdv) = 0$$

$$(1 - v \ln v + v \ln v)dx + x \ln v dv = 0$$

$$dx + x \ln v dv = 0$$

$$\int \frac{dx}{x} + \int \ln v dv = \int 0$$

$$\ln x + v \ln v - v = C$$

But: $v = y/x$

$$\ln x + \frac{y}{x} \ln \frac{y}{x} - \frac{y}{x} = C$$

$$x \ln x + y \ln \frac{y}{x} - y = Cx$$

$$x \ln x + y \ln y - y \ln x = Cx + y$$

$$(x - y) \ln x + y \ln y = Cx + y$$

17. $[x - y \operatorname{Arctan}(\frac{y}{x})]dx + x \operatorname{Arctan}(\frac{y}{x})dy = 0$

∴ Homogeneous, 1st degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$[x - vx \operatorname{Arctan}(v)]dx + x \operatorname{Arctan}(v) (vdx + xdv) = 0$$

$$(1 - v \operatorname{Arctan}(v))dx + \operatorname{Arctan}(v) (vdx + xdv) = 0$$

$$dx - v \operatorname{Arctan}(v) dx + v \operatorname{Arctan}(v) dx + x \operatorname{Arctan}(v) dv = 0$$

$$dx + x \operatorname{Arctan}(v) dv = 0$$

$$\frac{dx}{x} + \operatorname{Arctan}(v) dv = 0$$

$$\int \frac{dx}{x} + \int \operatorname{Arctan}(v) dv = \int 0$$

Considering

$$\int \operatorname{Arctan}(v) dv$$

Using Integration by Parts:

Let $u = \operatorname{Arctan}(v)$ $dv = dv$
 $du = \frac{dv}{1+v^2}$ $v = v$

then,

$$\int \operatorname{Arctan}(v) dv = v \operatorname{Arctan}(v) - \int \frac{v dv}{1+v^2}$$

$$= v \operatorname{Arctan}(v) - \frac{1}{2} \ln(1+v^2)$$

thus,

$$\ln x + v \operatorname{Arctan}(v) - \frac{1}{2} \ln(1+v^2) = \ln C$$

$$2 \ln x + 2v \operatorname{Arctan}(v) - \ln(1+v^2) = 2 \ln C$$

$$2v \operatorname{Arctan}(v) = 2 \ln C + \ln(1+v^2) - 2 \ln x$$

$$2v \operatorname{Arctan}(v) = \ln \frac{C^2(1+v^2)}{x^2}$$

But: $v = y/x$

$$2 \frac{y}{x} \operatorname{Arctan} \frac{y}{x} = \ln \frac{C^2(1+\frac{y^2}{x^2})}{x^2}$$

$$2 \frac{y}{x} \operatorname{Arctan} \frac{y}{x} = \ln \frac{C^2(x^2+y^2)}{x^4}$$

$$2y \operatorname{Arctan} \frac{y}{x} = x \ln \frac{C^2(x^2+y^2)}{x^4}$$

18. $y^2 dy = x(xdy - ydx)e^{x/y}$

∴ Homogeneous, 2nd degree!

Divide by y^2 ,

$$dy = \frac{-x(ydx - xdy)e^{x/y}}{y^2}$$

Let: $v = x/y$; $x = vy$
 $dv = \frac{ydx - xdy}{y^2}$

Substitute:

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$dy = -vydv e^v$$

$$dy + vydv e^v = 0$$

$$\frac{dy}{y} + ve^v dv = 0$$

$$\int \frac{dy}{y} + \int ve^v dv = \int 0$$

Considering:

$$\int ve^v dv$$

Using Integration by Parts:

$$\text{Let } u = v \quad dv = e^v dv \\ du = dv \quad v = e^v$$

then,

$$\int ve^v dv = ve^v - \int e^v dv \\ = ve^v - e^v$$

thus,

$$\ln y + ve^v - e^v = \ln C$$

$$\ln y + e^v(v-1) = \ln C$$

$$\ln y - \ln C = -e^v(v-1)$$

$$\ln(y/C) = e^v(1-v)$$

$$\text{But: } v = x/y$$

$$\ln(y/C) = e^{x/y} \left(1 - \frac{x}{y}\right)$$

$$\ln(y/C) = e^{x/y} \left(\frac{y-x}{y}\right)$$

$$y \ln(y/C) = e^{x/y} (y-x)$$

$$19. \quad t(s^2 + t^2)ds - s(s^2 - t^2)dt = 0$$

$\therefore$ Homogeneous, 3rd degree!

$$\text{Let: } s = vt \\ ds = vdt + t dv$$

Substitute:

$$t(v^2t^2 + t^2)(vdt + t dv) \\ - vt(v^2t^2 - t^2)dt = 0 \\ (v^2 + 1)(vdt + t dv) - (v^3 - v)dt = 0 \\ (v^3dt + v^2tdv + vdt + t dv) - (v^3 - v)dt = 0$$

$$(v^3 + v - v^3 + v)dt + (v^2t + t)dv = 0$$

$$(2v)dt + t(v^2 + 1)dv = 0$$

$$\frac{2dt}{t} + \frac{(v^2 + 1)dv}{v} = 0$$

$$\int \frac{2dt}{t} + \int \left(v + \frac{1}{v}\right)dv = \int 0$$

$$2 \ln t + \frac{v^2}{2} + \ln v = -\ln C$$

$$4 \ln t + v^2 + 2 \ln v = -2 \ln C$$

$$v^2 = -2 \ln C - 2 \ln v - 4 \ln t$$

$$v^2 = -2 \ln Cvt^2$$

$$\text{But: } v = s/t$$

$$\frac{s^2}{t^2} = -2 \ln C \frac{s}{t} t^2$$

$$\frac{s^2}{t^2} = -2 \ln Cst$$

$$s^2 = -2t^2 \ln Cst$$

$$20. \quad ydx = (x + \sqrt{y^2 - x^2})dy$$

$\therefore$ Homogeneous, 1st degree!

$$\text{Let: } x = vy \\ dx = vdy + ydv$$

Substitute:

$$y(vdy + ydv) = (vy + \sqrt{y^2 - v^2y^2})dy$$

$$(vdy + ydv) - (v + \sqrt{1 - v^2})dy = 0$$

$$(v - v - \sqrt{1 - v^2})dy + ydv = 0$$

$$-(\sqrt{1 - v^2})dy + ydv = 0$$

$$\frac{dy}{y} - \frac{dv}{\sqrt{1 - v^2}} = 0$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$\int \frac{dy}{y} - \int \frac{dv}{\sqrt{1 - v^2}} = \int 0$$

From the Table of Integration:

$$\int \frac{dv}{\sqrt{1 - v^2}} = \text{Arc sin } v$$

Thus,

$$\ln y - \text{Arc sin } v = \ln C$$

$$\text{Arc sin } v = \ln y - \ln C$$

$$\text{But: } v = x/y$$

$$\text{Arc sin } \frac{x}{y} = \ln(y/C)$$

$$21. \quad (3x^2 - 2xy + 3y^2)dx = 4xydy$$

$\therefore$ Homogeneous, 2nd degree!

$$\text{Let: } y = vx \\ dy = vdx + xdv$$

Substitute:

$$(3x^2 - 2vx^2 + 3v^2x^2)dx \\ = 4vx^2(vdx + xdv)$$

$$(3 - 2v + 3v^2)dx - 4v(vdx + xdv) = 0$$

$$(3 - 2v + 3v^2 - 4v^2)dx - 4vxdv = 0$$

$$(3 - 2v - v^2)dx - 4vxdv = 0$$

$$\frac{dx}{x} - \frac{4v dv}{(3 - 2v - v^2)} = 0$$

$$\frac{dx}{x} + \frac{4v dv}{(v^2 + 2v - 3)} = 0$$

$$\int \frac{dx}{x} + \int \frac{4v dv}{(v+3)(v-1)} = \int 0$$

Considering:

$$\int \frac{4v dv}{(v+3)(v-1)}$$

Using Integration by Partial Fraction:

$$\frac{4v}{(v+3)(v-1)} = \frac{A}{(v+3)} + \frac{B}{(v-1)} \\ 4v = A(v-1) + B(v+3)$$

$$\text{If } v = -3 \quad ; \quad A = 3 \\ v = 1 \quad ; \quad B = 1$$

Then,

$$\int \frac{4v dv}{(v+3)(v-1)} = 3 \int \frac{dv}{(v+3)} \\ + \int \frac{dv}{(v-1)} \\ = 3 \ln(v+3) + \ln(v-1)$$

Thus,

$$\ln x + 3 \ln(v+3) + \ln(v-1) = \ln C$$

$$\ln[x(v+3)^3(v-1)] = \ln C$$

$$x(v+3)^3(v-1) = C$$

$$\text{But: } v = y/x$$

$$x \left(\frac{y}{x} + 3\right)^3 \left(\frac{y}{x} - 1\right) = C$$

$$x \left(\frac{y+3x}{x}\right)^3 \left(\frac{y-x}{x}\right) = C$$

$$x \left(\frac{y+3x}{x}\right)^3 \left(\frac{y-x}{x}\right) = C$$

$$\frac{(y+3x)^3(y-x)}{x^3} = C$$

$$(y+3x)^3(y-x) = Cx^3$$

22. Prove that, with the aid of the substitution $y = vx$, you can solve any equation of the form

$$y^n f(x)dx + H(x,y)(ydx - xdy) = 0$$

Where $H(x,y)$ is homogeneous in x and y .

Divide by x^2 ,

$$\frac{y^n f(x)dx}{x^2} - H(x,y) \frac{(xdy - ydx)}{x^2} = 0$$

$$\text{Let: } v = y/x; \quad y = vx$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$dv = \frac{xdy - ydx}{x^2}$$

Substitute:

$$\frac{v^n x^n f(x) dx}{x^2} - H(x, vx) dv = 0$$

$$v^n x^{n-2} f(x) dx - xH(1, v) dv = 0$$

Divide by v^n ,

$$x^{n-2} f(x) dx - \frac{xH(1, v) dv}{v^n} = 0$$

Variable Separable

III. In exercises 23 through 35, find the particular solution indicated

23. $(x - y)dx + (3x + y)dy = 0$
when $x = 2, y = -1$

∴ Homogeneous, 1st degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$(x - vx)dx + (3x + vx)(vdx + xdv) = 0$$

$$(1 - v)dx + (3 + v)(vdx + xdv) = 0$$

$$(1 - v)dx + (3vdx + 3xdv + v^2dx + vx dv) = 0$$

$$(1 - v + 3v + v^2)dx + (3x + vx)dv = 0$$

$$(1 + 2v + v^2)dx + x(3 + v)dv = 0$$

$$\frac{dx}{x} + \frac{(3 + v)dv}{(1 + 2v + v^2)} = 0$$

$$\int \frac{dx}{x} + \int \frac{(3 + v)dv}{(v + 1)^2} = \int 0$$

Considering

$$\int \frac{(3 + v)dv}{(v + 1)^2}$$

Using Integration by Partial Fraction:

$$\frac{(3 + v)}{(v + 1)^2} = \frac{A}{(v + 1)} + \frac{B}{(v + 1)^2}$$

$$3 + v = A(v + 1) + B$$

Collecting v terms:

$$1 = A$$

Collecting constant terms:

$$3 = A + B$$

$$3 = 1 + B$$

$$B = 2$$

Then,

$$\int \frac{(3 + v)dv}{(v + 1)^2} = \int \frac{dv}{(v + 1)} + 2 \int \frac{dv}{(v + 1)^2}$$

$$= \ln(v + 1) - \frac{2}{(v + 1)}$$

Thus,

$$\ln x + \ln(v + 1) - \frac{2}{(v + 1)} = C$$

$$\ln x(v + 1) - \frac{2}{(v + 1)} = C$$

But: $v = y/x$

$$\ln x \left(\frac{y}{x} + 1 \right) - \frac{2}{\left(\frac{y}{x} + 1 \right)} = C$$

$$\ln x \left(\frac{y + x}{x} \right) - \frac{2}{\left(\frac{y + x}{x} \right)} = C$$

$$\ln(y + x) - \frac{2x}{(y + x)} = C$$

→ General Solution

when $x = 2, y = -1$

$$C = \ln(-1 + 2) - \frac{2(2)}{(-1 + 2)} = -4$$

Thus,

$$\ln(y + x) - \frac{2x}{(y + x)} = -4$$

$$(y + x) \ln(y + x) - 2x = -4(y + x)$$

$$(y + x) \ln(y + x) = -4(y + x) + 2x$$

$$(y + x) \ln(y + x) = -2x - 4y$$

$$(y + x) \ln(y + x) = -2(x + 2y)$$

$$2(x + 2y) + (y + x) \ln(y + x) = 0$$

→ Particular Solution

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

24.

$$(y - \sqrt{x^2 + y^2})dx - xdy = 0$$

when $x = \sqrt{3}, y = 1$

∴ Homogeneous, 1st degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$(vx - \sqrt{x^2 + v^2x^2})dx - x(vdx + xdv) = 0$$

$$= 0$$

$$(vx - x\sqrt{1 + v^2})dx - x(vdx + xdv) = 0$$

$$(v - \sqrt{1 + v^2})dx - (vdx + xdv) = 0$$

$$(v - \sqrt{1 + v^2} - v)dx - xdv = 0$$

$$(-\sqrt{1 + v^2})dx - xdv = 0$$

$$\frac{dx}{x} + \frac{dv}{\sqrt{1 + v^2}} = 0$$

$$\int \frac{dx}{x} + \int \frac{dv}{\sqrt{1 + v^2}} = \int 0$$

From Table of Integration

$$\int \frac{dv}{\sqrt{1 + v^2}} = \ln(v + \sqrt{1 + v^2})$$

Then,

$$\ln x + \ln(v + \sqrt{1 + v^2}) = \ln C$$

$$x(v + \sqrt{1 + v^2}) = C$$

But: $v = y/x$

$$x \left(\frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}} \right) = C$$

$$x \left(\frac{y}{x} + \sqrt{\frac{x^2 + y^2}{x^2}} \right) = C$$

$$x \left(\frac{y}{x} + \frac{\sqrt{x^2 + y^2}}{x} \right) = C$$

$$y + \sqrt{x^2 + y^2} = C \rightarrow \text{General Solution}$$

when $x = \sqrt{3}, y = 1$

$$C = 1 + \sqrt{(\sqrt{3})^2 + 1^2} = 3$$

Thus,

$$y + \sqrt{x^2 + y^2} = 3$$

$$\sqrt{x^2 + y^2} = 3 - y$$

Square both sides,

$$(\sqrt{x^2 + y^2})^2 = (3 - y)^2$$

$$x^2 + y^2 = 9 - 6y + y^2$$

$$x^2 = 9 - 6y \rightarrow \text{Particular Solution}$$

25.

$$(y + \sqrt{x^2 + y^2})dx - xdy = 0$$

when $x = \sqrt{3}, y = 1$

∴ Homogeneous, 1st degree!

Let: $y = vx$
 $dy = vdx + xdv$

Substitute:

$$(vx + \sqrt{x^2 + v^2x^2})dx - x(vdx + xdv) = 0$$

$$= 0$$

$$(vx + x\sqrt{1 + v^2})dx - x(vdx + xdv) = 0$$

$$(v + \sqrt{1 + v^2})dx - (vdx + xdv) = 0$$

$$(v + \sqrt{1 + v^2} - v)dx - xdv = 0$$

$$(\sqrt{1 + v^2})dx - xdv = 0$$

$$\frac{dx}{x} - \frac{dv}{\sqrt{1 + v^2}} = 0$$

$$\int \frac{dx}{x} - \int \frac{dv}{\sqrt{1 + v^2}} = \int 0$$

From Table of Integration

$$\int \frac{dv}{\sqrt{1 + v^2}} = \ln(v + \sqrt{1 + v^2})$$

Then,

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$\ln x - \ln(v + \sqrt{1+v^2}) = \ln C$$

$$\left(\frac{x}{v + \sqrt{1+v^2}}\right) = C$$

$$\text{But } v = y/x$$

$$\left(\frac{x}{\frac{y}{x} + \sqrt{1 + \frac{y^2}{x^2}}}\right) = C$$

$$\left(\frac{x}{\frac{y}{x} + \sqrt{\frac{x^2 + y^2}{x^2}}}\right) = C$$

$$\left(\frac{x}{\frac{y}{x} + \frac{1}{x}\sqrt{x^2 + y^2}}\right) = C$$

$$\frac{1}{\frac{y}{x^2} + \sqrt{x^2 + y^2}} = C$$

$$\frac{1}{(y + \sqrt{x^2 + y^2})} = C \rightarrow \text{General Solution}$$

$$\text{when } x = \sqrt{3}, y = 1$$

$$C = \left(\frac{(\sqrt{3})^2}{1 + \sqrt{(\sqrt{3})^2 + 1^2}}\right) = 1$$

Thus,

$$\frac{x^2}{(y + \sqrt{x^2 + y^2})} = 1$$

$$x^2 = (y + \sqrt{x^2 + y^2})$$

$$x^2 - y = \sqrt{x^2 + y^2}$$

Square both sides,

$$(x^2 - y)^2 = (\sqrt{x^2 + y^2})^2$$

$$x^4 - 2x^2y + y^2 = x^2 + y^2$$

$$x^4 - 2x^2y = x^2$$

$$x^2 - 2y = 1$$

$$x^2 = 2y + 1 \rightarrow \text{Particular Solution}$$

26.

$$[x \cos^2(y/x) - y]dx + xdy = 0$$

$$\text{when } x = 1, y = \pi/4$$

$\therefore$ Homogeneous, 1st degree!

$$x \cos^2(y/x) dx - ydx + xdy = 0$$

$$x \cos^2(y/x) dx + xdy - ydx = 0$$

Divide by x^2

$$\frac{\cos^2(y/x) dx}{x} + \frac{xdy - ydx}{x^2} = 0$$

$$\text{Let } v = y/x; \quad y = vx$$

$$dv = \frac{xdy - ydx}{x^2}$$

Substitute:

$$\frac{\cos^2(vx/x) dx}{x} + dv = 0$$

$$\frac{\cos^2 v dx}{x} + dv = 0$$

$$\frac{dx}{x} + \frac{dv}{\cos^2 v} = 0$$

$$\int \frac{dx}{x} + \int \sec^2 v dv = \int 0$$

$$\ln x + \tan v = C$$

$$\text{But } v = y/x$$

$$\ln x + \tan \frac{y}{x} = C \rightarrow \text{General Solution}$$

$$\text{when } x = 1, y = \pi/4$$

$$C = \ln 1 + \tan \frac{\pi/4}{1} = 1$$

Thus,

$$\ln x + \tan \frac{y}{x} = 1$$

$$\tan \frac{y}{x} = 1 - \ln x \rightarrow \text{Particular Solution}$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

27.

$$(y^2 + 7xy + 16x^2)dx + x^2dy = 0$$

$$\text{when } x = 1, y = 1$$

$\therefore$ Homogeneous, 2nd degree!

$$\text{Let } y = vx$$

$$dy = vdx + xdv$$

Substitute:

$$(v^2x^2 + 7vx^2 + 16x^2)dx + x^2(vdx + xdv) = 0$$

$$(v^2 + 7v + 16)dx + (vdx + xdv) = 0$$

$$(v^2 + 7v + 16 + v)dx + xdv = 0$$

$$(v^2 + 8v + 16)dx + xdv = 0$$

$$\frac{dx}{x} + \frac{dv}{(v^2 + 8v + 16)} = 0$$

$$\int \frac{dx}{x} + \int \frac{dv}{(v+4)^2} = \int 0$$

$$\ln x - \frac{1}{(v+4)} = C$$

$$\text{But } v = y/x$$

$$\ln x - \frac{1}{(\frac{y}{x} + 4)} = C$$

$$\ln x - \frac{1}{(\frac{y+4x}{x})} = C$$

$$\ln x - \frac{x}{(y+4x)} = C \rightarrow \text{General Solution}$$

$$\text{when } x = 1, y = 1$$

$$C = \ln 1 - \frac{1}{1+4(1)} = -\frac{1}{5}$$

Thus,

$$\ln x - \frac{x}{(y+4x)} = -\frac{1}{5}$$

$$5(y+4x) \ln x - 5x = -(y+4x)$$

$$5(y+4x) \ln x = 5x - (y+4x)$$

$$5(y+4x) \ln x = x - y$$

$$\rightarrow \text{Particular Solution}$$

28.

$$y^2dx + (x^2 + 3xy + 4y^2)dy = 0$$

$$\text{when } x = 2, y = 1$$

$\therefore$ Homogeneous, 2nd degree!

$$\text{Let } x = vy$$

$$dx = vdy + ydv$$

Substitute:

$$y^2(vdy + ydv) + (v^2y^2 + 3vy^2 + 4y^2)dy = 0$$

$$(vdy + ydv) + (v^2 + 3v + 4)dy = 0$$

$$(v^2 + 3v + 4 + v)dy + ydv = 0$$

$$(v^2 + 4v + 4)dy + ydv = 0$$

$$\frac{dy}{y} + \frac{dv}{(v+2)^2} = 0$$

$$\int \frac{dy}{y} + \int \frac{dv}{(v+2)^2} = \int 0$$

$$\ln y - \frac{1}{(v+2)} = C$$

$$\text{But } v = x/y$$

$$\ln y - \frac{1}{(\frac{x}{y} + 2)} = C$$

$$\ln y - \frac{1}{(\frac{x+2y}{y})} = C$$

$$\ln y - \frac{y}{(x+2y)} = C \rightarrow \text{General Solution}$$

$$\text{when } x = 2, y = 1$$

$$C = \ln 1 - \frac{1}{2+2(1)} = -\frac{1}{4}$$

Thus,

$$\ln y - \frac{y}{(x+2y)} = -\frac{1}{4}$$

$$4(x+2y) \ln y - 4y = -(x+2y)$$

$$4(x+2y) \ln y = 4y - (x+2y)$$

$$4(x+2y) \ln y = 2y - x$$

$$\rightarrow \text{Particular Solution}$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

29.

$$xydx + 2(x^2 + 2y^2)dy = 0$$

when $x = 0, y = 1$

∴ Homogeneous, 2nd degree!

$$\text{Let: } x = vy \\ dx = vdy + ydv$$

Substitute:

$$vy^2(vdy + ydv) + 2(v^2y^2 + 2y^2)dy = 0$$

$$v(vdy + ydv) + 2(v^2 + 2)dy = 0$$

$$v^2dy + vydv + (2v^2 + 4)dy = 0$$

$$(2v^2 + 4 + v^2)dy + vydv = 0$$

$$(3v^2 + 4)dy + vydv = 0$$

$$\frac{dy}{y} + \frac{vdv}{(3v^2 + 4)} = 0$$

$$\int \frac{dy}{y} + \frac{1}{6} \int \frac{6vdv}{(3v^2 + 4)} = \int 0$$

$$\ln y + \frac{1}{6} \ln(3v^2 + 4) = \ln C$$

$$6 \ln y + \ln(3v^2 + 4) = 6 \ln C$$

$$y^6(3v^2 + 4) = C^6$$

$$\text{But: } v = x/y$$

$$y^6 \left(3 \frac{x^2}{y^2} + 4 \right) = C^6$$

$$y^6 \left(\frac{3x^2 + 4y^2}{y^2} \right) = C^6$$

$$y^4(3x^2 + 4y^2) = C^6 \rightarrow \text{General Solution}$$

when $x = 0, y = 1$

$$C^6 = 1^4(3(0)^2 + 4(1)^2) = 4$$

Thus,

$$y^4(3x^2 + 4y^2) = 4 \rightarrow \text{Particular Solution}$$

30.

$$y(2x^2 - xy + y^2)dx - x^2(2x - y)dy = 0$$

when $x = 1, y = 1/2$

∴ Homogeneous, 3rd degree!

$$\text{Let: } y = vx \\ dy = vdx + xdv$$

Substitute:

$$vx(2x^2 - vx^2 + v^2x^2)dx - x^2(2x - vx)(vdx + xdv) = 0$$

$$v(2 - v + v^2)dx - (2 - v)(vdx + xdv) = 0$$

$$(2v - v^2 + v^3)dx - (2vdx + 2xdv - v^2dx - vxdv) = 0$$

$$(2v - v^2 + v^3 - 2v + v^2)dx - (2x - vx)dv = 0$$

$$v^3dx - x(2 - v)dv = 0$$

$$\frac{dx}{x} - \frac{(2 - v)dv}{v^3} = 0$$

$$\frac{dx}{x} - \frac{2}{v^2}dv + \frac{1}{v^3}dv = 0$$

$$\int \frac{dx}{x} - \int \frac{2}{v^2}dv + \int \frac{1}{v^3}dv = \int 0$$

$$\ln x + \frac{1}{v^2} - \frac{1}{v} = C$$

$$\text{But: } v = y/x$$

$$\ln x + \frac{1}{\frac{y^2}{x^2}} - \frac{1}{\frac{y}{x}} = C$$

$$\ln x + \frac{x^2}{y^2} - \frac{x}{y} = C \rightarrow \text{General Solution}$$

when $x = 1, y = 1/2$

$$C = \ln 1 + \frac{1^2}{(1/2)^2} - \frac{1}{1/2} = 2$$

Thus,

$$\ln x + \frac{x^2}{y^2} - \frac{x}{y} = 2$$

$$y^2 \ln x + x^2 - xy = 2y^2$$

$$y^2 \ln x = 2y^2 + xy - x^2$$

→ Particular Solution

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

31.

$$y(9x - 2y)dx - x(6x - y)dy = 0$$

when $x = 1, y = 1$

∴ Homogeneous, 2nd degree!

$$\text{Let: } x = vy \\ dx = vdy + ydv$$

Substitute:

$$y(9vy - 2y)(vdy + ydv) - vy(6vy - y)dy = 0$$

$$(9v - 2)(vdy + ydv) - v(6v - 1)dy = 0$$

$$(9v^2dy + 9vydv - 2vdy - 2ydv) - v(6v - 1)dy = 0$$

$$(9v^2 - 2v - 6v^2 + v)dy + (9vy - 2y)dv = 0$$

$$(3v^2 - v)dy + y(9v - 2)dv = 0$$

$$\frac{dy}{y} + \frac{(9v - 2)dv}{(3v^2 - v)} = 0$$

$$\int \frac{dy}{y} + \int \frac{(9v - 2)dv}{v(3v - 1)} = \int 0$$

Considering

$$\int \frac{(9v - 2)dv}{v(3v - 1)}$$

Using Integration by Partial Fraction:

$$\frac{9v - 2}{v(3v - 1)} = \frac{A}{v} + \frac{B}{(3v - 1)}$$

$$9v - 2 = A(3v - 1) + Bv$$

$$\text{If } v = 0 : A = 2 \\ v = 1/3 : B = 3$$

Then,

$$\int \frac{(9v - 2)dv}{v(3v - 1)} = 2 \int \frac{dv}{v} + 3 \int \frac{dv}{(3v - 1)} \\ = 2 \ln v + \ln(3v - 1)$$

Thus,

$$\ln y + 2 \ln v + \ln(3v - 1) = \ln C \\ yv^2(3v - 1) = C$$

$$\text{But: } v = x/y$$

$$y \frac{x^2}{y^2} \left(3 \frac{x}{y} - 1 \right) = C$$

$$\frac{x^2}{y} \left(\frac{3x - y}{y} \right) = C$$

$$\frac{x^2}{y^2} (3x - y) = C \rightarrow \text{General Solution}$$

when $x = 1, y = 1$;

$$C = \frac{1^2}{1^2} (3(1) - 1) = 2$$

Thus,

$$\frac{x^2}{y^2} (3x - y) = 2$$

$$\frac{x^2}{y^2} (3x - y) = 2$$

$$3x^3 - x^2y = 2y^2$$

$$3x^3 - x^2y - 2y^2 = 0$$

→ Particular Solution

32.

$$y(x^2 + y^2)dx + x(3x^2 - 5y^2)dy = 0$$

when $x = 2, y = 1$

∴ Homogeneous, 3rd degree!

$$\text{Let: } x = vy \\ dx = vdy + ydv$$

Substitute:

$$y(v^2y^2 + y^2)(vdy + ydv) + vy(3v^2y^2 - 5y^2)dy = 0$$

$$(v^2 + 1)(vdy + ydv) + v(3v^2 - 5)dy = 0$$

$$(v^3dy + v^2ydv + vdy + ydv) + v(3v^2 - 5)dy = 0$$

$$(v^3dy + v^2ydv + vdy + ydv) + v(3v^2 - 5)dy = 0$$

$$(v^3dy + v^2ydv + vdy + ydv) + v(3v^2 - 5)dy = 0$$

$$(v^3 + v + 3v^3 - 5v)dy + (v^2y + y)dv = 0$$

$$= 0$$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$(4v^3 - 4v)dy + y(v^2 + 1)dv = 0$$

$$4v(v^2 - 1)dy + y(v^2 + 1)dv = 0$$

$$\frac{4dy}{y} + \frac{(v^2 + 1)dv}{v(v^2 - 1)} = 0$$

$$4 \int \frac{dy}{y} + \int \frac{(v^2 + 1)dv}{v(v^2 - 1)(v + 1)} = 0$$

Considering

$$\int \frac{(v^2 + 1)dv}{v(v^2 - 1)(v + 1)}$$

Using Integration by Partial Fraction:

$$\frac{(v^2 + 1)}{v(v^2 - 1)(v + 1)} = \frac{A}{v} + \frac{B}{(v - 1)} + \frac{C}{(v + 1)}$$

$$(v^2 + 1) = A(v - 1)(v + 1) + Bv(v + 1) + Cv(v - 1)$$

If $\begin{matrix} v = 0 & : & A = -1 \\ v = 1 & : & B = 1 \\ v = -1 & : & C = 1 \end{matrix}$

Then,

$$\int \frac{(v^2 + 1)dv}{v(v^2 - 1)(v + 1)} = - \int \frac{dv}{v} + \int \frac{dv}{(v - 1)} + \int \frac{dv}{(v + 1)}$$

$$= -\ln v + \ln(v - 1) + \ln(v + 1)$$

Thus,

$$4\ln y - \ln v + \ln(v - 1) + \ln(v + 1) = \ln C$$

$$\frac{y^4(v - 1)(v + 1)}{v} = C$$

$$\frac{y^4(v^2 - 1)}{v} = C$$

But: $v = x/y$

$$y^4 \left(\frac{x^2}{y^2} - 1 \right) \frac{y}{x} = C$$

$$\frac{y^3(x^2 - y^2)}{x} = C$$

$$y^3(x^2 - y^2) = Cx \rightarrow \text{General Solution}$$

when $x = 2, y = 1$

$$C(2) = 1^3(2^2 - 1^2)$$

$$C = 3/2$$

Thus,

$$y^3(x^2 - y^2) = \frac{3}{2}x$$

$$2x^2y^3 - 2y^5 = 3x$$

$$2y^5 - 2x^2y^3 + 3x = 0$$

$\rightarrow$ Particular Solution

33. $(16x + 5y)dx + (3x + y)dy = 0$
The curve to pass through point $(1, -3)$

$\therefore$ Homogeneous, 1st degree!

Let $y = vx$
 $dy = vdx + xdv$

Substitute:

$$(16x + 5vx)dx + (3x + vx)(vdx + xdv) = 0$$

$$(16 + 5v)dx + (3 + v)(vdx + xdv) = 0$$

$$(16 + 5v)dx + (3vdx + 3xdv + v^2dx + vx dv) = 0$$

$$(16 + 5v + 3v + v^2)dx + (3x + vx)dv = 0$$

$$(16 + 8v + v^2)dx + x(3 + v)dv = 0$$

$$\frac{dx}{x} + \frac{(3 + v)dv}{(16 + 8v + v^2)} = 0$$

$$\int \frac{dx}{x} + \int \frac{(3 + v)dv}{(v + 4)^2} = 0$$

$$\ln x + \int \frac{(3 + v)dv}{(v + 4)^2}$$

$$\ln x + \frac{(3 + v)}{(v + 4)} = C$$

$$\ln x + \frac{(3 + v)}{(v + 4)} = C$$

Considering

$$\int \frac{(3 + v)dv}{(v + 4)^2}$$

Using Integration by Partial Fraction:

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$\frac{(3 + v)}{(v + 4)^2} = \frac{A}{(v + 4)} + \frac{B}{(v + 4)^2}$$

$$(3 + v) = A(v + 4) + B$$

Collecting v terms:

$$1 = A$$

Collecting constant terms:

$$3 = 4A + B$$

$$3 = 4(1) + B$$

$$B = -1$$

Then,

$$\int \frac{(3 + v)dv}{(v + 4)^2} = \int \frac{dv}{(v + 4)} - \int \frac{dv}{(v + 4)^2}$$

$$= \ln(v + 4) + \frac{1}{(v + 4)}$$

Thus,

$$\ln x + \ln(v + 4) + \frac{1}{(v + 4)} = C$$

$$\ln x(v + 4) + \frac{1}{(v + 4)} = C$$

But: $v = y/x$

$$\ln x \left(\frac{y}{x} + 4 \right) + \frac{1}{\left(\frac{y}{x} + 4 \right)} = C$$

$$\ln x \left(\frac{y + 4x}{x} \right) + \frac{1}{\left(\frac{y + 4x}{x} \right)} = C$$

$$\ln(y + 4x) + \frac{x}{(y + 4x)} = C$$

$\rightarrow$ General Solution

The curve to pass through point $(1, -3)$

$$C = \ln(-3 + 4(1)) + \frac{1}{(-3 + 4(1))} = 1$$

Thus,

$$\ln(y + 4x) + \frac{x}{(y + 4x)} = 1$$

$$(y + 4x) \ln(y + 4x) + x = (y + 4x)$$

$$(y + 4x) \ln(y + 4x) = y + 3x$$

$\rightarrow$ Particular Solution

34.

$$v(3x + 2v)dx - x^2dv = 0$$

when $x = 1, v = 2$

$\therefore$ Homogeneous, 2nd degree!

Let: $v = zx$

$$dv = zdx + xdz$$

Substitute:

$$zx(3x + 2zx)dx - x^2(zdx + xdz) = 0$$

$$x(3 + 2z)dx - (zdx + xdz) = 0$$

$$(3x + 2x^2 - x)dx - xdz = 0$$

$$(2x^2 + 2x)dx - xdz = 0$$

$$2(x^2 + x)dx - xdz = 0$$

$$\frac{2dx}{x} - \frac{dz}{z(z + 1)} = 0$$

$$2 \int \frac{dx}{x} - \int \frac{dz}{z(z + 1)} = 0$$

Considering

$$\int \frac{dz}{z(z + 1)}$$

Using Integration by Parts:

$$\frac{1}{z(z + 1)} = \frac{A}{z} + \frac{B}{(z + 1)}$$

$$1 = A(z + 1) + Bz$$

If $\begin{matrix} z = 0 & : & A = 1 \\ z = -1 & : & B = -1 \end{matrix}$

Then,

$$\int \frac{dz}{z(z + 1)} = \int \frac{dz}{z} - \int \frac{dz}{(z + 1)}$$

$$= \ln z - \ln(z + 1)$$

Thus,

$$2 \ln x - \ln z + \ln(z + 1) = \ln C$$

$$\frac{x^2(z + 1)}{z} = C$$

But: $z = v/x$

TYPE 2: EQUATIONS WITH HOMOGENEOUS COEFFICIENTS

$$x^2 \left(\frac{v}{x} + 1 \right) = C$$

$$x^2 \left(\frac{v+x}{x} \right) = C$$

$$\frac{x^2(v+x)}{x} = C \rightarrow \text{General Solution}$$

$$\text{when } x=1, v=2;$$

$$C = \frac{1^2(2+1)}{1} = \frac{3}{2}$$

Thus,

$$\frac{x^2(v+x)}{x} = \frac{3}{2}$$

$$2x^2(v+x) = 3x$$

$$2x^3 + 2x^2v - 3x = 0$$

$$\rightarrow \text{Particular Solution}$$

$$35. (3x^2 - 2y^2)y' = 2xy$$

$$\text{when } x=0, y=-1$$

$\therefore$ Homogeneous, 2nd degree!

$$(3x^2 - 2y^2) \frac{dy}{dx} = 2xy$$

$$2xydx - (3x^2 - 2y^2)dy = 0$$

$$\text{Let: } x = vy \\ dx = vdy + ydv$$

Substitute:

$$2vy^2(vdy + ydv) - (3v^3y^3 - 2y^3)dy = 0$$

$$2v(vdy + ydv) - (3v^3 - 2)dy = 0$$

$$(2v^2 - 3v^3 + 2)dy + 2vydv = 0$$

$$(-v^3 + 2)dy + 2vydv = 0$$

$$\frac{dy}{y} - \frac{2vdy}{(v^3 - 2)} = 0$$

$$\int \frac{dy}{y} - \int \frac{2vdy}{(v^3 - 2)} = \int 0$$

$$\ln y - \ln(v^3 - 2) = \ln C$$

$$\frac{y}{(v^3 - 2)} = C$$

$$\text{But: } v = x/y$$

$$\frac{y}{(x^3/y^3 - 2)} = C$$

$$\frac{y}{(x^3 - 2y^3)} = C$$

$$\frac{y^3}{(x^3 - 2y^3)} = C \rightarrow \text{General Solution}$$

$$\text{when } x=0, y=-1;$$

$$C = \frac{(-1)^3}{(0^3 - 2(-1)^3)} = \frac{1}{2}$$

$$\frac{y^3}{(x^3 - 2y^3)} = \frac{1}{2}$$

$$2y^3 = (x^3 - 2y^3)$$

$$2y^3 + 2y^3 - x^3 = 0$$

$$\rightarrow \text{Particular Solution}$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

2.3 TYPE 3: EXACT DIFFERENTIAL EQUATIONS

Consider some equations of the form

$$M(x, y)dx + N(x, y)dy = 0$$

in which separation of variables may not be possible. Suppose that a function $F(x, y)$ can be found that has for its differential the expression $Mdx + Ndy$; that is,

$$dF = Mdx + Ndy$$

Then certainly,

$$F(x, y) = c$$

$$dF = 0$$

$$Mdx + Ndy = 0$$

But, from calculus,

$$dF = \frac{\partial F}{\partial x} dx + \frac{\partial F}{\partial y} dy$$

so,

$$M = \frac{\partial F}{\partial x} \quad N = \frac{\partial F}{\partial y}$$

These two equations lead to,

$$\frac{\partial M}{\partial y} = \frac{\partial^2 F}{\partial x \partial y} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 F}{\partial y \partial x}$$

if $M, N, \partial M/\partial y$, and $\partial N/\partial x$ are continuous functions of x & y , then a necessary and sufficient condition that

$$M(x, y)dx + N(x, y)dy = 0$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \text{ then it is an EXACT DIFFERENTIAL EQUATION}$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

SOLUTIONS OF EXACT EQUATIONS

◆ FIRST METHOD

1. Let $M = \frac{\partial F}{\partial x}$ and let be equation (1)
2. Let $N = \frac{\partial F}{\partial y}$ and let be equation (2)
3. Integrate equation (1) partially with respect to x taking y as constant and let be equation (3)
4. Differentiate equation (3) partially with respect to y taking x as constant and let be equation (4)
5. Equation (4) is equal to Equation (2) and let be equation (5)
6. Integrate equation (5) and let be equation (6)
7. Substitute Equation (6) to Equation (3), $F = c$

◆ SECOND METHOD

$$\sum \int D.I + \int N.D.I = \int 0$$

- $D.I =$ Direct Integration (e.g., $f(x)dx$, $Q(x)dx$, $H(y)dy$, $\Phi(y)dy$)
 $N.D.I =$ Non-Direct Integrable (e.g., $f(x,y)dy$, $\Phi(x,y)dx$)

1. Arrange the equation into General Equation taking all of the direct integrable in M and N , choosing Non-Direct Integrable in M and N
2. Integrate $D.I$ and $N.D.I$. Integrate $N.D.I$ partially with respect to one particular variable taking one as constant.

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

1. Test each of the following equations for exactness and solve the equation. The equations that are not exact may, of course, be solved by methods discussed in the preceding sections.

1. $(x + 2y)dx + (2x + y)dy = 0$

$$M = x + 2y;$$

$$\frac{\partial M}{\partial y} = 2$$

$$N = 2x + y;$$

$$\frac{\partial N}{\partial x} = 2$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial x} = x + 2y \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = 2x + y \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (x + 2y) \partial x$$

$$F = \frac{x^2}{2} + 2xy + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = 2x + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$2x + \lambda'(y) = 2x + y$$

$$\lambda'(y) = y \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = y$$

$$\int d\lambda(y) = \int y dy$$

$$\lambda(y) = \frac{y^2}{2} + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$\frac{x^2}{2} + 2xy + \frac{y^2}{2} + C_2 = C_1$$

$$\frac{x^2}{2} + 2xy + \frac{y^2}{2} = C$$

$$x^2 + 4xy + y^2 = C$$

◆ Second Method

$$\sum \int D.I + \int N.D.I = \int 0$$

$D.I:$

$$x dx, y dy$$

$N.D.I:$

$$2y dx, 2x dy$$

Choose: $2y dx$

$$\int x dx + \int y dy + \int 2y dx = \int 0$$

$$\frac{x^2}{2} + \frac{y^2}{2} + 2xy = C$$

$$x^2 + y^2 + 4xy = C$$

$$x^2 + 4xy + y^2 = C$$

2. $(2xy - 3x^2)dx + (x^2 + 2y)dy = 0$

$$M = 2xy - 3x^2;$$

$$\frac{\partial M}{\partial y} = 2x$$

$$N = x^2 + 2y;$$

$$\frac{\partial N}{\partial x} = 2x$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial x} = 2xy - 3x^2 \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = x^2 + 2y \rightarrow (2)$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

Integrate Equation (1):

$$\int \partial F = \int (2xy - 3x^2) \partial x$$

$$F = x^2y - x^3 + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = x^2 + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$x^2 + \lambda'(y) = x^2 + 2y$$

$$\lambda'(y) = 2y \rightarrow (5)$$

Integrate Equation (5):

$$\lambda(y) = y^2$$

$$\int d\lambda(y) = \int 2y dy$$

$$\lambda(y) = y^2 + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$x^2y - x^3 + y^2 + C_2 = C_1$$

$$x^2y - x^3 + y^2 = C$$

Second Method

$$\sum \int D.I + \int N.D.I = 0$$

D.I:

$$-3x^2 dx, 2y dy$$

N.D.I:

$$2xy dx, x^2 dy$$

Choose: $2xy dx$

$$\int -3x^2 dx + \int 2y dy + \int 2xy dx = 0$$

$$-x^3 + y^2 + x^2y = C$$

$$x^2y - x^3 + y^2 = C$$

$$3. (6x + y^2)dx + y(2x - 3y)dy = 0$$

$$M = 6x + y^2;$$

$$\frac{\partial M}{\partial y} = 2y$$

$$N = 2xy - 3y^2;$$

$$\frac{\partial N}{\partial x} = 2y$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial x} = 6x + y^2 \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = 2xy - 3y^2 \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (6x + y^2) \partial x$$

$$F = 3x^2 + xy^2 + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = 2xy + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$2xy + \lambda'(y) = 2xy - 3y^2$$

$$\lambda'(y) = -3y^2 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda(y) = -y^3$$

$$\int d\lambda(y) = \int -3y^2 dy$$

$$\lambda(y) = -y^3 + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$F = 3x^2 + xy^2 - y^3 + C_2 = C_1$$

$$3x^2 + xy^2 - y^3 = C$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

Second Method

$$\sum \int D.I + \int N.D.I = 0$$

D.I:

$$6x dx, -3y^2 dy$$

N.D.I:

$$y^2 dx, 2xy dy$$

Choose: $y^2 dx$

$$\int 6x dx - \int 3y^2 dy + \int y^2 dx = 0$$

$$3x^2 - y^3 + xy^2 = C$$

$$3x^2 + xy^2 - y^3 = C$$

4.

$$(y^2 - 2xy + 6x)dx - (x^2 - 2xy + 2)dy = 0$$

$$M = y^2 - 2xy + 6x;$$

$$\frac{\partial M}{\partial y} = 2y - 2x$$

$$N = -x^2 + 2xy - 2;$$

$$\frac{\partial N}{\partial x} = -2x + 2y$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial x} = y^2 - 2xy + 6x \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = -x^2 + 2xy - 2 \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (y^2 - 2xy + 6x) \partial x$$

$$F = xy^2 - x^2y + 3x^2 + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = 2xy - x^2 + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$2xy - x^2 + \lambda'(y) = -x^2 + 2xy - 2$$

$$\lambda'(y) = -2 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda(y) = -2y$$

$$\int d\lambda(y) = \int -2 dy$$

$$\lambda(y) = -2y + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$xy^2 - x^2y + 3x^2 - 2y + C_2 = C_1$$

$$xy^2 - x^2y + 3x^2 - 2y = C$$

Second Method

$$\sum \int D.I + \int N.D.I = 0$$

D.I:

$$6x dx, -2 dy$$

N.D.I:

$$(y^2 - 2xy) dx, (-x^2 + 2xy) dy$$

Choose: $(y^2 - 2xy) dx$

$$\int 6x dx - \int 2 dy + \int (y^2 - 2xy) dx = 0$$

$$3x^2 - 2y + xy^2 - x^2y = C$$

$$xy^2 - x^2y + 3x^2 - 2y = C$$

5.

$$(2xy - y)dx + (x^2 + x)dy = 0$$

$$M = 2xy - y;$$

$$\frac{\partial M}{\partial y} = 2x - 1$$

$$N = x^2 + x;$$

$$\frac{\partial N}{\partial x} = 2x + 1$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \therefore \text{NOT EXACT DE!}$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

Try Variable Separable!

$$y(2x-1)dx + x(x+1)dy = 0$$

$$\frac{(2x-1)dx}{x(x+1)} + \frac{dy}{y} = 0$$

$$\int \frac{(2x-1)dx}{x(x+1)} + \int \frac{dy}{y} = \int 0$$

Considering:

$$\int \frac{(2x-1)dx}{x(x+1)}$$

Using Integration by Partial Fraction:

$$\frac{(2x-1)}{x(x+1)} = \frac{A}{x} + \frac{B}{(x+1)}$$

$$2x-1 = A(x+1) + Bx$$

$$\text{If } \begin{matrix} x=0 & : & A=-1 \\ x=-1 & : & B=3 \end{matrix}$$

Then,

$$\int \frac{(2x-1)dx}{x(x+1)} = -\int \frac{dx}{x} + 3 \int \frac{dx}{(x+1)} = -\ln x + 3 \ln(x+1)$$

Thus,

$$-\ln x + 3 \ln(x+1) + \ln y = \ln C$$

$$\frac{y(x+1)^3}{x} = C$$

$$y(x+1)^3 = Cx$$

6.

$$v(2uv^2-3)du + (3u^2v^2-3u+4v)dv = 0$$

$$M = 2uv^3 - 3v;$$

$$\frac{\partial M}{\partial v} = 6uv^2 - 3$$

$$N = 3u^2v^2 - 3u + 4v;$$

$$\frac{\partial N}{\partial u} = 6uv^2 - 3$$

$$\frac{\partial M}{\partial v} = \frac{\partial N}{\partial u} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial u} = 2uv^3 - 3v \rightarrow (1)$$

$$N = \frac{\partial F}{\partial v} = 3u^2v^2 - 3u + 4v \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (2uv^3 - 3v) \partial u$$

$$F = u^2v^3 - 3uv + \lambda(v) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial v} = 3u^2v^2 - 3u + \lambda'(v) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$3u^2v^2 - 3u + \lambda'(v) = 3u^2v^2 - 3u + 4v$$

$$\lambda'(v) = 4v \rightarrow (5)$$

Integrate Equation (5):

$$\lambda(v) = 4v$$

$$\int d\lambda(v) = \int 4v dv$$

$$\lambda(v) = 2v^2 + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$u^2v^3 - 3uv + 2v^2 + C_2 = C_1$$

$$u^2v^3 - 3uv + 2v^2 = C$$

Second Method

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:

$$4v dv$$

N.D.I:

$$(2uv^3 - 3v) du, (3u^2v^2 - 3u) dv$$

$$\text{Choose: } (2uv^3 - 3v) du$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

$$\int 4v dv + \int (2uv^3 - 3v) du = \int 0$$

$$2v^2 + u^2v^3 - 3uv = C$$

$$u^2v^3 - 3uv + 2v^2 = C$$

7.

$$(\cos 2y - 3x^2y^2)dx + (\cos 2y - 2x \sin 2y - 2x^3y)dy = 0$$

$$M = \cos 2y - 3x^2y^2;$$

$$\frac{\partial M}{\partial y} = -2 \sin 2y - 6x^2y$$

$$N = \cos 2y - 2x \sin 2y - 2x^3y;$$

$$\frac{\partial N}{\partial x} = -2 \sin 2y - 6x^2y$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial x} = \cos 2y - 3x^2y^2 \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = \cos 2y - 2x \sin 2y - 2x^3y \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (\cos 2y - 3x^2y^2) \partial x$$

$$F = x \cos 2y - x^3y^2 + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = -2x \sin 2y - 2x^3y + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$-2x \sin 2y - 2x^3y + \lambda'(y) = \cos 2y - 2x \sin 2y - 2x^3y$$

$$\lambda'(y) = \cos 2y \rightarrow (5)$$

Integrate Equation (5):

$$\lambda(y) = \sin 2y$$

$$\int d\lambda(y) = \int \cos 2y dy$$

$$\lambda(y) = \frac{1}{2} \sin 2y + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$x \cos 2y - x^3y^2 + \frac{1}{2} \sin 2y + C_2 = C_1$$

$$x \cos 2y - x^3y^2 + \frac{1}{2} \sin 2y = C$$

$$\frac{1}{2} \sin 2y + x \cos 2y - x^3y^2 = C$$

Second Method

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:

$$\cos 2y dy$$

N.D.I:

$$(\cos 2y - 3x^2y^2) dx, (-2x \sin 2y - 2x^3y) dy$$

Choose: $(\cos 2y - 3x^2y^2) dx$

$$\int \cos 2y dy + \int (\cos 2y - 3x^2y^2) dx = \int 0$$

$$\frac{1}{2} \sin 2y + x \cos 2y - x^3y^2 = C$$

8.

$$(1+y^2)dx + (x^2y+y)dy = 0$$

$$M = 1+y^2;$$

$$\frac{\partial M}{\partial y} = 2y$$

$$N = x^2y+y;$$

$$\frac{\partial N}{\partial x} = 2xy$$

$$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \therefore \text{NOT EXACT DE!}$$

Try Variable Separable!

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

$$(1+y^2)dx + y(x^2+1)dy = 0$$

$$\frac{dx}{(x^2+1)} + \frac{ydy}{(1+y^2)} = 0$$

$$\int \frac{dx}{(x^2+1)} + \int \frac{ydy}{(1+y^2)} = \int 0$$

$$\text{Arctan } x + \frac{1}{2} \ln(1+y^2) = C$$

$$2\text{Arctan } x + \ln(1+y^2) = C$$

9. $(1+y^2+xy^2)dx + (x^2y+y+2xy)dy = 0$

$$M = 1+y^2+xy^2;$$

$$\frac{\partial M}{\partial y} = 2xy+2y$$

$$N = x^2y+y+2xy;$$

$$\frac{\partial N}{\partial x} = 2xy+2y$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial x} = 1+y^2+xy^2 \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = x^2y+y+2xy \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (1+y^2+xy^2) dx$$

$$F = x + xy^2 + \frac{x^2y^2}{2} + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = 2xy + x^2y + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$2xy + x^2y + \lambda'(y) = x^2y + y + 2xy$$

$$\lambda'(y) = y \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = y$$

$$\int d\lambda(y) = \int y dy$$

$$\lambda(y) = \frac{y^2}{2} + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$x + xy^2 + \frac{x^2y^2}{2} + \frac{y^2}{2} + C_2 = C_1$$

$$x + xy^2 + \frac{x^2y^2}{2} + \frac{y^2}{2} = C$$

$$2x + 2xy^2 + x^2y^2 + y^2 = C$$

$$2x + y^2(2x + x^2 + 1) = C$$

$$2x + y^2(1+x)^2 = C$$

◆ Second Method

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:

$$1dx, ydy$$

N.D.I:

$$(y^2+xy^2)dx, (x^2y+2xy)dy$$

Choose: $(y^2+xy^2)dx$

$$\int dx + \int ydy + \int (y^2+xy^2)dx = \int 0$$

$$x + \frac{y^2}{2} + xy^2 + \frac{x^2y^2}{2} = C$$

$$2x + y^2 + 2xy^2 + x^2y^2 = C$$

$$2x + y^2(1+2x+x^2) = C$$

$$2x + y^2(1+x)^2 = C$$

10.

$$(w^3+wz^2-z)dw + (z^3+w^2z-w)dz = 0$$

$$M = w^3+wz^2-z;$$

$$\frac{\partial M}{\partial z} = 2wz-1$$

$$N = z^3+w^2z-w$$

$$\frac{\partial N}{\partial w} = 2wz-1$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

$$\frac{\partial M}{\partial z} = \frac{\partial N}{\partial w} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial w} = w^3+wz^2-z \rightarrow (1)$$

$$N = \frac{\partial F}{\partial z} = z^3+w^2z-w \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (w^3+wz^2-z)dw$$

$$F = \frac{w^4}{4} + \frac{w^2z^2}{2} - zw + \lambda(z) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial z} = w^2z - w + \lambda'(z) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$w^2z - w + \lambda'(z) = z^3 + w^2z - w$$

$$\lambda'(z) = z^3 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(z) = z^3$$

$$\int d\lambda(z) = \int z^3 dz$$

$$\lambda(z) = \frac{z^4}{4} + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$\frac{w^4}{4} + \frac{w^2z^2}{2} - zw + \frac{z^4}{4} + C_2 = C_1$$

$$\frac{w^4}{4} + \frac{w^2z^2}{2} - wz + \frac{z^4}{4} = C$$

$$w^4 + 2w^2z^2 - 4wz + z^4 = C$$

$$w^4 + 2w^2z^2 + z^4 = 4wz + C$$

$$(w^2 + z^2)^2 = 4wz + C$$

◆ Second Method

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:

$$w^3dw, z^3dz$$

N.D.I:

$$(wz^2-z)dw, (w^2z-w)dz$$

Choose: $(wz^2-z)dw$

$$\int w^3dw + \int z^3dz + \int (wz^2-z)dw = \int 0$$

$$\frac{w^4}{4} + \frac{z^4}{4} + \frac{w^2z^2}{2} - wz = C$$

$$w^4 + z^4 + 2w^2z^2 - 4wz = C$$

$$w^4 + 2w^2z^2 + z^4 = 4wz + C$$

$$(w^2 + z^2)^2 = 4wz + C$$

11. $(2xy - \tan y)dx + (x^2 - x \sec^2 y)dy = 0$

$$M = 2xy - \tan y;$$

$$\frac{\partial M}{\partial y} = 2x - \sec^2 y$$

$$N = x^2 - x \sec^2 y;$$

$$\frac{\partial N}{\partial x} = 2x - \sec^2 y$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial x} = 2xy - \tan y \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = x^2 - x \sec^2 y \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (2xy - \tan y) dx$$

$$F = x^2y - x \tan y + \lambda(y) = C_1 \rightarrow (3)$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = x^2 - x \sec^2 y + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$x^2 - x \sec^2 y + \lambda'(y) = x^2 - x \sec^2 y$$

$$\lambda'(y) = 0 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda(y) = 0$$

$$\int d\lambda(y) = \int 0 dy$$

$$\lambda(y) = C_1 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$x^2 y - x \tan y + C_1 = C_1$$

$$x^2 y - x \tan y = C$$

◆ Second Method

$$\sum \int D.I + \int N.D.I = 0$$

D.I:
None

N.D.I:

$$(2xy - \tan y)dx, (x^2 - x \sec^2 y)dy$$

Choose: $(2xy - \tan y)dx$

$$\int (2xy - \tan y)dx = \int 0$$

$$x^2 y - x \tan y = C$$

12. $(\cos x \cos y - \cot x)dx$
 $- \sin x \sin y dy = 0$

$$M = \cos x \cos y - \cot x;$$

$$\frac{\partial M}{\partial y} = -\cos x \sin y$$

$$N = -\sin x \sin y;$$

$$\frac{\partial N}{\partial x} = -\cos x \sin y$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial x} = \cos x \cos y - \cot x \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = -\sin x \sin y \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (\cos x \cos y - \cot x) \partial x$$

$$F = \sin x \cos y - \ln \sin x + \lambda(y) = C_1$$

$$\rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = -\sin x \sin y + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$-\sin x \sin y + \lambda'(y) = -\sin x \sin y$$

$$\lambda'(y) = 0 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda(y) = 0$$

$$\int d\lambda(y) = \int 0 dy$$

$$\lambda(y) = C_1 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$\sin x \cos y - \ln \sin x + C_1 = C_1$$

$$\sin x \cos y - \ln \sin x = \ln C$$

$$\sin x \cos y = \ln C + \ln \sin x$$

$$\sin x \cos y = \ln[C \sin x]$$

◆ Second Method

$$\sum \int D.I + \int N.D.I = 0$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

D.I:

$$-\cot x dx$$

N.D.I:

$$\cos x \cos y dx, -\sin x \sin y dy$$

Choose: $-\sin x \sin y dy$

$$\int -\cot x dx - \int \sin x \sin y dy = \int 0$$

$$-\ln \sin x + \sin x \cos y = \ln C$$

$$\sin x \cos y = \ln C + \ln \sin x$$

$$\sin x \cos y = \ln[C \sin x]$$

13. $(r + \sin \theta - \cos \theta)dr$
 $+ r(\sin \theta + \cos \theta)d\theta = 0$

$$M = r + \sin \theta - \cos \theta;$$

$$\frac{\partial M}{\partial \theta} = \cos \theta + \sin \theta$$

$$N = r \sin \theta + r \cos \theta;$$

$$\frac{\partial N}{\partial r} = \sin \theta + \cos \theta$$

$$\frac{\partial M}{\partial \theta} = \frac{\partial N}{\partial r} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial r} = r + \sin \theta - \cos \theta \rightarrow (1)$$

$$N = \frac{\partial F}{\partial \theta} = r \sin \theta + r \cos \theta \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (r + \sin \theta - \cos \theta) \partial r$$

$$F = \frac{r^2}{2} + r \sin \theta - r \cos \theta + \lambda(\theta) = C_1$$

$$\rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial \theta} = r \cos \theta + r \sin \theta + \lambda'(\theta) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$r \cos \theta + r \sin \theta + \lambda'(\theta)$$

$$= r \sin \theta + r \cos \theta$$

$$\lambda'(\theta) = 0 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(\theta) = 0$$

$$\int d\lambda(\theta) = \int 0 d\theta$$

$$\lambda(\theta) = C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$\frac{r^2}{2} + r \sin \theta - r \cos \theta + C_2 = C_1$$

$$\frac{r^2}{2} + r \sin \theta - r \cos \theta = C$$

$$r^2 + 2r \sin \theta - 2r \cos \theta = C$$

$$r^2 + 2r(\sin \theta - \cos \theta) = C$$

◆ Second Method

$$\sum \int D.I + \int N.D.I = 0$$

D.I:
 rdr

N.D.I:

$$(\sin \theta - \cos \theta)dr, (r \sin \theta + r \cos \theta)d\theta$$

Choose: $(\sin \theta - \cos \theta)dr$

$$\int r dr + \int (\sin \theta - \cos \theta)dr = \int 0$$

$$\frac{r^2}{2} + r \sin \theta - r \cos \theta = C$$

$$r^2 + 2r \sin \theta - 2r \cos \theta = C$$

$$r^2 + 2r(\sin \theta - \cos \theta) = C$$

14. $x(3xy - 4y^3 + 6)dx$
 $+ (x^3 - 6x^2y^2 - 1)dy = 0$

$$M = 3x^2y - 4xy^3 + 6x;$$

$$\frac{\partial M}{\partial y} = 3x^2 - 12xy^2$$

$$N = x^3 - 6x^2y^2 - 1;$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

$$\frac{\partial N}{\partial x} = 3x^2 - 12xy^2$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial x} = 3x^2y - 4xy^3 + 6x \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = x^3 - 6x^2y^2 - 1 \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (3x^2y - 4xy^3 + 6x) \partial x \\ F = x^3y - 2x^2y^3 + 3x^2 + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = x^3 - 6x^2y^2 + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$x^3 - 6x^2y^2 + \lambda'(y) = x^3 - 6x^2y^2 - 1 \\ \lambda'(y) = -1 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = -1 \\ \int d\lambda(y) = - \int dy \\ \lambda(y) = y + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$x^3y - 2x^2y^3 + 3x^2 - y + C_2 = C_1 \\ x^3y - 2x^2y^3 + 3x^2 - y = C$$

Second Method

$$\sum \int D.I + \int N.D.I = 0$$

$$D.I: \\ 6x dx, -dy$$

N.D.I:

$$(3x^2y - 4xy^3)dx, (x^3 - 6x^2y^2)dy$$

Choose: $(3x^2y - 4xy^3)dx$

$$\int 6x dx - \int dy + \int (3x^2y - 4xy^3) dx \\ = \int 0$$

$$3x^2 - y + x^3y - 2x^2y^3 = C \\ x^3y - 2x^2y^3 + 3x^2 - y = C$$

$$15. (\sin \theta - 2r \cos^2 \theta) dr \\ + r \cos \theta (2r \sin \theta + 1) d\theta = 0$$

$$M = \sin \theta - 2r \cos^2 \theta; \\ \frac{\partial M}{\partial \theta} = \cos \theta + 4r \sin \theta \cos \theta$$

$$N = 2r^2 \cos \theta \sin \theta + r \cos \theta; \\ \frac{\partial N}{\partial r} = 4r \sin \theta \cos \theta + \cos \theta$$

$$\frac{\partial M}{\partial \theta} = \frac{\partial N}{\partial r} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial r} = \sin \theta - 2r \cos^2 \theta \rightarrow (1)$$

$$N = \frac{\partial F}{\partial \theta} = 2r^2 \cos \theta \sin \theta + r \cos \theta \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (\sin \theta - 2r \cos^2 \theta) \partial r \\ F = r \sin \theta - r^2 \cos^2 \theta + \lambda(\theta) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial \theta} = r \cos \theta + 2r^2 \sin \theta \cos \theta + \lambda'(\theta) \\ \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$r \cos \theta + 2r^2 \sin \theta \cos \theta + \lambda'(\theta) \\ = 2r^2 \cos \theta \sin \theta + r \cos \theta$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

$$\lambda'(\theta) = 0 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(\theta) = 0$$

$$\int d\lambda(\theta) = \int 0 d\theta$$

$$\lambda(\theta) = C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$r \sin \theta - r^2 \cos^2 \theta + C_2 = C_1 \\ r \sin \theta - r^2 \cos^2 \theta = C$$

Second Method

$$\sum \int D.I + \int N.D.I = 0$$

D.I:

None

N.D.I:

$$(\sin \theta - 2r \cos^2 \theta) dr, (2r^2 \cos \theta \sin \theta \\ + r \cos \theta) d\theta$$

Choose: $(\sin \theta - 2r \cos^2 \theta) dr$

$$\int (\sin \theta - 2r \cos^2 \theta) dr = \int 0 \\ r \sin \theta - r^2 \cos^2 \theta = C$$

$$16. [2x + y \cos(xy)] dx + x \cos(xy) dy = 0$$

$$M = 2x + y \cos(xy);$$

$$\frac{\partial M}{\partial y} = -xy \sin(xy) + \cos(xy)$$

$$N = x \cos(xy);$$

$$\frac{\partial N}{\partial x} = -xy \sin(xy) + \cos(xy)$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial x} = 2x + y \cos(xy) \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = x \cos(xy) \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (2x + y \cos(xy)) \partial x \\ F = x^2 + \sin(xy) + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = x \cos(xy) + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$x \cos(xy) + \lambda'(y) = x \cos(xy) \\ \lambda'(y) = 0 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = 0 \\ \int d\lambda(y) = \int 0 dy \\ \lambda(y) = C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$x^2 + \sin(xy) + C_2 = C_1 \\ x^2 + \sin(xy) = C$$

Second Method

$$\sum \int D.I + \int N.D.I = 0$$

D.I:

$$2x dx$$

N.D.I:

$$y \cos(xy) dx, x \cos(xy) dy$$

Choose: $y \cos(xy) dx$

$$\int 2x dx + \int y \cos(xy) dx = \int 0 \\ x^2 + \sin(xy) = C$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

17. $2xydx + (y^2 + x^2)dy = 0$

$M = 2xy,$

$\frac{\partial M}{\partial y} = 2x$

$N = y^2 + x^2;$

$\frac{\partial N}{\partial x} = 2x$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$

◆ First Method

$M = \frac{\partial F}{\partial x} = 2xy \rightarrow (1)$

$N = \frac{\partial F}{\partial y} = y^2 + x^2 \rightarrow (2)$

Integrate Equation (1):

$\int \partial F = \int (2xy) \partial x$

$F = x^2y + \lambda(y) = C_1 \rightarrow (3)$

Differentiate Equation (3):

$\frac{\partial F}{\partial y} = x^2 + \lambda'(y) \rightarrow (4)$

Equate Equation (4) and Equation (2):

$x^2 + \lambda'(y) = y^2 + x^2$

$\lambda'(y) = y^2 \rightarrow (5)$

Integrate Equation (5):

$\lambda'(y) = y^2$

$\int d\lambda(y) = \int y^2 dy$

$\lambda(y) = \frac{y^3}{3} + C_2 \rightarrow (6)$

Substitute Equation (6) to Equation (3):

$x^2y + \frac{y^3}{3} + C_2 = C_1$

$x^2y + \frac{y^3}{3} = C$

$3x^2y + y^3 = C$

$y(3x^2 + y^2) = C$

◆ Second Method

$\sum \int D.I + \int N.D.I = \int 0$

D.I:

$y^2 dy$

N.D.I:

$2xydx, x^2dy$

Choose: x^2dy

$\int y^2 dy + \int x^2 dy = \int 0$

$\frac{y^3}{3} + x^2y = C$

$3x^2y + y^3 = C$

$y(3x^2 + y^2) = C$

18. $2xydx + (y^2 - x^2)dy = 0$

$M = 2xy,$

$\frac{\partial M}{\partial y} = 2x$

$N = y^2 - x^2,$

$\frac{\partial N}{\partial x} = -2x$

$\frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x} \therefore \text{NOT EXACT DE!}$

Try Homogeneous Equation

$\therefore$ Homogeneous, 2nd degree

Let: $x = vy$
 $dx = vdy + ydv$

Substitute:

$2vy^2(vdy + ydv) + (y^2 - v^2y^2)dy = 0$

$2v(vdy + ydv) + (1 - v^2)dy = 0$

$(1 - v^2 + 2v^2)dy + 2vydv = 0$

$(1 + v^2)dy + 2vydv = 0$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

$\frac{dy}{y} + \frac{2v dv}{(1 + v^2)} = 0$

$\int \frac{dy}{y} + \int \frac{2v dv}{(1 + v^2)} = \int 0$

$\ln y + \ln(1 + v^2) = \ln C$

$y(1 + v^2) = C$

But: $v = x/y$

$y\left(1 + \frac{x^2}{y^2}\right) = C$

$y\left(\frac{y^2 + x^2}{y^2}\right) = C$

$x^2 + y^2 = Cy$

19. $[2xy \cos(x^2) - 2xy + 1]dx + [\sin(x^2) - x^2]dy = 0$

$M = 2xy \cos(x^2) - 2xy + 1;$

$\frac{\partial M}{\partial y} = 2x \cos(x^2) - 2x$

$N = \sin(x^2) - x^2;$

$\frac{\partial N}{\partial x} = 2x \cos(x^2) - 2x$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$

◆ First Method

$M = \frac{\partial F}{\partial x} = 2xy \cos(x^2) - 2xy + 1 \rightarrow (1)$

$N = \frac{\partial F}{\partial y} = \sin(x^2) - x^2 \rightarrow (2)$

Integrate Equation (1):

$\int \partial F = \int (2xy \cos(x^2) - 2xy + 1) \partial x$

$F = y \sin(x^2) - x^2y + x + \lambda(y) = C_1$

$\rightarrow (3)$

Differentiate Equation (3):

$\frac{\partial F}{\partial y} = \sin(x^2) - x^2 + \lambda'(y) \rightarrow (4)$

Equate Equation (4) and Equation (2):

$\sin(x^2) - x^2 + \lambda'(y) = \sin(x^2) - x^2$

$\lambda'(y) = 0 \rightarrow (5)$

Integrate Equation (5):

$\lambda'(y) = 0$

$\int d\lambda(y) = \int 0 dy$

$\lambda(y) = C_2 \rightarrow (6)$

Substitute Equation (6) to Equation (3):

$y \sin(x^2) - x^2y + x + C_2 = C_1$

$y \sin(x^2) - x^2y + x = C$

$y[\sin(x^2) - x^2] = C - x$

◆ Second Method

$\sum \int D.I + \int N.D.I = \int 0$

D.I:

$1dx$

N.D.I:

$[2xy \cos(x^2) - 2xy]dx, [\sin(x^2) - x^2]dy$

Choose: $[\sin(x^2) - x^2]dy$

$\int dx + \int [\sin(x^2) - x^2]dy = \int 0$

$x + y \sin(x^2) - yx^2 = C$

$y \sin(x^2) - x^2y + x = C$

$y[\sin(x^2) - x^2] = C - x$

20. $(2x - 3y)dx + (2y - 3x)dy = 0$

$M = 2x - 3y;$

$\frac{\partial M}{\partial y} = -3$

$N = 2y - 3x;$

$\frac{\partial N}{\partial x} = -3$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$

◆ First Method

$$M = \frac{\partial F}{\partial x} = 2x - 3y \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = 2y - 3x \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (2x - 3y) \partial x$$

$$F = x^2 - 3xy + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = -3x + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$-3x + \lambda'(y) = 2y - 3x$$

$$\lambda'(y) = 2y \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = 2y$$

$$\int d\lambda(y) = \int 2y dy$$

$$\lambda(y) = y^2 + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$x^2 - 3xy + y^2 + C_2 = C_1$$

$$x^2 + y^2 - 3xy = C$$

◆ Second Method

$$\sum \int D.I + \int N.D.I = \int 0$$

$$D.I:$$

$$2x dx, 2y dy$$

$$N.D.I:$$

$$-3y dx, -3x dy$$

Choose: $-3y dx$

$$\int 2x dx + \int 2y dy - \int 3y dx = \int 0$$

$$x^2 + y^2 - 3xy = C$$

21. Do exercise 20 by a second method
 $(2x - 3y)dx + (2y - 3x)dy = 0$

Try Homogeneous Equation

∴ Homogeneous, 1st degree

Let: $y = vx$
 $dy = v dx + x dv$

Substitute:

$$(2x - 3vx)dx + (2vx - 3x)(v dx + x dv) = 0$$

$$(2 - 3v)dx + (2v - 3)(v dx + x dv) = 0$$

$$(2 - 3v)dx + (2v^2 dx + 2vxdv - 3vdx - 3xdv) = 0$$

$$(2 - 3v + 2v^2 - 3v)dx + (2vx - 3x)dv = 0$$

$$(2 - 6v + 2v^2)dx + x(2v - 3)dv = 0$$

$$\frac{dx}{x} + \frac{(2v - 3)dv}{(2 - 6v + 2v^2)} = 0$$

$$\int \frac{dx}{x} + \frac{1}{2} \int \frac{2(2v - 3)dv}{(2 - 6v + 2v^2)} = \int 0$$

$$\ln x + \frac{1}{2} \ln(2 - 6v + 2v^2) = \ln C$$

$$2 \ln x + \ln(2 - 6v + 2v^2) = \ln C$$

$$x^2(2 - 6v + 2v^2) = C$$

But: $v = y/x$

$$x^2 \left(2 - 6 \frac{y}{x} + 2 \frac{y^2}{x^2} \right) = C$$

$$2x^2 - 6xy + 2y^2 = C$$

$$x^2 - 3xy + y^2 = C$$

$$x^2 + y^2 - 3xy = C$$

22.

$$(xy^2 + y - x)dx + x(xy + 1)dy = 0$$

$$M = xy^2 + y - x;$$

$$\frac{\partial M}{\partial y} = 2xy + 1$$

$$N = x^2y + x;$$

$$\frac{\partial N}{\partial x} = 2xy + 1$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial x} = xy^2 + y - x \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = x^2y + x \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (xy^2 + y - x) \partial x$$

$$F = \frac{x^2y^2}{2} + xy - \frac{x^2}{2} + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = x^2y + x + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$x^2y + x + \lambda'(y) = x^2y + x$$

$$\lambda'(y) = 0 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = 0$$

$$\int d\lambda(y) = \int 0 dy$$

$$\lambda(y) = C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$\frac{x^2y^2}{2} + xy - \frac{x^2}{2} + C_2 = C_1$$

$$\frac{x^2y^2}{2} + xy - \frac{x^2}{2} = C$$

$$x^2y^2 + 2xy - x^2 = C$$

◆ Second Method

$$\sum \int D.I + \int N.D.I = \int 0$$

$$D.I:$$

$$-x dx$$

$$N.D.I:$$

$$(xy^2 + y)dx, (x^2y + x)dy$$

Choose: $(xy^2 + y)dx$

$$- \int x dx + \int (xy^2 + y) dx = \int 0$$

$$-\frac{x^2}{2} + \frac{x^2y^2}{2} + xy = C$$

$$x^2y^2 + 2xy - x^2 = C$$

23. $3y(x^2 - 1)dx + (x^3 + 8y - 3x)dy = 0$,
 when $x = 0, y = 1$

$$M = 3x^2y - 3y;$$

$$\frac{\partial M}{\partial y} = 3x^2 - 3$$

$$N = x^3 + 8y - 3x;$$

$$\frac{\partial N}{\partial x} = 3x^2 - 3$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial x} = 3x^2y - 3y \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = x^3 + 8y - 3x \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (3x^2y - 3y) \partial x$$

$$F = x^3y - 3xy + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = x^3 - 3x + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$x^3 - 3x + \lambda'(y) = x^3 + 8y - 3x$$

$$\lambda'(y) = 8y \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = 8y$$

$$\int d\lambda(y) = \int 8y dy$$

$$\lambda(y) = 4y^2 + C_2 \rightarrow (6)$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

Substitute Equation (6) to Equation (3):

$$x^3y - 3xy + 4y^2 + C_2 = C_1$$

$$x^3y - 3xy + 4y^2 = C \rightarrow \text{General Solution}$$

◆ Second Method

$$\sum \int D.I + \int N.D.I = 0$$

D.I:

$$8ydy$$

N.D.I:

$$(3x^2y - 3y)dx, (x^3 - 3x)dy$$

Choose: $(x^3 - 3x)dy$

$$\int 8ydy + \int (x^3 - 3x)dy = 0$$

$$4y^2 + x^3y - 3xy = C$$

$$x^3y - 3xy + 4y^2 = C \rightarrow \text{General Solution}$$

when $x = 0, y = 1$

$$C = 0^3(1) - 3(0)(1) + 4(1)^2 = 4$$

Thus,

$$x^3y - 3xy + 4y^2 = 4$$

$$xy(x^2 - 3) = 4(1 - y^2)$$

→ Particular Solution

24.

$$(1 - xy)^{-2} dx + [y^2 + x^2(1 - xy)^{-2}] dy = 0,$$

when $x = 4, y = 1$

$$M = (1 - xy)^{-2};$$

$$\frac{\partial M}{\partial y} = -2(1 - xy)^{-3}(-x)$$

$$\frac{\partial M}{\partial y} = 2x(1 - xy)^{-3}$$

$$N = y^2 + x^2(1 - xy)^{-2};$$

$$\frac{\partial N}{\partial x} = x^2[2y(1 - xy)^{-3}]$$

$$+ (1 - xy)^{-2}(2x)$$

$$\frac{\partial N}{\partial x} = 2x^2y(1 - xy)^{-3}$$

$$+ 2x(1 - xy)^{-2}$$

$$\frac{\partial N}{\partial x} = \frac{2x^2y}{(1 - xy)^3} + \frac{2x}{(1 - xy)^2}$$

$$\frac{\partial N}{\partial x} = \frac{2x^2y + 2x(1 - xy)}{(1 - xy)^3}$$

$$\frac{\partial N}{\partial x} = \frac{2x^2y + 2x - 2x^2y}{(1 - xy)^3}$$

$$\frac{\partial N}{\partial x} = \frac{2x}{(1 - xy)^3}$$

$$\frac{\partial N}{\partial x} = 2x(1 - xy)^{-3}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial x} = (1 - xy)^{-2} \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = y^2 + x^2(1 - xy)^{-2} \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (1 - xy)^{-2} \partial x$$

$$F = \frac{1}{y}(1 - xy)^{-1} + \lambda(y) = C_1$$

$$F = \frac{1}{y(1 - xy)} + \lambda(y) = C_1 \rightarrow (3)$$

$$F = \frac{1}{(y - xy^2)} + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = -1(y - xy^2)^{-2}(1 - 2xy) + \lambda'(y)$$

$$\rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$-1(y - xy^2)^{-2}(1 - 2xy) + \lambda'(y) = y^2 + x^2(1 - xy)^{-2}$$

$$\lambda'(y) = y^2 + x^2(1 - xy)^{-2}$$

$$+ (y - xy^2)^{-2}(1 - 2xy)$$

$$\lambda'(y) = \frac{y^4(1 - xy)^2 + x^2y^2 + (1 - 2xy)}{y^2(1 - xy)^2}$$

$$\lambda'(y) = \frac{y^4(1 - xy)^2 + x^2y^2 - 2xy + 1}{y^2(1 - xy)^2}$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

$$\lambda'(y) = \frac{y^4(1 - xy)^2 + (1 - xy)^2}{y^2(1 - xy)^2}$$

$$\lambda'(y) = \frac{(1 - xy)^2(y^4 + 1)}{y^2(1 - xy)^2}$$

$$\lambda'(y) = \frac{y^4 + 1}{y^2}$$

$$\lambda'(y) = y^2 + \frac{1}{y^2} \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = y^2 + \frac{1}{y^2}$$

$$\int d\lambda(y) = \int (y^2 + \frac{1}{y^2}) dy$$

$$\lambda(y) = \frac{y^3}{3} - \frac{1}{y} + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$\frac{1}{(y - xy^2)} + \frac{y^3}{3} - \frac{1}{y} + C_2 = C_1$$

$$\frac{1}{(y - xy^2)} + \frac{y^3}{3} - \frac{1}{y} = C$$

→ General Solution

when $x = 4, y = 1$

$$C = \frac{1}{(1 - 4(1)^2)} + \frac{(1)^3}{3} - \frac{1}{1} = -1$$

Thus,

$$\frac{1}{(y - xy^2)} + \frac{y^3}{3} - \frac{1}{y} = -1$$

$$3y + y^4(y - xy^2) - 3(y - xy^2) = -1[3y(y - xy^2)]$$

$$= -1[3y^2 - 3xy^3]$$

$$3y + y^5 - xy^6 - 3y + 3xy^3 = -3y^2 + 3xy^3$$

$$y^5 - xy^6 + 3xy^3 = -3y^2 + 3xy^3$$

$$y^3 - xy^4 + 3x = -3 + 3xy$$

$$xy^4 - y^3 + 3xy - 3x - 3 = 0$$

→ Particular Solution

25.

$$(ye^{xy} - 2y^3)dx + (xe^{xy} - 6xy^2 - 2y)dy = 0;$$

when $x = 0, y = 2$

$$M = ye^{xy} - 2y^3;$$

$$\frac{\partial M}{\partial y} = y(xe^{xy}) + e^{xy} - 6y^2$$

$$\frac{\partial M}{\partial y} = xye^{xy} + e^{xy} - 6y^2$$

$$N = xe^{xy} - 6xy^2 - 2y;$$

$$\frac{\partial N}{\partial x} = x(ye^{xy}) + e^{xy} - 6y^2$$

$$\frac{\partial N}{\partial x} = xye^{xy} + e^{xy} - 6y^2$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

◆ First Method

$$M = \frac{\partial F}{\partial x} = ye^{xy} - 2y^3 \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = xe^{xy} - 6xy^2 - 2y \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (ye^{xy} - 2y^3) \partial x$$

$$F = e^{xy} - 2xy^3 + \lambda(y) = C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = xe^{xy} - 6xy^2 + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$xe^{xy} - 6xy^2 + \lambda'(y) = xe^{xy} - 6xy^2 - 2y$$

$$\lambda'(y) = -2y \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = -2y$$

$$\int d\lambda(y) = - \int 2y dy$$

$$\lambda(y) = -y^2 + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$e^{xy} - 2xy^3 - y^2 + C_2 = C_1$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

$$e^{xy} - 2xy^3 - y^2 = C \rightarrow \text{General Solution}$$

Second Method

$$\int D.I. + \int N.D.I. = 0$$

D.I:

$$-2ydy$$

N.D.I:

$$(ye^{xy} - 2y^3)dx, (xe^{xy} - 6xy^2 - 2y)dy$$

$$\text{Choose: } (ye^{xy} - 2y^3)dx$$

$$-\int 2ydy + \int (ye^{xy} - 2y^3)dx = 0$$

$$-y^2 + e^{xy} - 2xy^3 = C$$

$$e^{xy} - 2xy^3 - y^2 = C \rightarrow \text{General Solution}$$

$$\text{when } x = 0, y = 2$$

$$C = e^{(0)(2)} - 2(0)(2)^3 - (2)^2 = -3$$

Thus,

$$e^{xy} - 2xy^3 - y^2 = -3$$

$$e^{xy} = 2xy^3 + y^2 - 3$$

$\rightarrow$ Particular Solution

$$26. (3 + y + 2y^2 \sin^2 x)dx + (x + 2xy - y \sin 2x)dy = 0;$$

$$M = 3 + y + 2y^2 \sin^2 x;$$

$$\frac{\partial M}{\partial y} = 1 + 4y \sin^2 x$$

$$N = x + 2xy - y \sin 2x;$$

$$\frac{\partial N}{\partial x} = 1 + 2y - 2y \cos 2x$$

Trigonometric Identity

$$\rightarrow \cos 2x = \cos^2 x - \sin^2 x$$

$$\rightarrow \cos 2x = 1 - 2\sin^2 x$$

$$\frac{\partial N}{\partial x} = 1 + 2y - 2y(1 - 2\sin^2 x)$$

$$\frac{\partial N}{\partial x} = 1 + 2y - 2y + 4y \sin^2 x$$

$$\frac{\partial N}{\partial x} = 1 + 4y \sin^2 x$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial x} = 3 + y + 2y^2 \sin^2 x \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = xe^{xy} - 6xy^2 - 2y \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (3 + y + 2y^2 \sin^2 x)dx$$

$$F = 3x + xy + 2y^2 \int \frac{(1 - \cos 2x)}{2} dx$$

$$+ \lambda(y) = C_1$$

$$F = 3x + xy + 2y^2 \left(\frac{x}{2} - \frac{\sin 2x}{4} \right) + \lambda(y)$$

$$= C_1$$

$$F = 3x + xy + xy^2 - \frac{1}{2}y^2 \sin 2x + \lambda(y)$$

$$= C_1 \rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = x + 2xy - y \sin 2x + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$x + 2xy - y \sin 2x + \lambda'(y) = x + 2xy - y \sin 2x$$

$$\lambda'(y) = 0 \rightarrow (5)$$

Integrate Equation (5):

$$\int d\lambda(y) = \int 0 dy$$

$$\lambda(y) = C_1 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$3x + xy + xy^2 - \frac{1}{2}y^2 \sin 2x + C_2 = C_1$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

$$3x + xy + xy^2 - \frac{1}{2}y^2 \sin 2x = C$$

$$6x + 2xy + 2xy^2 - y^2 \sin 2x = C$$

$$y^2 \sin 2x = 2x(3 + y + y^2) - C$$

Second Method

$$\int D.I. + \int N.D.I. = 0$$

D.I:

$$3dx$$

N.D.I:

$$(y + 2y^2 \sin^2 x)dx, (x + 2xy - y \sin 2x)dy$$

$$\text{Choose: } (x + 2xy - y \sin 2x)dy$$

$$\int 3dx + \int (x + 2xy - y \sin 2x)dy = 0$$

$$3x + xy + xy^2 - \frac{y^2}{2} \sin 2x = C$$

$$6x + 2xy + 2xy^2 - y^2 \sin 2x = C$$

$$y^2 \sin 2x = 2x(3 + y + y^2) - C$$

$$27. 2x[3x + y - ye^{-x^2}]dx + (x^2 + 3y^2 + e^{-x^2})dy = 0;$$

$$M = 6x^2 + 2xy - 2xye^{-x^2};$$

$$\frac{\partial M}{\partial y} = 2x - 2xe^{-x^2}$$

$$N = x^2 + 3y^2 + e^{-x^2};$$

$$\frac{\partial N}{\partial x} = 2x - 2xe^{-x^2}$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial x} = 6x^2 + 2xy - 2xye^{-x^2} \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = x^2 + 3y^2 + e^{-x^2} \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (6x^2 + 2xy - 2xye^{-x^2})dx$$

$$F = 2x^3 + x^2y + ye^{-x^2} + \lambda(y) = C_1$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = x^2 + e^{-x^2} + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$x^2 + e^{-x^2} + \lambda'(y) = x^2 + 3y^2 + e^{-x^2}$$

$$\lambda'(y) = 3y^2 \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = 3y^2$$

$$\int d\lambda(y) = \int 3y^2 dy$$

$$\lambda(y) = y^3 + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$2x^3 + x^2y + ye^{-x^2} + y^3 + C_2 = C_1$$

$$2x^3 + x^2y + ye^{-x^2} + y^3 = C$$

$$x^2y + y^3 + 2x^3 + ye^{-x^2} = C$$

Second Method

$$\int D.I. + \int N.D.I. = 0$$

D.I:

$$6x^2 dx, 3y^2 dy$$

N.D.I:

$$(2xy - 2xye^{-x^2})dx, (x^2 + e^{-x^2})dy$$

$$\text{Choose: } (x^2 + e^{-x^2})dy$$

$$\int 6x^2 dx + \int 3y^2 dy + \int (x^2 + e^{-x^2})dy = 0$$

$$2x^3 + y^3 + x^2y + ye^{-x^2} = C$$

$$2x^3 + x^2y + ye^{-x^2} + y^3 = C$$

$$x^2y + y^3 + 2x^3 + ye^{-x^2} = C$$

TYPE 3: EXACT DIFFERENTIAL EQUATIONS

28. $(xy^2 + x - 2y + 3)dx + x^2ydy = 2(x + y)dy$,
when $x = 1, y = 1$

$$M = xy^2 + x - 2y + 3;$$

$$\frac{\partial M}{\partial y} = 2xy - 2$$

$$N = x^2y - 2x - 2y;$$

$$\frac{\partial N}{\partial x} = 2xy - 2$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

First Method

$$M = \frac{\partial F}{\partial x} = xy^2 + x - 2y + 3 \rightarrow (1)$$

$$N = \frac{\partial F}{\partial y} = x^2y - 2x - 2y \rightarrow (2)$$

Integrate Equation (1):

$$\int \partial F = \int (xy^2 + x - 2y + 3)dx$$

$$F = \frac{x^2y^2}{2} + \frac{x^2}{2} - 2xy + 3x + \lambda(y) = C_1$$

$$\rightarrow (3)$$

Differentiate Equation (3):

$$\frac{\partial F}{\partial y} = x^2y - 2x + \lambda'(y) \rightarrow (4)$$

Equate Equation (4) and Equation (2):

$$x^2y - 2x + \lambda'(y) = x^2y - 2x - 2y$$

$$\lambda'(y) = -2y \rightarrow (5)$$

Integrate Equation (5):

$$\lambda'(y) = -2y$$

$$\int d\lambda(y) = -\int 2ydy$$

$$\lambda(y) = -y^2 + C_2 \rightarrow (6)$$

Substitute Equation (6) to Equation (3):

$$\frac{x^2y^2}{2} + \frac{x^2}{2} - 2xy + 3x - y^2 + C_2 = C_1$$

$$\frac{x^2y^2}{2} + \frac{x^2}{2} - 2xy + 3x - y^2 = C$$

$$x^2y^2 + x^2 - 4xy + 6x - 2y^2 = C$$

→ General Solution

Second Method

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:
 $(x + 3)dx, -2ydy$

N.D.I:
 $(xy^2 - 2y)dx, (x^2y - 2x)dy$

Choose: $(xy^2 - 2y)dx$

$$\int (x + 3)dx - \int 2ydy + \int (xy^2 - 2y)dx$$

$$= \int 0$$

$$\frac{x^2}{2} + 3x - y^2 + \frac{x^2y^2}{2} - 2xy = C$$

$$\frac{x^2y^2}{2} + \frac{x^2}{2} - 2xy + 3x - y^2 = C$$

$$x^2y^2 + x^2 - 4xy + 6x - 2y^2 = C$$

→ General Solution

when $x = 1, y = 1$

$$C = 1^2 \cdot 1^2 + 1^2 - 4(1)(1) + 6(1) - 2(1)^2$$

$$= 2$$

Thus,

$$x^2y^2 + x^2 - 4xy + 6x - 2y^2 = 2$$

$$(x^2y^2 - 4xy + 4) + (x^2 + 6x + 9)$$

$$= 2y^2 + 2 + 13$$

$$(xy - 2)^2 + (x + 3)^2 = 2y^2 + 15$$

→ Particular Solution

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

2.4 TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

If an equation is not exact, it is natural to attempt to make it exact by the introduction of an appropriate factor, which is then called the integrating factor.

STANDARD FORM:

$$\frac{dy}{dx} + P(x)y = Q(x), \quad \text{where } P(x) \text{ and } Q(x) \text{ are functions of } x$$

use Integrating Factor = $e^{\int P(x)dx}$

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$\frac{dx}{dy} + P(y)x = Q(y), \quad \text{where } P(y) \text{ and } Q(y) \text{ are functions of } y$$

use Integrating Factor = $e^{\int P(y)dy}$

$$xe^{\int P(y)dy} = \int Q(y)e^{\int P(y)dy} dy + C$$

Steps involved in solving linear differential equations

- Put the equation in standard form
- Obtain the integrating factor
- Multiply both sides of the equation (in standard form) by the integrating factor
- Continue solving using Exact Equation.

I. In exercises 1 through 25, find the general solution

1. $(x^4 + 2y)dx - xdy = 0$

Divide by $-xdx$:

$$-\frac{(x^4 + 2y)}{x} + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} - \frac{(x^4 + 2y)}{x} = 0$$

$$\frac{dy}{dx} - \frac{2}{x}y = x^3$$

$$P(x) = -\frac{2}{x}$$

$$Q(x) = x^3$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int (2/x)dx} = e^{-2 \ln x} = e^{\ln x^{-2}}$$

$$= x^{-2}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$yx^{-2} = \int x^3x^{-2} dx + C$$

$$yx^{-2} = \int x^3x^{-2} dx + C$$

$$\frac{y}{x^2} = \int x dx + C$$

$$\frac{y}{x^2} = \frac{x^2}{2} + C$$

$$2y = x^4 + Cx^2$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

2. $(3xy + 3y - 4)dx + (x + 1)^2 dy = 0$

Divide by $(x + 1)^2 dx$,

$$\frac{(3xy + 3y - 4)}{(x + 1)^2} + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} + \frac{3y + 3y - 4}{(x + 1)^2} = 0$$

$$\frac{dy}{dx} + \frac{3xy + 3y}{(x + 1)^2} - \frac{4}{(x + 1)^2} = 0$$

$$\frac{dy}{dx} + \frac{3(x + 1)}{(x + 1)^2} y = \frac{4}{(x + 1)^2}$$

$$\frac{dy}{dx} + \frac{3}{(x + 1)} y = \frac{4}{(x + 1)^2}$$

$$P(x) = \frac{3}{(x + 1)}$$

$$Q(x) = \frac{4}{(x + 1)^2}$$

For integrating factor:

$$e^{\int P(x) dx} = e^{\int \frac{3}{(x+1)} dx} = e^{3 \ln(x+1)} = e^{\ln(x+1)^3} = (x+1)^3$$

Thus,

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + C$$

$$y(x+1)^3 = \int \frac{4}{(x+1)^2} (x+1)^3 dx + C$$

$$y(x+1)^3 = \int 4(x+1) dx + C$$

$$\frac{y(x+1)^3}{(x+1)^3} = \frac{2(x+1)^2 + C}{(x+1)^3}$$

$$y = 2(x+1)^{-1} + C(x+1)^{-3}$$

3. $y' = \csc x - y \cot x$

$$\frac{dy}{dx} + y \cot x = \csc x$$

$$P(x) = \cot x$$

$$Q(x) = \csc x$$

For integrating factor:

$$e^{\int P(x) dx} = e^{\int \cot x dx} = e^{\int \frac{\cos x}{\sin x} dx} = e^{\ln \sin x} = \sin x$$

Thus,

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + C$$

$$y \sin x = \int \csc x \sin x dx + C$$

$$y \sin x = \int \frac{1}{\sin x} \sin x dx + C$$

$$y \sin x = \int dx + C$$

$$y \sin x = x + C$$

4. $t(dx/dt) = 6te^{2t} + x(2t - 1)$

Divide by t :

$$\frac{dx}{dt} = 6e^{2t} + \frac{x}{t}(2t - 1)$$

$$\frac{dx}{dt} - \frac{1}{t}(2t - 1)x = 6e^{2t}$$

$$\frac{dx}{dt} - \left(2 - \frac{1}{t}\right)x = 6e^{2t}$$

$$P(t) = \left(-2 + \frac{1}{t}\right)$$

$$Q(t) = 6e^{2t}$$

For integrating factor:

$$e^{\int P(t) dt} = e^{\int \left(-2 + \frac{1}{t}\right) dt} = e^{-2t + \ln t} = e^{-2t} \ln t$$

Thus,

$$xe^{\int P(t) dt} = \int Q(t)e^{\int P(t) dt} dt + C$$

$$xe^{-2t + \ln t} = \int 6e^{2t} e^{-2t + \ln t} dt + C$$

$$xe^{\ln t} e^{-2t} = \int 6e^{\ln t} dt + C$$

$$xt e^{-2t} = \int 6t dt + C$$

$$xt e^{-2t} = 3t^2 + C$$

$$xt = (3t^2 + C)e^{2t}$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

5. $dy = (x - 3y)dx$

$$\frac{dy}{dx} = x - 3y$$

$$\frac{dy}{dx} + 3y = x$$

$$P(x) = 3$$

$$Q(x) = x$$

For integrating factor:

$$e^{\int P(x) dx} = e^{\int 3 dx} = e^{3x}$$

Thus,

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + C$$

$$ye^{3x} = \int xe^{3x} dx + C$$

Considering

$$\int xe^{3x} dx$$

Using Integration by Parts:

$$\text{Let } u = x \quad dv = e^{3x} dx$$

$$du = dx \quad v = \frac{1}{3}e^{3x}$$

thus,

$$\int xe^{3x} dx = \frac{1}{3}xe^{3x} - \int \frac{1}{3}e^{3x} dx$$

$$= \frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x}$$

Then,

$$ye^{3x} = \frac{1}{3}xe^{3x} - \frac{1}{9}e^{3x} + C$$

$$9ye^{3x} = 3xe^{3x} - e^{3x} + C$$

$$9y = 3x - 1 + Ce^{-3x}$$

6. $(3x - 1)y' = 6y - 10(3x - 1)^{1/3}$

Divide by $(3x - 1)$:

$$\frac{dy}{dx} - \frac{6y}{(3x - 1)} = -\frac{10(3x - 1)^{1/3}}{(3x - 1)}$$

$$\frac{dy}{dx} - \frac{6y}{(3x - 1)} = -\frac{10}{(3x - 1)^{2/3}}$$

$$P(x) = -\frac{6}{(3x - 1)}$$

$$Q(x) = -\frac{10}{(3x - 1)^{2/3}}$$

For integrating factor:

$$e^{\int P(x) dx} = e^{-\int \frac{6}{(3x-1)} dx} = e^{-2 \ln(3x-1)} = (3x-1)^{-2}$$

Thus,

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + C$$

$$y(3x - 1)^{-2}$$

$$= -\int \frac{10}{(3x - 1)^{2/3}} (3x - 1)^{-2} dx + C$$

$$y(3x - 1)^{-2} = -\int \frac{10}{(3x - 1)^{8/3}} dx + C$$

$$y(3x - 1)^{-2} = -10 \int (3x - 1)^{-8/3} dx + C$$

$$y(3x - 1)^{-2} = \left(-\frac{10}{3}\right) \int (3x - 1)^{-8/3} 3 dx + C$$

$$y(3x - 1)^{-2}$$

$$= \left(-\frac{10}{3}\right) \left(-\frac{3}{5}\right) (3x - 1)^{-5/3} + C$$

$$y = 2(3x - 1)^{1/3} + C(3x - 1)^2$$

7. $(y - 2)dx + (3x - y)dy = 0$

Divide by $(y - 2)dy$:

$$\frac{dx}{dy} + \frac{(3x - y)}{(y - 2)} = 0$$

$$\frac{dx}{dy} + \frac{3x}{(y - 2)} - \frac{y}{(y - 2)} = 0$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

$$\frac{dx}{dy} + \frac{3x}{(y-2)} = \frac{y}{(y-2)}$$

$$P(y) = \frac{3}{(y-2)}$$

$$Q(y) = \frac{y}{(y-2)}$$

For integrating factor:

$$e^{\int P(y)dy} = e^{\int \frac{3}{(y-2)} dy} = e^{3 \ln(y-2)} = (y-2)^3$$

Thus,

$$xe^{\int P(y)dy} = \int Q(y)e^{\int P(y)dy} dy + C$$

$$x(y-2)^3 = \int \frac{y}{(y-2)} (y-2)^3 dy + C$$

$$x(y-2)^3 = \int y(y-2)^2 dy + C$$

Considering:

$$\int y(y-2)^2 dy$$

Using Integration by Parts:

$$\text{Let } u = y \quad dv = (y-2)^2 dy$$

$$du = dy \quad v = \frac{(y-2)^3}{3}$$

thus,

$$\int y(y-2)^2 dy = \frac{y(y-2)^3}{3} - \int \frac{(y-2)^3}{3} dy$$

$$= \frac{y(y-2)^3}{3} - \frac{(y-2)^4}{12}$$

Then,

$$x(y-2)^3 = \frac{y(y-2)^3}{3} - \frac{(y-2)^4}{12} + C$$

Multiply by $12(y-2)^{-3}$:

$$12x = 4y - (y-2) + C(y-2)^{-3}$$

$$12x = 3y + 2 + C(y-2)^{-3}$$

$$8. (2xy + x^2 + x^4)dx - (1 + x^2)dy = 0$$

Divide by $-(1+x^2)dx$:

$$-\frac{(2xy + x^2 + x^4)}{(1+x^2)} + \frac{dy}{dx} = 0$$

$$\frac{dx}{dy} - \frac{(2xy + x^2 + x^4)}{(1+x^2)} = 0$$

$$\frac{dx}{dy} - \frac{2xy}{(1+x^2)} - \frac{(x^2 + x^4)}{(1+x^2)} = 0$$

$$\frac{dx}{dy} - \frac{2xy}{(1+x^2)} - \frac{x^2(1+x^2)}{(1+x^2)} = 0$$

$$\frac{dx}{dy} - \frac{2xy}{(1+x^2)} = x^2$$

$$P(x) = -\frac{2x}{(1+x^2)}$$

$$Q(x) = x^2$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int \frac{2x}{(1+x^2)} dx} = e^{-\ln(1+x^2)} = (1+x^2)^{-1}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y(1+x^2)^{-1} = \int x^2(1+x^2)^{-1} dx + C$$

$$y(1+x^2)^{-1} = \int \frac{x^2}{(1+x^2)} dx + C$$

$$y(1+x^2)^{-1} = \int \frac{(1+x^2) - 1}{(1+x^2)} dx + C$$

$$y(1+x^2)^{-1} = \int \left(1 - \frac{1}{(1+x^2)}\right) dx + C$$

$$y(1+x^2)^{-1} = x - \text{Arctan } x + C$$

$$\text{Multiply by } (1+x^2);$$

$$y = (1+x^2)(C + x - \text{Arctan } x)$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

$$9. y' = x - 2xy. \text{ Solve by two methods}$$

◇ First Method

$$\frac{dy}{dx} + 2xy = x$$

$$P(x) = 2x$$

$$Q(x) = x$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int 2x dx} = e^{x^2}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$ye^{x^2} = \int xe^{x^2} dx + C$$

$$ye^{x^2} = \frac{1}{2}e^{x^2} + C$$

Multiply by $2e^{-x^2}$:

$$2y = 1 + Ce^{-x^2}$$

◇ Second Method

Separation of Variables

$$\frac{dy}{dx} = x(1-2y)$$

$$xdx - \frac{dy}{(1-2y)} = 0$$

$$\int xdx - \int \frac{dy}{(1-2y)} = \int 0$$

$$\frac{x^2}{2} + \frac{1}{2} \ln(1-2y) = \ln C$$

$$x^2 + \ln(1-2y) = \ln C$$

$$x^2 = \ln C - \ln(1-2y)$$

$$x^2 = \ln \left(\frac{C}{1-2y} \right)$$

$$e^{x^2} = \frac{C}{1-2y}$$

$$(1-2y)e^{x^2} = C$$

Multiply by e^{-x^2}

$$2y = 1 + Ce^{-x^2}$$

$$10. (y - \cos^2 x)dx + \cos x dy = 0$$

Divide by $\cos x dx$:

$$\frac{(y - \cos^2 x)}{\cos x} + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} + \frac{y}{\cos x} - \frac{\cos^2 x}{\cos x} = 0$$

$$\frac{dy}{dx} + \frac{1}{\cos x} y = \cos x$$

$$\frac{dy}{dx} + \sec x (y) = \cos x$$

$$P(x) = \sec x$$

$$Q(x) = \cos x$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \sec x dx} = e^{\ln(\sec x + \tan x)} = \sec x + \tan x$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y(\sec x + \tan x) = \int \cos x (\sec x + \tan x) dx + C$$

$$y(\sec x + \tan x) = \int \left[\cos x \left(\frac{1}{\cos x} \right) + \cos x \left(\frac{\sin x}{\cos x} \right) \right] dx + C$$

$$y(\sec x + \tan x) = \int (1 + \sin x) dx + C$$

$$y(\sec x + \tan x) = x - \cos x + C$$

$$y(\sec x + \tan x) = x + C - \cos x$$

$$11. y' = x - 2y \cot 2x$$

$$\frac{dy}{dx} = x - 2y \cot 2x$$

$$\frac{dy}{dx} + 2y \cot 2x = x$$

$$P(x) = 2 \cot 2x$$

$$Q(x) = x$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int 2 \cot 2x dx} = e^{\ln \sin 2x} = \sin 2x$$

Thus,

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y \sin 2x = \int x \sin 2x dx + C$$

Considering:

$$\int x \sin 2x dx$$

Using Integration by Parts:

$$\text{Let } u = x \quad dv = \sin 2x dx \\ du = dx \quad v = -\frac{\cos 2x}{2}$$

thus,

$$\int x \sin 2x dx = -x \frac{\cos 2x}{2} + \int \frac{\cos 2x}{2} dx \\ = -\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x$$

Then,

$$y \sin 2x = -\frac{x}{2} \cos 2x + \frac{1}{4} \sin 2x + C$$

$$4y \sin 2x = C + \sin 2x - 2x \cos 2x$$

$$12. (y - x + xy \cot x) dx + x dy = 0$$

Divide by $x dx$;

$$\frac{dy}{dx} + \frac{(y - x + xy \cot x)}{x} = 0$$

$$\frac{dy}{dx} + \frac{y(1 + x \cot x) - x}{x} = 0$$

$$\frac{dy}{dx} + \frac{y(1 + x \cot x)}{x} - 1 = 0$$

$$\frac{dy}{dx} + \frac{(1 + x \cot x)}{x} y = 1$$

$$\frac{dy}{dx} + \left(\frac{1}{x} + \cot x\right) y = 1$$

$$P(x) = \left(\frac{1}{x} + \cot x\right) \\ Q(x) = 1$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \left(\frac{1}{x} + \cot x\right) dx} = e^{\ln x + \ln \sin x} \\ = x \sin x$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$yx \sin x = \int x \sin x dx + C$$

Considering:

$$\int x \sin x dx$$

Using Integration by Parts:

$$\text{Let } u = x \quad dv = \sin x dx \\ du = dx \quad v = -\cos x$$

thus,

$$\int x \sin x dx = -x \cos x + \int \cos x dx \\ = -x \cos x + \sin x$$

Then,

$$yx \sin x = -x \cos x + \sin x + C$$

$$xy \sin x = C + \sin x - x \cos x$$

$$13. \left(\frac{dy}{dx}\right) - my = C_1 e^{mx}; \\ \text{where } C_1 \text{ and } m \text{ are constants}$$

$$\frac{dy}{dx} - my = C_1 e^{mx}$$

$$P(x) = -m \\ Q(x) = C_1 e^{mx}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int m dx} = e^{-mx}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

$$ye^{-mx} = \int C_1 e^{mx} e^{-mx} dx + C_2$$

$$ye^{-mx} = \int C_1 dx + C_2$$

$$ye^{-mx} = C_1 x + C_2$$

$$y = (C_1 x + C_2) e^{mx}$$

$$14. \left(\frac{dy}{dx}\right) - m_2 y = C_1 e^{m_1 x}; \\ \text{where } C_1, m_1, \text{ and } m_2 \text{ are constants} \\ m_1 \neq m_2$$

$$\frac{dy}{dx} - m_2 y = C_1 e^{m_1 x}$$

$$P(x) = -m_2 \\ Q(x) = C_1 e^{m_1 x}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int m_2 dx} = e^{-m_2 x}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$ye^{-m_2 x} = \int C_1 e^{m_1 x} e^{-m_2 x} dx + C_2$$

$$ye^{-m_2 x} = \int C_1 e^{x(m_1 - m_2)} dx + C_2$$

$$ye^{-m_2 x} = \int C_1 e^{x(m_1 - m_2)} dx + C_2$$

$$ye^{-m_2 x} = \frac{C_1}{(m_1 - m_2)} e^{x(m_1 - m_2)} + C_2$$

$$(m_1 - m_2) ye^{-m_2 x} = C_1 e^{m_1 x} e^{-m_2 x} + C_2$$

Multiply by $e^{m_2 x}$;

$$(m_1 - m_2) y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$15. v dx + (2x + 1 - vx) dv = 0;$$

Divide by vdv ;

$$\frac{dx}{dv} + \frac{(2x + 1 - vx)}{v} = 0$$

$$\frac{dx}{dv} + \frac{x(2 - v)}{v} + \frac{1}{v} = 0$$

$$\frac{dx}{dv} + \frac{(2 - v)}{v} x = -\frac{1}{v}$$

$$\frac{dx}{dv} + \left(\frac{2}{v} - 1\right) x = -\frac{1}{v}$$

$$P(v) = \left(\frac{2}{v} - 1\right)$$

$$Q(v) = -\frac{1}{v}$$

For integrating factor:

$$e^{\int P(v)dv} = e^{\int \left(\frac{2}{v} - 1\right) dv} = e^{2 \ln v - v} = v^2 e^{-v}$$

Thus,

$$xe^{\int P(v)dv} = \int Q(v)e^{\int P(v)dv} dv + C$$

$$xv^2 e^{-v} = -\int \frac{1}{v} v^2 e^{-v} dv + C_2$$

$$xv^2 e^{-v} = -\int ve^{-v} dv + C_2$$

Considering:

$$\int ve^{-v} dv$$

Using Integration by Parts:

$$\text{Let } u = v \quad dv = e^{-v} dv \\ du = dv \quad v = -e^{-v}$$

thus,

$$\int ve^{-v} dv = -ve^{-v} + \int e^{-v} dv \\ = -ve^{-v} - e^{-v}$$

Then,

$$xv^2 e^{-v} = ve^{-v} + e^{-v} + C_2$$

Divide by e^{-v} ;

$$xv^2 = v + 1 + C_2 e^v$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

$$16. \quad x(x^2 + 1)y' + 2y = (x^2 + 1)^3$$

$$x(x^2 + 1) \frac{dy}{dx} + 2y = (x^2 + 1)^3$$

Divide by $x(x^2 + 1)$:

$$\frac{dy}{dx} + \frac{2y}{x(x^2 + 1)} = \frac{(x^2 + 1)^2}{x}$$

$$P(v) = \frac{2}{x(x^2 + 1)}$$

$$Q(v) = \frac{(x^2 + 1)^2}{x}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{2}{x(x^2 + 1)} dx}$$

Considering:

$$\int \frac{2}{x(x^2 + 1)} dx$$

Using Integration by Partial Fraction:

$$\frac{2}{x(x^2 + 1)} = \frac{A}{x} + \frac{2xB}{(x^2 + 1)}$$

$$2 = A(x^2 + 1) + 2Bx^2$$

$$\text{If } \begin{matrix} x = 0 & ; & A = 2 \\ x = 1 & ; & B = -1 \end{matrix}$$

Then,

$$\int \frac{2}{x(x^2 + 1)} dx = 2 \int \frac{dx}{x} - \int \frac{2x dx}{(x^2 + 1)}$$

$$= 2 \ln x - \ln(x^2 + 1)$$

$$= \ln x^2 - \ln(x^2 + 1)$$

$$e^{\int P(x)dx} = e^{\int \frac{2}{x(x^2 + 1)} dx} = e^{\ln x^2 - \ln(x^2 + 1)}$$

$$= e^{\ln x^2} e^{-\ln(x^2 + 1)}$$

$$= \frac{x^2}{(x^2 + 1)}$$

thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y \frac{x^2}{(x^2 + 1)} = \int \frac{(x^2 + 1)^2}{x} \cdot \frac{x^2}{(x^2 + 1)} dx + C$$

$$y \frac{x^2}{(x^2 + 1)} = \int x(x^2 + 1) dx + C$$

$$y \frac{x^2}{(x^2 + 1)} = \frac{1}{4}(x^2 + 1)^2 + C$$

$$x^2 y = \frac{1}{4}(x^2 + 1)^3 + C(x^2 + 1)$$

$$17. \quad 2x(y - x^2)dx + dy = 0$$

$$(2xy - 2x^3)dx + dy = 0$$

Divide by dx :

$$\frac{dy}{dx} + (2xy - 2x^3) = 0$$

$$\frac{dy}{dx} + 2xy = 2x^3$$

$$P(x) = 2x$$

$$Q(x) = 2x^3$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int 2x dx} = e^{x^2}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$ye^{x^2} = \int 2x^3 e^{x^2} dx + C$$

$$ye^{x^2} = 2 \int x^3 e^{x^2} dx + C$$

Considering:

$$\int x^3 e^{x^2} dx$$

Using Integration by Parts:

$$\text{Let } \begin{matrix} u = x^2 & dv = xe^{x^2} dx \\ du = 2x dx & v = \frac{1}{2}e^{x^2} \end{matrix}$$

thus,

$$\int x^3 e^{x^2} dx = x^2 \cdot \frac{1}{2}e^{x^2} - \int xe^{x^2} dx$$

$$= \frac{1}{2}x^2 e^{x^2} - \frac{1}{2}e^{x^2}$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

Then,

$$ye^{x^2} = 2 \left(\frac{1}{2}x^2 e^{x^2} - \frac{1}{2}e^{x^2} \right) + C$$

Divide by e^{x^2} :

$$y = x^2 - 1 + Ce^{-x^2}$$

$$18. \quad (1 + xy)dx - (1 + x^2)dy = 0$$

Divide by $(1 + x^2)dx$:

$$\frac{dy}{dx} - \frac{(1 + xy)}{(1 + x^2)} = 0$$

$$\frac{dy}{dx} - \frac{1}{(1 + x^2)} - \frac{xy}{(1 + x^2)} = 0$$

$$\frac{dy}{dx} - \frac{x}{(1 + x^2)} y = \frac{1}{(1 + x^2)}$$

$$P(x) = -\frac{x}{(1 + x^2)}$$

$$Q(x) = \frac{1}{(1 + x^2)}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int \frac{x}{(1 + x^2)} dx} = e^{-\frac{1}{2} \ln(1 + x^2)}$$

$$= (1 + x^2)^{-1/2}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y(1 + x^2)^{-1/2} = \int \frac{1}{(1 + x^2)} (1 + x^2)^{-1/2} dx + C$$

$$y(1 + x^2)^{-1/2} = \int \frac{1}{(1 + x^2)^{3/2}} dx + C$$

Considering:

$$\int \frac{1}{(1 + x^2)^{3/2}} dx$$

Perform Trigonometric substitution:

$$\text{Let } \begin{matrix} x = \tan u \\ dx = \sec^2 u \\ u = \arctan x \end{matrix}$$

Then,

$$\int \frac{1}{(1 + x^2)^{3/2}} dx = \int \frac{\sec^2 u}{(\tan^2 u + 1)^{3/2}} du$$

$$= \int \frac{\sec^2 u}{(\sec^2 u)^{3/2}} du$$

$$= \int \frac{1}{\sec u} du$$

$$= \int \cos u du$$

$$= \sin u = \frac{x}{\sqrt{x^2 + 1}}$$

Thus,

$$y(1 + x^2)^{-1/2} = \frac{x}{\sqrt{x^2 + 1}} + C$$

Divide by $(1 + x^2)^{-1/2}$:

$$y = x + C(1 + x^2)^{1/2}$$

$$19. \quad 2ydx = (x^2 - 1)(dx - dy)$$

Divide by $(x^2 - 1)dx$:

$$\frac{2y}{(x^2 - 1)} = 1 - \frac{dy}{dx}$$

$$\frac{dy}{dx} + \frac{2}{(x^2 - 1)} y = 1$$

$$\frac{dy}{dx} + \frac{2}{(x - 1)(x + 1)} y = 1$$

$$P(x) = \frac{2}{(x - 1)(x + 1)}$$

$$Q(x) = 1$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{2}{(x - 1)(x + 1)} dx}$$

Considering:

$$\int \frac{2}{(x - 1)(x + 1)} dx$$

Using Integration by Partial Fraction:

$$\frac{2}{(x - 1)(x + 1)} = \frac{A}{(x + 1)} + \frac{B}{(x - 1)}$$

$$2 = A(x - 1) + B(x + 1)$$

$$\text{If } \begin{matrix} x = -1 & ; & A = -1 \\ x = 1 & ; & B = 1 \end{matrix}$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

Then,

$$\int \frac{2}{(x-1)(x+1)} dx = -\int \frac{dx}{(x-1)} + \int \frac{dx}{(x+1)}$$

$$= -\ln(x-1) + \ln(x+1)$$

$$= \ln \frac{(x+1)}{(x-1)}$$

$$e^{\int P(x)dx} = e^{\int \frac{2}{(x-1)(x+1)} dx} = e^{\ln \frac{(x+1)}{(x-1)}}$$

$$= \frac{(x+1)}{(x-1)}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y \frac{(x+1)}{(x-1)} = \int \frac{(x-1)}{(x+1)} dx + C$$

$$y \frac{(x+1)}{(x-1)} = \int \frac{(x+1)-2}{(x+1)} dx + C$$

$$y \frac{(x+1)}{(x-1)} = \int \left(1 - \frac{2}{x+1}\right) dx + C$$

$$y \frac{(x+1)}{(x-1)} = x - 2\ln(x+1) + C$$

$$y(x-1) = (x+1)[C + x - 2\ln(x+1)]$$

20.

$$dx - (1 + 2x \tan y) dy = 0$$

Divide by dy ,

$$\frac{dx}{dy} - (1 + 2x \tan y) = 0$$

$$\frac{dx}{dy} - 2x \tan y = 1$$

$$P(y) = -2 \tan y$$

$$Q(y) = 1$$

For integrating factor:

$$e^{\int P(y)dy} = e^{-\int 2 \tan y dy} = e^{2 \ln \cos y}$$

$$= \cos^2 y$$

Thus,

$$xe^{\int P(y)dy} = \int Q(y)e^{\int P(y)dy} dy + C$$

$$x \cos^2 y = \int \cos^2 y dy + C$$

$$x \cos^2 y = \int \frac{1 + \cos 2y}{2} dy + C$$

$$x \cos^2 y = \frac{1}{2}y + \frac{1}{4} \sin 2y + C$$

$$x \cos^2 y = \frac{1}{2}y + \frac{1}{4} (2 \sin y \cos y) + C$$

$$2x \cos^2 y = y + \sin y \cos y + C$$

$$2x \cos^2 y = y + C + \sin y \cos y$$

21.

$$(1 + \cos x)y' = \sin x (\sin x + \sin x \cos x - y)$$

$$(1 + \cos x) \frac{dy}{dx} = \sin x (\sin x + \sin x \cos x - y)$$

Divide by $(1 + \cos x)$;

$$\frac{dy}{dx} = \frac{\sin x (\sin x + \sin x \cos x - y)}{(1 + \cos x)}$$

$$\frac{dy}{dx} + \frac{\sin x}{(1 + \cos x)} y = \frac{\sin^2 x (1 + \cos x)}{(1 + \cos x)}$$

$$\frac{dy}{dx} + \frac{\sin x}{(1 + \cos x)} y = \sin^2 x$$

$$P(x) = \frac{\sin x}{(1 + \cos x)}$$

$$Q(x) = \sin^2 x$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{\sin x}{(1 + \cos x)} dx} = e^{-\ln(1 + \cos x)}$$

$$= (1 + \cos x)^{-1}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y(1 + \cos x)^{-1} = \int \sin^2 x (1 + \cos x)^{-1} dx + C$$

$$y(1 + \cos x)^{-1} = \int \frac{\sin^2 x}{(1 + \cos x)} dx + C$$

$$y(1 + \cos x)^{-1} = \int \frac{(1 - \cos^2 x)}{(1 + \cos x)} dx + C$$

$$y(1 + \cos x)^{-1} = \int \frac{(1 + \cos x)(1 - \cos x)}{(1 + \cos x)} dx + C$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

$$y(1 + \cos x)^{-1} = \int (1 - \cos x) dx + C$$

$$y(1 + \cos x)^{-1} = x - \sin x + C$$

Multiply by $(1 + \cos x)$;

$$y = (1 + \cos x)(C + x - \sin x)$$

22.

$$y' = 1 + 3y \tan x$$

$$\frac{dy}{dx} = 1 + 3y \tan x$$

$$\frac{dy}{dx} - 3y \tan x = 1$$

$$P(x) = -3 \tan x$$

$$Q(x) = 1$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int 3 \tan x dx} = e^{3 \ln \cos x}$$

$$= \cos^3 x$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y \cos^3 x = \int \cos^3 x dx + C$$

$$y \cos^3 x = \int \cos x (1 - \sin^2 x) dx + C$$

$$y \cos^3 x = \sin x - \frac{\sin^3 x}{3} + C$$

$$3y \cos^3 x = 3 \sin x - \sin^3 x + C$$

$$3y \cos^3 x = C + 3 \sin x - \sin^3 x$$

23.

$$(x^2 + a^2)dy = 2x[(x^2 + a^2)^2 + 3y]dx;$$

a is constant

Divide by $(x^2 + a^2)dx$;

$$\frac{dy}{dx} = \frac{2x[(x^2 + a^2)^2 + 3y]}{(x^2 + a^2)}$$

$$\frac{dy}{dx} = \frac{2x(x^2 + a^2)^2}{(x^2 + a^2)} + \frac{6xy}{(x^2 + a^2)}$$

$$\frac{dy}{dx} - \frac{6x}{(x^2 + a^2)} y = 2x(x^2 + a^2)$$

$$P(x) = -\frac{6x}{(x^2 + a^2)}$$

$$Q(x) = 2x(x^2 + a^2)$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int \frac{6x}{(x^2 + a^2)} dx} = e^{-3 \ln(x^2 + a^2)}$$

$$= (x^2 + a^2)^{-3}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y(x^2 + a^2)^{-3} = \int 2x(x^2 + a^2)(x^2 + a^2)^{-3} dx + C$$

$$y(x^2 + a^2)^{-3} = \int 2x(x^2 + a^2)^{-2} dx + C$$

$$y(x^2 + a^2)^{-3} = -(x^2 + a^2)^{-1} + C$$

Multiply by $(x^2 + a^2)^3$;

$$y = -(x^2 + a^2)^2 + C(x^2 + a^2)^3$$

$$y = (x^2 + a^2)^2 [C(x^2 + a^2) - 1]$$

24.

$$(x + a)y' = bx - ny;$$

a, b, n are constants with $n \neq 0, n \neq -1$

$$(x + a) \frac{dy}{dx} = bx - ny$$

$$\frac{dy}{dx} = \frac{bx - ny}{(x + a)}$$

$$\frac{dy}{dx} + \frac{n}{(x + a)} y = \frac{bx}{(x + a)}$$

$$P(x) = \frac{n}{(x + a)}$$

$$Q(x) = \frac{bx}{(x + a)}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{n}{(x+a)} dx} = e^{n \ln(x+a)}$$

$$= (x + a)^n$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$y(x + a)^n = \int \frac{bx}{(x+a)} (x + a)^n dx + C$$

$$y(x + a)^n = b \int x(x + a)^{n-1} dx + C$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

Considering:

$$\int x(x+a)^{n-1} dx$$

Using Integration by Parts:

$$\begin{aligned} \text{Let } u &= x & dv &= (x+a)^{n-1} dx \\ du &= dx & v &= \frac{(x+a)^n}{n} \end{aligned}$$

thus,

$$\begin{aligned} \int x(x+a)^{n-1} dx &= x \frac{(x+a)^n}{n} - \int \frac{(x+a)^n}{n} dx \\ &= x \frac{(x+a)^n}{n} - \frac{(x+a)^{n+1}}{n(n+1)} \end{aligned}$$

Then,

$$\begin{aligned} y(x+a)^n &= b \left(x \frac{(x+a)^n}{n} - \frac{(x+a)^{n+1}}{n(n+1)} \right) + C \\ y(x+a)^n &= \frac{bx(x+a)^n}{n} - \frac{b(x+a)^{n+1}}{n(n+1)} + C \end{aligned}$$

Multiply by $n(n+1)$;

→ constant C multiply by constant $n(n+1)$ is equal to constant

$$ny(n+1)(x+a)^n = bx(n+1)(x+a)^n - b(x+a)^{n+1} + C$$

Divide by $(x+a)^n$;

$$\begin{aligned} n(n+1)y &= bx(n+1) - b(x+a) + C(x+a)^{-n} \\ n(n+1)y &= bnx + bx - bx - ba + C(x+a)^{-n} \\ n(n+1)y &= b(n+1)x - ba + C(x+a)^{-n} \end{aligned}$$

25. Solve the equation of exercise 24 for the exceptional cases $n = 0$ and $n = -1$

◇ For $n = 0$

$$(x+a) \frac{dy}{dx} = bx - (0)y$$

$$(x+a) \frac{dy}{dx} = bx$$

$$dy = \frac{bx}{(x+a)} dx$$

$$dy = b \left(1 - \frac{a}{x+a} \right) dx$$

$$\int dy = b \int \left(1 - \frac{a}{x+a} \right) dx$$

$$\begin{aligned} y &= bx - ab \ln(x+a) + C \\ y &= bx + C - ab \ln(x+a) \end{aligned}$$

◇ For $n = -1$

$$(x+a) \frac{dy}{dx} = bx - (-1)y$$

$$(x+a) \frac{dy}{dx} = bx + y$$

$$\frac{dy}{dx} = \frac{bx+y}{(x+a)}$$

$$\frac{dy}{dx} - \frac{1}{(x+a)} y = \frac{bx}{(x+a)}$$

$$\begin{aligned} P(x) &= -\frac{1}{(x+a)} \\ Q(x) &= \frac{bx}{(x+a)} \end{aligned}$$

For integrating factor:

$$\begin{aligned} e^{\int P(x) dx} &= e^{-\int \frac{1}{(x+a)} dx} = e^{-\ln(x+a)} \\ &= (x+a)^{-1} \end{aligned}$$

Thus,

$$ye^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + C$$

$$y(x+a)^{-1} = \int \frac{bx}{(x+a)} (x+a)^{-1} dx + C$$

$$y(x+a)^{-1} = \int \frac{bx}{(x+a)^2} dx + C$$

Considering:

$$\int \frac{bx}{(x+a)^2} dx$$

Using Integration by Parts:

$$\begin{aligned} \text{Let } u &= x & dv &= (x+a)^{-2} dx \\ du &= dx & v &= -(x+a)^{-1} \end{aligned}$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

thus,

$$\begin{aligned} \int \frac{bx}{(x+a)^2} dx &= -bx(x+a)^{-1} \\ &\quad + b \int (x+a)^{-1} dx \\ &= -bx(x+a)^{-1} + b \ln(x+a) \end{aligned}$$

Then,

$$y(x+a)^{-1} = -bx(x+a)^{-1} + b \ln(x+a) + C$$

Multiply by $(x+a)$:

$$y = -bx + b(x+a) \ln(x+a) + C(x+a)$$

II. In exercises 27 through 33, find the particular solution indicated.

27. $(2x+3)y' = y + (2x+3)^{1/2}$;
when $x = -1, y = 0$

$$(2x+3) \frac{dy}{dx} = y + (2x+3)^{1/2}$$

Divide by $(2x+3)$;

$$\begin{aligned} \frac{dy}{dx} - \frac{y}{(2x+3)} &= (2x+3)^{-1/2} \\ P(x) &= -\frac{1}{(2x+3)} \\ Q(x) &= (2x+3)^{-1/2} \end{aligned}$$

For integrating factor:

$$\begin{aligned} e^{\int P(x) dx} &= e^{-\int \frac{1}{(2x+3)} dx} = e^{-\frac{1}{2} \ln(2x+3)} \\ &= (2x+3)^{-1/2} \end{aligned}$$

Thus,

$$\begin{aligned} ye^{\int P(x) dx} &= \int Q(x)e^{\int P(x) dx} dx + C \\ y(2x+3)^{-1/2} &= \int (2x+3)^{-1/2} (2x+3)^{-1/2} dx + C \\ y(2x+3)^{-1/2} &= \int \frac{1}{(2x+3)} dx + C \end{aligned}$$

$$\begin{aligned} y(2x+3)^{-1/2} &= \frac{1}{2} \ln(2x+3) + C \\ 2y(2x+3)^{-1/2} - \ln(2x+3) &= C \\ &\rightarrow \text{General Solution} \end{aligned}$$

when $x = -1, y = 0$

$$C = 2(0)(2(-1)+3)^{-1/2} - \ln(2(-1)+3) = 0$$

Thus,

$$\begin{aligned} 2y(2x+3)^{-1/2} - \ln(2x+3) &= 0 \\ 2y(2x+3)^{-1/2} &= \ln(2x+3) \\ y &= (2x+3)^{1/2} \ln(2x+3) \\ &\rightarrow \text{Particular Solution} \end{aligned}$$

28. $y' = x^3 - 2xy$;
when $x = 1, y = 1$

$$\frac{dy}{dx} + 2xy = x^3$$

$$\begin{aligned} P(x) &= 2x \\ Q(x) &= x^3 \end{aligned}$$

For integrating factor:

$$e^{\int P(x) dx} = e^{\int 2x dx} = e^{x^2}$$

Thus,

$$\begin{aligned} ye^{\int P(x) dx} &= \int Q(x)e^{\int P(x) dx} dx + C \\ ye^{x^2} &= \int x^3 e^{x^2} dx + C \end{aligned}$$

Considering:

$$\int x^3 e^{x^2} dx$$

Using Integration by Parts:

$$\begin{aligned} \text{Let } u &= x^2 & dv &= x e^{x^2} dx \\ du &= 2x dx & v &= \frac{1}{2} e^{x^2} \end{aligned}$$

thus,

$$\begin{aligned} \int x^3 e^{x^2} dx &= x^2 \frac{1}{2} e^{x^2} - \int x e^{x^2} dx \\ &= \frac{1}{2} x^2 e^{x^2} - \frac{1}{2} e^{x^2} \end{aligned}$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

Then,

$$ye^{x^2} = \left(\frac{1}{2}x^2 e^{x^2} - \frac{1}{2}e^{x^2}\right) + C$$

$$ye^{x^2} = \left(\frac{1}{2}x^2 e^{x^2} - \frac{1}{2}e^{x^2}\right) + C$$

$$2ye^{x^2} = x^2 e^{x^2} - e^{x^2} + C$$

$$2y = (x^2 - 1) + Ce^{-x^2} \rightarrow \text{General Solution}$$

when $x = 1, y = 1$

$$2(1) = ((1)^2 - 1) + Ce^{-(1)^2}$$

$$C = 2e^1$$

Thus,

$$2y = (x^2 - 1) + 2e^1 e^{-x^2}$$

$$2y = (x^2 - 1) + 2e^{1-x^2}$$

$\rightarrow$ Particular Solution

29. $L(di/dt) + Ri = E$;
Where $L, R,$ and E are constants
when $t = 0, i = 0$

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L}$$

$$P(t) = \frac{R}{L}$$

$$Q(t) = \frac{E}{L}$$

For integrating factor:

$$e^{\int P(t)dt} = e^{\int \frac{R}{L}dt} = e^{\frac{R}{L}t}$$

Thus,

$$ie^{\frac{R}{L}t} = \int Q(t)e^{\frac{R}{L}t} dt + C$$

$$ie^{\frac{R}{L}t} = \int \frac{E}{L} e^{\frac{R}{L}t} dt + C$$

$$ie^{\frac{R}{L}t} = \frac{E}{R} e^{\frac{R}{L}t} + C \rightarrow \text{General Solution}$$

when $t = 0, i = 0$

$$ie^{\frac{R}{L}(0)} = \frac{E}{R} e^{\frac{R}{L}(0)} + C$$

$$C = -\frac{E}{R}$$

Thus,

$$ie^{\frac{R}{L}t} = \frac{E}{R} e^{\frac{R}{L}t} - \frac{E}{R}$$

Divide by $e^{\frac{R}{L}t}$;

$$i = \frac{E}{R} - \frac{E}{R} e^{-\frac{R}{L}t}$$

$$i = \frac{E}{R} (1 - e^{-\frac{R}{L}t}) \rightarrow \text{Particular Solution}$$

30. $L(di/dt) + Ri = E \sin \omega t$;
when $t = 0, i = 0$

$$\frac{di}{dt} + \frac{R}{L}i = \frac{E}{L} \sin \omega t$$

$$P(t) = \frac{R}{L}$$

$$Q(t) = \frac{E}{L} \sin \omega t$$

For integrating factor:

$$e^{\int P(t)dt} = e^{\int \frac{R}{L}dt} = e^{\frac{R}{L}t}$$

Thus,

$$ie^{\frac{R}{L}t} = \int Q(t)e^{\frac{R}{L}t} dt + C$$

$$ie^{\frac{R}{L}t} = \int \frac{E}{L} \sin \omega t e^{\frac{R}{L}t} dt + C$$

$$ie^{\frac{R}{L}t} = \frac{E}{L} \int e^{\frac{R}{L}t} \sin \omega t dt + C$$

Considering:

$$\int e^{\frac{R}{L}t} \sin \omega t dt$$

Using Integration by Parts 1:

$$\text{Let } u = e^{\frac{R}{L}t} \quad dv = \sin \omega t dt$$

$$du = \frac{R}{L} e^{\frac{R}{L}t} dt \quad v = -\frac{1}{\omega} \cos \omega t$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

$$\int e^{\frac{R}{L}t} \sin \omega t dt$$

$$= -\frac{1}{\omega} \cos \omega t e^{\frac{R}{L}t} + \frac{R}{\omega L} \int e^{\frac{R}{L}t} \cos \omega t dt$$

$$\rightarrow (1)$$

Considering:

$$\int e^{\frac{R}{L}t} \cos \omega t dt$$

Using Integration by Parts 2:

$$\text{Let } u = e^{\frac{R}{L}t} \quad dv = \cos \omega t dt$$

$$du = \frac{R}{L} e^{\frac{R}{L}t} dt \quad v = \frac{1}{\omega} \sin \omega t$$

$$\int e^{\frac{R}{L}t} \cos \omega t dt$$

$$= \frac{1}{\omega} \sin \omega t e^{\frac{R}{L}t} - \frac{R}{\omega L} \int e^{\frac{R}{L}t} \sin \omega t dt \rightarrow (2)$$

Combining (1) and (2):

$$\int e^{\frac{R}{L}t} \sin \omega t dt$$

$$= -\frac{1}{\omega} \cos \omega t e^{\frac{R}{L}t} + \frac{R}{\omega L} \left(\frac{1}{\omega} \sin \omega t e^{\frac{R}{L}t} - \frac{R}{\omega L} \int e^{\frac{R}{L}t} \sin \omega t dt \right)$$

$$\int e^{\frac{R}{L}t} \sin \omega t dt$$

$$= -\frac{1}{\omega} \cos \omega t e^{\frac{R}{L}t} + \frac{R}{\omega^2 L} \sin \omega t e^{\frac{R}{L}t}$$

$$- \frac{R^2}{\omega^2 L^2} \int e^{\frac{R}{L}t} \sin \omega t dt$$

$$\int e^{\frac{R}{L}t} \sin \omega t dt + \frac{R^2}{\omega^2 L^2} \int e^{\frac{R}{L}t} \sin \omega t dt$$

$$= -\frac{1}{\omega} \cos \omega t e^{\frac{R}{L}t} + \frac{R}{\omega^2 L} \sin \omega t e^{\frac{R}{L}t}$$

$$\left(1 + \frac{R^2}{\omega^2 L^2}\right) \int e^{\frac{R}{L}t} \sin \omega t dt$$

$$= -\frac{1}{\omega} \cos \omega t e^{\frac{R}{L}t} + \frac{R}{\omega^2 L} \sin \omega t e^{\frac{R}{L}t}$$

$$\left(\frac{\omega^2 L^2 + R^2}{\omega^2 L^2}\right) \int e^{\frac{R}{L}t} \sin \omega t dt$$

$$= \frac{1}{\omega} e^{\frac{R}{L}t} \left(\frac{R}{\omega L} \sin \omega t - \cos \omega t\right)$$

$$\int e^{\frac{R}{L}t} \sin \omega t dt$$

$$= \frac{1}{\omega} e^{\frac{R}{L}t} \left(\frac{R}{\omega L} \sin \omega t - \cos \omega t\right)$$

$$= \frac{R \sin \omega t - \omega \cos \omega t}{\omega^2 L^2 + R^2} e^{\frac{R}{L}t}$$

$$\int e^{\frac{R}{L}t} \sin \omega t dt$$

$$= \frac{\omega^2 L^2}{\omega} e^{\frac{R}{L}t} \left(\frac{R}{\omega L} \sin \omega t - \cos \omega t\right)$$

$$= \frac{R \sin \omega t - \omega \cos \omega t}{\omega^2 L^2 + R^2} e^{\frac{R}{L}t}$$

$$\int e^{\frac{R}{L}t} \sin \omega t dt$$

$$= \frac{L^2 e^{\frac{R}{L}t} (R \sin \omega t - \omega \cos \omega t)}{\omega^2 L^2 + R^2}$$

Then,

$$ie^{\frac{R}{L}t} = \frac{E}{L} \left(\frac{L^2 e^{\frac{R}{L}t} (R \sin \omega t - \omega \cos \omega t)}{\omega^2 L^2 + R^2} \right)$$

$$+ C$$

$$ie^{\frac{R}{L}t} = \left(\frac{EL e^{\frac{R}{L}t} (R \sin \omega t - \omega \cos \omega t)}{\omega^2 L^2 + R^2} \right)$$

$$+ C$$

$$\rightarrow \text{General Solution}$$

when $t = 0, i = 0$

$$(0)e^{\frac{R}{L}(0)}$$

$$= \left(\frac{EL e^{\frac{R}{L}(0)} (R \sin \omega(0) - \omega \cos \omega(0))}{\omega^2 L^2 + R^2} \right)$$

$$+ C$$

$$C = \frac{\omega EL}{\omega^2 L^2 + R^2}$$

Thus,

$$ie^{\frac{R}{L}t} = \left(\frac{EL e^{\frac{R}{L}t} (R \sin \omega t - \omega \cos \omega t)}{\omega^2 L^2 + R^2} \right)$$

$$+ \frac{\omega EL}{\omega^2 L^2 + R^2}$$

$$\text{Let } z^2 = R^2 + \omega^2 L^2$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

$$ie^{\frac{k}{L}t} = \left(\frac{ELe^{\frac{k}{L}t} \left(\frac{R}{L} \sin \omega t - \omega \cos \omega t \right)}{z^2} + \frac{\omega EL}{z^2} \right)$$

$$ie^{\frac{k}{L}t} = \frac{ELe^{\frac{k}{L}t} \left(\frac{R}{L} \sin \omega t - \omega \cos \omega t \right) + \omega EL}{z^2}$$

$$ie^{\frac{k}{L}t} = Ez^{-2}e^{\frac{k}{L}t} \left(R \sin \omega t - \omega L \cos \omega t + \omega EL e^{-\frac{k}{L}t} \right)$$

Divide by $e^{\frac{k}{L}t}$:

$$i = Ez^{-2} \left(R \sin \omega t - \omega L \cos \omega t + \omega EL e^{-\frac{k}{L}t} \right)$$

→ Particular Solution

31. Find that solution of $y' = 2(2x - y)$; which passes through the point $(0, -1)$

$$\frac{dy}{dx} = 2(2x - y)$$

$$\frac{dy}{dx} + 2y = 4x$$

$$P(x) = 2$$

$$Q(x) = 4x$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int 2dx} = e^{2x}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ye^{2x} = \int 4xe^{2x}dx + C$$

Considering:

$$\int xe^{2x}dx$$

Using Integration by Parts:

$$\text{Let } u = x \quad dv = e^{2x}dx$$

$$du = dx \quad v = \frac{1}{2}e^{2x}$$

thus,

$$\int xe^{2x}dx = x \frac{1}{2}e^{2x} - \frac{1}{2} \int e^{2x}dx$$

$$= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x}$$

Then,

$$ye^{2x} = 4 \left(\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} \right) + C$$

$$ye^{2x} = 2xe^{2x} - e^{2x} + C$$

→ General Solution

which passes through the point $(0, -1)$

$$(-1)e^{2(0)} = 2(0)e^{2(0)} - e^{2(0)} + C$$

$$C = 0$$

Thus,

$$ye^{2x} = 2xe^{2x} - e^{2x}$$

$$y = 2x - 1 \rightarrow \text{Particular Solution}$$

32. Find that solution of $y' = 2(2x - y)$; which passes through the point $(0, 1)$

$$\frac{dy}{dx} = 2(2x - y)$$

$$\frac{dy}{dx} + 2y = 4x$$

$$P(x) = 2$$

$$Q(x) = 4x$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int 2dx} = e^{2x}$$

Thus,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ye^{2x} = \int 4xe^{2x}dx + C$$

TYPE 4: LINEAR DIFFERENTIAL EQUATIONS

Considering:

$$\int xe^{2x}dx$$

Using Integration by Parts:

$$\text{Let } u = x \quad dv = e^{2x}dx$$

$$du = dx \quad v = \frac{1}{2}e^{2x}$$

thus,

$$\int xe^{2x}dx = x \frac{1}{2}e^{2x} - \frac{1}{2} \int e^{2x}dx$$

$$= \frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x}$$

Then,

$$ye^{2x} = 4 \left(\frac{1}{2}xe^{2x} - \frac{1}{4}e^{2x} \right) + C$$

$$ye^{2x} = 2xe^{2x} - e^{2x} + C$$

→ General Solution

which passes through the point $(0, 1)$:

$$(1)e^{2(0)} = 2(0)e^{2(0)} - e^{2(0)} + C$$

$$C = 2$$

Thus,

$$ye^{2x} = 2xe^{2x} - e^{2x} + 2$$

$$y = 2x - 1 + 2e^{-2x}$$

→ Particular Solution

33. $(1 + t^2)ds + 2t[st^2 - 3(1 + t^2)^2]dt = 0$; when $t = 0, s = 2$

Divide by $(1 + t^2)dt$:

$$\frac{ds}{dt} + \frac{2t[st^2 - 3(1 + t^2)^2]}{(1 + t^2)} = 0$$

$$\frac{ds}{dt} + \frac{2t^3}{(1 + t^2)}s = 6t(1 + t^2)$$

$$P(t) = \frac{2t^3}{(1 + t^2)}$$

$$Q(t) = 6t(1 + t^2)$$

For integrating factor:

$$e^{\int P(t)dt} = e^{\int \frac{2t^3}{(1 + t^2)}dt}$$

Considering:

$$\int \frac{2t^3}{(1 + t^2)}dt$$

Using Integration by Substitution:

$$\text{Let } u = t^2$$

$$du = 2tdt$$

$$\int \frac{2t^3}{(1 + t^2)}dt = \int \frac{udu}{1 + u}$$

$$= \int \left(1 - \frac{1}{1 + u} \right) du$$

$$= u - \ln(1 + u)$$

$$= t^2 - \ln(1 + t^2)$$

thus,

$$e^{\int P(t)dt} = e^{t^2 - \ln(1 + t^2)} = e^{t^2} \cdot e^{-\ln(1 + t^2)}$$

$$= e^{t^2} (1 + t^2)^{-1}$$

Then,

$$se^{\int P(t)dt} = \int Q(t)e^{\int P(t)dt}dt + C$$

$$se^{t^2} (1 + t^2)^{-1} = \int 6t(1 + t^2)e^{t^2} (1 + t^2)^{-1}dt + C$$

$$se^{t^2} (1 + t^2)^{-1} = \int 6te^{t^2}dt + C$$

$$se^{t^2} (1 + t^2)^{-1} = 3e^{t^2} + C$$

→ General Solution

when $t = 0, s = 2$

$$(2)e^{(0)^2} (1 + (0)^2)^{-1} = 3e^{(0)^2} + C$$

$$C = -1$$

Thus,

$$se^{t^2} (1 + t^2)^{-1} = 3e^{t^2} - 1$$

Multiply by $e^{-t^2} (1 + t^2)$

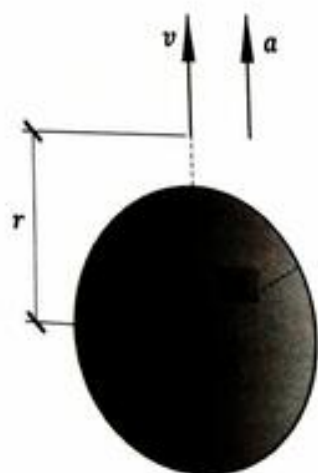
$$s = 3(1 + t^2) - (1 + t^2)e^{-t^2}$$

$$s = (1 + t^2)(3 - e^{-t^2})$$

→ Particular Solution

CHAPTER 3: ELEMENTARY DIFFERENTIAL APPLICATIONS

3.1 VELOCITY OF ESCAPE FROM THE EARTH



Let:

r = variable distance

R = radius of the Earth

t = time

v = velocity of the particle

a = acceleration

k = constant of proportionality in the Newton Law

$$v^2 = \frac{2gR^2}{r} + v_0^2 - 2gR$$

If $v_0^2 - 2gR < 0$, there will be a critical value of r for which the right member of the equation is $v = 0$. For this condition the body will stop and reverse its direction, that is back to earth. For no return, the velocity should always be positive but $v > 0$.

If $v_0^2 - 2gR \geq 0$, thus, $v_0 \geq \sqrt{2gR}$, the body will escape.

where:

$$g = 32.2 \text{ ft/sec}^2 = 6.09 \times 10^{-3} \text{ miles/sec}^2$$

APPLICATIONS: VELOCITY OF ESCAPE FROM THE EARTH

EXERCISES:

1. The radius of the moon is roughly 1080 miles. The acceleration of gravity at the surface of the moon is about $0.165g$, where g is the acceleration of gravity at the surface of the earth. Determine the velocity of escape for the moon.

Given:

$$g_m = 0.165g_e$$

$$g_m = 0.165(6.09 \times 10^{-3} \text{ mi/sec}^2) = 1.00485 \times 10^{-3} \text{ mile/sec}^2$$

$$R_m = 1080 \text{ miles}$$

Solution:

$$V_e = \sqrt{2gR}$$

$$V_e = \sqrt{2(1.00485 \times 10^{-3})(1080)}$$

$$V_e = 1.473 \text{ miles/sec}$$

2. Determine, to two significant figures, the velocity of escape for each of the celestial bodies listed below. The data given are rough and g may be taken to be $6.1 \times 10^{-3} \text{ miles/sec}^2$

	Acceleration of gravity at surface	Radius (miles)	Answer (miles/sec)
Venus	0.85g	3,800	6.3
Mars	0.38g	2,100	3.1
Jupiter	2.6g	43,000	37
Sun	28g	432,000	380
Ganymede	0.12g	1,780	1.6

Solutions:

❖ For Venus,

$$V_e = \sqrt{2gR}$$

$$V_e = \sqrt{2(0.85)(6.1 \times 10^{-3})(3800)}$$

$$V_e = 6.30 \text{ miles/sec}$$

❖ For Mars,

$$V_e = \sqrt{2gR}$$

$$V_e = \sqrt{2(0.38)(6.1 \times 10^{-3})(2100)}$$

$$V_e = 3.12 \text{ miles/sec}$$

❖ For Jupiter,

$$V_e = \sqrt{2gR}$$

$$V_e = \sqrt{2(2.6)(6.1 \times 10^{-3})(43000)}$$

$$V_e = 36.93 \text{ miles/sec}$$

❖ For Sun,

$$V_e = \sqrt{2gR}$$

$$V_e = \sqrt{2(28)(6.1 \times 10^{-3})(432000)}$$

$$V_e = 384.15 \text{ miles/sec}$$

❖ For Ganymede,

$$V_e = \sqrt{2gR}$$

$$V_e = \sqrt{2(0.12)(6.1 \times 10^{-3})(1780)}$$

$$V_e = 1.61 \text{ miles/sec}$$

3.2 NEWTON'S LAW OF COOLING

Experiment has shown that under certain conditions, a good approximation to the temperature of an object can be obtained by using Newton's Law of Cooling: The temperature of a body changes at a rate that is proportional to the difference in temperature between the outside medium and the body itself. We shall assume here that the constant of proportionality is the same whether the temperature is increasing or decreasing.

$$\frac{du}{dt} \propto u - u_a$$

$$\frac{du}{dt} = -k(u - u_a)$$

$$\frac{du}{(u - u_a)} = -k dt$$

$$\int \frac{du}{(u - u_a)} = \int -k dt$$

$$\ln(u - u_a) = -kt + \ln C$$

$$\ln \frac{(u - u_a)}{C} = -kt$$

$$\frac{(u - u_a)}{C} = e^{-kt}$$

$$(u - u_a) = C e^{-kt}$$

$$u = u_a + C e^{-kt}$$

where,

$\frac{du}{dt}$ = time rate of change of temperature

u = temperature of the body at time t

u_a = temperature of the medium at time t

EXERCISES:

3. A thermometer reading 18°F is brought into a room where the temperature is 70°F . One (1) minute later, the thermometer reading is 31°F . Determine the temperature reading as a function of time and, in particular, find the temperature reading 5 minutes after the thermometer is first brought into the room.

Given:

t	u	u_a
0	18°F	70°F
1	31°F	70°F

Solution:

- (a) Temperature Reading as a function of time
From the general Equation

$$u = u_a + C e^{-kt}$$

$$u = 70 + C e^{-kt}$$

When $t = 0, u = 18^\circ\text{F}$

$$18 = 70 + C e^{-k(0)}$$

$$C = -52$$

$$u = 70 - 52 e^{-kt}$$

When $t = 1, u = 31^\circ\text{F}$

$$31 = 70 - 52 e^{-k(1)}$$

$$k = 0.28768$$

Thus,

$$u = 70 - 52 e^{-0.28768t}$$

(b) when $t = 5 \text{ min}, u = ?$

$$u = 70 - 52 e^{-0.28768(5)} = 57.66^\circ\text{F} \approx 58^\circ\text{F}$$

4. A thermometer reading 75°F is taken out where the temperature is 20°F . The reading is 30°F 4 minutes later. Find (a) the thermometer reading 7 minutes after the thermometer was brought outside, and (b) the time taken for the reading to drop from 75°F to within a half degree of the air temperature.

Given:

t	u	u_a
0	75°F	20°F
4	30°F	20°F
7	?	20°F
?	20.5°F	20°F

Solution:

$$u = u_a + C e^{-kt}$$

$$u = 20 + C e^{-kt}$$

When $t = 0, u = 75^\circ\text{F}$

$$75 = 20 + C e^{-k(0)}$$

$$C = 55$$

$$u = 20 + 55 e^{-kt}$$

When $t = 4, u = 30^\circ\text{F}$

$$30 = 20 + 55 e^{-k(4)}$$

$$k = 0.42619$$

Thus,

$$u = 20 + 55 e^{-0.42619t}$$

(a) when $t = 7 \text{ min}, u = ?$

$$u = 20 + 55 e^{-0.42619(7)} = 22.78^\circ\text{F} \approx 23^\circ\text{F}$$

(b) when $u = 20.5^\circ\text{F}, t = ?$

$$20.5 = 20 + 55 e^{-0.42619(t)}$$

$$t = 11.03 \text{ min} \approx 11.5 \text{ min}$$

5. At 1:00 P.M., a thermometer reading 70°F is taken outside where the air temperature is -10°F (ten below zero). At 1:02 P.M., the reading is 26°F . At 1:05 P.M., the thermometer is taken back indoors where the air is at 70°F . What is the thermometer reading at 1:09 P.M.?

Given:

Outside	t	u	u_a
1:00 P.M.	0	70°F	-10°F
1:02 P.M.	2	26°F	-10°F
1:05 P.M.	5	?	-10°F

Inside

	t	u	u_a
1:05 P.M.	0	?	70°F
1:09 P.M.	4	?	70°F

Solution:

✦ For Outside:

$$u = u_a + C e^{-kt}$$

$$u = -10 + C e^{-kt}$$

When $t = 0, u = 70^\circ\text{F}$

$$70 = -10 + C e^{-k(0)}$$

$$C = 80$$

$$u = -10 + 80 e^{-kt}$$

When $t = 2, u = 26^\circ\text{F}$

$$26 = -10 + 80 e^{-k(2)}$$

$$k = 0.39925$$

Thus,

$$u = -10 + 80 e^{-0.39925t}$$

When $t = 5, u = ?$

$$u = -10 + 80 e^{-0.39925(5)} = 0.86750^\circ\text{F}$$

APPLICATIONS: NEWTON'S LAW OF COOLING

◆ For Inside:

$$u = u_a + Ce^{-kt}$$

$$u = 70 + Ce^{-kt}$$

For the same temperature,

$$k = 0.39925$$

Then,

$$u = 70 + Ce^{-0.39925t}$$

When $t = 0$, $u = 0.86750^\circ\text{F}$ @ 1:05 P.M.

$$0.86750 = 70 + Ce^{-0.39925(0)}$$

$$C = -69.1343$$

$$u = 70 - 69.1343e^{-0.39925t}$$

When $t = 4$, $u = ?$ @ 1:09 P.M.

$$u = 70 - 69.1343e^{-0.39925(4)}$$

$$u = 56.00^\circ\text{F}$$

6. At 9 A.M., a thermometer reading 70°F is taken outdoors where the temperature is 15°F . At 9:05 A.M., the thermometer reading is 45°F . At 9:10 A.M., the thermometer is taken back indoors where the temperature is fixed at 70°F . Find (a) the reading at 9:20 A.M. and (b) when the reading, to the nearest degree, will show the correct (70°F) indoor temperature.

Given:

Outside

	t	u	u_a
9:00 A.M.	0	70°F	15°F
9:05 A.M.	5	45°F	15°F
9:10 A.M.	10	u_{10}	15°F

Inside

	t	u	u_a
9:10 A.M.	0	u_{10}	70°F
9:20 A.M.	10	?	70°F
	?	69.5	70°F

Solution:

◆ For Outdoors:

$$u = u_a + Ce^{-kt}$$

$$u = 15 + Ce^{-kt}$$

When $t = 0$, $u = 70^\circ\text{F}$

$$70 = 15 + Ce^{-k(0)}$$

$$C = 55$$

$$u = 15 + 55e^{-kt}$$

When $t = 5$, $u = 45^\circ\text{F}$

$$45 = 15 + 55e^{-k(5)}$$

$$k = 0.12123$$

Thus,

$$u = 15 + 55e^{-0.12123t}$$

When $t = 10$, $u_{10} = ?$

$$u_{10} = 15 + 55e^{-0.12123(10)} = 31.36317^\circ\text{F}$$

◆ For Indoors:

$$u = u_a + Ce^{-kt}$$

$$u = 70 + Ce^{-kt}$$

For the same temperature, $k = 0.12123$

Then,

$$u = 70 + Ce^{-0.12123t}$$

When $t = 0$, $u = 31.36317^\circ\text{F}$ @ 9:10 A.M.

$$31.36317 = 70 + Ce^{-0.12123(0)}$$

$$C = -38.63683$$

$$u = 70 - 38.63683e^{-0.12123t}$$

When $t = 10$, $u = ?$ @ 9:20 P.M.

$$u = 70 - 38.63683e^{-0.12123(10)}$$

$$u = 58.50^\circ\text{F} \approx 58^\circ\text{F}$$

APPLICATIONS: NEWTON'S LAW OF COOLING

When $t = ?$, $u = 69.5^\circ\text{F}$

$$69.5 = 70 - 38.63683e^{-0.12123t}$$

$$t = 35.86 \text{ min} \approx 36 \text{ min}$$

Thus,

$$T = 9:10 \text{ A.M.} + 36 \text{ min}$$

$$T = 9:46 \text{ A.M.}$$

7. At 2:00 P.M., a thermometer reading 80°F is taken outside where the air temperature is 20°F . At 2:03 P.M., the temperature reading yielded by the thermometer is 42°F . Later, the thermometer is brought inside where the air is at 80°F . At 2:10 P.M., the reading is 71°F . When was the thermometer brought indoors?

Given:

Outside

	t	u	u_a
2:00 P.M.	0	80°F	20°F
2:03 P.M.	3	42°F	20°F
2:00 + x	x	u_x	20°F

Inside

	t	u	u_a
2:00 + x	0	u_x	80°F
2:10 P.M.	$10 - x$	71°F	80°F

Required:

x time when the thermometer brought inside.

Solution:

◆ For Outside:

$$u = u_a + Ce^{-kt}$$

$$u = 20 + Ce^{-kt}$$

When $t = 0$, $u = 80^\circ\text{F}$

$$80 = 20 + Ce^{-k(0)}$$

$$C = 60$$

$$u = 20 + 60e^{-kt}$$

When $t = 3$, $u = 42^\circ\text{F}$

$$42 = 20 + 60e^{-k(3)}$$

$$k = 0.33443$$

Thus,

$$u = 20 + 60e^{-0.33443t}$$

When $t = x$, $u = u_x$

$$u_x = 20 + 60e^{-0.33443x} \rightarrow (1)$$

◆ For Inside:

$$u = u_a + Ce^{-kt}$$

$$u = 80 + Ce^{-kt}$$

For the same temperature, $k = 0.33443$

Then,

$$u = 80 + Ce^{-0.33443t}$$

When $t = 0$, $u = u_x$

$$u_x = 80 + Ce^{-0.33443(0)}$$

$$C = u_x - 80$$

$$u = 80 + (u_x - 80)e^{-0.33443t} \rightarrow (2)$$

When $t = 10 - x$, $u = 71^\circ\text{F}$

$$71 = 80 + (u_x - 80)e^{-0.33443(10-x)}$$

$$71 = 80 + u_x e^{-0.33443(10-x)} - 80e^{-0.33443(10-x)}$$

$$u_x = \frac{71 - 80 + 80e^{-0.33443(10-x)}}{e^{-0.33443(10-x)}} \rightarrow (2)$$

Equate Equation (1) and (2):

$$20 + 60e^{-0.33443x} = \frac{71 - 80 + 80e^{-0.33443(10-x)}}{e^{-0.33443(10-x)}}$$

$$x = 4.96 \approx 5 \text{ min}$$

Thus,

$$T = 2:00 \text{ P.M.} + 5 \text{ min}$$

$$T = 2:05 \text{ P.M.}$$

3.2 SIMPLE CHEMICAL CONVERSION

It is known from the results of chemical experimentation that, in certain reactions in which a substance A is being converted into another substance, the time rate of change of the amount x of unconverted substance is proportional to x .

$$\frac{dx}{dt} \propto x$$

$$\frac{dx}{dt} = -kx$$

$$\frac{dx}{x} = -k dt$$

$$\int \frac{dx}{x} = -k \int dt$$

$$\ln x = -kt + \ln C$$

$$\ln \frac{x}{C} = -kt$$

$$\frac{x}{C} = e^{-kt}$$

$$x = Ce^{-kt}$$

where:

x = the amount of unconverted substance

$\frac{dx}{dt}$ = time rate of conversion

EXERCISES:

8. Suppose that a chemical reaction proceeds according to the law given in Section 15 above. If half the substance A has been converted at the end of 10 sec, find when nine-tenths of the substance will have been converted.

Solution:

$$x = Ce^{-kt}$$

When $t = 0$, $x = A_0$

$$\begin{aligned} A_0 &= Ce^{-k(0)} \\ C &= A_0 \\ \rightarrow x &= A_0 e^{-kt} \end{aligned}$$

When $t = 10$ sec, $x = \frac{1}{2}A_0$

$$\begin{aligned} \frac{1}{2}A_0 &= A_0 e^{-k(10)} \\ k &= 0.06931 \end{aligned}$$

Thus,

$$x = A_0 e^{-0.06931t}$$

When $x = \frac{1}{10}A_0$, $t = ?$

$$\begin{aligned} \frac{1}{10}A_0 &= A_0 e^{-0.06931t} \\ t &= 33.22 \text{ sec} \approx 33 \text{ sec} \end{aligned}$$

9. The conversion of a substance B follows the law in Section 15 above. If only a fourth of the substance has been converted at the end of 10 sec, find when nine-tenths of the substance will have been converted.

Solution:

$$x = Ce^{-kt}$$

When $t = 0$, $x = B_0$

$$\begin{aligned} B_0 &= Ce^{-k(0)} \\ C &= B_0 \\ \rightarrow x &= B_0 e^{-kt} \end{aligned}$$

When $t = 10$ sec, $x = \frac{3}{4}B_0$

$$\begin{aligned} \frac{3}{4}B_0 &= B_0 e^{-k(10)} \\ k &= 0.02877 \end{aligned}$$

Thus,

$$x = B_0 e^{-0.02877t}$$

When $x = \frac{1}{10}B_0$, $t = ?$

$$\begin{aligned} \frac{1}{10}B_0 &= B_0 e^{-0.02877t} \\ t &= 80.03 \text{ sec} \approx 80 \text{ sec} \end{aligned}$$

10. For a substance C , the time rate of conversion is proportional to the square of the amount x of unconverted substance. Let k be the numerical value of the constant of proportionality and let the amount of unconverted substance be x_0 at time $t = 0$. Determine x for all $t \geq 0$.

Solution:

$$\frac{dx}{dt} \propto x^2$$

$$\frac{dx}{dt} = -kx^2$$

$$\int \frac{dx}{x^2} = -k \int dt$$

$$-\frac{1}{x} = -kt - C$$

$$\frac{1}{x} = kt + C$$

When $t = 0$, $x = x_0$

$$\begin{aligned} \frac{1}{x_0} &= k(0) + C \\ C &= \frac{1}{x_0} \end{aligned}$$

Then,

$$\frac{1}{x} = kt + \frac{1}{x_0}$$

$$\frac{1}{x} = \frac{x_0 kt + 1}{x_0}$$

$$x = \frac{x_0}{1 + x_0 kt}$$

13. Radium decomposes at a rate proportional to the quantity of radium present. Suppose that it is found that in 25 years approximately 1.1% of a certain quantity of radium has decomposed. Determine approximately how long it will take for one-half the original amount of radium to decompose.

Solution:

$$x = Ce^{-kt}$$

When $t = 0$, $x = x_0$

$$x_0 = Ce^{-k(0)}$$

$$C = x_0$$

$$\rightarrow x = x_0 e^{-kt}$$

When $t = 25$ years, $x = (1 - 1.1\%)x_0$

$$0.989x_0 = x_0 e^{-k(25)}$$

$$k = 0.000442$$

Thus,

$$x = x_0 e^{-0.000442t}$$

When $x = 0.5x_0$, $t = ?$

$$0.5x_0 = x_0 e^{-0.000442t}$$

$$t = 1568.21 \text{ years}$$

$$\approx 1600 \text{ years}$$

14. A certain radioactive substance has a half-life of 38 hours. Find how long it takes for 90% of the radioactivity to be dissipated.

Solution:

$$x = Ce^{-kt}$$

When $t = 0$, $x = x_0$

$$x_0 = Ce^{-k(0)}$$

$$C = x_0$$

$$\rightarrow x = x_0 e^{-kt}$$

When $t = 38$ hours, $x = \frac{1}{2}x_0$

$$\frac{1}{2}x_0 = x_0 e^{-k(38)}$$

$$k = 0.01824$$

Thus,

$$x = x_0 e^{-0.01824t}$$

When $x = 0.10x_0$, $t = ?$

$$0.10x_0 = x_0 e^{-0.01824t}$$

$$t = 126.24 \text{ hours} \approx 126 \text{ hours}$$

15. A bacterial population B is known to have a rate of growth proportional to B itself. If between noon and 2 P.M., the population triples, at what time, no controls being exerted, should B become 100 times what it was at noon?

Solution:

$$B = Ce^{-kt}$$

When $t = 0$, $B = B_0$

$$B_0 = Ce^{-k(0)}$$

$$C = B_0$$

$$\rightarrow B = B_0 e^{-kt}$$

When $t = 2$ hours, $B = 3B_0$

$$3B_0 = B_0 e^{-k(2)}$$

$$k = -0.54931$$

Thus,

$$B = B_0 e^{0.54931t}$$

When $B = 100B_0$, $t = ?$

$$100B_0 = B_0 e^{0.54931t}$$

$$t = 8.38 \text{ hours}$$

$$0.38 \text{ hours} \approx 22 \text{ mins}$$

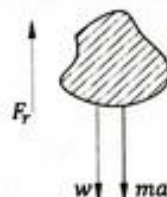
$$\therefore \text{Time} = 8:22 \text{ P.M.}$$

16. In the motion of an object through a certain medium (air at certain pressures is an example), the medium furnishes a resisting force proportional to the square of the velocity of the moving object. Suppose a body falls, due to the action of gravity, through such a medium. Let t represent time, and v represent velocity, positive downward. Let g be the usual constant acceleration of gravity and let w be the weight of the body. Use Newton's Law, force equals mass times acceleration, to conclude that the differential equation of the motion is

$$\frac{w}{g} \frac{dv}{dt} = w - kv^2$$

where kv^2 is the magnitude of the resisting force furnished by the medium.

Solution:



$$F_{\text{net}} = w - F_r$$

By Newton's Second Law:

$$F = ma = \frac{w}{g} \frac{dv}{dt}$$

Then,

$$\frac{w}{g} \frac{dv}{dt} = w - F_r$$

Since,

$$F_r \propto v^2$$

$$F_r = kv^2$$

Therefore,

$$w - kv^2 = \frac{w}{g} \frac{dv}{dt}$$

17. Solve the differential equation of exercises #16 with the initial condition that $v = v_0$ when $t = 0$. Introduce the constant $a^2 = w/k$ to simplify formulas.

Solution:

$$w - kv^2 = \frac{w}{g} \frac{dv}{dt}$$

$$\frac{g}{w} dt = \frac{dv}{w - kv^2}$$

$$\frac{g}{w} dt = \frac{dv}{k \left(\frac{w}{k} - v^2 \right)}$$

$$\frac{k}{w} g dt = \frac{dv}{\frac{w}{k} - v^2}$$

Let: $a^2 = w/k$

$$a^{-2} g dt = \frac{dv}{a^2 - v^2}$$

$$\int a^{-2} g dt = \int \frac{dv}{a^2 - v^2}$$

$$a^{-2} g t = \frac{1}{2a} \ln \frac{a+v}{a-v} + C$$

When $t = 0$, $v = v_0$:

$$[a^{-2} g t]_0^t = \left[\frac{1}{2a} \ln \frac{a+v}{a-v} \right]_{v_0}^v$$

$$a^{-2} g t - 0 = \frac{1}{2a} \ln \frac{a+v}{a-v} - \frac{1}{2a} \ln \frac{a+v_0}{a-v_0}$$

$$a^{-2} g t = \frac{1}{2a} \ln \left[\frac{a+v}{a-v} \cdot \frac{a-v_0}{a+v_0} \right]$$

$$2a^{-1} g t = \ln \left[\frac{a+v}{a-v} \cdot \frac{a-v_0}{a+v_0} \right]$$

$$e^{2a^{-1} g t} = \frac{a+v}{a-v} \cdot \frac{a-v_0}{a+v_0}$$

$$\frac{a+v_0}{a-v_0} e^{2a^{-1} g t} = \frac{a+v}{a-v}$$

$$\frac{a+v}{a-v} = \frac{a+v_0}{a-v_0} e^{2g t/a}$$

APPLICATIONS: MOTION THROUGH A MEDIUM

18. There are mediums that resist motion through them with a force proportional to the first power of the velocity. For such a medium, state and solve problems analogous to exercises #16 through #18 above, except that for a convenience a constant $b = w/k$ may be introduced to replace the a^2 of exercise #17. Show that b has the dimensions of a velocity.

Solution:

$$F_r \propto v$$

$$F_r = kv$$

$$F_{net} = w - F_r$$

By Newton's Second Law:

$$F = ma = \frac{w}{g} \frac{dv}{dt}$$

Then,

$$\frac{w}{g} \frac{dv}{dt} = w - kv$$

$$\frac{g}{w} dt = \frac{dv}{w - kv}$$

$$\frac{g}{w} dt = \frac{dv}{-k \left(\frac{-w}{k} + v \right)}$$

$$-\frac{k}{w} g dt = \frac{dv}{v - \frac{w}{k}}$$

$$\text{Let } b = w/k;$$

$$-b^{-1} g dt = \frac{dv}{v - b}$$

$$\int -\frac{g}{b} dt = \int \frac{dv}{v - b}$$

$$\left[-\frac{g}{b} t \right]_0^t = \left[\ln(v - b) \right]_{v_0}^v$$

$$-\frac{g}{b} t = \ln(v - b) - \ln(v_0 - b)$$

$$-\frac{g}{b} t = \ln \frac{(v - b)}{(v_0 - b)}$$

$$e^{-gt/b} = \frac{(v - b)}{(v_0 - b)}$$

$$v = b + (v_0 - b)e^{-gt/b}$$

APPLICATIONS: MIXING OF SOLUTIONS

24. A tank contains 80 gallons (gal) of pure water. A brine solution with 2 lb/gal of salt enters at 2 gal/min, and the well-stirred mixtures leaves at the same rate. Find (a) the amount of salt in the tank at any time, and (b) the time at which the brine leaving will contain 1 lb/gal of salt.

Solution:

Brine solution:

$$C_i = 2 \text{ lb/gal}$$

$$V_i = 2 \text{ gal/min}$$

(in)



Mixture:

$$C_0 = \frac{m}{80} \text{ lb/gal}$$

$$V_0 = 2 \text{ gal/min}$$

(out)

Let:

$$C = \text{concentration, } \frac{\text{lb salt}}{\text{gal}}$$

$$V = \text{volume of flow rate, gal/min}$$

$$m = \text{mass of salt in tank at any time, lb}$$

Then,

$$\frac{dm}{dt} = C_i V_i - C_0 V_0$$

$$\frac{dm}{dt} = 2(2) - \frac{m}{80}(2)$$

$$\frac{dm}{dt} = 4 - \frac{m}{40}$$

$$\frac{dm}{4 - \frac{m}{40}} = dt$$

$$\int \frac{40 dm}{160 - m} = \int dt$$

$$-40 \ln(160 - m) = t + C$$

$$\text{When } t = 0, m = 0;$$

$$-40 \ln(160 - 0) = 0 + C$$

$$C = -40 \ln 160$$

Thus,

$$-40 \ln(160 - m) = t - 40 \ln 160$$

$$-t = 40 \ln(160 - m) - 40 \ln 160$$

$$-t = 40 \ln \frac{160 - m}{160}$$

$$-\frac{t}{40} = \ln \frac{160 - m}{160}$$

$$e^{-t/40} = \frac{160 - m}{160}$$

$$m = 160 - 160e^{-t/40}$$

$$m = 160(1 - e^{-t/40})$$

(b)

$$\text{When } C_0 = 1 \frac{\text{lb salt}}{\text{gal}}$$

$$C_0 = \frac{m}{80} = 1$$

$$m = 80 \text{ lbs}$$

Then,

$$80 = 160(1 - e^{-t/40})$$

$$80 - 160 = -160e^{-t/40}$$

$$e^{-t/40} = \frac{1}{2}$$

$$\ln e^{-t/40} = \ln \frac{1}{2}$$

$$-\frac{t}{40} = \ln 1 - \ln 2$$

$$t = 40 \ln 2$$

25. For a tank in exercise #24, determine the limiting value for the amount of salt in the tank after a long time. How much time must pass for the amount of salt in the tank reaches 80% of this limiting value?

Solution:

(a) After a long time,

$$\frac{dm}{dt} \rightarrow 0$$

Thus,

$$\frac{dm}{dt} = 4 - \frac{m}{40} = 0$$

$$m = 160 \text{ lbs}$$

(b) When $m = 80\%$ of 160 lbs

$$m = 160(1 - e^{-t/40})$$

$$0.8(160) = 160(1 - e^{-t/40})$$

$$t = 64.38 \text{ min} \approx 64 \text{ min}$$

APPLICATIONS: ORTHOGONAL TRAJECTORIES

3.6 ORTHOGONAL TRAJECTORIES

Suppose that we have a family of curves given by

$$f(x, y, C)$$

One curve corresponding to each C in some range of values of the parameter C . In certain applications it is found desirable to know what curves have the property of intersecting a curve of the family at right angles whenever they do intersect.

Consider the Equation

$$Mdx + Ndy = 0$$

$$M + N \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{M}{N}$$

The differential equation of the orthogonal trajectories is,

$$\frac{dy}{dx} = \frac{N}{M}$$

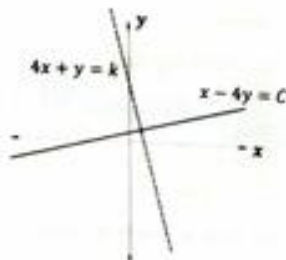
or

$$Ndx - Mdy = 0$$

EXERCISES:

In each exercise, find the orthogonal trajectories of the given family of curves. Draw a few representative curves of each family whenever a figure is requested.

1. $x - 4y = C$. Draw the figure



Differentiate,

$$1 - 4y' = 0$$

$$dx - 4dy = 0$$

$$\frac{dy}{dx} = \frac{1}{4}$$

The orthogonal trajectories,

$$\frac{dy}{dx} = -4$$

$$dy = -4dx$$

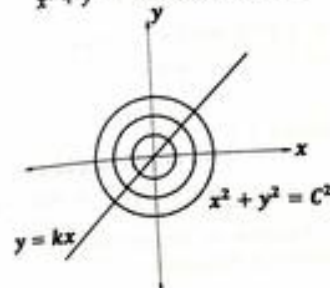
$$4dx + dy = 0$$

$$4 \int dx + \int dy = \int 0$$

$$4x + y = k \text{ ANS.}$$

APPLICATIONS: ORTHOGONAL TRAJECTORIES

2. $x^2 + y^2 = C^2$. Draw the figure



Differentiate,

$$2x + 2yy' = 0$$

$$x + y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = -\frac{x}{y}$$

The orthogonal trajectories,

$$\frac{dy}{dx} = \frac{y}{x}$$

$$\frac{dy}{y} = \frac{dx}{x}$$

$$\int \frac{dy}{y} = \int \frac{dx}{x}$$

$$\ln y = \ln x + \ln k$$

$$y = kx \text{ ANS.}$$

3. $x^2 - y^2 = C_1$.

$$2x - 2yy' = 0$$

$$x - y \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{x}{y}$$

The orthogonal trajectories,

$$\frac{dy}{dx} = -\frac{y}{x}$$

$$\frac{dy}{y} = -\frac{dx}{x}$$

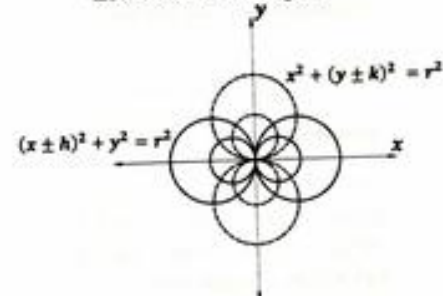
$$\frac{dy}{y} + \frac{dx}{x} = 0$$

$$\int \frac{dy}{y} + \int \frac{dx}{x} = \int 0$$

$$\ln y + \ln x = \ln C_2$$

$$xy = C_2 \text{ ANS.}$$

4. Circles through the origin with centers on the x -axis. Draw the figure



General Equation of a Circle:

$$(x - h)^2 + (y - k)^2 = r^2$$

With center on the x -axis @ $(h, 0)$;

$$(x - h)^2 + y^2 = r^2$$

But $h \pm r$

$$(x \pm r)^2 + y^2 = r^2$$

$$(x^2 \pm 2rx + r^2) + y^2 = r^2$$

$$x^2 \pm 2rx + y^2 = 0$$

$$x^2 + y^2 = \pm 2rx$$

$$\frac{x^2 + y^2}{x} = \pm 2r$$

Differentiate,

$$\frac{x[2xdx + 2ydy] - (x^2 + y^2)dx}{x^2} = 0$$

$$2x^2dx + 2xydy - x^2dx - y^2dx = 0$$

$$x^2dx - y^2dx + 2xydy = 0$$

$$(x^2 - y^2)dx + 2xydy = 0$$

$$\frac{dy}{dx} = -\frac{(x^2 - y^2)}{2xy}$$

The orthogonal trajectories,

$$\frac{dy}{dx} = \frac{2xy}{(x^2 - y^2)}$$

$$(x^2 - y^2)dy = 2xydx$$

$$2xydx - (x^2 - y^2)dy = 0$$

∴ Homogeneous, 2nd degree!

Let: $x = vy$
 $dx = vdy + ydv$

Substitute:

$$2vy^2(vdy + ydv) - (v^2y^2 - y^2)dy = 0$$

$$2v(vdy + ydv) - (v^2 - 1)dy = 0$$

$$2v^2dy + 2vydv - v^2dy + dy = 0$$

$$(v^2 + 1)dy + 2vydv = 0$$

$$\frac{dy}{y} + \frac{2v dv}{(v^2 + 1)} = 0$$

$$\int \frac{dy}{y} + \int \frac{2v dv}{(v^2 + 1)} = \int 0$$

$$\ln y + \ln(v^2 + 1) = \ln C$$

$$y(v^2 + 1) = C$$

But $v = x/y$,

$$y\left(\frac{x^2}{y^2} + 1\right) = C$$

$$y\left(\frac{x^2 + y^2}{y^2}\right) = C$$

$$x^2 + y^2 = Cy$$

When $x = h = \pm r$
 $y = k = \pm r$

$$r^2 + r^2 = C(\pm r)$$

$$C = \pm 2r$$

Then,

$$x^2 + y^2 = \pm 2ry$$

$$x^2 + y^2 \pm 2ry = 0$$

Adding r^2 on both sides,

$$x^2 + y^2 \pm 2ry + r^2 = 0 + r^2$$

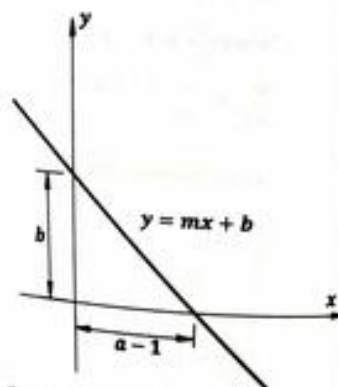
$$x^2 + (y \pm r)^2 = 0 + r^2$$

Since $k \pm r$

$$x^2 + (y \pm k)^2 = r^2 \text{ ANS.}$$

∴ Equation of circles through origin with centers on the y-axis.

5. Straight lines with slope and y-intercept equal. Draw the figure



General Equation of a Line:

$$y - y_1 = m(x - x_1)$$

$$\text{@ } (0, b);$$

$$y - b = m(x - 0)$$

$$y = mx + b$$

$$\frac{dy}{dx} = m = b$$

Then,

$$y = \frac{dy}{dx}x + \frac{dy}{dx}$$

$$ydx = (x + 1)dy$$

$$\frac{dy}{dx} = \frac{y}{(x + 1)}$$

The orthogonal trajectories,

$$\frac{dy}{dx} = -\frac{(x + 1)}{y}$$

$$ydy = -(x + 1)dx$$

$$ydy + (x + 1)dx = 0$$

$$\int ydy + \int (x + 1)dx = \int 0$$

$$\frac{y^2}{2} + \frac{(x + 1)^2}{2} = \frac{C}{2}$$

$$y^2 + (x + 1)^2 = C$$

When $y = 0, x = a - 1$

$$0^2 + (a - 1 + 1)^2 = C$$

$$C = a^2$$

Thus,

$$y^2 + (x + 1)^2 = a^2 \text{ ANS.}$$

6. $y^2 = Cx^3$

$$C = \frac{y^2}{x^3}$$

Differentiate,

$$0 = \frac{x^3(2ydy) - y^2(3x^2dx)}{x^6}$$

$$2x^3ydy - 3x^2y^2dx = 0$$

Divide by x^2y ,

$$2xdy - 3ydx = 0$$

$$\frac{dy}{dx} = \frac{3y}{2x}$$

The orthogonal trajectories,

$$\frac{dy}{dx} = -\frac{2x}{3y}$$

$$2xdx + 3ydy = 0$$

$$\int 2xdx + \int 3ydy = \int 0$$

$$x^2 + \frac{3y^2}{2} = k$$

$$2x^2 + 3y^2 = k^2 \text{ ANS.}$$

7. $e^x + e^{-y} = C_1$

Differentiate,

$$e^x dx - e^{-y} dy = 0$$

$$\frac{dy}{dx} = \frac{e^x}{e^{-y}}$$

The orthogonal trajectories,

$$\frac{dy}{dx} = -\frac{e^{-y}}{e^x}$$

$$\frac{dy}{e^{-y}} = -\frac{dx}{e^x}$$

$$e^y dy + e^{-x} dx = 0$$

$$e^y dy + e^{-x} dx = 0$$

$$\int e^y dy + \int e^{-x} dx = \int 0$$

$$e^y - e^{-x} = C_2 \text{ ANS.}$$

8. $y = C_1(\sec x + \tan x)$

$$C_1 = \frac{y}{\sec x + \tan x}$$

Differentiate,

$$0 = \frac{(\sec x + \tan x)dy - y(\sec x \tan x dx + \sec^2 x dx)}{(\sec x + \tan x)^2}$$

$$(\sec x + \tan x)dy - y(\sec x \tan x dx + \sec^2 x dx) = 0$$

$$(\sec x + \tan x)dy - y \sec x (\tan x + \sec x)dx = 0$$

Divide by $(\sec x + \tan x)$;

$$dy - y \sec x dx = 0$$

$$\frac{dy}{dx} = y \sec x$$

The orthogonal trajectories,

$$\frac{dy}{dx} = -\frac{1}{y \sec x}$$

$$\frac{dy}{dx} = -\frac{\cos x}{y}$$

$$y dy + \cos x dx = 0$$

$$\int y dy + \int \cos x dx = \int 0$$

$$\frac{y^2}{2} + \sin x = C_1$$

$$y^2 + 2 \sin x = 2C_1$$

$$y^2 = 2(C_1 - \sin x) \text{ ANS.}$$

$$9. \quad x^3 = 3(y - C)$$

$$x^3 = 3y - 3C$$

$$x^3 - 3y = -3C$$

Differentiate,

$$3x^2 dx - 3dy = 0$$

$$x^2 dx - dy = 0$$

$$\frac{dy}{dx} = x^2$$

The orthogonal trajectories,

$$\frac{dy}{dx} = -\frac{1}{x^2}$$

$$dy = -\frac{dx}{x^2}$$

$$dy + \frac{dx}{x^2} = 0$$

$$\int dy + \int \frac{dx}{x^2} = \int 0$$

$$y - \frac{1}{x} = k$$

$$x(y - k) = 1 \text{ ANS.}$$

$$10. \quad x = Ce^{y^2}$$

$$C = \frac{x}{e^{y^2}}$$

Differentiate,

$$0 = \frac{e^{y^2} dx - x(2ye^{y^2})dy}{e^{2y^2}}$$

$$e^{y^2} dx - 2xye^{y^2} dy = 0$$

Divide by e^{y^2}

$$dx - 2xydy = 0$$

$$\frac{dy}{dx} = \frac{1}{2xy}$$

The orthogonal trajectories,

$$\frac{dy}{dx} = -2xy$$

$$\frac{dy}{y} = -2xdx$$

$$\frac{dy}{y} + 2xdx = 0$$

$$\int \frac{dy}{y} + \int 2xdx = \int 0$$

$$\ln y + x^2 = \ln C_1$$

$$\ln y - \ln C_1 = -x^2$$

$$\frac{y}{C_1} = e^{-x^2}$$

$$y = C_1 e^{-x^2} \text{ ANS.}$$

$$11. \quad y = Ce^{-mx}; \text{ with } m \text{ held fixed.}$$

$$C = \frac{y}{e^{-mx}}$$

$$C = ye^{mx}$$

Differentiate,

$$0 = mye^{mx} dx + e^{mx} dy$$

$$mydx + dy = 0$$

$$\frac{dy}{dx} = -my$$

The orthogonal trajectories,

$$\frac{dy}{dx} = \frac{1}{my}$$

$$mydy = dx$$

$$\int mydy = \int dx$$

$$m \frac{y^2}{2} = x - C_1$$

$$my^2 = 2(x - C_1) \text{ ANS.}$$

CHAPTER 4: ADDITIONAL TOPICS ON EQUATIONS OF ORDER ONE

4.1 TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

We are concerned with equations that are simple enough to enable us to find integrating factors by inspection. The ability to do this depends largely upon recognition of certain common exact differentials and upon experience.

Below are four exact differential that occur frequently:

$$(1) \quad d(xy) = xdy + ydx$$

$$(2) \quad d\left(\frac{x}{y}\right) = \frac{ydx - xdy}{y^2}$$

$$(3) \quad d\left(\frac{y}{x}\right) = \frac{xdy - ydx}{x^2}$$

$$(4) \quad d\left(\arctan \frac{y}{x}\right) = \frac{xdy - ydx}{x^2 + y^2}$$

Note the homogeneity of the coefficients of dx and dy in each of these differentials.

A differential involving only one variable, such as $x^{-2}dx$, is an *exact differential*.

1. Except when the exercise indicates otherwise, find a set of solutions.

$$1. \quad y(2xy + 1)dx - xdy = 0$$

$$2xy^2 dx + ydx - xdy = 0$$

Divide by y^2 ;

$$2xdx + \frac{ydx - xdy}{y^2} = 0$$

$$2xdx + d\left(\frac{x}{y}\right) = 0$$

$$2 \int xdx + \int d\left(\frac{x}{y}\right) = \int 0$$

$$x^2 + \frac{x}{y} = C$$

$$x^2 y + x = Cy$$

$$x(xy + 1) = Cy$$

$$2. \quad y(y^3 - x)dx + x(y^3 + x)dy = 0$$

$$y^4 dx - xydx + y^3 xdy + x^2 dy = 0$$

$$y^3(ydx + xdy) - x(ydx - xdy) = 0$$

Divide by y^3 ;

$$(ydx + xdy) - \frac{x}{y} \left(\frac{ydx - xdy}{y^2} \right) = 0$$

$$d(xy) - \frac{x}{y} d\left(\frac{x}{y}\right) = 0$$

$$\int d(xy) - \int \frac{x}{y} d\left(\frac{x}{y}\right) = \int 0$$

$$xy - \frac{1}{2} \left(\frac{x}{y}\right)^2 = \frac{C}{2}$$

$$xy - \frac{x^2}{2y^2} = \frac{C}{2}$$

Multiply by $2y^2$;

$$2xy^3 - x^2 = Cy^2$$

TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

$$3. (x^3y^3 + 1)dx + x^4y^2dy = 0$$

$$x^3y^3dx + dx + x^4y^2dy = 0$$

$$x^3y^2(ydx + xdy) + dx = 0$$

Divide by x ;

$$x^2y^2(ydx + xdy) + \frac{dx}{x} = 0$$

$$(xy)^2d(xy) + \frac{dx}{x} = 0$$

$$\int (xy)^2d(xy) + \int \frac{dx}{x} = \int 0$$

$$\frac{(xy)^3}{3} + \ln x = -\ln C$$

$$(xy)^3 + 3\ln x = -3\ln C$$

$$(xy)^3 = -3\ln C - 3\ln x$$

$$(xy)^3 = -3\ln Cx$$

$$4. 2tds + s(2 + s^2t)dt = 0$$

$$2tds + 2sdt + s^3tdt = 0$$

$$2(tds + sdt) + s^3tdt = 0$$

Divide by s^3t^3 ;

$$\frac{2d(st)}{s^3t^3} + \frac{dt}{t^2} = 0$$

$$2 \int \frac{d(st)}{(st)^3} + \int \frac{dt}{t^2} = \int 0$$

$$2 \left(-\frac{1}{2} \right) \frac{1}{(st)^2} - \frac{1}{t} = -C$$

$$-\frac{1}{(st)^2} - \frac{1}{t} = -C$$

Multiply by $-(st)^2$;

$$1 + s^2t = Cs^2t^2$$

$$5. y(x^4 - y^2)dx + x(x^4 + y^2)dy = 0$$

$$x^4ydx - y^3dx + x^5dy + xy^2dy = 0$$

$$x^4(ydx + xdy) + y^2(xdy - ydx) = 0$$

Divide by x^4 ;

$$(ydx + xdy) + \frac{y^2}{x^2} \left(\frac{xdy - ydx}{x^2} \right) = 0$$

$$d(xy) + \left(\frac{y}{x} \right)^2 d \left(\frac{x}{y} \right) = 0$$

$$\int d(xy) + \int \left(\frac{y}{x} \right)^2 d \left(\frac{x}{y} \right) = \int 0$$

$$xy + \frac{1}{3} \left(\frac{y}{x} \right)^3 = \frac{C}{3}$$

$$3xy + \left(\frac{y}{x} \right)^3 = C$$

Multiply by x^3 ;

$$3x^4y + y^3 = Cx^3$$

$$y(3x^4 + y^2) = Cx^3$$

$$6. y(y^2 + 1)dx + x(y^2 - 1)dy = 0$$

$$y^3dx + ydx + xy^2dy - xdy = 0$$

$$y^2(ydx + xdy) + (ydx - xdy) = 0$$

Divide by y^2 ;

$$(ydx + xdy) + \left(\frac{ydx - xdy}{y^2} \right) = 0$$

$$d(xy) + d \left(\frac{x}{y} \right) = 0$$

$$\int d(xy) + \int d \left(\frac{x}{y} \right) = \int 0$$

$$xy + \frac{x}{y} = C$$

$$xy^2 + x = Cy$$

$$x(y^2 + 1) = Cy$$

Do exercise 6 by a second method

$$y(y^2 + 1)dx + x(y^2 - 1)dy = 0$$

Try Variable Separable:

$$\frac{dx}{x} + \frac{(y^2 - 1)dy}{y(y^2 + 1)} = 0$$

$$\int \frac{dx}{x} + \int \frac{(y^2 - 1)dy}{y(y^2 + 1)} = \int 0$$

TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

Considering:

$$\int \frac{(y^2 - 1)dy}{y(y^2 + 1)}$$

Using Integration by Partial Fractions:

$$\frac{(y^2 - 1)}{y(y^2 + 1)} = \frac{A}{y} + \frac{2yB}{(y^2 + 1)}$$

$$(y^2 - 1) = A(y^2 + 1) + 2y^2B$$

Collecting constant terms:

$$A = -1$$

Collecting y^2 terms:

$$1 = A + 2B$$

$$1 = -1 + 2B$$

$$B = 1$$

Then,

$$\int \frac{(y^2 - 1)dy}{y(y^2 + 1)} = - \int \frac{dy}{y} + \int \frac{2ydy}{(y^2 + 1)}$$

$$= -\ln y + \ln(y^2 + 1)$$

Thus,

$$\ln x - \ln y + \ln(y^2 + 1) = \ln C$$

$$\frac{x(y^2 + 1)}{y} = C$$

$$x(y^2 + 1) = Cy$$

$$8. y(x^3 - y^5)dx - x(x^3 + y^5)dy = 0$$

$$x^3ydx - y^6dx - x^4dy - xy^5dy = 0$$

$$x^3(ydx - xdy) - y^5(ydx + xdy) = 0$$

Divide by y^5 ;

$$\frac{x^3}{y^3} \left(\frac{ydx - xdy}{y^2} \right) - (ydx + xdy) = 0$$

$$\left(\frac{x}{y} \right)^3 d \left(\frac{x}{y} \right) - d(xy) = 0$$

$$\int \left(\frac{x}{y} \right)^3 d \left(\frac{x}{y} \right) - \int d(xy) = \int 0$$

$$\frac{1}{4} \left(\frac{x}{y} \right)^4 - xy = C$$

Multiply by $4y^4$;

$$x^4 - 4xy^5 = Cy^4$$

$$x^4 = Cy^4 + 4xy^5$$

$$x^4 = y^4(C + 4xy)$$

$$9. y(x^2 - y^2 + 1)dx - x(x^2 - y^2 - 1)dy = 0$$

$$x^2ydx - y^3dx + ydx - x^3dy + xy^2dy + xdy = 0$$

$$(x^2ydx - x^3dy) + (xy^2dy - y^3dx) + (ydx + xdy) = 0$$

$$x^2(ydx - xdy) + y^2(xdy - ydx) + (ydx + xdy) = 0$$

Divide by $(xy)^2$;

$$\frac{(ydx - xdy)}{y^2} + \frac{(xdy - ydx)}{x^2} + \frac{(ydx + xdy)}{(xy)^2} = 0$$

$$d \left(\frac{x}{y} \right) + d \left(\frac{y}{x} \right) + \frac{d(xy)}{(xy)^2} = 0$$

$$\int d \left(\frac{x}{y} \right) + \int d \left(\frac{y}{x} \right) + \int \frac{d(xy)}{(xy)^2} = \int 0$$

$$\frac{x}{y} + \frac{y}{x} - \frac{1}{xy} = -C$$

Multiply by xy ;

$$x^2 + y^2 - 1 = -Cxy$$

$$x^2 + Cxy + y^2 = 1$$

$$10. (x^3 + xy^2 + y)dx + (y^3 + x^2y + x)dy = 0$$

$$x^3dx + xy^2dx + ydx + y^3dy + x^2ydy + xdy = 0$$

$$x^3dx + y^3dy + xy(ydx + xdy) + (ydx + xdy) = 0$$

$$x^3dx + y^3dy + xyd(xy) + d(xy) = 0$$

TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

$$\int x^3 dx + \int y^3 dy + \int xyd(xy) + \int d(xy) = \int 0$$

$$\frac{x^4}{4} + \frac{y^4}{4} + \frac{(xy)^2}{2} + xy = \frac{C}{4}$$

$$x^4 + y^4 + 2x^2y^2 + 4xy = C$$

$$x^4 + 2x^2y^2 + y^4 = C - 4xy$$

$$(x^2 + y^2)^2 = C - 4xy$$

11. $y(x^2 + y^2 - 1)dx + x(x^2 + y^2 + 1)dy = 0$

$$x^2ydx + y^3dx - ydx + x^3dy + xy^2dy + xdy = 0$$

$$(x^2ydx + x^3dy) + (y^3dy + xy^2dy) + (x^2ydx - ydx) = 0$$

$$(x^2 + y^2)(ydx + xdy) + (xdy - ydx) = 0$$

Divide by $(x^2 + y^2)$:

$$(ydx + xdy) + \frac{(xdy - ydx)}{(x^2 + y^2)} = 0$$

$$d(xy) + d\left(\text{Arctan}\frac{y}{x}\right) = 0$$

$$\int d(xy) + \int d\left(\text{Arctan}\frac{y}{x}\right) = \int 0$$

$$xy + \text{Arctan}\frac{y}{x} = C$$

12. $(x^3 + xy^2 - y)dx + (y^3 + x^2y + x)dy = 0$

$$x^3dx + xy^2dx - ydx + y^3dy + x^2ydy + xdy = 0$$

$$(x^3dx + x^2ydy) + (y^3dy + xy^2dy) + (x^2ydy - ydx) = 0$$

$$x^2(xdx + ydy) + y^2(ydy + xdx) + (xdy - ydx) = 0$$

$$(x^2 + y^2)(xdx + ydy) + (xdy - ydx) = 0$$

Divide by $(x^2 + y^2)$:

$$(xdx + ydy) + \frac{(xdy - ydx)}{(x^2 + y^2)} = 0$$

$$xdx + ydy + d\left(\text{Arctan}\frac{y}{x}\right) = 0$$

$$\int xdx + \int ydy + \int d\left(\text{Arctan}\frac{y}{x}\right) = \int 0$$

$$\frac{x^2}{2} + \frac{y^2}{2} + \text{Arctan}\frac{y}{x} = \frac{C}{2}$$

$$x^2 + y^2 + 2\text{Arctan}\frac{y}{x} = C$$

$$2\text{Arctan}\frac{y}{x} = C - x^2 - y^2$$

13. $y(x^3e^{xy} - y)dx + x(y + x^3e^{xy})dy = 0$

$$x^3ye^{xy}dx - y^2dx + xydy + x^4e^{xy}dy = 0$$

$$(x^3ye^{xy}dx + x^4e^{xy}dy) + (xydy - y^2dx) = 0$$

$$x^3e^{xy}(ydx + xdy) + y(xdy - ydx) = 0$$

Divide by x^3 :

$$e^{xy}(ydx + xdy) + \frac{y}{x}\left(\frac{xdy - ydx}{x^2}\right) = 0$$

$$e^{xy}d(xy) + \frac{y}{x}d\left(\frac{y}{x}\right) = 0$$

$$\int e^{xy}d(xy) + \int \frac{y}{x}d\left(\frac{y}{x}\right) = \int 0$$

$$e^{xy} + \frac{1}{2}\left(\frac{y}{x}\right)^2 = \frac{C}{2}$$

Multiply by $2x^2$:

$$2x^2e^{xy} + y^2 = Cx^2$$

14. $xy(y^2 + 1)dx + (x^2y^2 - 2)dy = 0$;
when $x = 1, y = 1$

$$xy^3dx + xydx + x^2y^2dy - 2dy = 0$$

$$(xy^3dx + x^2y^2dy) + (xydx - 2dy) = 0$$

$$xy^2(ydx + xdy) + xydx - 2dy = 0$$

Divide by y :

$$xy(ydx + xdy) + xdx - \frac{2dy}{y} = 0$$

$$xyd(xy) + xdx - \frac{2dy}{y} = 0$$

TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

$$\int xyd(xy) + \int xdx - 2\int \frac{dy}{y} = \int 0$$

$$\frac{x^2y^2}{2} + \frac{x^2}{2} - 2\ln y = C$$

$$x^2y^2 + x^2 - 4\ln y = C \rightarrow \text{General Solution}$$

When $x = 1, y = 1$

$$C = (1)^2(1)^2 + (1)^2 - 4\ln(1) = 2$$

Thus,

$$x^2y^2 + x^2 - 4\ln y = 2$$

$$x^2(y^2 + 1) = 2 + 4\ln y$$

$\rightarrow$ Particular Solution

15. $y^2(1 - x^2)dx + x(x^2y + 2x + y)dy = 0$

$$y^2dx - x^2y^2dx + x^3ydy + 2x^2dy + xydy = 0$$

$$(y^2dx + xydy) + (x^3ydy - x^2y^2dx) + 2x^2dy = 0$$

$$y(ydx + xdy) - x^2y(ydx - xdy) + 2x^2dy = 0$$

Divide by x^2y^3 :

$$\frac{(ydx + xdy)}{x^2y^2} - \frac{(ydx - xdy)}{y^2} + \frac{2dy}{y^3} = 0$$

$$\frac{d(xy)}{(xy)^2} - d\left(\frac{x}{y}\right) + 2\frac{dy}{y^3} = 0$$

$$\int \frac{d(xy)}{(xy)^2} - \int d\left(\frac{x}{y}\right) + 2\int \frac{dy}{y^3} = \int 0$$

$$-\frac{1}{xy} - \frac{x}{y} - \frac{1}{y^2} = -C$$

Multiply by $-xy^2$:

$$y + x^2y + x = Cxy^2$$

$$x^2y + x + y = Cxy^2$$

16. $y(x^2y^2 - 1)dx + x(x^2y^2 + 1)dy = 0$

$$x^2y^3dx - ydx + x^3y^2dy + xdy = 0$$

$$(x^2y^3dx + x^3y^2dy) - (ydx - xdy) = 0$$

$$x^2y^2(ydx + xdy) - (ydx - xdy) = 0$$

Multiply by y/xy^2 :

$$xy(ydx + xdy) - \frac{y}{x}\left(\frac{ydx - xdy}{y^2}\right) = 0$$

$$xyd(xy) - \frac{y}{x}d\left(\frac{x}{y}\right) = 0$$

$$\int xyd(xy) - \int \left(\frac{x}{y}\right)^{-1}d\left(\frac{x}{y}\right) = \int 0$$

$$\frac{(xy)^2}{2} - \ln\left(\frac{x}{y}\right) = \ln C$$

$$\frac{(xy)^2}{2} = \ln C + \ln\left(\frac{x}{y}\right)$$

$$x^2y^2 = 2\ln\left(C\frac{x}{y}\right)$$

17. $x^4y' = -x^3y - \csc(xy)$

$$x^4\frac{dy}{dx} = -x^3y - \csc(xy)$$

$$x^4dy = -x^3ydx - \csc(xy)dx$$

$$x^4dy + x^3ydx + \csc(xy)dx = 0$$

$$x^3(xdy + ydx) + \csc(xy)dx = 0$$

Divide by $x^3 \csc(xy)$:

$$\frac{(xdy + ydx)}{\csc(xy)} + \frac{dx}{x^3} = 0$$

$$\sin(xy)d(xy) + \frac{dx}{x^3} = 0$$

$$\int \sin(xy)d(xy) + \int \frac{dx}{x^3} = \int 0$$

$$-\cos(xy) - \frac{1}{2x^2} = -\frac{C}{2}$$

Multiply by $-2x^2$:

$$2x^2 \cos(xy) + 1 = Cx^2$$

$$2x^2 \cos(xy) = Cx^2 - 1$$

TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

18. $[1 + y \tan(xy)]dx + x \tan(xy) dy = 0$

$$dx + y \tan(xy) dx + x \tan(xy) dy = 0$$

$$dx + \tan(xy) (y dx + x dy) = 0$$

$$dx + \tan(xy) d(xy) = 0$$

$$\int dx + \int \tan(xy) d(xy) = \int 0$$

$$x - \ln \cos(xy) = -\ln C$$

$$x = \ln \cos(xy) - \ln C$$

$$x = \ln \frac{\cos(xy)}{C}$$

$$e^x = \frac{\cos(xy)}{C}$$

$$\cos(xy) = Ce^x$$

19. $y(x^2y^2 - m)dx + x(x^2y^2 + n)dy = 0$

$$x^2y^3dx - mydx + x^3y^2dy + nx dy = 0$$

$$(x^2y^3dx + x^3y^2dy) + nx dy - mydx = 0$$

$$x^2y^2(y dx + x dy) + nx dy - mydx = 0$$

Divide by xy ,

$$xyd(xy) + \frac{ndy}{y} - \frac{mdx}{x} = 0$$

$$\int xyd(xy) + n \int \frac{dy}{y} - m \int \frac{dx}{x} = \int 0$$

$$\frac{x^2y^2}{2} + n \ln y - m \ln x = \ln C$$

$$\frac{x^2y^2}{2} = \ln C + m \ln x - n \ln y$$

$$\frac{x^2y^2}{2} = \ln \frac{Cx^m}{y^n}$$

$$x^2y^2 = 2 \ln \frac{Cx^m}{y^n}$$

20. $x(x^2 - y^2 - x)dx - y(x^2 - y^2)dy = 0$;
when $x = 2, y = 0$

$$x^3dx - xy^2dx - x^2dx - x^2ydy + y^3dy = 0$$

$$x^3dx - x^2dx + y^3dy - (xy^2dx + x^2ydy) = 0$$

$$x^3dx - x^2dx + y^3dy - xy(ydx + xdy) = 0$$

$$x^3dx - x^2dx + y^3dy - xyd(xy) = 0$$

$$\int x^3dx - \int x^2dx + \int y^3dy - \int xyd(xy) = \int 0$$

$$\frac{x^4}{4} - \frac{x^3}{3} + \frac{y^4}{4} - \frac{x^2y^2}{2} = \frac{C}{12}$$

$$3x^4 - 4x^3 + 3y^4 - 6x^2y^2 = C$$

→ General Solution

when $x = 2, y = 0$

$$C = 3(2)^4 - 4(2)^3 + 3(0)^4 - 6(2)^2(0)^2 = 16$$

Thus,

$$3x^4 - 4x^3 + 3y^4 - 6x^2y^2 = 16$$

$$3x^4 - 6x^2y^2 + 3y^4 = 16 + 4x^3$$

$$3(x^4 - 2x^2y^2 + y^4) = 4(4 + x^3)$$

$$3(x^2 - y^2)^2 = 4(4 + x^3) \rightarrow$$

Particular Solution

21. $y(x^2 + y)dx + x(x^2 - 2y)dy = 0$;
when $x = 1, y = 2$

$$(x^2y + y^2)dx + (x^3 - 2xy)dy = 0$$

Using the method of Determination of the Integrating factors:

$$M = x^2y + y^2$$

$$\frac{\partial M}{\partial y} = x^2 + 2y$$

$$N = x^3 - 2xy$$

$$\frac{\partial N}{\partial x} = 3x^2 - 2y$$

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{x^2 + 2y - 3x^2 + 2y}{x^3 - 2xy}$$

$$= \frac{-2x^2 + 4y}{x(x^2 - 2y)}$$

$$= \frac{-2(x^2 - 2y)}{x(x^2 - 2y)} = -\frac{2}{x}$$

TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

For Integrating factor:

$$e^{\int f dx} = e^{-\int \frac{2dx}{x}} = e^{-2 \ln x} = x^{-2}$$

Then,

$$x^{-2}y(x^2 + y)dx + x^{-2}x(x^2 - 2y)dy = 0$$

$$ydx + x^{-2}y^2dx + xdy - 2xydy = 0$$

Exact Equation (Test for Exactness)!

$$\int D.I + \int N.D.I = \int 0$$

D.I:

None

N.D.I:

$$ydx + x^{-2}y^2dx, xdy - 2xydy$$

Choose: $ydx + x^{-2}y^2dx$

$$\int ydx + \int x^{-2}y^2dx = \int 0$$

$$xy - \frac{y^2}{x} = C$$

$$x^2y - y^2 = Cx \rightarrow \text{General Solution}$$

when $x = 1, y = 2$

$$1^2(2) - (2)^2 = C(1)$$

$$C = -2$$

Then,

$$x^2y - y^2 = -2x$$

$$x^2y - y^2 + 2x = 0 \rightarrow \text{Particular Solution}$$

22. $y(x^3y^3 + 2x^2 - y)dx + x^3(xy^3 - 2)dy = 0$;

when $x = 1, y = 1$

$$x^3y^4dx + 2x^2ydx - y^2dx + x^4y^3dy - 2x^3dy = 0$$

$$(x^3y^4dx + x^4y^3dy) + (2x^2ydx - 2x^3dy) - y^2dx = 0$$

$$x^3y^3(ydx + xdy) + 2x^2(ydx - xdy) - y^2dx = 0$$

Divide by x^2y^3 ;

$$xy(ydx + xdy) + 2 \left(\frac{ydx - xdy}{y^2} \right) - \frac{dx}{x^2} = 0$$

$$xyd(xy) + 2d\left(\frac{x}{y}\right) - \frac{dx}{x^2} = 0$$

$$\int xyd(xy) + 2 \int d\left(\frac{x}{y}\right) - \int \frac{dx}{x^2} = \int 0$$

$$\frac{x^2y^2}{2} + \frac{2x}{y} + \frac{1}{x} = \frac{C}{2}$$

Multiply by $2xy$;

$$x^3y^3 + 4x^2 + 2y = Cxy \rightarrow \text{General Solution}$$

when $x = 1, y = 1$

$$1^3(1) + 4(1)^2 + 2(1) = C(1)(1)$$

$$C = 7$$

Thus,

$$x^3y^3 + 4x^2 + 2y = 7xy$$

$$x^3y^3 + 4x^2 - 7xy + 2y = 0$$

→ Particular Solution

23. $y(2 - 3xy)dx - xdy = 0$

$$2ydx - 3xy^2dx - xdy = 0$$

Multiply by y^{-2} ;

$$2y^{-1}dx - 3xdx - xy^{-2}dy = 0$$

Bernoulli's Equation:

Let

$$u = y^{-1}$$

$$du = -y^{-2}dy$$

Substitute:

$$2udu - 3xdx + xdu = 0$$

Divide by xdu ;

$$\frac{2u}{x} - 3 + \frac{du}{dx} = 0$$

TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

$$\frac{du}{dx} + \frac{2u}{x} = 3$$

Linear Equation!

$$P(x) = \frac{2}{x}$$

$$Q(x) = 3$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{2}{x}dx} = e^{2 \ln x} = x^2$$

Then,

$$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ux^2 = \int 3x^2 dx + C$$

$$ux^2 = x^3 + C$$

But $u = y^{-1}$

$$y^{-1}x^2 = x^3 + C$$

$$x^2 = x^3y + Cy$$

$$x^2 - x^3y = Cy$$

$$x^2(1 - xy) = Cy$$

24. $y(2x + y^2)dx + x(y^2 - x)dy = 0$

Using the method of Determination of the Integrating factors:

$$M = 2xy + y^3;$$

$$\frac{\partial M}{\partial y} = 2x + 3y^2$$

$$N = xy^2 - x^2;$$

$$\frac{\partial N}{\partial x} = y^2 - 2x$$

$$\frac{1}{M} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{2x + 3y^2 - y^2 + 2x}{2xy + y^3}$$

$$= \frac{4x + 2y^2}{2xy + y^3}$$

$$= \frac{2(2x + y^2)}{y(2x + y^2)}$$

$$= \frac{2}{y}$$

For Integrating factor:

$$e^{-\int \frac{2}{y}dy} = e^{-2 \ln y} = e^{-2 \ln y} = y^{-2}$$

Then,

$$y^{-2}y(2x + y^2)dx + y^{-2}x(y^2 - x)dy = 0$$

$$(2xy^{-1} + y)dx + (x - x^2y^{-2})dy = 0$$

Exact Equation (Test for Exactness):

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:
None

N.D.I:
 $(2xy^{-1} + y)dx, (x - x^2y^{-2})dy$

Choose: $(2xy^{-1} + y)dx$

$$\int (2xy^{-1} + y)dx = \int 0$$

$$\frac{x^2}{y} + xy = C$$

$$\frac{x^2}{y} + xy^2 = Cy$$

$$x(x + y^2) = Cy$$

25. $ydx + 2(y^4 - x)dy = 0$

Using the method of Determination of the Integrating factors:

$$M = y;$$

$$\frac{\partial M}{\partial y} = 1$$

$$N = 2y^4 - 2x;$$

$$\frac{\partial N}{\partial x} = -2$$

$$\frac{1}{M} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{1 + 2}{y} = \frac{3}{y}$$

For Integrating factor:

$$e^{-\int \frac{3}{y}dy} = e^{-3 \ln y} = e^{-3 \ln y} = y^{-3}$$

Then,

$$y^{-3}ydx + y^{-3}2(y^4 - x)dy = 0$$

$$y^{-2}dx + (2y - 2xy^{-2})dy = 0$$

TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

Exact Equation:

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:
2ydy

N.D.I:
 $y^{-2}dx, (-2xy^{-3})dy$

Choose: $y^{-2}dx$

$$\int 2ydy + \int y^{-2}dx = \int 0$$

$$y^2 + \frac{x}{y^2} = C$$

$$y^4 + x = Cy^2$$

27. $2x^2y' = y(3x^4 + y^2)$

$$2x^2 \frac{dy}{dx} = y(3x^4 + y^2)$$

$$2x^2 dy = 3x^4 y dx + y^3 dx$$

Divide by y^3 :
 $2x^2 y^{-3} dy - 3x^4 y^{-2} dx - dx = 0$

Bernoulli's Equation:

Let $u = -y^{-2}$
 $du = 2y^{-3}dy$

Substitute:

$$x^2 du + 3x^4 u dx - dx = 0$$

$$\frac{du}{dx} + \frac{3u}{x} = \frac{1}{x^2}$$

$$\frac{du}{dx} + \frac{3u}{x} = \frac{1}{x^2}$$

Linear Equation!

$$P(x) = \frac{3}{x}$$

$$Q(x) = \frac{1}{x^2}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{3}{x}dx} = e^{3 \ln x} = x^3$$

Then,

$$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ux^3 = \int \frac{1}{x^2} x^3 dx + C$$

$$ux^3 = \int \frac{1}{x^2} dx + C$$

$$ux^3 = -\frac{1}{x} + C$$

$$ux^4 + 1 = Cx$$

But $u = -y^{-2}$

$$-y^{-2}x^4 + 1 = Cx$$

$$-x^4 + y^2 = Cxy^2$$

$$x^4 = y^2 - Cxy^2$$

$$x^4 = y^2(1 - Cx)$$

28. $(x^n y^{n+1} + ay)dx + (x^{n+1} y^n + bx)dy = 0$

(a) If $n \neq 0$

$$x^n y^{n+1} dx + ay dx + x^{n+1} y^n dy + bx dy = 0$$

$$x^n y^n (y dx + x dy) + ay dx + bx dy = 0$$

Divide by xy :

$$x^{n-1} y^{n-1} d(xy) + \frac{adx}{x} + \frac{bdy}{y} = 0$$

$$\int x^{n-1} y^{n-1} d(xy) + a \int \frac{dx}{x} + b \int \frac{dy}{y} = \int 0$$

$$\frac{x^n y^n}{n} + a \ln x + b \ln y = \ln C$$

$$\frac{x^n y^n}{n} = \ln C - a \ln x - b \ln y$$

$$\frac{x^n y^n}{n} = \ln \frac{C}{x^a y^b}$$

$$\frac{x^n y^n}{n} = \ln C x^{-a} y^{-b}$$

$$x^n y^n = n \ln C x^{-a} y^{-b}$$

(b) If $n = 0$

$$(x^n y^{n+1} + ay)dx + (x^{n+1} y^n + bx)dy = 0$$

$$(y + ay)dx + (x + bx)dy = 0$$

TYPE 5: INTEGRATING FACTORS FOUND BY INSPECTION

$$y(1+a)dx + x(1+b)dy = 0$$

Divide by xy ;

$$\begin{aligned} \frac{(1+a)dx}{x} + \frac{(1+b)dy}{y} &= 0 \\ \int \frac{(1+a)dx}{x} + \int \frac{(1+b)dy}{y} &= \int 0 \\ (1+a)\ln x + (1+b)\ln y &= \ln C_1 \\ \ln x + a\ln x + \ln y + b\ln y &= \ln C_1 \\ \ln x + \ln y &= \ln C_1 - a\ln x - b\ln y \\ \ln xy &= \ln C_1 - a\ln x - b\ln y \\ \ln xy &= \ln C_1 x^{-a} y^{-b} \\ xy &= C_1 x^{-a} y^{-b} \end{aligned}$$

$$29. (x^{n+1}y^n + ay)dx + (x^ny^{n+1} + ax)dy = 0$$

(a) If $n \neq 0 \neq 1$

$$\begin{aligned} x^{n+1}y^n dx + ay dx + x^ny^{n+1} dy + ax dy &= 0 \\ x^ny^n(xdx + ydy) + a(ydx + xdy) &= 0 \end{aligned}$$

Multiply by $(xy)^{-n}$;

$$\begin{aligned} (x dx + y dy) + a(xy)^{-n}(y dx + x dy) &= 0 \\ \int x dx + \int y dy + \int a(xy)^{-n}d(xy) &= \int 0 \\ \frac{x^2}{2} + \frac{y^2}{2} + a \frac{(xy)^{-n+1}}{-n+1} &= C \\ x^2 + y^2 - \frac{2a(xy)^{-n+1}}{n-1} &= C \\ x^2 + y^2 - C &= \frac{2a(xy)^{-n+1}}{n-1} \end{aligned}$$

Multiply by $(n-1)(xy)^{n-1}$;

$$(n-1)(xy)^{n-1}(x^2 + y^2 - C) = 2a$$

(b) If $n = 1$

$$\begin{aligned} (x^{n+1}y^n + ay)dx + (x^ny^{n+1} + ax)dy &= 0 \\ x^2y dx + ay dx + xy^2 dy + ax dy &= 0 \\ xy(xdx + ydy) + a(ydx + xdy) &= 0 \end{aligned}$$

Divide by xy ;

$$(x dx + y dy) + \frac{a(y dx + x dy)}{xy} = 0$$

$$x dx + y dy + \frac{ad(xy)}{xy} = 0$$

$$\int x dx + \int y dy + \int \frac{ad(xy)}{xy} = \int 0$$

$$\frac{x^2}{2} + \frac{y^2}{2} + a \ln(xy) = C$$

$$x^2 + y^2 + 2a \ln(xy) = C$$

$$x^2 + y^2 - C = -2a \ln(xy)$$

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

4.2 TYPE 6: THE DETERMINATION OF INTEGRATING FACTORS

Consider some equations of the form

$$M dx + N dy = 0$$

- If $\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = f(x)$, a function of x alone, then use Integrating Factor $= e^{\int f(x) dx}$
- If $\frac{1}{M} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = g(y)$, a function of y alone, then use Integrating Factor $= e^{-\int g(y) dy}$

Steps:

1. Multiply Integrating factor to the original equation
2. Tests for its exactness
3. Solve using method of exact equations.

EXERCISES

1. Solve each of the following equations.

$$1. (x^2 + y^2 + 1)dx + x(x - 2y)dy = 0$$

$$M = x^2 + y^2 + 1;$$

$$\frac{\partial M}{\partial y} = 2y$$

$$N = x^2 - 2xy;$$

$$\frac{\partial N}{\partial x} = 2x - 2y$$

For Integrating factor:

$$\frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = \frac{2y - 2x + 2y}{x^2 - 2xy}$$

$$= \frac{4y - 2x}{x(x - 2y)}$$

$$= \frac{-2(x - 2y)}{x(x - 2y)}$$

$$= -\frac{2}{x}$$

$$e^{\int f(x) dx} = e^{-\int \frac{2dx}{x}} = e^{-2 \ln x} = x^{-2}$$

Multiply x^{-2} to original equation:

$$\begin{aligned} x^{-2}(x^2 + y^2 + 1)dx + x^{-2}x(x - 2y)dy &= 0 \\ (1 + x^{-2}y^2 + x^{-2})dx + (1 - 2x^{-1}y)dy &= 0 \end{aligned}$$

Exact Equation! (Test for exactness)

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:

$$(1 + x^{-2})dx, dy$$

N.D.I:

$$x^{-2}y^2 dx, (1 - 2x^{-1}y)dy$$

Choose: $x^{-2}y^2 dx$

$$\int (1 + x^{-2})dx + \int dy + \int x^{-2}y^2 dx = \int 0$$

$$x - \frac{1}{x} + y - \frac{1}{x}y^2 = C$$

$$x^2 - 1 + xy - y^2 = Cx$$

$$x^2 - y^2 + xy - 1 = Cx$$

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

2. $2y(x^2 - y + x)dx + (x^2 - 2y)dy = 0$

$$M = 2yx^2 - 2y^2 + 2xy;$$

$$\frac{\partial M}{\partial y} = 2x^2 - 4y + 2x$$

$$N = x^2 - 2y;$$

$$\frac{\partial N}{\partial x} = 2x$$

For Integrating factor:

$$\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{2x^2 - 4y + 2x - 2x}{x^2 - 2y}$$

$$= \frac{2x^2 - 4y}{x^2 - 2y}$$

$$= \frac{2(x^2 - 2y)}{(x^2 - 2y)}$$

$$= 2$$

$$e^{\int f(x)dx} = e^{\int 2dx} = e^{2x}$$

Multiply e^{2x} to original equation:

$$2ye^{2x}(x^2 - y + x)dx + e^{2x}(x^2 - 2y)dy = 0$$

$$(2yx^2e^{2x} - 2y^2e^{2x} + 2xye^{2x})dx + e^{2x}(x^2 - 2y)dy = 0$$

$$(2yx^2e^{2x} - 2y^2e^{2x} + 2xye^{2x})dx + e^{2x}(x^2 - 2y)dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

D.I:

None

N.D.I:

$$(2yx^2e^{2x} - 2y^2e^{2x} + 2xye^{2x})dx, (x^2e^{2x} - 2ye^{2x})dy$$

Choose: $(x^2e^{2x} - 2ye^{2x})dy$

$$\int (x^2e^{2x} - 2ye^{2x})dy = \int 0$$

$$x^2e^{2x}y - y^2e^{2x} = C$$

$$e^{2x}y(x^2 - y) = C$$

$$y(x^2 - y) = Ce^{-2x}$$

3. $y(2x - y + 1)dx + x(3x - 4y + 3)dy = 0$

$$M = 2xy - y^2 + y;$$

$$\frac{\partial M}{\partial y} = 2x - 2y + 1$$

$$N = 3x^2 - 4xy + 3x;$$

$$\frac{\partial N}{\partial x} = 6x - 4y + 3$$

For Integrating factor:

$$\frac{1}{M} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{2x - 2y + 1 - 6x + 4y - 3}{2xy - y^2 + y}$$

$$= \frac{-4x + 2y - 2}{2xy - y^2 + y}$$

$$= \frac{-4x + 2y - 2}{y(2x - y + 1)}$$

$$= \frac{-2(2x - y + 1)}{y(2x - y + 1)}$$

$$= -\frac{2}{y}$$

$$e^{-\int \frac{2}{y}dy} = e^{-2\ln y} = y^{-2}$$

Multiply y^{-2} to original equation:

$$y^2y(2x - y + 1)dx + y^2x(3x - 4y + 3)dy = 0$$

$$(2xy^3 - y^4 + y^3)dx + (3x^2y^2 - 4xy^3 + 3xy^2)dy = 0$$

$$(2xy^3 - y^4 + y^3)dx + (3x^2y^2 - 4xy^3 + 3xy^2)dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

D.I:

None

N.D.I:

$$(2xy^3 - y^4 + y^3)dx, (3x^2y^2 - 4xy^3 + 3xy^2)dy$$

Choose: $(2xy^3 - y^4 + y^3)dx$

$$\int (2xy^3 - y^4 + y^3)dx = \int 0$$

$$x^2y^3 - xy^4 + xy^3 = C$$

$$xy^3(x - y + 1) = C$$

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

4. $y(4x + y)dx - 2(x^2 - y)dy = 0$

$$M = 4xy + y^2;$$

$$\frac{\partial M}{\partial y} = 4x + 2y$$

$$N = -2x^2 + 2y;$$

$$\frac{\partial N}{\partial x} = -4x$$

For Integrating factor:

$$\frac{1}{M} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{4x + 2y + 4x}{4xy + y^2}$$

$$= \frac{8x + 2y}{y(4x + y)}$$

$$= \frac{2(4x + y)}{y(4x + y)}$$

$$= \frac{2}{y}$$

$$e^{-\int \frac{2}{y}dy} = e^{-2\ln y} = e^{-2\ln y} = y^{-2}$$

Multiply y^{-2} to original equation:

$$y^{-2}y(4x + y)dx - 2y^{-2}(x^2 - y)dy = 0$$

$$(4xy^{-1} + 1)dx - (2x^2y^{-2} - 2y^{-1})dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

D.I:

$$dx, 2y^{-1}dy$$

N.D.I:

$$4xy^{-1}dx, -2x^2y^{-2}dy$$

Choose: $4xy^{-1}dx$

$$\int dx + \int 2y^{-1}dy + \int 4xy^{-1}dx = \int 0$$

$$x + 2\ln y + 2x^2y^{-1} = C$$

Multiply by y :

$$xy + 2y\ln y + 2x^2 = Cy$$

$$2x^2 + xy + 2y\ln y = Cy$$

5. $(xy + 1)dx + x(x + 4y - 2)dy = 0$

$$M = xy + 1;$$

$$\frac{\partial M}{\partial y} = x$$

$$N = x^2 + 4xy - 2x;$$

$$\frac{\partial N}{\partial x} = 2x + 4y - 2$$

For Integrating factor:

$$\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{x - 2x - 4y + 2}{x^2 + 4xy - 2x}$$

$$= \frac{-x - 4y + 2}{x(x + 4y - 2)}$$

$$= \frac{-1(x + 4y - 2)}{x(x + 4y - 2)}$$

$$= -\frac{1}{x}$$

$$e^{\int f(x)dx} = e^{-\int \frac{1}{x}dx} = e^{-\ln x} = x^{-1}$$

Multiply x^{-1} to original equation:

$$x^{-1}(xy + 1)dx + x^{-1}x(x + 4y - 2)dy = 0$$

$$(y + x^{-1})dx + (x + 4y - 2)dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

D.I:

$$x^{-1}dx, (4y - 2)dy$$

N.D.I:

$$ydx, xdy$$

Choose: ydx

$$\int x^{-1}dx + \int (4y - 2)dy + \int ydx = \int 0$$

$$\ln x + 2y^2 - 2y + xy = C$$

$$xy + \ln x + 2y^2 - 2y = C$$

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

$$6. (2y^2 + 3xy - 2y + 6x)dx + x(x + 2y - 1)dy = 0$$

$$M = 2y^2 + 3xy - 2y + 6x;$$

$$\frac{\partial M}{\partial y} = 4y + 3x - 2$$

$$N = x^2 + 2xy - x;$$

$$\frac{\partial N}{\partial x} = 2x + 2y - 1$$

For Integrating factor:

$$\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{4y + 3x - 2 - 2x - 2y + 1}{x(x + 2y - 1)}$$

$$= \frac{x + 2y - 1}{x(x + 2y - 1)}$$

$$= \frac{1}{x}$$

$$e^{\int f(x)dx} = e^{\int \frac{1}{x} dx} = e^{\ln x} = x$$

Multiply x to original equation:

$$x(2y^2 + 3xy - 2y + 6x)dx + x^2(x + 2y - 1)dy = 0$$

$$(2xy^2 + 3x^2y - 2xy + 6x^2)dx + (x^3 + 2x^2y - x^2)dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

$$D.I.: 6x^2 dx$$

$$N.D.I.: (2xy^2 + 3x^2y - 2xy)dx, (x^3 + 2x^2y - x^2)dy$$

$$\text{Choose: } (2xy^2 + 3x^2y - 2xy)dx$$

$$\int 6x^2 dx + \int (2xy^2 + 3x^2y - 2xy)dx = \int 0$$

$$2x^3 + x^2y^2 + x^3y - x^2y = C$$

$$x^2y^2 + x^3y - x^2y + 2x^3 = C$$

$$x^2(y^2 + xy - y + 2x) = C$$

$$7. y(y + 2x - 2)dx - 2(x + y)dy = 0$$

$$M = y^2 + 2xy - 2y;$$

$$\frac{\partial M}{\partial y} = 2y + 2x - 2$$

$$N = -2x - 2y;$$

$$\frac{\partial N}{\partial x} = -2$$

For Integrating factor:

$$\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{2y + 2x - 2 + 2}{-2(x + y)}$$

$$= \frac{2x + 2y}{-2(x + y)}$$

$$= \frac{2(x + y)}{-2(x + y)} = -1$$

$$e^{\int f(x)dx} = e^{-\int 1 dx} = e^{-x}$$

Multiply e^{-x} to original equation:

$$e^{-x}y(y + 2x - 2)dx - 2e^{-x}(x + y)dy = 0$$

$$(e^{-x}y^2 + 2e^{-x}xy - 2e^{-x}y)dx - (2e^{-x}x + 2e^{-x}y)dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

$$D.I.: \text{None}$$

$$N.D.I.: (e^{-x}y^2 + 2e^{-x}xy - 2e^{-x}y)dx, (-2e^{-x}x - 2e^{-x}y)dy$$

$$\text{Choose: } (-2e^{-x}x - 2e^{-x}y)dy$$

$$\int (-2e^{-x}x - 2e^{-x}y)dy = \int 0$$

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

$$9. 2y(x + y + 2)dx + (y^2 - x^2 - 4x - 1)dy = 0$$

$$-2e^{-x}xy - y^2e^{-x} = -C$$

$$e^{-x}y(-2x - y) = -C$$

$$y(2x + y) = Ce^x$$

$$y^2dx + (3xy + y^2 - 1)dy = 0$$

$$M = y^2;$$

$$\frac{\partial M}{\partial y} = 2y$$

$$N = 3xy + y^2 - 1;$$

$$\frac{\partial N}{\partial x} = 3y$$

For Integrating factor:

$$\frac{1}{M} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{2y - 3y}{y^2}$$

$$= \frac{-y}{y^2}$$

$$= -\frac{1}{y}$$

$$e^{-\int f(y)dy} = e^{\int \frac{1}{y} dy} = e^{\ln y} = y$$

Multiply y to original equation:

$$yy^2dx + y(3xy + y^2 - 1)dy = 0$$

$$y^3dx + (3xy^2 + y^3 - y)dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

$$D.I.: (y^3 - y)dy$$

$$N.D.I.: y^3dx, 3xy^2dy$$

$$\text{Choose: } y^3dx$$

$$\int (y^3 - y)dy + \int y^3dx = \int 0$$

$$\frac{y^4}{4} - \frac{y^2}{2} + xy^3 = C$$

$$y^4 - 2y^2 + 4xy^3 = C$$

$$y^2(y^2 + 4xy - 2) = C$$

$$9. 2y(x + y + 2)dx + (y^2 - x^2 - 4x - 1)dy = 0$$

$$M = 2xy + 2y^2 + 4y;$$

$$\frac{\partial M}{\partial y} = 2x + 4y + 4$$

$$N = y^2 - x^2 - 4x - 1;$$

$$\frac{\partial N}{\partial x} = -2x - 4$$

For Integrating factor:

$$\frac{1}{M} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{2x + 4y + 4 + 2x + 4}{2y(x + y + 2)}$$

$$= \frac{4x + 4y + 8}{2y(x + y + 2)}$$

$$= \frac{4(x + y + 2)}{2y(x + y + 2)} = \frac{2}{y}$$

$$e^{-\int f(x)dx} = e^{-\int \frac{2}{y} dy} = e^{-2 \ln y} = y^{-2}$$

Multiply y^{-2} to original equation:

$$y^{-2}(2xy + 2y^2 + 4y)dx + y^{-2}(y^2 - x^2 - 4x - 1)dy = 0$$

$$(2xy^{-1} + 2 + 4y^{-1})dx + (1 - x^2y^{-2} - 4xy^{-2} - y^{-2})dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

$$D.I.: 2dx, (1 - y^{-2})dy$$

$$N.D.I.: (2xy^{-1} + 4y^{-1})dx, (-x^2y^{-2} - 4xy^{-2})dy$$

$$\text{Choose: } (2xy^{-1} + 4y^{-1})dx$$

$$\int 2dx + \int (1 - y^{-2})dy + \int (2xy^{-1} + 4y^{-1})dx = \int 0$$

$$2x + y + \frac{1}{y} + \frac{x^2}{y} + \frac{4x}{y} = C$$

$$2xy + y^2 + 1 + x^2 + 4x = Cy$$

$$x^2 + 2xy + y^2 + 4x + 1 = Cy$$

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

$$10. \quad 2(2y^2 + 5xy - 2y + 4)dx + x(2x + 2y - 1)dy = 0$$

$$M = 4y^2 + 10xy - 4y + 8;$$

$$\frac{\partial M}{\partial y} = 8y + 10x - 4$$

$$N = 2x^2 + 2xy - x;$$

$$\frac{\partial N}{\partial x} = 4x + 2y - 1$$

For Integrating factor:

$$\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{8y + 10x - 4 - 4x - 2y - 1}{x(2x + 2y - 1)}$$

$$= \frac{6x + 6y - 3}{x(2x + 2y - 1)}$$

$$= \frac{3(2x + 2y - 1)}{x(2x + 2y - 1)} = \frac{3}{x}$$

$$e^{\int f(x)dx} = e^{\int \frac{3dx}{x}} = e^{3 \ln x} = x^3$$

Multiply x^3 to original equation:

$$2x^3(2y^2 + 5xy - 2y + 4)dx + x^4(2x + 2y - 1)dy = 0$$

$$(4x^3y^2 + 10x^4y - 4x^3y + 8x^3)dx + (2x^5 + 2x^4y - x^4)dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

$$D.I.: 8x^3dx$$

$$N.D.I.: (4x^3y^2 + 10x^4y - 4x^3y + 8x^3)dx, (2x^5 + 2x^4y - x^4)dy$$

$$\text{Choose: } (2x^5 + 2x^4y - x^4)dy$$

$$\int 8x^3dx + \int (2x^5 + 2x^4y - x^4)dy = \int 0$$

$$2x^4 + 2x^5y + x^4y^2 - x^4y = C$$

$$x^4y^2 + 2x^5y - x^4y + 2x^4 = C$$

$$x^4(y^2 + 2xy - y + 2) = C$$

$$11. \quad 3(x^2 + y^2)dx + x(x^2 + 3y^2 + 6y)dy = 0$$

$$M = 3x^2 + 3y^2;$$

$$\frac{\partial M}{\partial y} = 6y$$

$$N = x^3 + 3xy^2 + 6xy;$$

$$\frac{\partial N}{\partial x} = 3x^2 + 3y^2 + 6y$$

For Integrating factor:

$$\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{6y - 3x^2 - 3y^2 - 6y}{x(x^2 + 3y^2 + 6y)}$$

$$= \frac{-3x^2 - 3y^2}{3(x^2 + y^2)}$$

$$= \frac{-3(x^2 + y^2)}{3(x^2 + y^2)}$$

$$= -1$$

$$e^{\int f(y)dy} = e^{\int -1 dy} = e^{-y}$$

Multiply e^{-y} to original equation:

$$3e^{-y}(x^2 + y^2)dx + xe^{-y}(x^2 + 3y^2 + 6y)dy = 0$$

$$(3x^2e^{-y} + 3y^2e^{-y})dx + (x^3e^{-y} + 3xy^2e^{-y} + 6xye^{-y})dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

$$D.I.: \text{None}$$

$$N.D.I.: (3x^2e^{-y} + 3y^2e^{-y})dx, (x^3e^{-y} + 3xy^2e^{-y} + 6xye^{-y})dy$$

$$\text{Choose: } (3x^2e^{-y} + 3y^2e^{-y})dx$$

$$\int (3x^2e^{-y} + 3y^2e^{-y})dx = \int 0$$

$$x^3e^{-y} + 3xy^2e^{-y} = C$$

$$xe^{-y}(x^2 + 3y^2) = C$$

$$x(x^2 + 3y^2) = Ce^{-y}$$

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

$$12. \quad y(8x - 9y)dx + 2x(x - 3y)dy = 0$$

$$M = 8xy - 9y^2;$$

$$\frac{\partial M}{\partial y} = 8x - 18y$$

$$N = 2x^2 - 6xy$$

$$\frac{\partial N}{\partial x} = 4x - 6y$$

For Integrating factor:

$$\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{8x - 18y - 4x + 6y}{2x(x - 3y)}$$

$$= \frac{4x - 12y}{2x(x - 3y)}$$

$$= \frac{4(x - 3y)}{2x(x - 3y)}$$

$$= \frac{2}{x}$$

$$e^{\int f(x)dx} = e^{\int \frac{2dx}{x}} = e^{2 \ln x} = x^2$$

Multiply x^2 to original equation:

$$x^2y(8x - 9y)dx + 2x^3(x - 3y)dy = 0$$

$$(8x^3y - 9x^2y^2)dx + (2x^4 - 6x^3y)dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

$$D.I.: \text{None}$$

$$N.D.I.: (8x^3y - 9x^2y^2)dx, (2x^4 - 6x^3y)dy$$

$$\text{Choose: } (2x^4 - 6x^3y)dy$$

$$\int (2x^4 - 6x^3y)dy = \int 0$$

$$2x^4y - 3x^3y^2 = C$$

$$x^3y(2x - 3y) = C$$

13. Do exercise 12 by another method.

$$y(8x - 9y)dx + 2x(x - 3y)dy = 0$$

Homogeneous, 2nd degree!

$$\text{Let: } y = vx$$

$$dy = vdx + xdv$$

Substitute:

$$vx(8x - 9vx)dx + 2x(x - 3vx)(vdx + xdv) = 0$$

$$v(8 - 9v)dx + 2(1 - 3v)(vdx + xdv) = 0$$

$$(8v - 9v^2)dx + (2 - 6v)(vdx + xdv) = 0$$

$$(8v - 9v^2)dx + (2vdv + 2xdv - 6v^2dx - 6vxdv) = 0$$

$$(8v - 9v^2 + 2v - 6v^2)dx + (2x - 6vx)dv = 0$$

$$(10v - 15v^2)dx + 2x(1 - 3v)dv = 0$$

$$5v(2 - 3v)dx + 2x(1 - 3v)dv = 0$$

$$\frac{5dx}{x} + \frac{2(1 - 3v)dv}{v(2 - 3v)} = 0$$

$$5 \int \frac{dx}{x} + \int \frac{2(1 - 3v)dv}{v(2 - 3v)} = \int 0$$

Considering

$$\int \frac{2(1 - 3v)dv}{v(2 - 3v)}$$

Using Integration by Partial Fraction:

$$\frac{2(1 - 3v)}{v(2 - 3v)} = \frac{A}{v} + \frac{B}{(2 - 3v)}$$

$$2(1 - 3v) = A(2 - 3v) + Bv$$

$$\text{If } v = 0 : A = 1$$

$$v = 2/3 : B = -3$$

Then,

$$\int \frac{2(1 - 3v)dv}{v(2 - 3v)} = \int \frac{dv}{v} - 3 \int \frac{dv}{(2 - 3v)}$$

$$= \ln v + \ln(2 - 3v)$$

Thus,

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

$$\begin{aligned} 5 \ln x + \ln v + \ln(2-3v) &= \ln C \\ \ln x^5 + \ln v + \ln(2-3v) &= \ln C \\ \ln x^5 v(2-3v) &= \ln C \\ x^5 v(2-3v) &= C \end{aligned}$$

$$\text{But } v = y/x$$

$$\begin{aligned} x^5 \frac{y}{x} \left(2 - 3 \frac{y}{x} \right) &= C \\ x^4 y \left(\frac{2x-3y}{x} \right) &= C \\ x^3 y(2x-3y) &= C \end{aligned}$$

$$14. \quad y(2x^2 - xy + 1)dx + (x-y)dy = 0$$

$$\begin{aligned} M &= 2x^2y - xy^2 + y; \\ \frac{\partial M}{\partial y} &= 2x^2 - 2xy + 1 \end{aligned}$$

$$\begin{aligned} N &= x - y \\ \frac{\partial N}{\partial x} &= 1 \end{aligned}$$

For Integrating factor:

$$\begin{aligned} \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) &= \frac{2x^2 - 2xy + 1 - 1}{(x-y)} \\ &= \frac{2x^2 - 2xy}{(x-y)} \\ &= \frac{2x(x-y)}{(x-y)} \\ &= 2x \end{aligned}$$

$$e^{\int f(x)dx} = e^{\int 2x dx} = e^{x^2}$$

Multiply e^{x^2} to original equation:

$$\begin{aligned} ye^{x^2}(2x^2 - xy + 1)dx \\ + e^{x^2}(x-y)dy = 0 \\ (2x^2ye^{x^2} - xy^2e^{x^2} + ye^{x^2})dx \\ + (xe^{x^2} - ye^{x^2})dy = 0 \end{aligned}$$

Exact Equation! (Test for exactness)

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:
None

$$\begin{aligned} \text{N.D.I:} \\ (2x^2ye^{x^2} - xy^2e^{x^2} + ye^{x^2})dx, (xe^{x^2} \\ - ye^{x^2})dy \end{aligned}$$

$$\text{Choose: } (xe^{x^2} - ye^{x^2})dy$$

$$\int (xe^{x^2} - ye^{x^2})dy = \int 0$$

$$yxe^{x^2} - \frac{y^2}{2}e^{x^2} = \frac{C}{2}$$

$$2yxe^{x^2} - y^2e^{x^2} = C$$

$$ye^{x^2}(2x - y) = C$$

$$y(2x - y) = Ce^{-x^2}$$

II. Use the integrating factor to solve each of the equations in exercises 17 through 20.

If M and N are homogeneous functions of the same degree, and if $Mx + Ny \neq 0$, then

Use Integrating factor:

$$\frac{1}{Mx + Ny}$$

$$17. \quad xydx - (x^2 + 2y^2)dy = 0$$

$$\begin{aligned} M &= xy \\ N &= -x^2 - 2y^2 \end{aligned}$$

Δ Homogeneous, 2nd degree!

For Integrating factor:

$$\begin{aligned} \frac{1}{Mx + Ny} &= \frac{1}{(xy)x + (-x^2 - 2y^2)y} \\ &= \frac{1}{x^2y - x^2y - 2y^3} \\ &= -\frac{1}{2y^3} \end{aligned}$$

Multiply $-\frac{1}{2y^3}$ to original equation:

$$-\frac{1}{2y^3} xydx + \frac{1}{2y^3} (x^2 + 2y^2)dy = 0$$

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

$$-\frac{xdx}{2y^2} + \left(\frac{x^2}{2y^3} + \frac{1}{y} \right) dy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:
 $\frac{1}{y} dy$

$$\begin{aligned} \text{N.D.I:} \\ -\frac{xdx}{2y^2}, \frac{x^2}{2y^3} dy \end{aligned}$$

$$\text{Choose: } -\frac{xdx}{2y^2}$$

$$\int \frac{1}{y} dy + \int -\frac{xdx}{2y^2} = \int 0$$

$$\ln y - \frac{x^2}{4y^2} = \ln C$$

$$\ln y - \ln C = \frac{x^2}{4y^2}$$

$$\ln \frac{y}{C} = \frac{x^2}{4y^2}$$

$$x^2 = 4y^2 \ln \frac{y}{C}$$

$$18. \quad v^2 dx + x(x+v)dv = 0$$

$$\begin{aligned} M &= v^2 \\ N &= x^2 + vx \end{aligned}$$

Δ Homogeneous, 2nd degree!

For Integrating factor:

$$\begin{aligned} \frac{1}{Mx + Nv} &= \frac{1}{v^2x + (x^2 + vx)v} \\ &= \frac{1}{v^2x + vx^2 + v^2x} \\ &= \frac{1}{2v^2x + vx^2} \\ &= \frac{1}{vx(2v+x)} \end{aligned}$$

Multiply $\frac{1}{vx(2v+x)}$ to original equation:

$$\begin{aligned} \left(\frac{1}{vx(2v+x)} \right) v^2 dx \\ + \left(\frac{1}{vx(2v+x)} \right) x(x+v)dv = 0 \end{aligned}$$

$$\frac{v}{x(2v+x)} dx + \frac{x+v}{v(2v+x)} dv = 0$$

$$\frac{v}{x(2v+x)} dx + \frac{(2v+x)-v}{v(2v+x)} dv = 0$$

$$\frac{v}{x(2v+x)} dx + \left(\frac{1}{v} - \frac{1}{(2v+x)} \right) dv = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:
 $\frac{1}{v} dv$

$$\begin{aligned} \text{N.D.I:} \\ \frac{v}{x(2v+x)} dx, \frac{1}{(2v+x)} dv \end{aligned}$$

$$\text{Choose: } \frac{v}{x(2v+x)} dx$$

$$\int \frac{1}{v} dv + \int \frac{v}{x(2v+x)} dx = \int 0$$

Considering:

$$\int \frac{v}{x(2v+x)} dx$$

Using Integration by Partial Fraction,

$$\frac{v}{x(2v+x)} = \frac{A}{x} + \frac{B}{(2v+x)}$$

$$\text{If } \begin{matrix} x=0 & ; & A=1/2 \\ x=-2v & ; & B=-1/2 \end{matrix}$$

Then,

$$\begin{aligned} \int \frac{v}{x(2v+x)} dx &= \frac{1}{2} \int \frac{dx}{x} - \frac{1}{2} \int \frac{dx}{(2v+x)} \\ &= \frac{1}{2} \ln x - \frac{1}{2} \ln(2v+x) \end{aligned}$$

TYPE 6: DETERMINATION OF INTEGRATING FACTORS

Thus,

$$\ln v + \frac{1}{2} \ln x - \frac{1}{2} \ln(2v+x) = \ln C$$

$$2 \ln v + \ln x - \ln(2v+x) = \ln C$$

$$\frac{xv^2}{(2v+x)} = C$$

$$xv^2 = C(2v+x)$$

19. $v(u^2 + v^2)du - u(u^2 + 2v^2)dv = 0$

$$M = vu^2 + v^3$$

$$N = -u^3 - 2uv^2$$

∴ Homogeneous, 3rd degree!

For Integrating factor:

$$\begin{aligned} \frac{1}{Mu + Nv} &= \frac{1}{(vu^2 + v^3)u + (-u^3 - 2uv^2)v} \\ &= \frac{1}{(vu^2 + v^3)u + (-u^3 - 2uv^2)v} \\ &= \frac{1}{vu^3 + uv^3 - vu^3 - 2uv^3} = \frac{1}{-uv^3} \end{aligned}$$

Multiply $\frac{1}{-uv^3}$ to original equation:

$$\begin{aligned} \left(\frac{1}{-uv^3}\right)v(u^2 + v^2)du &- \left(\frac{1}{-uv^3}\right)u(u^2 + 2v^2)dv = 0 \\ \left(\frac{1}{-uv^2}\right)(u^2 + v^2)du + \left(\frac{1}{v^3}\right)(u^2 + 2v^2)dv &= 0 \\ \left(\frac{u}{-v^2} - \frac{1}{u}\right)du + \left(\frac{u^2}{v^3} + \frac{2}{v}\right)dv &= 0 \end{aligned}$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

D.I:
 $-\frac{1}{u} du, \frac{2}{v} dv$

N.D.I:
 $\frac{u}{-v^2} du, \frac{u^2}{v^3} dv$

Choose: $\frac{u}{-v^2} du$

$$\int -\frac{1}{u} du + \int \frac{2}{v} dv - \int \frac{u}{v^2} du = \int 0$$

$$-\ln u + 2 \ln v - \frac{u^2}{2v^2} = -\ln C$$

$$-2v^2 \ln u + 2v^2 \ln v^2 - u^2 = -2v^2 \ln C$$

$$u^2 = 2v^2 \ln C + 2v^2 \ln v^2 - 2v^2 \ln u$$

$$u^2 = 2v^2 \ln(Cv^2/u)$$

20. $(x^2 + y^2)dx - xydy = 0$

$$M = x^2 + y^2$$

$$N = -xy$$

∴ Homogeneous, 2nd degree!

For Integrating factor:

$$\begin{aligned} \frac{1}{Mx + Ny} &= \frac{1}{(x^2 + y^2)x - (xy)y} \\ &= \frac{1}{x^3 + xy^2 - xy^2} = \frac{1}{x^3} \end{aligned}$$

Multiply $\frac{1}{x^3}$ to original equation:

$$\begin{aligned} \frac{1}{x^3}(x^2 + y^2)dx - \frac{1}{x^3}xydy &= 0 \\ \left(\frac{1}{x} + \frac{y^2}{x^3}\right)dx - \frac{y}{x^2}dy &= 0 \end{aligned}$$

Exact Equation! (Test for exactness)

$$\sum \int D.I. + \int N.D.I. = \int 0$$

D.I:
 $(1/x)dx$
N.D.I:
 $\frac{y^2}{x^3} dx, -\frac{y}{x^2} dy$
Choose: $-\frac{y}{x^2} dy$

$$\int \frac{1}{x} dx - \int \frac{y}{x^2} dy = \int 0$$

$$\ln x - \frac{y^2}{2x^2} = \ln C$$

$$\ln(x/C) = \frac{y^2}{2x^2}$$

$$y^2 = 2x^2 \ln(x/C)$$

TYPE 7: SUBSTITUTION SUGGESTED BY THE EQUATION TYPE 8: BERNOULLI'S EQUATION

4.3 TYPE 7: SUBSTITUTION SUGGESTED BY THE EQUATION

Consider some equations of the form

$$M dx + N dy = 0$$

May not yield at once (or at all) to the methods of Chapter 2. Even then the usefulness of those methods is not exhausted. It may be possible by some change of variables to transform the equation into a type that we know how to solve.

A natural source of suggestions for useful transformations is the differential equation itself. If a particular function of one or both variables stands out in the equation, then it is worthwhile to examine the equation after that function has been introduced as a new variable.

4.4 TYPE 8: BERNOULLI'S EQUATION

Standard Form:

$$\frac{dy}{dx} + P(x)y = Q(x)y^n$$

If $n = 1 \rightarrow$ Variable Separable

If $n = 0 \rightarrow$ Linear Equation

If $n \neq 1 \rightarrow$ Bernoulli's Equation

Divide by y^n

$$y'y^{-n} + P(x)y^{-n+1} = Q(x)$$

Put:

$$y^{-n+1} = u$$

$$y'y^{-n} = du$$

Substitute:

Then continue solving into linear or variable separable

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$$\frac{dx}{dy} + P(y)x = Q(y)x^n$$

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Put:

$$x^{-n+1} = u$$

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Substitute:

Then continue solving into linear or variable separable

TYPE 7: SUBSTITUTION SUGGESTED BY THE EQUATION
TYPE 8: BERNOULLI'S EQUATION

I. In exercises 1 through 21, solve the equation.

1. $(3x - 2y + 1)dx + (3x - 2y + 3)dy = 0$

Let:

$$\begin{aligned} u &= 3x - 2y \\ du &= 3dx - 2dy \\ dy &= \frac{3dx - du}{2} \end{aligned}$$

Substitute,

$$(u + 1)dx + (u + 3)\left(\frac{3dx - du}{2}\right) = 0$$

$$(u + 1)dx + \frac{3}{2}(u + 3)dx$$

$$-\frac{1}{2}(u + 3)du = 0$$

$$\left[(u + 1) + \frac{3}{2}(u + 3)\right]dx$$

$$-\frac{1}{2}(u + 3)du = 0$$

$$\left(\frac{5}{2}u + \frac{11}{2}\right)dx - \frac{1}{2}(u + 3)du = 0$$

$$\frac{1}{2}(5u + 11)dx - \frac{1}{2}(u + 3)du = 0$$

Divide by $\frac{(5u + 11)}{2}$,

$$dx - \frac{(u + 3)du}{(5u + 11)} = 0$$

$$dx - \left[\frac{1}{5}du + \frac{4}{5}\left(\frac{du}{5u + 11}\right)\right] = 0$$

$$dx - \frac{1}{5}du - \frac{4}{5}\left(\frac{du}{5u + 11}\right) = 0$$

$$\int dx - \frac{1}{5} \int du - \frac{4}{5} \int \frac{du}{5u + 11} = \int 0$$

$$x - \frac{u}{5} - \frac{4}{25} \ln(5u + 11) = C$$

$$25x - 5u - 4 \ln(5u + 11) = C$$

But $u = 3x - 2y$

$$25x - 5(3x - 2y)$$

$$-4 \ln[5(3x - 2y) + 11] = C$$

$$25x - 15x + 10y$$

$$-4 \ln(15x - 10y + 11) = C$$

$$25x - 15x + 10y$$

$$-4 \ln(15x - 10y + 11) = C$$

$$10x + 10y - 4 \ln(15x - 10y + 11) = C$$

Divide by 2,

$$5x + 5y - 2 \ln(15x - 10y + 11) = -5C$$

$$5x + 5y + 5C = 2 \ln(15x - 10y + 11)$$

$$5(x + y + C) = 2 \ln(15x - 10y + 11)$$

2. $\sin y (x + \sin y)dx + 2x^2 \cos y dy = 0$

Let:

$$\begin{aligned} u &= \sin y \\ du &= \cos y dy \end{aligned}$$

Substitute,

$$u(x + u)dx + 2x^2 du = 0$$

∴ Homogeneous, 2nd degree!

Let:

$$\begin{aligned} u &= vx \\ du &= vdx + xdv \end{aligned}$$

Substitute,

$$vx(x + vx)dx + 2x^2(vdx + xdv) = 0$$

$$v(1 + v)dx + 2(vdx + xdv) = 0$$

$$(v + v^2 + 2v)dx + 2xdv = 0$$

$$(v^2 + 3v)dx + 2xdv = 0$$

$$\frac{dx}{x} + \frac{2dv}{(v^2 + 3v)} = 0$$

$$\int \frac{dx}{x} + \int \frac{2dv}{v(v + 3)} = \int 0$$

Considering:

$$\int \frac{2dv}{v(v + 3)}$$

Using Integration by Partial Fraction:

$$\frac{2}{v(v + 3)} = \frac{A}{v} + \frac{B}{(v + 3)}$$

$$2 = A(v + 3) + Bv$$

TYPE 7: SUBSTITUTION SUGGESTED BY THE EQUATION
TYPE 8: BERNOULLI'S EQUATION

If $v = 0$; $A = 2/3$
 $v = -3$; $B = -2/3$

Then,

$$\begin{aligned} \int \frac{2dv}{v(v + 3)} &= \frac{2}{3} \int \frac{dv}{v} - \frac{2}{3} \int \frac{dv}{(v + 3)} \\ &= \frac{2}{3} \ln v - \frac{2}{3} \ln(v + 3) \end{aligned}$$

Thus,

$$\ln x + \frac{2}{3} \ln v - \frac{2}{3} \ln(v + 3) = \ln C$$

$$3 \ln x + 2 \ln v - 2 \ln(v + 3) = \ln C$$

$$\frac{x^3 v^2}{(v + 3)^2} = C$$

But $v = \frac{u}{x} = \frac{\sin y}{x}$

$$\frac{x^3 \left(\frac{\sin y}{x}\right)^2}{\left(\frac{\sin y}{x} + 3\right)^2} = C$$

$$\frac{x^3 \left(\frac{\sin y}{x}\right)^2}{\left(\frac{\sin y + 3x}{x}\right)^2} = C$$

$$\frac{x^3 \sin^2 y}{(\sin y + 3x)^2} = C$$

$$x^3 \sin^2 y = C(\sin y + 3x)^2$$

3. $dy/dx = (9x + 4y + 1)^2$

Let:

$$\begin{aligned} u &= 9x + 4y + 1 \\ du &= 9dx + 4dy \\ dy &= \frac{du - 9dx}{4} \end{aligned}$$

Substitute,

$$\frac{du - 9dx}{4} = u^2$$

$$du - 9dx = 4u^2 dx$$

$$du = 4u^2 dx + 9dx$$

$$du = (4u^2 + 9)dx$$

$$\frac{du}{(4u^2 + 9)} = dx$$

$$\int \frac{du}{(4u^2 + 9)} = \int dx$$

$$\int \frac{du}{(2u)^2 + (3)^2} = x + C$$

From Integration Formulas (Appendix A):

$$\int \frac{du}{a^2 + u^2} = \frac{1}{a} \operatorname{Arctan} \frac{u}{a}$$

Then,

$$\frac{1}{6} \operatorname{Arctan} \frac{2u}{3} = x + C$$

$$\operatorname{Arctan} \frac{2u}{3} = 6x + C$$

From:
 $\operatorname{Arctan} \theta = A$
 $\theta = \tan A$

Thus,

$$\frac{2u}{3} = \tan(6x + C)$$

$$2u = 3 \tan(6x + C)$$

But $u = 9x + 4y + 1$;

$$2(9x + 4y + 1) = 3 \tan(6x + C)$$

4. $y' = y - xy^3 e^{-2x}$

Multiply by y^{-3} ,

$$y^{-3} \frac{dy}{dx} = y^{-2} - x e^{-2x}$$

$$y^{-3} dy = y^{-2} dx - x e^{-2x} dx$$

Bernoulli's Equation:

Let:

$$\begin{aligned} u &= y^{-2} \\ du &= -2y^{-3} dy \end{aligned}$$

Substitute,

$$-\frac{1}{2} du = u dx - x e^{-2x} dx$$

Multiply by $-2/dx$,

$$\frac{du}{dx} = -2u + 2e^{-2x}x$$

$$\frac{du}{dx} + 2u = 2e^{-2x}x$$

Linear Equation!

$$P(x) = 2$$

$$Q(x) = 2e^{-2x}x$$

For integrating factor,

$$e^{\int P(x) dx} = e^{\int 2 dx} = e^{2x}$$

Then,

$$ue^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + C$$

$$ue^{2x} = \int 2e^{-2x}xe^{2x} dx + C$$

$$ue^{2x} = \int 2x dx + C$$

$$ue^{2x} = x^2 + C$$

But $u = y^{-2}$

$$y^{-2}e^{2x} = x^2 + C$$

$$\frac{e^{2x}}{y^2} = x^2 + C$$

$$e^{2x} = y^2(x^2 + C)$$

5. $dy/dx = \sin(x+y)$

Let:

$$\begin{aligned} u &= x+y \\ du &= dx+dy \\ dy &= du-dx \end{aligned}$$

Substitute,

$$\frac{du-dx}{dx} = \sin u$$

$$du-dx = \sin u dx$$

$$du = \sin u dx + dx$$

$$du = (1 + \sin u) dx$$

$$\frac{du}{(1 + \sin u)} = dx$$

$$\frac{du}{(1 + \sin u)} \left(\frac{(1 - \sin u)}{(1 - \sin u)} \right) = dx$$

$$\frac{(1 - \sin u) du}{1 - \sin^2 u} = dx$$

$$\frac{(1 - \sin u) du}{\cos^2 u} = dx$$

$$\frac{du}{\cos^2 u} - \frac{\sin u du}{\cos^2 u} = dx$$

$$\int \sec^2 u du - \int \sec u \tan u du = \int dx$$

$$\tan u - \sec u = x + C$$

But $u = x + y$

$$\begin{aligned} \tan(x+y) - \sec(x+y) &= x + C \\ x + C &= \tan(x+y) - \sec(x+y) \end{aligned}$$

6. $xydx + (x^2 - 3y)dy = 0$

Let:

$$\begin{aligned} u &= x^2 \\ du &= 2x dx \end{aligned}$$

Substitute,

$$\frac{1}{2} y du + (u - 3y) dy = 0$$

Multiply by $\frac{2}{y dy}$

$$\frac{du}{dy} + \frac{2(u - 3y)}{y} = 0$$

$$\frac{du}{dy} + \frac{2}{y}u - 6 = 0$$

$$\frac{du}{dy} + \frac{2}{y}u = 6$$

Linear Equation!

$$P(y) = 2/y$$

$$Q(y) = 6$$

For integrating factor:

$$e^{\int P(y) dy} = e^{\int \frac{2 dy}{y}} = e^{2 \ln y} = y^2$$

Then,

$$ye^{\int P(y) dy} = \int Q(y)e^{\int P(y) dy} dy + C$$

$$uy^2 = \int 6y^2 dy + C$$

$$uy^2 = 2y^3 + C$$

But $u = x^2$

$$x^2y^2 = 2y^3 + C$$

7. $(3 \tan x - 2 \cos y) \sec^2 x dx + \tan x \sin y dy = 0$

Let:

$$\begin{aligned} u &= \tan x \\ du &= \sec^2 x dx \end{aligned}$$

Substitute,

$$(3u - 2 \cos y) du + u \sin y dy = 0$$

Let:

$$\begin{aligned} v &= \cos y \\ dv &= -\sin y dy \end{aligned}$$

Substitute,

$$(3u - 2v) du - u dv = 0$$

Divide by $-u dv$,

$$\frac{(3u - 2v)}{-u} + \frac{dv}{du} = 0$$

$$\frac{dv}{du} - \frac{(3u - 2v)}{u} = 0$$

$$\frac{dv}{du} - 3 + \frac{2v}{u} = 0$$

$$\frac{dv}{du} + \frac{2}{u}v = 3$$

Linear Equation!

$$P(u) = 2/u$$

$$Q(u) = 3$$

For integrating factor:

$$e^{\int P(u) du} = e^{\int \frac{2 du}{u}} = e^{2 \ln u} = u^2$$

Then,

$$ue^{\int P(u) du} = \int Q(u)e^{\int P(u) du} du + C$$

$$vu^2 = \int 3u^2 du + C$$

$$vu^2 = u^3 + C$$

But:

$$\begin{aligned} u &= \tan x \\ v &= \cos y \end{aligned}$$

$$\cos y \tan^2 x = \tan^3 x + C$$

8. $(x + 2y - 1)dx + (2x + 4y - 3)dy = 0$.
Solve by two methods.

♦ First Method

$$(x + 2y - 1)dx + (2(x + 2y) - 3)dy = 0$$

Let:

$$\begin{aligned} u &= x + 2y \\ du &= dx + 2dy \\ dx &= du - 2dy \end{aligned}$$

Substitute,

$$\begin{aligned} (u - 1)(du - 2dy) + (2u - 3)dy &= 0 \\ (u - 1)du - 2udy + 2udy + 2udy - 3dy &= 0 \\ &= 0 \end{aligned}$$

$$(u - 1)du - dy = 0$$

$$\int (u - 1)du - \int dy = \int 0$$

$$\frac{(u - 1)^2}{2} - y = C$$

$$(u - 1)^2 - 2y = C$$

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TYPE 8: BERNOULLI'S EQUATION

But $u = x + 2y$

$(x + 2y - 1)^2 - 2y = C$

$(x + 2y - 1)^2 = 2y + C$

◆ Second Method

$M = x + 2y - 1;$

$\frac{\partial M}{\partial y} = 2$

$N = 2x + 4y - 3$

$\frac{\partial N}{\partial x} = 2$

$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$

$\int D.I + \int N.D.I = 0$

D.I:

$(x - 1)dx, (4y - 3)dy$

N.D.I:

$2ydx, 2xdy$

Choose: $2ydx$

$\int (x - 1)dx + \int (4y - 3)dy$
 $+ \int 2ydx = 0$

$\frac{x^2}{2} - x + 2y^2 - 3y + 2xy = C$

$x^2 - 2x + 4y^2 - 6y + 4xy + 1 = C + 1$

$(x^2 + 4xy - 2x + 4y^2 - 4y + 1) - 2y = C_1$

$(x + 2y - 1)^2 = 2y + C_1$

9.

$6y^2 dx - x(2x^3 + y)dy = 0$

$6y^2 dx - 2x^4 dy - xydy = 0$

$6y^2 dx - xydy - 2x^4 dy = 0$

Bernoulli's Equation!

Multiply by x^{-4}

$6y^2 x^{-4} dx - x^{-3} y dy - 2dy = 0$

Let:

$u = x^{-3}$

$du = -3x^{-4} dx$

Substitute,

$-2y^2 du - u y dy - 2dy = 0$

Divide by $-2y^2 dy$,

$\frac{du}{dy} + \frac{1}{2y} u + \frac{1}{y^2} = 0$

$\frac{du}{dy} + \frac{1}{2y} u = -\frac{1}{y^2}$

Linear Equation!

$P(y) = 1/2y$

$Q(y) = -1/y^2$

For integrating factor:

$e^{\int P(y)dy} = e^{\int \frac{1}{2y} dy} = e^{\frac{1}{2} \ln y} = y^{1/2}$

Then,

$ue^{\int P(y)dy} = \int Q(y)e^{\int P(y)dy} dy + C$

$uy^{1/2} = \int -\frac{1}{y^2} y^{1/2} dy + C$

$uy^{1/2} = \int -y^{-3/2} dy + C$

$uy^{1/2} = 2y^{-1/2} + C$

Multiply by $y^{1/2}$,

$uy^2 = -2y + Cy^{3/2}$

But $u = x^{-3}$

$x^{-3} y^2 = -2y + Cy^{3/2}$

Multiply by x^3 ,

$y^2 = -2x^3 y + Cx^3 y^{3/2}$

$y^2 = x^3 y(Cy^{1/2} - 2)$

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TYPE 8: BERNOULLI'S EQUATION

$2x^3 y' = y(y^2 + 3x^2).$
Solve by two methods.

◆ First Method

$2x^3 \frac{dy}{dx} = y(y^2 + 3x^2)$

$2x^3 dy = y^3 dx + 3x^2 y dx$

$2x^3 dy - 3x^2 y dx - y^3 dx = 0$

Bernoulli's Equation:

Multiply by y^{-3}

$2x^3 y^{-3} dy - 3x^2 y^{-2} dx - dx = 0$

Let:

$u = y^{-2}$

$du = -2y^{-3} dy$

Substitute,

$-x^3 du - 3ux^2 dx - dx = 0$

Multiply by $\frac{-1}{x^3 dx}$:

$\frac{du}{dx} + \frac{3}{x} u + \frac{1}{x^3} = 0$

$\frac{du}{dx} + \frac{3}{x} u = -\frac{1}{x^3}$

Linear Equation!

$P(x) = 3/x$

$Q(x) = -1/x^3$

For integrating factor:

$e^{\int P(x)dx} = e^{\int \frac{3}{x} dx} = e^{3 \ln x} = x^3$

Then,

$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$

$ux^3 = \int -\frac{1}{x^3} x^3 dx + C$

$ux^3 = \int -dx + C$

$ux^3 = -x + C$

But $u = y^{-2}$

$y^{-2} x^3 = -x + C$

$y^{-2} x^3 + x = C$

Multiply by y^2 ,

$x^3 + xy^2 = Cy^2$

$x^3 = y^2(C - x)$

◆ Second Method

$2x^3 dy = (y^3 + 3x^2 y) dx$

∴ Homogeneous, 3rd degree

Let:

$y = vx$

$dy = vdx + xdv$

Substitute,

$2x^3(vdx + xdv) = (v^3 x^3 + 3vx^3) dx$

$2(vdx + xdv) = (v^3 + 3v) dx$

$2vdx + 2xdv = (v^3 + 3v) dx$

$2xdv = (v^3 + 3v - 2v) dx$

$2xdv = (v^3 + v) dx$

$\frac{2dv}{v(v^2 + 1)} = \frac{dx}{x}$

$\frac{dx}{x} - \frac{2dv}{v(v^2 + 1)} = 0$

$\int \frac{dx}{x} - \int \frac{2dv}{v(v^2 + 1)} = \int 0$

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$\int \frac{dx}{x} - \int \frac{2dv}{v(v^2 + 1)} = 0$

TYPE 7: SUBSTITUTION SUGGESTED BY THE EQUATION TYPE 8: BERNOULLI'S EQUATION

Collecting constant terms,
 $2 = A$

Collecting v^2 terms,
 $0 = A + 2B$
 $B = -1$

$$\int \frac{2dv}{v(v^2+1)} = 2 \int \frac{dv}{v} - \int \frac{2vdv}{(v^2+1)}$$

$$= 2 \ln v - \ln(v^2+1)$$

Thus,

$$\ln x - 2 \ln v + \ln(v^2+1) = \ln C$$

$$\frac{x(v^2+1)}{v^2} = C$$

But $v = y/x$

$$\frac{x\left(\frac{y^2}{x^2} + 1\right)}{\frac{y^2}{x^2}} = C$$

$$x\left(\frac{y^2+x^2}{x^2}\right) \frac{x^2}{y^2} = C$$

$$x(y^2+x^2) = Cy^2$$

$$x^3 + xy^2 = Cy^2$$

$$x^3 = y^2(C-x)$$

11. $(3 \sin y - 5x)dx + 2x^2 \cot y dy = 0$

$$(3 \sin y - 5x)dx + 2x^2 \frac{\cos y}{\sin y} dy = 0$$

Let:

$$u = \sin y$$

$$du = \cos y dy$$

Substitute,

$$(3u - 5x)dx + 2x^2 \frac{du}{u} = 0$$

∴ Homogeneous, 1st degree

Let:

$$u = vx$$

$$du = vdx + xdv$$

Substitute,

$$(3vx - 5x)dx + 2x^2 \frac{(vdx + xdv)}{vx} = 0$$

$$x(3v - 5)dx + 2x \frac{(vdx + xdv)}{v} = 0$$

$$(3v - 5)dx + 2 \frac{(vdx + xdv)}{v} = 0$$

$$(3v - 5)dx + 2dx + \frac{2xdv}{v} = 0$$

$$(3v - 3)dx + \frac{2xdv}{v} = 0$$

Divide by $x(3v - 3)$

$$\frac{dx}{x} + \frac{2dv}{v(3v-3)} = 0$$

$$\int \frac{dx}{x} + \int \frac{2dv}{3v(v-1)} = \int 0$$

Considering,

$$\int \frac{2dv}{3v(v-1)}$$

Using Integration by Partial Fraction,

$$\frac{2/3}{v(v-1)} = \frac{A}{v} + \frac{B}{(v-1)}$$

$$\frac{2}{3} = A(v-1) + Bv$$

If $v = 0$; $A = -2/3$
 $v = 1$; $B = 2/3$

Then,

$$\int \frac{2dv}{3v(v-1)} = -\frac{2}{3} \int \frac{dv}{v} + \frac{2}{3} \int \frac{dv}{(v-1)}$$

$$= -\frac{2}{3} \ln v + \frac{2}{3} \ln(v-1)$$

Thus,

$$\ln x - \frac{2}{3} \ln v + \frac{2}{3} \ln(v-1) = \ln C$$

$$3 \ln x - 2 \ln v + 2 \ln(v-1) = \ln C$$

$$\frac{x^3(v-1)^2}{v^2} = C$$

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But $v = \frac{u}{x} = \frac{\sin y}{x}$

$$x^3 \left(\frac{\sin y}{x} - 1 \right)^2 = C$$

$$x^3 \left(\frac{\sin y - x}{x} \right)^2 \frac{x^2}{\sin^2 y} = C$$

$$x^3 (\sin y - x)^2 = C \sin^2 y$$

12. $y' = 1 + 6xe^{x-y}$

$$\frac{dy}{dx} = 1 + 6xe^{x-y}$$

$$\frac{dy}{dx} = 1 + 6xe^x e^{-y}$$

$$dy = dx + 6xe^x e^{-y} dx$$

Divide by e^{-y} ,

$$e^y dy = e^y dx + 6xe^x dx$$

Let:

$$u = e^y$$

$$du = e^y dy$$

Substitute,

$$du = u dx + 6xe^x dx$$

$$\frac{du}{dx} = u + 6xe^x$$

$$\frac{du}{dx} - u = 6xe^x$$

Linear Equation!

$$P(x) = -1$$

$$Q(x) = 6xe^x$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int dx} = e^{-x}$$

Then,

$$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx} dx + C$$

$$ue^{-x} = \int 6xe^x e^{-x} dx + C$$

$$ue^{-x} = \int 6x dx + C$$

$$ue^{-x} = 3x^2 + C$$

But $u = e^y$

$$e^y e^{-x} = 3x^2 + C$$

$$e^{y-x} = 3x^2 + C$$

13. $dv/du = (u-v)^2 - 2(u-v) - 2$

Let:

$$w = u - v$$

$$dw = du - dv$$

$$dv = du - dw$$

Substitute,

$$\frac{du - dw}{du} = w^2 - 2w - 2$$

$$du - dw = (w^2 - 2w - 2)du$$

$$-dw = (w^2 - 2w - 2)du - du$$

$$-dw = (w^2 - 2w - 3)du$$

$$-dw = (w+1)(w-3)du$$

$$du + \frac{dw}{(w+1)(w-3)} = 0$$

$$\int du + \int \frac{dw}{(w+1)(w-3)} = \int 0$$

Considering:

$$\int \frac{dw}{(w+1)(w-3)}$$

Using Integration by Partial Fraction,

$$\frac{1}{(w+1)(w-3)} = \frac{A}{(w+1)} + \frac{B}{(w-3)}$$

$$1 = A(w-3) + B(w+1)$$

If $w = -1$; $A = -1/4$
 $w = 3$; $B = 1/4$

Then,

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$$\begin{aligned} \int \frac{dw}{(w+1)(w-3)} &= -\frac{1}{4} \int \frac{dw}{(w+1)} + \frac{1}{4} \int \frac{dw}{(w-3)} \\ &= -\frac{1}{4} \ln(w+1) + \frac{1}{4} \ln(w-3) \end{aligned}$$

Then,

$$u - \frac{1}{4} \ln(w+1) + \frac{1}{4} \ln(w-3) = \ln C$$

$$4u - \ln(w+1) + \ln(w-3) = \ln C$$

$$4u = \ln C + \ln(w+1) - \ln(w-3)$$

$$4u = \ln \frac{C(w+1)}{(w-3)}$$

$$e^{4u} = \frac{C(w+1)}{(w-3)}$$

$$\text{But } w = u - v$$

$$e^{4u} = \frac{C(u-v+1)}{(u-v-3)}$$

$$e^{4u}(u-v-3) = C(u-v+1)$$

$$14. \quad 2ydx + x(x^2 \ln y - 1)dy = 0$$

$$2ydx + x^3 \ln y dy - xdy = 0$$

$$2ydx - xdy + x^3 \ln y dy = 0$$

Bernoulli's Equation!

Divide by x^3

$$2x^{-3}ydx - x^{-2}dy + \ln y dy = 0$$

Let

$$u = x^{-2} \\ du = -2x^{-3}dx$$

Substitute,

$$-ydu - udy + \ln y dy = 0$$

Divide by $-ydy$,

$$\frac{du}{dy} + \frac{u}{y} - \frac{\ln y}{y} = 0$$

$$\frac{du}{dy} + \frac{1}{y}u = \frac{\ln y}{y}$$

Linear Equation!

$$P(y) = 1/y$$

$$Q(y) = \frac{\ln y}{y}$$

For integrating factor:

$$e^{\int P(y)dy} = e^{\int \frac{1}{y}dy} = e^{\ln y} = y$$

Then,

$$ue^{\int P(y)dy} = \int Q(y)e^{\int P(y)dy}dy + C$$

$$uy = \int \frac{\ln y}{y} y dy + C$$

$$uy = \int \ln y dy + C$$

$$uy = y \ln y - y + C$$

$$\text{But } u = x^{-2}$$

$$x^{-2}y = y \ln y - y + C$$

Multiply by x^2 ,

$$y = x^2 y \ln y - x^2 y + Cx^2$$

$$y - x^2 y \ln y + x^2 y = Cx^2$$

$$y(1 - x^2 \ln y + x^2) = Cx^2$$

$$15. \quad \cos y \sin 2x dx + (\cos^2 y - \cos^2 x)dy = 0$$

Since,

$$\sin 2x = 2 \cos x \sin x$$

Thus,

$$2 \cos y \cos x \sin x dx + (\cos^2 y - \cos^2 x)dy = 0$$

Let

$$u = \cos^2 x \\ du = -2 \cos x \sin x dx$$

Substitute,

TYPE 7: SUBSTITUTION SUGGESTED BY THE EQUATION
TYPE 8: BERNOULLI'S EQUATION

$$-\cos y du + \cos^2 y - udy = 0$$

Divide by $-\cos y dy$,

$$\frac{du}{dy} - \cos y + \frac{u}{\cos y} = 0$$

$$\frac{du}{dy} + \frac{1}{\cos y}u = \cos y$$

$$\frac{du}{dy} + \sec y u = \cos y$$

Linear Equation!

$$P(y) = \sec y$$

$$Q(y) = \cos y$$

For integrating factor:

$$e^{\int P(y)dy} = e^{\int \sec y dy} = e^{\ln(\sec y + \tan y)} \\ = \sec y + \tan y$$

Then,

$$ue^{\int P(y)dy} = \int Q(y)e^{\int P(y)dy}dy + C$$

$$u(\sec y + \tan y) = \int \cos y (\sec y + \tan y)dy + C$$

$$u\left(\frac{1}{\cos y} + \frac{\sin y}{\cos y}\right) = \int \cos y \left(\frac{1}{\cos y} + \frac{\sin y}{\cos y}\right)dy + C$$

$$u\left(\frac{\cos y + \sin y \cos y}{\cos^2 y}\right) = \int (1 + \sin y)dy + C$$

$$u\left(\frac{\cos y (1 + \sin y)}{\cos^2 y}\right) = y - \cos y + C$$

$$u\frac{(1 + \sin y)}{\cos y} = y - \cos y + C$$

$$u(1 + \sin y) = \cos y (y + C - \cos y)$$

$$\text{But } u = \cos^2 x$$

$$\cos^2 x (1 + \sin y) = \cos y (y + C - \cos y)$$

$$16. \quad (ke^{2x} - u)du = 2e^{2x}(e^{2x} + ku)dv$$

Let

$$w = e^{2x} \\ dw = 2e^{2x}dx$$

Substitute,

$$(kw - u)du = dw(w + ku)$$

$\therefore$ Homogeneous, 1st degree!

Let

$$w = zu \\ dw = zdu + udz$$

Substitute,

$$(kzu - u)du = (zdu + udz)(zu + ku)$$

$$u(kz - 1)du = (zdu + udz)(z + k)u$$

$$(kz - 1)du = (zdu + udz)(z + k)$$

$$(kz - 1)du = z^2 du + zudx + kzd u + kudx$$

$$(kz - 1 - z^2 - kz)du = zudx + kudx$$

$$(-1 - z^2)du = u(z + k)dz$$

$$-1(z^2 + 1)du = u(z + k)dz$$

$$-\frac{du}{u} = \frac{(z + k)dz}{z^2 + 1}$$

$$-\frac{du}{u} - \frac{z}{z^2 + 1}dz - \frac{k}{z^2 + 1}dz = 0$$

$$-\int \frac{du}{u} - \int \frac{z}{z^2 + 1}dz - k \int \frac{1}{z^2 + 1}dz = 0$$

$$-\ln u - \frac{1}{2} \ln(z^2 + 1) - k \text{Arctan } z = \ln C$$

$$-2 \ln u - \ln(z^2 + 1) - 2k \text{Arctan } z = \ln C$$

$$-\ln u^2 - \ln(z^2 + 1) - 2k \text{Arctan } z = \ln C$$

$$2k \text{Arctan } z = \ln C + \ln u^2 + \ln(z^2 + 1)$$

$$\text{But } z = \frac{w}{u} = \frac{e^{2x}}{u}$$

$$2k \text{Arctan } \frac{e^{2x}}{u} = \ln C + \ln u^2 + \ln \left(\frac{e^{4x}}{u^2} + 1 \right)$$

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$$2k \operatorname{Arctan} \frac{e^{2y}}{u} = \ln C + \ln u^2 + \ln \left(\frac{e^{4y} + u^2}{u^2} \right)$$

$$2k \operatorname{Arctan} \frac{e^{2y}}{u} = \ln C u^2 \left(\frac{e^{4y} + u^2}{u^2} \right)$$

$$2k \operatorname{Arctan} \frac{e^{2y}}{u} = \ln [C(e^{4y} + u^2)]$$

17. $y' \tan x \sin 2y = \sin^2 x + \cos^2 y$

$$\frac{dy}{dx} \tan x \sin 2y = \sin^2 x + \cos^2 y$$

Since,

$$\sin 2y = 2 \sin y \cos y$$

$$2 \frac{dy}{dx} \tan x \sin y \cos y = \sin^2 x + \cos^2 y$$

$$2 \tan x \sin y \cos y dy = (\sin^2 x + \cos^2 y) dx$$

Let:

$$u = \cos^2 y$$

$$du = -2 \sin y \cos y dy$$

Substitute,

$$-\tan x du = (\sin^2 x + u) dx$$

$$-\frac{\sin x}{\cos x} du = (\sin^2 x + u) dx$$

$$-\sin x du = \cos x (\sin^2 x + u) dx$$

Let:

$$w = \sin x$$

$$dw = \cos x dx$$

Substitute,

$$-w du = (w^2 + u) dw$$

$$\frac{du}{dw} = \frac{(w^2 + u)}{w}$$

$$\frac{du}{dw} - \frac{1}{w} u = w$$

$$\frac{du}{dw} + \frac{1}{w} u = -w$$

Linear Equation!

$$P(w) = 1/w$$

$$Q(w) = -w$$

For integrating factor:

$$e^{\int P(w) dw} = e^{\int \frac{1}{w} dw} = e^{\ln w} = w$$

Then,

$$ue^{\int P(w) dw} = \int Q(w)e^{\int P(w) dw} dw + C$$

$$uw = \int (-w)w dw + C$$

$$uw = \int -w^2 dw + C$$

$$uw = -\frac{w^3}{3} + C$$

$$3uw + w^3 = C$$

$$w(3u + w^2) = C$$

But $w = \sin x$, $u = \cos^2 y$

$$\sin x (3 \cos^2 y + \sin^2 x) = C$$

18. $(x + 2y - 1)dx - (x + 2y - 5)dy = 0$

Let:

$$u = x + 2y$$

$$du = dx + 2dy$$

$$dx = du - 2dy$$

Substitute,

$$(u - 1)(du - 2dy) - (u - 5)dy = 0$$

$$(u - 1)du - 2(u - 1)dy - (u - 5)dy = 0$$

$$(u - 1)du - [(2u - 2) + (u - 5)]dy = 0$$

$$(u - 1)du - (3u - 7)dy = 0$$

$$\frac{(u - 1)du}{(3u - 7)} - dy = 0$$

$$\frac{1}{3} du + \frac{4}{3} \left(\frac{du}{(3u - 7)} \right) - dy = 0$$

$$\frac{1}{3} \int du + \frac{4}{3} \int \frac{du}{(3u - 7)} - \int dy = \int 0$$

$$\frac{u}{3} + \frac{4}{9} \ln(3u - 7) - y = C$$

$$3u + 4 \ln(3u - 7) - 9y = C$$

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But $u = x + 2y$

$$3(x + 2y) + 4 \ln(3(x + 2y) - 7) - 9y = C$$

$$3x + 6y + 4 \ln(3x + 6y - 7) - 9y = C$$

$$3x - 3y + 4 \ln(3x + 6y - 7) = C$$

$$3(x - y) + 4 \ln(3x + 6y - 7) = C$$

19. $y(x \tan x + \ln y)dx + \tan x dy = 0$

$$(xy \tan x + y \ln y)dx + \tan x dy = 0$$

$$xy \tan x dx + y \ln y dx + \tan x dy = 0$$

Divide by $y \tan x$,

$$x dx + \frac{\ln y}{\tan x} dx + \frac{dy}{y} = 0$$

$$(x + \cot x \ln y)dx + \frac{dy}{y} = 0$$

Let:

$$u = \ln y$$

$$du = dy/y$$

Substitute,

$$(x + \cot x u)dx + du = 0$$

$$(x + \cot x u) + \frac{du}{dx} = 0$$

$$\frac{du}{dx} + \cot x u = -x$$

Linear Equation!

$$P(x) = \cot x$$

$$Q(x) = -x$$

For integrating factor:

$$e^{\int P(x) dx} = e^{\int \cot x dx} = e^{\ln \sin x} = \sin x$$

Then,

$$ue^{\int P(x) dx} = \int Q(x)e^{\int P(x) dx} dx + C$$

$$u \sin x = - \int x \sin x dx + C$$

Considering:

$$\int x \sin x dx$$

Using Integration by Parts:

Let $u = x$ $dv = \sin x dx$
 $du = dx$ $v = -\cos x$

$$\int x \sin x dx = -x \cos x + \int \cos x dx$$

$$= -x \cos x + \sin x$$

thus,

$$u \sin x = x \cos x - \sin x + C$$

But $u = \ln y$

$$\sin x \ln y = x \cos x - \sin x + C$$

20. $xy' - y = x^k y^n$,
 Where $n \neq 1$ and $k + n \neq 1$

$$x \frac{dy}{dx} - y = x^k y^n$$

$$x dy - y dx = x^k y^n dx$$

$$x dy - y dx - x^k y^n dx = 0$$

Bernoulli's Equation!

Multiply by y^{-n} ,

$$xy^{-n} dy - y^{(1-n)} dx - x^k dx = 0$$

Let:

$$u = y^{(1-n)}$$

$$du = (1-n)y^{-n} dy$$

Substitute,

$$\frac{x du}{1-n} - u dx - x^k dx = 0$$

Multiply by $\frac{(1-n)}{dx}$,

$$\frac{du}{dx} - \frac{u(1-n)}{x} - \frac{x^k(1-n)}{x} = 0$$

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$$\frac{du}{dx} - \frac{(1-n)}{x}u = x^{k-1}(1-n)$$

$$\frac{du}{dx} + \frac{n-1}{x}u = (1-n)x^{k-1}$$

Linear Equation!

$$P(x) = \frac{n-1}{x}$$

$$Q(x) = (1-n)x^{k-1}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{n-1}{x}dx} = e^{(n-1)\ln x} = e^{\ln(x^{n-1})}$$

$$= \frac{x^{n-1}}{x} = x^{n-2}$$

Then,

$$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ux^{n-1} = \int (1-n)x^{k-1}x^{n-1}dx + C$$

$$ux^{n-1} = \int (1-n)x^{k+n-2}dx + C$$

$$ux^{n-1} = \frac{(1-n)}{k+n-1}x^{k+n-1} + \frac{C}{k+n-1}$$

Multiply by $(k+n-1)x^{-n+1}$

$$(k+n-1)u = (1-n)x^{k+n-1}x^{-n+1} + Cx^{-n+1}$$

$$(k+n-1)u = (1-n)x^k + Cx^{1-n}$$

But $u = y^{1-n}$

$$(k+n-1)y^{(1-n)} = (1-n)x^k + Cx^{1-n}$$

21. Solve the equation of exercise 20 for the values of k and n not include there.

$$xy' - y = x^k y^n$$

- (a) $n=1; k \neq 0$
(b) $n=1; k=0$
(c) $n \neq 1; k+n=1$

Solution:

- (a) $n=1; k \neq 0$

$$x \frac{dy}{dx} - y = x^k y^1$$

$$x \frac{dy}{dx} = x^k y + y$$

$$xdy = (x^k y + y)dx$$

$$xdy = y(x^k + 1)dx$$

Divide by xy ,

$$\frac{dy}{y} = \frac{(x^k + 1)}{x} dx$$

$$\frac{dy}{y} = x^{k-1} dx + \frac{dx}{x}$$

$$x^{k-1} dx + \frac{dx}{x} - \frac{dy}{y} = 0$$

$$\int x^{k-1} dx + \int \frac{dx}{x} - \int \frac{dy}{y} = \int 0$$

$$\frac{x^k}{k} + \ln x - \ln y = \ln C$$

$$\frac{x^k}{k} = \ln C + \ln y - \ln x$$

$$x^k = k \ln \left| \frac{Cy}{x} \right|$$

- (b) $n=1; k=0$

$$x \frac{dy}{dx} - y = x^0 y^1$$

$$x \frac{dy}{dx} - y = y$$

$$x \frac{dy}{dx} = 2y$$

$$xdy = 2ydx$$

$$\frac{dy}{y} = \frac{2dx}{x}$$

$$\int \frac{dy}{y} = 2 \int \frac{dx}{x}$$

$$\ln y = 2 \ln x + \ln C$$

$$y = Cx^2$$

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- (c) $n \neq 1; k+n=1$

$$x \frac{dy}{dx} - y = x^{(1-n)} y^n$$

$$xdy - ydx = x^{(1-n)} y^n dx$$

$$xdy - ydx - x^{(1-n)} y^n dx = 0$$

Bernoulli's Equation!

Multiply by y^{-n} ,

$$xy^{-n} dy - y^{1-n} dx - x^{(1-n)} dx = 0$$

Let:

$$u = y^{1-n}$$

$$du = (1-n)y^{-n} dy$$

Substitute,

$$\frac{x}{(1-n)} du - udx - x^{(1-n)} dx = 0$$

Multiply by $\frac{(1-n)}{x}$,

$$\frac{du}{dx} - \frac{(1-n)u}{x} - (1-n)x^{-n} = 0$$

$$\frac{du}{dx} - \frac{(1-n)}{x}u = (1-n)x^{-n}$$

$$\frac{du}{dx} + \frac{(n-1)}{x}u = (1-n)x^{-n}$$

Linear Equation!

$$P(x) = \frac{n-1}{x}$$

$$Q(x) = (1-n)x^{-n}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{n-1}{x}dx} = e^{(n-1)\ln x} = e^{\ln(x^{n-1})}$$

$$= \frac{x^{n-1}}{x} = x^{n-2}$$

Then,

$$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx$$

$$ux^{n-1} = \int (1-n)x^{-n}x^{n-1}dx$$

$$ux^{n-1} = \int \frac{(1-n)}{x} dx$$

$$ux^{n-1} = (1-n)\ln x + (1-n)\ln C$$

$$ux^{n-1} = (1-n)\ln(Cx)$$

But $u = y^{1-n}$

$$y^{1-n}x^{n-1} = (1-n)\ln(Cx)$$

Multiply by x^{-n+1}

$$y^{1-n} = (1-n)x^{-n+1}\ln(Cx)$$

$$y^{1-n} = (1-n)x^{1-n}\ln(Cx)$$

II. In exercises 22 through 27, find the particular solution required.

22. $4(3x+y-2)dx - (3x+y)dy = 0$
when $x=1, y=0$

Let:

$$u = 3x + y$$

$$du = 3dx + dy$$

$$dy = du - 3dx$$

Substitute,

$$4(u-2)dx - u(du-3dx) = 0$$

$$4udx - 8dx - udu + 3udx = 0$$

$$-udu + (7u-8)dx = 0$$

$$-\frac{udu}{(7u-8)} + dx = 0$$

$$-\left(\frac{1}{7}du + \frac{8}{7} \frac{du}{(7u-8)}\right) + dx = 0$$

$$-\frac{1}{7}du - \frac{8}{7} \frac{du}{(7u-8)} + dx = 0$$

$$-\frac{1}{7} \int du - \frac{8}{7} \int \frac{du}{(7u-8)} + \int dx = \int 0$$

$$-\frac{1}{7}u - \frac{8}{49} \ln(7u-8) + x = \frac{C}{49}$$

$$-7u - 8 \ln(7u-8) + 49x = C$$

But $u = 3x + y$

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$$-7(3x + y) - 8 \ln(7(3x + y) - 8) + 49x = C$$

$$-21x - 7y - 8 \ln(21x + 7y - 8) + 49x = C$$

$$28x - 7y - 8 \ln(21x + 7y - 8) = C \rightarrow \text{General Solution}$$

$$\text{when } x = 1, y = 0;$$

$$28(1) - 7(0) - 8 \ln(21(1) + 7(0) - 8) = C$$

$$C = 28 - 8 \ln 13$$

Thus,

$$28x - 7y - 8 \ln(21x + 7y - 8) = 28 - 8 \ln 13$$

$$28x - 7y - 28 = 8 \ln(21x + 7y - 8) - 8 \ln 13$$

$$7(4x - y - 4) = 8 \ln \frac{21x + 7y - 8}{13} \rightarrow \text{Particular Solution}$$

$$23. \quad y' = 2(3x + y)^2 - 1; \text{ when } x = 0, y = 1$$

$$\frac{dy}{dx} = 2(3x + y)^2 - 1$$

Let:

$$u = 3x + y \\ du = 3dx + dy \\ dy = du - 3dx$$

Substitute,

$$\frac{du - 3dx}{dx} = 2u^2 - 1$$

$$du - 3dx = (2u^2 - 1)dx$$

$$du = 3dx + 2u^2dx - dx$$

$$du = 2dx + 2u^2dx$$

$$du = 2(u^2 + 1)dx$$

$$\frac{du}{(u^2 + 1)} = 2dx$$

$$\int \frac{du}{(u^2 + 1)} = 2 \int dx$$

$$\text{Arctan } u = 2x + C$$

$$C = \text{Arctan } u - 2x$$

$$\text{But } u = 3x + y$$

$$C = \text{Arctan}(3x + y) - 2x \rightarrow \text{General Solution}$$

$$\text{when } x = 0, y = 1;$$

$$C = \text{Arctan}(3(0) + 1) - 2(0) = \text{Arctan}(1) \\ C = \frac{\pi}{4}$$

Thus,

$$\frac{\pi}{4} = \text{Arctan}(3x + y) - 2x$$

$$\pi = 4 \text{Arctan}(3x + y) - 8x$$

$$4 \text{Arctan}(3x + y) = 8x + \pi \rightarrow \text{Particular Solution}$$

24.

$$2xyy' = y^2 - 2x^3;$$

Find the solution that passes through the point (1, 2)

$$2xy \frac{dy}{dx} = y^2 - 2x^3$$

$$2xydy = (y^2 - 2x^3)dx$$

Let:

$$u = y^2 \\ du = 2ydy$$

Substitute,

$$xdx = (u - 2x^3)dx$$

$$\frac{dx}{dx} = \frac{(u - 2x^3)}{x}$$

$$\frac{du}{dx} = \frac{u}{x} - 2x^2$$

$$\frac{du}{dx} - \frac{u}{x} = -2x^2$$

Linear Equation!

$$P(x) = -1/x \\ Q(x) = -2x^2$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int \frac{1}{x}dx} = e^{-\ln x} = x^{-1}$$

Then,

$$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ux^{-1} = -\int 2x^2x^{-1}dx + C$$

$$ux^{-1} = -\int 2xdx + C$$

$$ux^{-1} = -x^2 + C$$

Multiply by x ,

$$u = -x^3 + Cx$$

$$\text{But } u = y^2$$

$$y^2 = -x^3 + Cx$$

$$y^2 = x(C - x^2) \rightarrow \text{General Solution}$$

When the solution that passes through the point (1, 2);

$$(2)^2 = (1)(C - (1)^2) \\ C = 5$$

Thus,

$$y^2 = x(5 - x^2) \rightarrow \text{Particular Solution}$$

25.

$$(y^4 - 2xy)dx + 3x^2dy = 0; \text{ when } x = 2, y = 1$$

$$y^4dx - 2xydx + 3x^2dy = 0 \\ 3x^2dy - 2xydx + y^4dx = 0$$

Bernoulli's Equation!

Divide by y^4 ,

$$3x^2y^{-4}dy - 2xy^{-3}dx + dx = 0$$

Let:

$$u = y^{-3} \\ du = -3y^{-4}dy$$

PROBLEMS AND SOLUTIONS IN ELEMENTARY EQUATIONS

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Substitute,

$$-x^2du - 2xudx + dx = 0$$

Divide by $-x^2dx$,

$$\frac{du}{dx} + \frac{2u}{x} - \frac{1}{x^2} = 0 \\ \frac{du}{dx} + \frac{2u}{x} = \frac{1}{x^2}$$

Linear Equation!

$$P(x) = 2/x \\ Q(x) = \frac{1}{x^2}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{2}{x}dx} = e^{2 \ln x} = x^2$$

Then,

$$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ux^2 = \int \frac{1}{x^2}x^2dx + C$$

$$ux^2 = \int dx + C$$

$$ux^2 = x + C$$

$$\text{But } u = y^{-3}$$

$$y^{-3}x^2 = x + C$$

$$x^2 = xy^3 + Cy^3$$

$$x^2 = y^3(x + C) \rightarrow \text{General Solution}$$

$$\text{when } x = 2, y = 1$$

$$(2)^2 = 1^3(2 + C) \\ C = 2$$

Thus,

$$x^2 = y^3(x + 2) \rightarrow \text{Particular Solution}$$

TYPE 7: SUBSTITUTION SUGGESTED BY THE EQUATION TYPE 8: BERNOULLI'S EQUATION

26. $(2y^3 - x^2)dx + 3xy^2dy = 0$
when $x = 1, y = 1$
Solve by two methods.

First Method

Let:
 $u = y^3$
 $du = 3y^2dy$

Substitute,

$$(2u - x^2)dx + xdu = 0$$

Divide by xdu ,

$$\frac{du}{dx} + \frac{2u - x^2}{x} = 0$$

$$\frac{du}{dx} + \frac{2}{x}u = x^2$$

Linear Equation!

$$P(x) = 2/x$$

$$Q(x) = x^2$$

For integrating factor:

$$e^{\int P(x)dx} = e^{\int \frac{2}{x}dx} = e^{2\ln x} = x^2$$

Then,

$$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ux^2 = \int x^2x^2dx + C$$

$$ux^2 = \int x^4dx + C$$

$$ux^2 = \frac{x^5}{5} + C$$

$$5ux^2 = x^5 + C$$

$$\text{But } u = y^3$$

$$5x^2y^3 = x^5 + C \rightarrow \text{General Solution}$$

when $x = 1, y = 1$
 $5(1)^2(1)^3 = (1)^5 + C$
 $C = 4$

Thus,

$$5x^2y^3 = x^5 + 4 \rightarrow \text{Particular Solution}$$

Second Method

$$(2y^3 - x^2)dx + 3xy^2dy = 0$$

Homogeneous, 3rd degree!

Let:

$$y = vx$$

$$dy = vdx + xdv$$

Substitute,

$$(2v^3x^3 - x^2)dx + 3xv^2x^2(vdx + xdv) = 0$$

$$x^2(2v^3 - 1)dx + 3v^2x^3(vdx + xdv) = 0$$

$$(2v^3 - 1)dx + 3v^2(vdx + xdv) = 0$$

$$(2v^3 - 1 + 3v^3)dx + 3v^2xdv = 0$$

$$(5v^3 - 1)dx + 3v^2xdv = 0$$

$$\frac{dx}{x} + \frac{3v^2dv}{(5v^3 - 1)} = 0$$

$$\int \frac{dx}{x} + \frac{3}{15} \int \frac{15v^2dv}{(5v^3 - 1)} = \int 0$$

$$\ln x + \frac{1}{5} \ln(5v^3 - 1) = \ln C$$

$$5 \ln x + \ln(5v^3 - 1) = \ln C$$

$$x^5(5v^3 - 1) = C$$

$$\text{But } v = y/x$$

$$x^5 \left(5 \frac{y^3}{x^3} - 1 \right) = C$$

$$x^5 \left(\frac{5y^3 - x^3}{x^3} \right) = C$$

$$x^2(5y^3 - x^3) = C$$

$$5x^2y^3 - x^5 = C$$

$$5x^2y^3 = x^5 + C \rightarrow \text{General Solution}$$

when $x = 1, y = 1$

$$5(1)^2(1)^3 = (1)^5 + C$$

$$C = 4$$

Thus,

$$5x^2y^3 = x^5 + 4 \rightarrow \text{Particular Solution}$$

TYPE 7: SUBSTITUTION SUGGESTED BY THE EQUATION TYPE 8: BERNOULLI'S EQUATION

27. $(x^2 + 6y^2)dx - 4xydy = 0$
when $x = 1, y = 1$
Solve by three methods.

First Method

Let:
 $u = y^2$
 $du = 2ydy$

Substitute,

$$(x^2 + 6u)dx - 2xdu = 0$$

Divide by $-2xdu$,

$$-\frac{(x^2 + 6u)}{2x} + \frac{du}{dx} = 0$$

$$\frac{du}{dx} - \frac{(x^2 + 6u)}{2x} = 0$$

$$\frac{du}{dx} - \frac{x}{2} - \frac{3u}{x} = 0$$

$$\frac{du}{dx} - \frac{3}{x}u = \frac{x}{2}$$

Linear Equation!

$$P(x) = -3/x$$

$$Q(x) = x/2$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int \frac{3}{x}dx} = e^{-3\ln x} = x^{-3}$$

Then,

$$ue^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ux^{-3} = \int \frac{x}{2}x^{-3}dx + C$$

$$ux^{-3} = \int \frac{1}{x^2}dx + C$$

$$ux^{-3} = -\frac{1}{2x} + C$$

$$2ux^{-2} = -1 + Cx$$

Multiply by x^2

$$2u = -x^2 + Cx^3$$

$$2u = Cx^3 - x^2$$

$$2u = x^2(Cx - 1)$$

But y^2

$$2y^2 = x^2(Cx - 1) \rightarrow \text{General Solution}$$

when $x = 1, y = 1$

$$2(1)^2 = (1)^2(C(1) - 1)$$

$$C = 3$$

Thus,

$$2y^2 = x^2(3x - 1) \rightarrow \text{Particular Solution}$$

Second Method

$$(x^2 + 6y^2)dx - 4xydy = 0$$

Homogeneous, 2nd degree!

Let:

$$y = vx$$

$$dy = vdx + xdv$$

Substitute,

$$(x^2 + 6v^2x^2)dx - 4xvx(vdx + xdv) = 0$$

$$x^2(1 + 6v^2)dx - 4v^2x^2(vdx + xdv) = 0$$

$$(1 + 6v^2)dx - 4v(vdx + xdv) = 0$$

$$(1 + 6v^2 - 4v^2)dx - 4vxdv = 0$$

$$(1 + 2v^2)dx - 4vxdv = 0$$

$$\frac{dx}{x} - \frac{4vxdv}{(1 + 2v^2)} = 0$$

$$\int \frac{dx}{x} - \int \frac{4vxdv}{(1 + 2v^2)} = \int 0$$

$$\ln x - \ln(1 + 2v^2) = \ln C$$

$$\frac{x}{(1 + 2v^2)} = C$$

$$\frac{x}{(1 + 2 \frac{y^2}{x^2})} = C$$

$$\text{But } v = y/x$$

$$\frac{x}{(1 + 2 \frac{y^2}{x^2})} = C$$

TYPE 7: SUBSTITUTION SUGGESTED BY THE EQUATION

TYPE 8: BERNOULLI'S EQUATION

$$\frac{x}{(x^2+2y^2)} = C$$

$$\frac{x^3}{(x^2+2y^2)^2} = C$$

$$x^3 = C(x^2+2y^2)^2$$

$$\text{when } x=1, y=1$$

$$(1)^3 = C(1^2+2(1)^2)$$

$$C = 1/3$$

Thus,

$$x^3 = \frac{1}{3}(x^2+2y^2)^2$$

$$3x^3 = x^2+2y^2$$

$$2y^2 = 3x^3 - x^2$$

$$2y^2 = x^2(3x-1) \rightarrow \text{Particular Solution}$$

Third Method

The Determination of Integrating Factor:

$$(x^2+6y^2)dx - 4xydy = 0$$

$$M = x^2+6y^2;$$

$$\frac{\partial M}{\partial y} = 12y$$

$$N = -4xy;$$

$$\frac{\partial N}{\partial x} = -4y$$

For Integrating factor:

$$\frac{1}{N} \left[\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right] = \frac{12y+4y}{-4xy} = \frac{16y}{-4xy} = -\frac{4}{x}$$

$$e^{\int f(x)dx} = e^{-\int \frac{4}{x}dx} = e^{-4 \ln x} = x^{-4}$$

Multiply x^{-4} to original equation:

$$x^{-4}(x^2+6y^2)dx - x^{-4}4xydy = 0$$

$$(x^{-2}+6x^{-4}y^2)dx - 4x^{-3}ydy = 0$$

Exact Equation! (Test for exactness)

$$\sum \int D.I + \int N.D.I = \int 0$$

D.I:

$$x^{-2}dx$$

N.D.I:

$$6x^{-4}y^2dx, -4x^{-3}ydy$$

Choose: $6x^{-4}y^2dx$

$$\int x^{-2}dx + \int 6x^{-4}y^2dx = \int 0$$

$$-\frac{1}{x} - 2x^{-3}y^2 = C$$

$$-1 - 2x^{-2}y^2 = Cx$$

Multiply by x^2

$$-x^2 - 2y^2 = Cx^3$$

when $x=1, y=1$

$$-(1)^2 - 2(1)^2 = C(1)^3$$

$$C = -3$$

Thus,

$$-x^2 - 2y^2 = -3x^3$$

$$3x^3 = x^2+2y^2$$

$$2y^2 = 3x^3 - x^2$$

$$2y^2 = x^2(3x-1) \rightarrow \text{Particular Solution}$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

4.5 TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

Consider the equation:

$$(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0$$

$$\text{Case 1: if } \frac{a_1}{a_2} \neq \frac{b_1}{b_2} \rightarrow \text{Type 9}$$

$$\text{Case 2: if } \frac{a_1}{a_2} = \frac{b_1}{b_2} \rightarrow \text{Type 7}$$

$$\text{if } \frac{a_1}{a_2} \neq \frac{b_1}{b_2}, \text{ assume the value } h, k \text{ are constants}$$

$$x = u + h$$

$$dx = du$$

$$y = v + k$$

$$dy = dv$$

$$[a_1(u+h) + b_1(v+k) + c_1]du + [a_2(u+h) + b_2(v+k) + c_2]dv = 0$$

$$[(a_1u + b_1v) + (a_1h + b_1k + c_1)]du + [(a_2u + b_2v) + (a_2h + b_2k + c_2)]dv = 0$$

where:

$$a_1h + b_1k + c_1 = 0$$

$$a_2h + b_2k + c_2 = 0$$

$$(a_1u + b_1v) + (a_1h + b_1k + c_1)du + (a_2u + b_2v) + (a_2h + b_2k + c_2)dv = 0$$

$$[(a_1u + b_1v) + 0]du + [(a_2u + b_2v) + 0]dv = 0$$

$$(a_1u + b_1v)du + (a_2u + b_2v)dv = 0$$

Thus: reduces to homogeneous functions of degree 1

Steps:

1. Let M and N equal to zero
2. Equate to find the value of x and y or the point of intersection (h, k)
3. Let

$$x = u + h$$

$$dx = du$$

$$y = v + k$$

$$dy = dv$$

4. Substitute then continue solving

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

1. In exercises 1 through 18, solve the equations.

$$1. (y-2)dx - (x-y-1)dy = 0$$

Let:

$$M = y - 2 = 0 \quad ; \quad y = 2$$

$$N = -x + y + 1 = 0 \quad ; \quad x = 3$$

$(h, k) = (3, 2)$ point of intersection.

Let:

$$x = u + h$$

$$x = u + 3$$

$$dx = du$$

$$y = v + k$$

$$y = v + 2$$

$$dy = dv$$

Substitute,

$$[(v+2)-2]du - [(u+3)-(v+2)-1]dv = 0$$

$$vdu - (u-v)dv = 0$$

$$vdu - u dv + v dv = 0$$

Divide by v^2 ,

$$\frac{vdu - u dv}{v^2} + \frac{dv}{v} = 0$$

$$d\left(\frac{u}{v}\right) + \frac{dv}{v} = 0$$

$$\int d\left(\frac{u}{v}\right) + \int \frac{dv}{v} = \int 0$$

$$\frac{u}{v} + \ln v = -\ln C$$

$$u = -v \ln C - v \ln v$$

$$u = -v \ln Cv$$

But

$$u = x - 3$$

$$v = y - 2$$

$$x - 3 = -(y - 2) \ln[C(y - 2)]$$

$$x - 3 = (2 - y) \ln[C(y - 2)]$$

$$2. (x - 4y - 9)dx + (4x + y - 2)dy = 0$$

Let:

$$M = x - 4y - 9 = 0$$

$$N = 4x + y - 2 = 0$$

Multiply N by 4 then add M :

$$4N + M = 4(4x + y - 2) + x - 4y - 9$$

$$= 0$$

$$17x - 17 = 0$$

$$x = 1$$

Substituting $x = 1$ in M :

$$1 - 4y - 9 = 0$$

$$y = -2$$

$(h, k) = (1, -2)$ point of intersection.

Let:

$$x = u + h$$

$$x = u + 1$$

$$dx = du$$

$$y = v + k$$

$$y = v - 2$$

$$dy = dv$$

Substitute,

$$[(u+1)-4(v-2)-9]du + [4(u+1)+(v-2)-2]dv = 0$$

$$[(u+1)-4(v-2)-9]du + [4(u+1)+(v-2)-2]dv = 0$$

$$(u-4v)du + (4u+v)dv = 0$$

Δ Homogeneous, 1st degree

Let:

$$u = wv$$

$$du = w dv + v dw$$

Substitute,

$$(wv-4v)(w dv + v dw) + (4wv+v)dv = 0$$

$$v(w-4)(w dv + v dw) + v(4w+1)dv = 0$$

$$= 0$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

Divide by v .

$$(w-4)(w dv + v dw) + (4w+1)dv = 0$$

$$(w^2 dv + w v dw - 4w dv - 4v dw) + (4w+1)dv = 0$$

$$(w^2 dv + w v dw - 4w dv - 4v dw) + (4w+1)dv = 0$$

$$(4w+1+w^2-4w)dv + (wv-4v)dw = 0$$

$$(w^2+1)dv + v(w-4)dw = 0$$

$$\frac{dv}{v} + \frac{(w-4)dw}{(w^2+1)} = 0$$

$$\frac{dv}{v} + \frac{w}{(w^2+1)}dw - \frac{4}{(w^2+1)}dw = 0$$

$$\int \frac{dv}{v} + \int \frac{w}{(w^2+1)}dw - 4 \int \frac{dw}{(w^2+1)}$$

$$= \int 0$$

$$\ln v + \frac{1}{2} \ln(w^2+1) - 4 \operatorname{Arctan} w = \frac{C}{2}$$

$$2 \ln v + \ln(w^2+1) - 8 \operatorname{Arctan} w = C$$

$$\ln[v^2(w^2+1)] - 8 \operatorname{Arctan} w = C$$

But,

$$u = x - 1$$

$$v = y + 2$$

$$w = u/v$$

$$\ln \left| (y+2)^2 \left(\left(\frac{x-1}{y+2} \right)^2 + 1 \right) \right| - 8 \operatorname{Arctan} \frac{x-1}{y+2} = C$$

$$\ln[(x-1)^2 + (y+2)^2] - 8 \operatorname{Arctan} \frac{x-1}{y+2} = C$$

3.

$$(2x - y)dx + (4x + y - 6)dy = 0$$

Let:

$$M = 2x - y = 0$$

$$N = 4x + y - 6 = 0$$

Add M and N :

$$M + N = 2x - y + 4x + y - 6 = 0$$

$$6x - 6 = 0$$

$$x = 1$$

Substituting $x = 1$ in M :

$$2(1) - y = 0$$

$$y = 2$$

$(h, k) = (1, 2)$ point of intersection.

Let:

$$x = u + h$$

$$x = u + 1$$

$$dx = du$$

$$y = v + k$$

$$y = v + 2$$

$$dy = dv$$

Substitute,

$$[2(u+1)-(v+2)]du + [4(u+1)+(v+2)-6]dv = 0$$

$$(2u-v)du + (4u+v)dv = 0$$

Δ Homogeneous, 1st degree

Let:

$$u = wv$$

$$du = w dv + v dw$$

$$(2wv-v)(w dv + v dw) + (4wv+v)dv = 0$$

$$v(2w-1)(w dv + v dw) + v(4w+1)dv = 0$$

Divide by v .

$$(2w-1)(w dv + v dw) + (4w+1)dv = 0$$

$$(2w^2 dv + 2w v dw - w dv - v dw) + (4w+1)dv = 0$$

$$(4w+1+2w^2-w)dv + (2wv-v)dw = 0$$

$$(2w^2+3w+1)dv + v(2w-1)dw = 0$$

$$\frac{dv}{v} + \frac{(2w-1)dw}{(2w^2+3w+1)} = 0$$

$$\frac{dv}{v} + \frac{(2w-1)dw}{(2w^2+3w+1)} = 0$$

$$\int \frac{dv}{v} + \int \frac{(2w-1)dw}{(2w+1)(w+1)} = \int 0$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

Considering:

$$\int \frac{(2w-1)dw}{(2w+1)(w+1)}$$

Using Integration by Partial Fraction,

$$\frac{(2w-1)}{(2w+1)(w+1)} = \frac{A}{(2w+1)} + \frac{B}{(w+1)}$$

$$(2w-1) = A(w+1) + B(2w+1)$$

If $w = -1/2$; $A = -4$
 $w = -1$; $B = 3$

$$\int \frac{(2w-1)dw}{(2w+1)(w+1)}$$

$$= -4 \int \frac{dw}{(2w+1)} + 3 \int \frac{dw}{(w+1)}$$

$$= -2 \ln(2w+1) + 3 \ln(w+1)$$

Then,

$$\ln v - 2 \ln(2w+1) + 3 \ln(w+1) = \ln C$$

$$\frac{v(w+1)^3}{(2w+1)^2} = C$$

But,

$$u = x-1$$

$$v = y-2$$

$$w = u/v$$

$$(y-2) \left(\frac{(x-1)}{(y-2)} + 1 \right)^3$$

$$\left(2 \frac{(x-1)}{(y-2)} + 1 \right)^3 = C$$

$$(y-2) \left(\frac{(x-1)}{(y-2)} + \frac{(y-2)}{(y-2)} \right)^3$$

$$\left(\frac{2(x-1) + (y-2)}{(y-2)} \right)^3 = C$$

$$\frac{((x-1) + (y-2))^3}{(2(x-1) + (y-2))^3} = C$$

$$\frac{(x-1+y-2)^3}{(2(x-1)+y-2)^3} = C$$

$$\frac{(x+y-3)^3}{(2x+y-4)^3} = C$$

$$(x+y-3)^3 = C(2x+y-4)^3$$

4. $(x-4y-3)dx - (x-6y-5)dy = 0$

Let:

$$M = x-4y-3 = 0$$

$$N = -x+6y+5 = 0$$

Add M and N:

$$M+N = x-4y-3-x+6y+5 = 0$$

$$2y+2 = 0$$

$$y = -1$$

Substituting $y = -1$ in M:

$$x-4(-1)-3 = 0$$

$$x = -1$$

$(h, k) = (-1, -1)$ point of intersection.

Let:

$$x = u+h$$

$$x = u-1$$

$$dx = du$$

$$y = v+k$$

$$y = v-1$$

$$dy = dv$$

Substitute,

$$[(u-1)-4(v-1)-3]du$$

$$-[(u-1)-6(v-1)-5]dv = 0$$

$$(u-4v)du - (u-6v)dv = 0$$

∴ Homogeneous, 1st degree

Let:

$$u = wv$$

$$du = wdv + vdw$$

Substitute,

$$(wv-4v)(wdv+vdw) - (wv-6v)dv = 0$$

$$v(w-4)(wdv+vdw) - v(w-6)dv = 0$$

Divide by v,

$$(w-4)(wdv+vdw) - (w-6)dv = 0$$

$$(w^2 dv + wvdw - 4wdv - 4vdw) - (w-6)dv = 0$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

$$(-w+6+w^2-4w)dv + (wv-4v)dw = 0$$

$$(w^2-5w+6)dv + v(w-4)dw = 0$$

$$\frac{dv}{v} + \frac{(w-4)dw}{(w^2-5w+6)} = 0$$

$$\int \frac{dv}{v} + \int \frac{(w-4)dw}{(w-2)(w-3)} = \int 0$$

Considering:

$$\int \frac{(w-4)dw}{(w-2)(w-3)}$$

Using Integration by Partial Fraction,

$$\frac{(w-4)}{(w-2)(w-3)} = \frac{A}{(w-2)} + \frac{B}{(w-3)}$$

$$(w-4) = A(w-3) + B(w-2)$$

If $w = 2$; $A = 2$
 $w = 3$; $B = -1$

$$\int \frac{(w-4)dw}{(w-2)(w-3)}$$

$$= 2 \int \frac{dw}{(w-2)} - \int \frac{dw}{(w-3)}$$

$$= 2 \ln(w-2) - \ln(w-3)$$

Then,

$$\ln v + 2 \ln(w-2) - \ln(w-3) = \ln C$$

$$\frac{v(w-2)^2}{(w-3)} = C$$

But,

$$u = x+1$$

$$v = y+1$$

$$w = u/v$$

$$(y+1) \left(\frac{(x+1)}{(y+1)} - 2 \right)^2$$

$$\left(\frac{(x+1)}{(y+1)} - 3 \right)^2 = C$$

$$(y+1) \left(\frac{(x+1)-2(y+1)}{(y+1)} \right)^2 = C$$

$$\left(\frac{(x+1)-3(y+1)}{(y+1)} \right)^2 = C$$

$$\frac{((x+1)-2(y+1))^2}{((x+1)-3(y+1))} = C$$

$$\frac{(x-2y-1)^2}{(x-3y-2)} = C$$

$$(x-2y-1)^2 = C(x-3y-2)$$

5. $(2x+3y-5)dx + (3x-y-2)dy = 0$
 Solve by two methods.

◇ First Method

Let:

$$M = 2x+3y-5 = 0$$

$$N = 3x-y-2 = 0$$

Multiply N by 3 then add M:

$$3N + M = 3(3x-y-2) + 2x+3y-5$$

$$= 0$$

$$9x-3y-6+2x+3y-5 = 0$$

$$11x-11 = 0$$

$$x = 1$$

Substituting $x = 1$ in M:

$$2(1)+3y-5 = 0$$

$$y = 1$$

$(h, k) = (1, 1)$ point of intersection.

Let:

$$x = u+h$$

$$x = u+1$$

$$dx = du$$

$$y = v+k$$

$$y = v+1$$

$$dy = dv$$

Substitute,

$$[2(u+1)+3(v+1)-5]du$$

$$+[3(u+1)-(v+1)-2]dv = 0$$

$$(2u+3v)du + (3u-v)dv = 0$$

$$2u du + 3v du + 3u dv - v dv = 0$$

$$2u du + 3(v du + u dv) - v dv = 0$$

$$2u du + 3d(vu) - v dv = 0$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

$$2 \int u du + 3 \int d(vu) - \int v dv = \int 0$$

$$u^2 + 3vu - \frac{v^2}{2} = C$$

$$2u^2 + 6vu - v^2 = C$$

But,

$$u = x - 1$$

$$v = y - 1$$

$$2(x-1)^2 + 6(y-1)(x-1) - (y-1)^2 = C$$

$$2(x^2 - 2x + 1) + 6(xy - x - y + 1) - (y^2 - 2y + 1) = C$$

$$2x^2 - 4x + 2 + 6xy - 6x - 6y + 6 - y^2 + 2y - 1 = C$$

$$2x^2 - 10x - y^2 + 6xy - 4y + 7 = C$$

Second Method

Integrating Factor Found by Inspection:

$$(2x + 3y - 5)dx + (3x - y - 2)dy = 0$$

$$2xdx + 3ydy - 5dx + 3xdy - ydy - 2dy = 0$$

$$2xdx + 3(ydx + xdy) - 5dx - ydy - 2dy = 0$$

$$2xdx + 3d(xy) - 5dx - ydy - 2dy = 0$$

$$2 \int xdx + 3 \int d(xy) - 5 \int dx - \int ydy - 2 \int dy = \int 0$$

$$x^2 + 3xy - 5x - \frac{y^2}{2} - 2y = C$$

$$2x^2 + 6xy - 10x - y^2 - 4y = C$$

$$2x^2 - 10x - y^2 + 6xy - 4y = C$$

Third Method

$$(2x + 3y - 5)dx + (3x - y - 2)dy = 0$$

$$M = 2x + 3y - 5;$$

$$\frac{\partial M}{\partial y} = 3$$

$$N = 3x - y - 2;$$

$$\frac{\partial N}{\partial x} = 3$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

$$\int \int D.I + \int N.D.I = \int 0$$

D.I:

$$(2x - 5)dx, (-y - 2)dy$$

N.D.I:

$$3ydx, 3xdy$$

Choose: $3ydx$

$$\int (2x - 5)dx - \int (y + 2)dy + \int 3ydx = \int 0$$

$$x^2 - 5x - \frac{y^2}{2} - 2y + 3xy = C$$

$$2x^2 - 10x - y^2 - 4y + 6xy = C$$

$$2x^2 - 10x - y^2 + 6xy - 4y = C$$

6.

$$2dx + (2x - y + 3)dy = 0$$

Use a change of variable

Let:

$$x = \alpha_1 u + \alpha_2 v$$

$$dx = \alpha_1 du + \alpha_2 dv$$

$$y = u + v$$

$$dy = du + dv$$

From exercise #23 page

$$a_1 a^2 + (a_2 + b_1)a + b_2 = 0$$

$$a_1 = 0; b_1 = 0$$

$$a_2 = 2; b_2 = -1$$

$$(0)a^2 + (2 + 0)a + (-1) = 0$$

$$2a - 1 = 0$$

$$a_2 = 1/2$$

Then,

$$x = \alpha u + \alpha_2 v$$

$$x = (0)u + \frac{1}{2}v$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

$$x = \frac{1}{2}v$$

$$dx = \frac{1}{2}dv$$

Substitute,

$$2\left(\frac{1}{2}dv\right) + \left[2\left(\frac{1}{2}dv\right) - (u + v) + 3\right](du + dv) = 0$$

$$dv + (-u + 3)(du + dv) = 0$$

$$dv - udu - udu + 3du + 3dv = 0$$

$$(3 - u)du + (4 - u)dv = 0$$

$$\frac{(3 - u)}{(4 - u)} du + dv = 0$$

$$\left(1 - \frac{1}{4 - u}\right) du + dv = 0$$

$$\int \left(1 - \frac{1}{4 - u}\right) du + \int dv = \int 0$$

$$u + \ln(4 - u) + v = -C$$

But,

$$v = 2x$$

$$u = y - v$$

$$u = y - 2x$$

$$y - 2x + \ln(4 - y + 2x) + 2x = -C$$

$$y + C = -\ln(2x - y + 4)$$

7.

Solve the equation of exercise #6 by using the fact that the equation is linear in x

$$2dx + (2x - y + 3)dy = 0$$

Divide by $2dy$,

$$\frac{dx}{dy} + \frac{(2x - y + 3)}{2} = 0$$

$$\frac{dx}{dy} + x - \frac{y}{2} + \frac{3}{2} = 0$$

$$\frac{dx}{dy} + x = \frac{y}{2} - \frac{3}{2}$$

$$\frac{dx}{dy} + x = \frac{y - 3}{2}$$

$$\frac{dx}{dy} + x = \frac{y - 3}{2}$$

Linear Equation!

$$P(y) = 1$$

$$Q(y) = \frac{y - 3}{2}$$

For integrating factor:

$$e^{\int P(y)dy} = e^{\int 1 dy} = e^y$$

Then,

$$xe^{\int P(y)dy} = \int Q(y)e^{\int P(y)dy} dy + C$$

$$xe^y = \int \frac{y - 3}{2} e^y dy + C$$

$$xe^y = \frac{1}{2} \int ye^y dy - \frac{3}{2} \int e^y dy + C$$

Considering:

$$\int ye^y dy$$

Using Integration by Parts:

$$\text{Let } u = y \quad dv = e^y dy$$

$$du = dy$$

$$dv = e^y dy$$

$$v = e^y$$

then,

$$\int ye^y dy = ye^y - \int e^y dy = ye^y - e^y$$

thus,

$$xe^y = \frac{1}{2}(ye^y - e^y) - \frac{3}{2}e^y + C$$

$$2xe^y = (ye^y - e^y) - 3e^y + C$$

$$2xe^y = ye^y - e^y - 3e^y + C$$

$$2xe^y = ye^y - 4e^y + C$$

Divide by e^y ,

$$2x = y - 4 + Ce^{-y}$$

$$2x - y + 4 = Ce^{-y}$$

$$\ln(2x - y + 4) = \ln Ce^{-y}$$

$$\ln(2x - y + 4) = \ln C + \ln e^{-y}$$

$$\ln(2x - y + 4) = \ln C - y \ln e$$

$$\ln(2x - y + 4) = -C - y$$

$$-C = -C = \ln C$$

$$y + C = -\ln(2x - y + 4)$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

8. $(x - y + 2)dx + 3dy = 0$

Using change of variable:

Let

$$x = a_1u + \delta_1v$$

$$dx = a_1du + \delta_1dv$$

$$y = u + v$$

$$dy = du + dv$$

From exercise #23 page

$$a_1a^2 + (a_2 + b_1)a + b_2 = 0$$

$$a_1 = 1; \quad b_1 = -1$$

$$a_2 = 0; \quad b_2 = 0$$

$$(1)a^2 + (0 - 1)a + (0) = 0$$

$$a^2 - a = 0$$

$$a(a - 1) = 0$$

$$a_1 = 0, a_2 = 1$$

Then,

$$x = a_1u + a_2v$$

$$x = (0)u + (1)v$$

$$x = v$$

$$dx = dv$$

Substitute,

$$[v - (u + v) + 2]dv + 3(du + dv) = 0$$

$$(-u + 2)dv + 3du + 3dv = 0$$

$$(-u + 5)dv + 3du = 0$$

Divide by $(-5 + u)$,

$$dv + \frac{3du}{(-5 + u)} = 0$$

$$dv - \frac{3du}{(5 - u)} = 0$$

$$\int dv - \int \frac{3du}{(5 - u)} = \int 0$$

$$v - 3 \ln(5 - u) = -C$$

But,

$$v = x$$

$$u = y - v$$

$$u = y - x$$

$$x - 3 \ln(5 - y + x) = -C$$

$$x - 3 \ln(x - y + 5) = -C$$

$$x + C = 3 \ln(x - y + 5)$$

9. Solve exercise #8 by another method.

$$(x - y + 2)dx + 3dy = 0$$

Divide by $3dx$,

$$\frac{(x - y + 2)}{3} + \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} + \frac{x}{3} - \frac{y}{3} + \frac{2}{3} = 0$$

$$\frac{dy}{dx} - \frac{1}{3}y = -\frac{x}{3} - \frac{2}{3}$$

$$\frac{dy}{dx} - \frac{1}{3}y = \frac{(-x - 2)}{3}$$

$$\frac{dy}{dx} - \frac{1}{3}y = -\frac{(x + 2)}{3}$$

Linear Equation!

$$P(x) = -1/3$$

$$Q(x) = -\frac{(x + 2)}{3}$$

For integrating factor:

$$e^{\int P(x)dx} = e^{-\int \frac{1}{3}dx} = e^{-\frac{1}{3}x}$$

Then,

$$ye^{\int P(x)dx} = \int Q(x)e^{\int P(x)dx}dx + C$$

$$ye^{-\frac{1}{3}x} = \int -\frac{(x + 2)}{3}e^{-\frac{1}{3}x}dx + C$$

$$ye^{-\frac{1}{3}x} = -\frac{1}{3} \int xe^{-\frac{1}{3}x}dx - \frac{2}{3} \int e^{-\frac{1}{3}x}dx + C$$

Considering:

$$\int xe^{-\frac{1}{3}x}dx$$

Using Integration by Parts:

Let $u = x$ $dv = e^{-\frac{1}{3}x}dx$

$du = dx$ $v = -3e^{-\frac{1}{3}x}$

then,

$$\int xe^{-\frac{1}{3}x}dx = -3xe^{-\frac{1}{3}x} + 3 \int e^{-\frac{1}{3}x}dx$$

$$= -3xe^{-\frac{1}{3}x} - 9e^{-\frac{1}{3}x}$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

thus,

$$ye^{-\frac{1}{3}x} = -\frac{1}{3}(-3xe^{-\frac{1}{3}x} - 9e^{-\frac{1}{3}x}) + 2e^{-\frac{1}{3}x} + C$$

$$3ye^{-\frac{1}{3}x} = -(-3xe^{-\frac{1}{3}x} - 9e^{-\frac{1}{3}x}) + 6e^{-\frac{1}{3}x} + C$$

$$3ye^{-\frac{1}{3}x} = 3xe^{-\frac{1}{3}x} + 9e^{-\frac{1}{3}x} + 6e^{-\frac{1}{3}x} + C$$

$$3ye^{-\frac{1}{3}x} = 3xe^{-\frac{1}{3}x} + 15e^{-\frac{1}{3}x} + C$$

Divide by $-3e^{-\frac{1}{3}x}$,

$$-y = -x - 5 + Ce^{\frac{1}{3}x}$$

$$x - y + 5 = Ce^{\frac{1}{3}x}$$

$$\ln(x - y + 5) = \ln Ce^{\frac{1}{3}x}$$

$$\ln(x - y + 5) = \ln C + \ln e^{\frac{1}{3}x}$$

$$\ln(x - y + 5) = \ln C + \ln e^{\frac{1}{3}x}$$

$$\ln(x - y + 5) = \ln C + \frac{1}{3}x \ln e$$

$$3 \ln(x - y + 5) = \ln C + x$$

$$\rightarrow C = -C = \ln C$$

$$3 \ln(x - y + 5) = x + C$$

$$x + C = 3 \ln(x - y + 5)$$

10.

$$(x + y - 1)dx + (2x + 2y + 1)dy = 0$$

$$(x + y - 1)dx + [2(x + y) + 1]dy = 0$$

Let,

$$u = x + y$$

$$du = dx + dy$$

$$dy = du - dx$$

Substitute,

$$(u - 1)dx + (2u + 1)(du - dx) = 0$$

$$(u - 1)dx + (2u + 1)du - (2u + 1)dx = 0$$

$$(-u - 2)dx + (2u + 1)du = 0$$

$$-(u + 2)dx + (2u + 1)du = 0$$

Divide by $-(u + 2)$,

$$dx - \frac{(2u + 1)}{(u + 2)}du = 0$$

$$\int dx - \int \left(2 - \frac{3}{u + 2}\right)du = \int 0$$

$$x - 2u + \ln(u + 2) = C$$

But $u = x + y$

$$x - 2(x + y) + 3 \ln(x + y + 2) = C$$

$$-x - 2y + 3 \ln(x + y + 2) = C$$

$$x + 2y + C = 3 \ln(x + y + 2)$$

11. $(3x + 2y + 7)dx + (2x - y)dy = 0$
Solve by two methods.

◆ First Method

Let:

$$M = 3x + 2y + 7 = 0$$

$$N = 2x - y = 0$$

Multiply N by 2 then add M :

$$2N + M = 2(2x - y) + 3x + 2y + 7 = 0$$

$$4x - 2y + 3x + 2y + 7 = 0$$

$$7x + 7 = 0$$

$$x = -1$$

Substituting $x = -1$ in N :

$$2(-1) - y = 0$$

$$y = -2$$

$(h, k) = (-1, -2)$ point of intersection.

Let:

$$x = u + h$$

$$x = u - 1$$

$$dx = du$$

$$y = v + k$$

$$y = v - 2$$

$$dy = dv$$

Substitute,

$$[3(u-1) + 2(v-2) + 7]du + [2(u-1) - (v-2)]dv = 0$$

$$(3u + 2v)du + (2u - v)dv = 0$$

$$3u du + 2v du + 2u dv - v dv = 0$$

$$3u du + 2(v du + u dv) - v dv = 0$$

$$3u du + 2d(uv) - v dv = 0$$

$$3 \int u du + 2 \int d(uv) - \int v dv = \int 0$$

$$\frac{3u^2}{2} + 2uv - \frac{v^2}{2} = C$$

$$3u^2 + 4uv - v^2 = C$$

But

$$u = x + 1$$

$$v = y + 2$$

$$3(x+1)^2 + 4(y+2)(x+1) - (y+2)^2 = C$$

$$3(x^2 + 2x + 1) + 4(xy + 2x + y + 2) - (y^2 + 4y + 4) = C$$

$$3x^2 + 6x + 3 + 4xy + 8x + 4y + 8 - (y^2 + 4y + 4) = C$$

$$3x^2 - y^2 + 4xy + 14x + 7 = C$$

$$3x^2 - y^2 + 4xy + 14x = C$$

Second Method

Type 5: Integrating Factor found by Inspection

$$(3x + 2y + 7)dx + (2x - y)dy = 0$$

$$3x dx + 2y dx + 7dx + 2x dy - y dy = 0$$

$$3x dx + 7dx - y dy + 2(y dx + x dy) = 0$$

$$3x dx + 7dx - y dy + 2d(xy) = 0$$

$$3 \int x dx + 7 \int dx - \int y dy + 2 \int d(xy) = \int 0$$

$$\frac{3x^2}{2} + 7x - \frac{y^2}{2} + 2xy = C$$

$$3x^2 + 14x - y^2 + 4xy = C$$

$$3x^2 - y^2 + 4xy + 14x = C$$

Third Method

Type 3: Exact Equation

$$(3x + 2y + 7)dx + (2x - y)dy = 0$$

$$M = 3x + 2y + 7;$$

$$\frac{\partial M}{\partial y} = 2$$

$$N = 2x - y;$$

$$\frac{\partial N}{\partial x} = 2$$

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x} \therefore \text{EXACT DE!}$$

$$\int D.I + \int N.D.I = \int 0$$

D.I:

$$(3x + 7)dx, (-y)dy$$

N.D.I:

$$2y dx, 2x dy$$

Choose: $2y dx$

$$\int (3x + 7)dx - \int y dy + \int 2y dx = \int 0$$

$$\frac{3x^2}{2} + 7x - \frac{y^2}{2} + 2xy = C$$

$$3x^2 + 14x - y^2 + 4xy = C$$

$$3x^2 - y^2 + 4xy + 14x = C$$

12.

$$(x - 2)dx + 4(x + y - 1)dy = 0$$

Let:

$$M = x - 2 = 0$$

$$N = 4x + 4y - 4 = 0$$

From M:

$$x = 2$$

Substituting $x = 2$ in N:

$$4(2) + 4y - 4 = 0$$

$$y = -1$$

$(h, k) = (2, -1)$ point of intersection.
Let:

$$x = u + h$$

$$x = u + 2$$

$$dx = du$$

$$y = v + k$$

$$y = v - 1$$

$$dy = dv$$

Substitute,

$$[(u - 2) - 2]du$$

$$+ [4(u + 2) + 4(v - 1) - 4]dv = 0$$

$$u du + (4u + 4v)dv = 0$$

$$u du + 4(u + v)dv = 0$$

$\therefore$ Homogeneous, 1st degree

Let:

$$u = wv$$

$$du = w dv + v dw$$

Substitute,

$$wv(w dv + v dw) + 4(wv + v)dv = 0$$

$$wv(w dv + v dw) + 4v(w + 1)dv = 0$$

Divide by v ,

$$w(w dv + v dw) + 4(w + 1)dv = 0$$

$$(w^2 + 4w + 4)dv + wv dw = 0$$

$$\frac{dv}{v} + \frac{wdw}{(w^2 + 4w + 4)} = 0$$

$$\frac{dv}{v} + \frac{wdw}{(w + 2)(w + 2)} = 0$$

$$\frac{dv}{v} + \frac{(w + 2) - 2}{(w + 2)(w + 2)} dw = 0$$

$$\int \frac{dv}{v} + \int \frac{1}{(w + 2)} dw - \int \frac{2}{(w + 2)^2} dw = \int 0$$

$$\ln v + \ln(w + 2) + 2 \frac{1}{(w + 2)} = -\ln C$$

$$\ln v + \ln(w + 2) + \ln C = -\frac{2}{(w + 2)}$$

$$\ln[Cv(w + 2)] = -\frac{2}{(w + 2)}$$

But,

$$u = x - 2$$

$$v = y + 1$$

$$w = u/v$$

$$\ln[C(y + 1)\left(\frac{x - 2}{y + 1} + 2\right)] = -\frac{2}{\left(\frac{x - 2}{y + 1} + 2\right)}$$

$$\ln\left[C(y + 1)\left(\frac{x - 2 + 2(y + 1)}{y + 1}\right)\right]$$

$$= -\frac{2}{\left(\frac{x - 2 + 2(y + 1)}{y + 1}\right)}$$

$$\ln[C(x + 2y)] = -\frac{2}{\left(\frac{x + 2y}{y + 1}\right)}$$

$$2(y + 1) = -(x + 2y)\ln[C(x + 2y)]$$

13.

$$(x - 3y + 2)dx + 3(x + 3y - 4)dy = 0$$

Let:

$$M = x - 3y + 2 = 0$$

$$N = 3x + 9y - 12 = 0$$

Multiply M by 3 then subtract N:

$$3M - N = 3(x - 3y + 2) - (3x + 9y - 12) = 0$$

$$3x - 9y + 6 - 3x - 9y + 12 = 0$$

$$-18y + 18 = 0$$

$$y = 1$$

Substituting $y = 1$ in M:

$$x - 3(1) + 2 = 0$$

$$x = 1$$

$(h, k) = (1, 1)$ point of intersection.

Let:

$$x = u + h$$

$$x = u + 1$$

$$dx = du$$

$$y = v + k$$

$$y = v + 1$$

$$dy = dv$$

Substitute,

$$[(u + 1) - 3(v + 1) + 2]du + [3(u + 1) + 9(v + 1) - 12]dv = 0$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

$$(u - 3v)du + (3u + 9v)dv = 0$$

$$(u - 3v)du + 3(u + 3v)dv = 0$$

Homogeneous, 1st degree

Let:

$$u = wv$$

$$du = wdv + vdw$$

Substitute,

$$(wv - 3v)(wdv + vdw) + 3(wv + 3v)dv = 0$$

$$v(w - 3)(wdv + vdw) + 3v(w + 3)dv = 0$$

Divide by v ,

$$(w - 3)(wdv + vdw) + 3(w + 3)dv = 0$$

$$(w^2 dv + wvdw - 3wdv - 3vdw) + (3w + 9)dv = 0$$

$$(3w + 9 + w^2 - 3w)dv + (wv - 3v)dw = 0$$

$$(w^2 + 9)dv + v(w - 3)dw = 0$$

$$\frac{dv}{v} + \frac{(w - 3)dw}{(w^2 + 9)} = 0$$

$$\frac{dv}{v} + \frac{w}{(w^2 + 9)}dw - \frac{3}{(w^2 + 9)}dw = 0$$

$$\int \frac{dv}{v} + \int \frac{w}{(w^2 + 9)}dw - \int \frac{3}{(w^2 + 9)}dw = \int 0$$

$$\ln v + \frac{1}{2} \ln(w^2 + 9) - \text{Arctan} \frac{w}{3} = C$$

$$2 \ln v + \ln(w^2 + 9) - 2 \text{Arctan} \frac{w}{3} = C$$

$$\ln[v^2(w^2 + 9)] - 2 \text{Arctan} \frac{w}{3} = C$$

But,

$$u = x - 1$$

$$v = y - 1$$

$$w = u/v$$

$$\ln \left[(y - 1)^2 \left(\left(\frac{x - 1}{y - 1} \right)^2 + 9 \right) \right]$$

$$-2 \text{Arctan} \frac{(x - 1)}{3} = C$$

$$\ln \left[(y - 1)^2 \left(\frac{(x - 1)^2 + 9(y - 1)^2}{(y - 1)^2} \right) \right]$$

$$-2 \text{Arctan} \frac{x - 1}{3(y - 1)} = C$$

$$\ln[(x - 1)^2 + 9(y - 1)^2]$$

$$-2 \text{Arctan} \frac{x - 1}{3(y - 1)} = C$$

14.

$$(6x - 3y + 2)dx - (2x - y - 1)dy = 0$$

$$[3(2x - y) + 2]dx - (2x - y - 1)dy = 0$$

Let:

$$u = 2x - y$$

$$du = 2dx - dy$$

$$dy = 2dx - du$$

Substitute,

$$(3u + 2)dx - (u - 1)(2dx - du) = 0$$

$$(3u + 2)dx - 2(u - 1)dx + (u - 1)du = 0$$

$$(3u + 2 - 2u + 2)dx + (u - 1)du = 0$$

$$(u + 4)dx + (u - 1)du = 0$$

$$dx + \frac{(u - 1)du}{(u + 4)} = 0$$

$$dx + \frac{(u + 4) - 5}{(u + 4)} du = 0$$

$$dx + \left(1 - \frac{5}{u + 4} \right) du = 0$$

$$\int dx + \int \left(1 - \frac{5}{u + 4} \right) du = \int 0$$

$$x + u - 5 \ln(u + 4) = -C$$

$$\text{But } u = 2x - y$$

$$x + 2x - y - 5 \ln[2x - y + 4] = -C$$

$$3x - y + C = 5 \ln[2x - y + 4]$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

$$(9x - 4y + 4)dx - (2x - y + 1)dy = 0$$

Let:

$$M = 9x - 4y + 4 = 0$$

$$N = -2x + y - 1 = 0$$

Multiply N by 4 then add M :

$$4N + M = 4(-2x + y - 1) + (9x - 4y + 4) = 0$$

$$-8x + 4y - 4 + 9x - 4y + 4 = 0$$

$$x = 0$$

$$\text{Substituting } x = 0 \text{ in } M:$$

$$9(0) - 4y + 4 = 0$$

$$y = 1$$

$$(h, k) = (0, 1) \text{ point of intersection.}$$

Let:

$$x = u + h$$

$$x = u$$

$$dx = du$$

$$y = v + k$$

$$y = v + 1$$

$$dy = dv$$

Substitute,

$$[9(u) - 4(v + 1) + 4]du - [2(u) - (v + 1) + 1]dv = 0$$

$$(9u - 4v)du - (2u - v)dv = 0$$

Homogeneous, 1st degree

Let:

$$u = wv$$

$$du = wdv + vdw$$

Substitute,

$$(9wv - 4v)(wdv + vdw) - (2wv - v)dv = 0$$

$$v(9w - 4)(wdv + vdw) - v(2w - 1)dv = 0$$

Divide by v ,

$$(9w - 4)(wdv + vdw) - (2w - 1)dv = 0$$

$$(9w^2 dv + 9wvdw - 4wdv - 4vdw) - (2w - 1)dv = 0$$

$$(-2w + 1 + 9w^2 - 4w)dv + (9wv - 4v)dw = 0$$

$$(9w^2 - 6w + 1)dv + v(9w - 4)dw = 0$$

$$\frac{dv}{v} + \frac{(9w - 4)dw}{(9w^2 - 6w + 1)} = 0$$

$$\frac{dv}{v} + \frac{(9w - 3 - 1)dw}{(3w - 1)^2} = 0$$

$$\frac{dv}{v} + \frac{9w - 3}{(9w^2 - 6w + 1)}dw - \frac{1}{(3w - 1)^2}dw = 0$$

$$\int \frac{dv}{v} + \frac{1}{2} \int \frac{2(9w - 3)}{(9w^2 - 6w + 1)}dw - \int \frac{1}{(3w - 1)^2}dw = \int 0$$

$$\ln v + \frac{1}{2} \ln(9w^2 - 6w + 1) + \frac{1/3}{3w - 1} = -\ln C$$

$$\ln v + \frac{1}{2} \ln(3w - 1)^2 + \frac{1}{3w - 1} = -\ln C$$

$$\ln v + \ln(3w - 1) + \frac{1/3}{3w - 1} = -\ln C$$

$$\ln v + \ln(3w - 1) + \ln C = \frac{-1/3}{3w - 1}$$

$$\ln[Cv(3w - 1)] = \frac{-1/3}{3w - 1}$$

$$3 \ln[Cv(3w - 1)] = \frac{-1}{3w - 1}$$

But,

$$u = x$$

$$v = y - 1$$

$$w = u/v$$

$$3 \ln \left[C(y - 1) \left(3 \left(\frac{x}{y - 1} \right) - 1 \right) \right]$$

$$= \frac{-1}{3 \left(\frac{x}{y - 1} \right) - 1}$$

$$3 \ln \left[C(y - 1) \left(\frac{3x - y + 1}{(y - 1)} \right) \right]$$

$$= \frac{-1}{\frac{3x - y + 1}{(y - 1)}}$$

$$3(y - 3x - 1) \ln[C(3x - y + 1)] = (y - 1)$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

$$16. (x+3y-4)dx + (x+4y-5)dy = 0$$

Let

$$M = x+3y-4 = 0$$

$$N = x+4y-5 = 0$$

Subtract M and N :

$$M - N = (x+3y-4) - (x+4y-5) = 0$$

$$-y+1 = 0$$

$$y = 1$$

Substituting $y = 1$ in M :

$$x+3(1)-4 = 0$$

$$x = 1$$

$(h, k) = (1, 1)$ point of intersection.

Let

$$x = u + h$$

$$x = u + 1$$

$$dx = du$$

$$y = v + k$$

$$y = v + 1$$

$$dy = dv$$

Substitute,

$$[(u+1)+3(v+1)-4]du$$

$$+[(u+1)+4(v+1)-5]dv = 0$$

$$(u+3v)du + (u+4v)dv = 0$$

$\therefore$ Homogeneous, 1st degree

Let

$$u = wv$$

$$du = wdv + vdw$$

Substitute,

$$(wv+3v)(wdv+vdw) + (wv+4v)dv = 0$$

$$v(w+3)(wdv+vdw) + v(w+4)dv = 0$$

Divide by v ,

$$(w+3)(wdv+vdw) + (w+4)dv = 0$$

$$(w^2dv + wvdw + 3wdv + 3vdw) + (w+4)dv = 0$$

$$(w+4+w^2+3w)dv + (wv+3v)dw = 0$$

$$(w^2+4w+4)dv + v(w+3)dw = 0$$

$$\frac{dv}{v} + \frac{(w+3)dw}{(w^2+4w+4)} = 0$$

$$\frac{dv}{v} + \frac{(w+2+1)dw}{(w+2)^2} = 0$$

$$\frac{dv}{v} + \frac{1}{(w+2)}dw + \frac{1}{(w+2)^2}dw = 0$$

$$\int \frac{dv}{v} + \int \frac{1}{(w+2)}dw + \int \frac{1}{(w+2)^2}dw = 0$$

$$= \int 0$$

$$\ln v + \ln(w+2) - \frac{1}{(w+2)} = -\ln C$$

$$\ln v + \ln(w+2) + \ln C = \frac{1}{(w+2)}$$

$$\ln[Cv(w+2)] = \frac{1}{(w+2)}$$

But,

$$u = x-1$$

$$v = y-1$$

$$w = u/v$$

$$\ln \left[C(y-1) \left(\frac{x-1}{y-1} + 2 \right) \right] = \frac{1}{\left(\frac{x-1}{y-1} + 2 \right)}$$

$$\ln \left[C(y-1) \left(\frac{x-1+2(y-1)}{y-1} \right) \right]$$

$$= \frac{1}{\left(\frac{x-1+2(y-1)}{y-1} \right)}$$

$$\ln[C(x+2y-3)] = \frac{1}{\left(\frac{x+2y-3}{y-1} \right)}$$

$$(x+2y-3) \ln[C(x+2y-3)] = y-1$$

17.

$$(x+2y-1)dx - (2x+y-5)dy = 0$$

Let

$$M = x+2y-1 = 0$$

$$N = -2x-y+5 = 0$$

Multiply M by 2 then add N :

$$2M + N = 2(x+2y-1) + (-2x-y+5)$$

$$= 0$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

$$3y+3=0$$

$$y=-1$$

$$\text{Substituting } y = -1 \text{ in } M:$$

$$x+2(-1)-1=0$$

$$x=3$$

$(h, k) = (3, -1)$ point of intersection.

Let

$$x = u + h$$

$$x = u + 3$$

$$dx = du$$

$$y = v + k$$

$$y = v - 1$$

$$dy = dv$$

Substitute,

$$[(u+3)+2(v-1)-1]du$$

$$-[2(u+3)+(v-1)-5]dv = 0$$

$$(u+2v)du - (2u+v)dv = 0$$

$\therefore$ Homogeneous, 1st degree

Let

$$u = wv$$

$$du = wdv + vdw$$

Substitute,

$$(wv+2v)(wdv+vdw) - (2wv+v)dv = 0$$

$$v(w+2)(wdv+vdw) - v(2w+1)dv = 0$$

Divide by v ,

$$(w+2)(wdv+vdw) - (2w+1)dv = 0$$

$$(w^2dv + wvdw + 2wdv + 2vdw) - (2w+1)dv = 0$$

$$(-2w-1+w^2+2w)dv + (wv+2v)dw = 0$$

$$(w^2-1)dv + v(w+2)dw = 0$$

$$\frac{dv}{v} + \frac{(w+2)dw}{(w^2-1)} = 0$$

$$\int \frac{dv}{v} + \int \frac{(w+2)dw}{(w+1)(w-1)} = \int 0$$

Considering:

$$\int \frac{(w+2)dw}{(w+1)(w-1)}$$

Using Integration by Partial Fraction,

$$\frac{(w+2)}{(w+1)(w-1)} = \frac{A}{(w+1)} + \frac{B}{(w-1)}$$

$$(w+2) = A(w-1) + B(w+1)$$

$$\text{If } w = -1 ; A = -1/2$$

$$w = 1 ; B = 3/2$$

$$\int \frac{(w+2)}{(w+1)(w-1)}$$

$$= -\frac{1}{2} \int \frac{dw}{(w+1)} + \frac{3}{2} \int \frac{dw}{(w-1)}$$

$$= -\frac{1}{2} \ln(w+1) + \frac{3}{2} \ln(w-1)$$

Then,

$$\ln v - \frac{1}{2} \ln(w+1) + \frac{3}{2} \ln(w-1) = \ln C$$

$$2 \ln v - \ln(w+1) + 3 \ln(w-1) = \ln C$$

$$\frac{v^2(w-1)^3}{(w+1)} = C$$

But,

$$u = x-3$$

$$v = y+1$$

$$w = u/v$$

$$\frac{(y+1)^2 \left(\frac{x-3}{y+1} - 1 \right)^3}{\left(\frac{x-3}{y+1} + 1 \right)} = C$$

$$\frac{(y+1)^2 \left(\frac{x-3-y-1}{y+1} \right)^3}{\left(\frac{x-3+y+1}{y+1} \right)} = C$$

$$\frac{(x-y-4)^3}{(x+y-2)} = C$$

$$(x-y-4)^3 = C(x+y-2)$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

18. $(x-1)dx - (3x-2y-5)dy = 0$

Let:

$$M = x - 1 = 0$$

$$N = -3x + 2y + 5 = 0$$

From M;

$$x = 1$$

Substituting $x = 1$ in N:

$$-3(1) + 2y + 5 = 0$$

$$y = -1$$

$(h, k) = (1, -1)$ point of intersection.

Let:

$$x = u + h$$

$$x = u + 1$$

$$dx = du$$

$$y = v + k$$

$$y = v - 1$$

$$dy = dv$$

Substitute,

$$[(u+1)-1]du$$

$$- [3(u+1) - 2(v-1) - 5]dv = 0$$

$$udu - (3u - 2v)dv = 0$$

∴ Homogeneous, 1st degree

Let:

$$u = wv$$

$$du = wdv + vdw$$

Substitute,

$$wv(wdv + vdw) - (3wv - 2v)dv = 0$$

$$wv(wdv + vdw) - v(3w - 2)dv = 0$$

Divide by v ,

$$w(wdv + vdw) - (3w - 2)dv = 0$$

$$(-3w + 2 + w^2)dv + wv dw = 0$$

$$(w^2 - 3w + 2)dv + wv dw = 0$$

$$\frac{dv}{v} + \frac{wdw}{(w^2 - 3w + 2)} = 0$$

$$\int \frac{dv}{v} + \int \frac{wdw}{(w-1)(w-2)} = \int 0$$

Considering:

$$\int \frac{wdw}{(w-1)(w-2)}$$

Using Integration by Partial Fraction,

$$\frac{w}{(w-1)(w-2)} = \frac{A}{(w-1)} + \frac{B}{(w-2)}$$

$$\text{If } \begin{matrix} w=1 & ; & A=-1 \\ w=2 & ; & B=2 \end{matrix}$$

$$\begin{aligned} \int \frac{wdw}{(w-1)(w-2)} &= -\int \frac{dw}{(w-1)} + 2\int \frac{dw}{(w-2)} \\ &= -\ln(w-1) + 2\ln(w-2) \end{aligned}$$

Then,

$$\ln v - \ln(w-1) + 2\ln(w-2) = \ln C$$

But,

$$u = x - 1$$

$$v = y + 1$$

$$w = u/v$$

$$\frac{(y+1)\left(\frac{x-1}{y+1}-2\right)^2}{\left(\frac{x-1}{y+1}-1\right)} = C$$

$$\frac{(y+1)\left(\frac{x-1-2y-2}{y+1}\right)^2}{\left(\frac{x-1-y-1}{y+1}\right)} = C$$

$$\frac{(x-2y-3)^2}{(x-y-2)} = C$$

$$(x-2y-3)^2 = C(x-y-2)$$

Multiply by -1,

$$(2y-x+3)^2 = C(y-x+2)$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

In exercises 19 through 22, obtain the particular solution indicated.

19. $(2x-3y+4)dx + 3(x-1)dy = 0$
when $x = 3, y = 2$

Let:

$$M = 2x - 3y + 4 = 0$$

$$N = 3x - 3 = 0$$

From N;

$$3x - 3 = 0$$

$$x = 1$$

Substituting $x = 1$ in M:

$$2(1) - 3y + 4 = 0$$

$$y = 2$$

$(h, k) = (1, 2)$ point of intersection.

Let:

$$x = u + h$$

$$x = u + 1$$

$$dx = du$$

$$y = v + k$$

$$y = v + 2$$

$$dy = dv$$

Substitute,

$$[2(u+1) - 3(v+2) + 4]du + [3(u+1) - 3]dv = 0$$

$$(2u - 3v)du + 3udv = 0$$

Divide by $3udu$,

$$\frac{(2u-3v)}{3u} + \frac{dv}{du} = 0$$

$$\frac{dv}{du} + \frac{2}{3} - \frac{v}{u} = 0$$

$$\frac{dv}{du} + \frac{2}{3} - \frac{v}{u} = 0$$

$$\frac{dv}{du} - \frac{v}{u} = -\frac{2}{3}$$

$$\frac{dv}{du} - \frac{v}{u} = -\frac{2}{3}$$

Linear Equation!

$$P(u) = -1/u$$

$$Q(u) = -2/3$$

For integrating factor:

$$e^{\int P(u)du} = e^{-\int \frac{1}{u}du} = e^{-\ln u} = u^{-1}$$

Then,

$$ve^{\int P(u)du} = \int Q(u)e^{\int P(u)du} du + C$$

$$vu^{-1} = -\int \frac{2}{3}u^{-1}du + C$$

$$vu^{-1} = -\frac{2}{3}\ln u + C$$

$$3vu^{-1} = -2\ln u + C$$

But

$$u = x - 1$$

$$v = y - 2$$

$$3\left(\frac{y-2}{x-1}\right) = -2\ln(x-1) + C$$

$$3(y-2) = -2(x-1)\ln(x-1)$$

$$+ C(x-1)$$

→ General Solution

when $x = 3, y = 2$

$$3(2-2) = -2(3-1)\ln(3-1)$$

$$+ C(3-1)$$

$$C = 2\ln 2$$

Thus,

$$3(y-2) = -2(x-1)\ln(x-1) + 2(x-1)\ln 2$$

$$3(y-2) = -2(x-1)[\ln(x-1) - \ln 2]$$

$$3(y-2) = -2(x-1)\ln \frac{x-1}{2}$$

20.

Solve the equation of exercises #19 but with the condition that when $x = -1, y = 2$

$$3(y-2) = -2(x-1)\ln(x-1) + C(x-1) \rightarrow \text{General Solution}$$

when $x = -1, y = 2$

$$3(2-2) = -2(-1-1)\ln(-1-1) + C(-1-1)$$

$$0 = -2(-2)\ln(-2) + C(-2)$$

$$C = 2\ln(-2)$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

Thus,

$$3(y-2) = -2(x-1)\ln(x-1) + 2(x-1)\ln(-2)$$

$$3(y-2) = -2(x-1)[\ln(x-1) - \ln(-2)]$$

$$3(y-2) = -2(x-1)\ln\frac{x-1}{-2}$$

$$3(y-2) = -2(x-1)\ln\frac{1-x}{2}$$

21. $(x+y-4)dx - (3x-y-4)dy = 0$
when $x=4, y=1$

Let

$$M = x+y-4 = 0$$

$$N = -3x+y-4 = 0$$

Subtract M and N:

$$M - N = x+y-4 - (-3x+y-4) = 0$$

$$4x - 8 = 0$$

$$x = 2$$

Substituting $x=2$ in M:

$$2+y-4=0$$

$$y=2$$

$(h, k) = (2, 2)$ point of intersection.

Let

$$x = u + h$$

$$x = u + 2$$

$$dx = du$$

$$y = v + k$$

$$y = v + 2$$

$$dy = dv$$

Substitute,

$$[(u+2) + (v+2) - 4]du$$

$$- [3(u+2) - (v+2) - 4]dv = 0$$

$$(u+v)du - (3u-v)dv = 0$$

$\therefore$ Homogeneous, 1st degree

Let

$$u = wv$$

$$du = wdv + vdw$$

Substitute,

$$(wv + v)(wdv + vdw) - (3wv - v)dv = 0$$

$$v(w+1)(wdv + vdw) - v(3w-1)dv = 0$$

Divide by v ,

$$(w+1)(wdv + vdw) - (3w-1)dv = 0$$

$$(w^2dv + wvdw + wdv + vdw) - (3w-1)dv = 0$$

$$(-3w+1+w^2+w)dv + (wv+v)dw = 0$$

$$(w^2-2w+1)dv + v(w+1)dw = 0$$

$$\frac{dv}{v} + \frac{(w+1)dw}{(w^2-2w+1)} = 0$$

$$\frac{dv}{v} + \frac{(w+1)dw}{(w-1)^2} = 0$$

$$\frac{dv}{v} + \frac{(w-1+2)dw}{(w-1)^2} = 0$$

$$\frac{dv}{v} + \frac{1}{(w-1)}dw + \frac{2}{(w-1)^2}dw = 0$$

$$\int \frac{dv}{v} + \int \frac{1}{(w-1)}dw + \int \frac{2}{(w-1)^2}dw$$

$$= \int 0$$

$$\ln v + \ln(w-1) - \frac{2}{(w-1)} = C$$

$$\ln[v(w-1)] - \frac{2}{(w-1)} = C$$

$$\ln[v(w-1)] - \frac{2}{(w-1)} = C$$

$$\ln[v(w-1)] - \frac{2}{(w-1)} = C$$

$$\ln[v(w-1)] - \frac{2}{(w-1)} = C$$

$$\ln[v(w-1)] - \frac{2}{(w-1)} = C$$

$$\ln[(y-2)\left(\frac{x-2}{y-2}-1\right)]$$

$$-\frac{2}{\left(\frac{x-2}{y-2}-1\right)} = C$$

TYPE 9: COEFFICIENTS LINEAR IN THE TWO VARIABLES

Thus,

$$(x-y)\ln(x-y) - 2y + 4 = C(x-y)$$

$$(x-y)\ln(x-y) - 2y + 4 = C(x-y)$$

$$2(x-y)\ln(x-y) - 4y + 8 = 2(x-y)\ln(-4) + 5(x-y)$$

$$2(x-y)\ln(x-y) - 2(x-y)\ln(-4) = 5x - 5y + 4y - 8$$

$$2(x-y)[\ln(x-y) - \ln(-4)] = 5x - 5y + 4y - 8$$

$$2(x-y)\ln\frac{x-y}{-4} = 5x - y - 8$$

$$2(x-y)\ln\frac{y-x}{4} = 5x - y - 8$$

$$2(y-x)\ln\frac{y-x}{4} = y - 5x + 8$$

23. Change of Variables

$$x = a_1u + a_2v$$

$$y = u + v$$

will transform the equation

$$(a_1x + b_1y + c_1)dx + (a_2x + b_2y + c_2)dy = 0 \rightarrow (A)$$

into an equation in which the variables u and v are separable, if a_1 and a_2 are roots of the equation

$$a_1a^2 + (a_2 + b_1)a + b_2 = 0 \rightarrow (B)$$

and if $a_1 \neq a_2$

Note that this method of solution of (A) is not practical for us unless the roots of equation (B) are real and distinct.

$$\ln[(y-2)\left(\frac{x-2-y+2}{y-2}\right)] - \frac{2}{\left(\frac{x-2-y+2}{y-2}\right)} = C$$

$$\ln(x-y) - \frac{2(y-2)}{(x-y)} = C$$

$$(x-y)\ln(x-y) - 2y + 4 = C(x-y)$$

$$\rightarrow \text{General Solution}$$

when $x=4, y=1$

$$(4-1)\ln(4-1) - 2(1) + 4 = C(4-1)$$

$$C = \ln 3 + \frac{2}{3}$$

Thus,

$$(x-y)\ln(x-y) - 2y + 4 = \left(\ln 3 + \frac{2}{3}\right)(x-y)$$

$$(x-y)\ln(x-y) - 2y + 4 = (x-y)\ln 3 + \frac{2}{3}(x-y)$$

$$3(x-y)\ln(x-y) - 6y + 12 = 3(x-y)\ln 3 + 2(x-y)$$

$$3(x-y)\ln(x-y) - 3(x-y)\ln 3 = 2x - 2y + 6y - 12$$

$$3(x-y)[\ln(x-y) - \ln 3] = 2x - 2y + 6y - 12$$

$$3(x-y)\ln\frac{x-y}{3} = 2x + 4y - 12$$

$$3(x-y)\ln\frac{x-y}{3} = 2(x+2y-6)$$

$$(x-y)\ln(x-y) - 2y + 4 = C(x-y)$$

$$\rightarrow \text{General Solution}$$

$$(3-7)\ln(3-7) - 2(7) + 4 = C(3-7)$$

$$(-4)\ln(-4) - 10 = C(-4)$$

$$C = \ln(-4) + \frac{5}{2}$$

CHAPTER 5: LINEAR DIFFERENTIAL EQUATION OF ORDER n

5.1 STANDARD FORM OF A LINEAR DIFFERENTIAL EQUATION OF ORDER n

The general linear differential equation of order n is an equation that can be written

$$b_0(x) \frac{d^n y}{dx^n} + b_1(x) \frac{d^{n-1} y}{dx^{n-1}} + \cdots + b_{n-1}(x) \frac{dy}{dx} + b_n(x)y = R(x)$$

The functions $R(x)$ and $b_i(x)$; $i = 0, 1, \dots, n$, are too independent of the variable y . If $R(x)$ is identically zero, equation is said to be linear and homogeneous; if $R(x)$ is not identically zero, equation is called linear and nonhomogeneous.

5.2 LINEAR INDEPENDENCE OF A SET OF FUNCTIONS

Given the functions $f_1, f_2, \dots, f_n$

If $c_1, c_2, c_3, \dots, c_n$, not all zero, exist such that

$$c_1 f_1(x) + c_2 f_2(x) + \cdots + c_n f_n(x) = 0 \rightarrow (1)$$

Identically in some interval $a \leq x \leq b$, then the functions, $f_1, f_2, \dots, f_n$ are said to be linearly dependent. If no such relation exists, the functions are said to be linearly independent. That is, the functions $f_1, f_2, \dots, f_n$ are linearly independent when equation (1) implies that

$$c_1 = c_2 = \cdots = c_n = 0$$

5.3 THE DIFFERENTIAL OPERATORS (D - OPERATORS)

Let D denote differentiation with respect to x , D^2 differentiation twice with respect to x , and so on; that is, for positive integral k ,

$$D^k y = \frac{d^k y}{dx^k}$$

LINEAR DIFFERENTIAL EQUATION OF ORDER n

The expression

$$A = a_0 D^n + a_1 D^{n-1} + \cdots + a_{n-1} D + a_n$$

is called a differential operator of order n . It may be defined as that operator which, when applied to any function y , yields the result

$$Ay = a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \cdots + a_{n-1} \frac{dy}{dx} + a_n y$$

The coefficients $\{a_0, a_1, \dots, a_n\}$ in the operator A may be a function of x .

Two operators A and B are said to be equal if and only if, the same result is produced when each act upon the function y . That is, $A = B$ if, and only if, $Ay = By$ for all functions y possessing the derivatives necessary for the operations involved.

The product AB of two operators A and B is defined as that operator which produces the same result as obtained by using operator B followed by the operator A . Thus $AB y = A(By)$. The product of two differential operators always exists and is a differential operator. For operators with constant coefficients, but not usually for those with variable coefficients, it is true that $AB = BA$.

CHAPTER 6: HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION WITH CONSTANTS COEFFICIENTS

6.1 GENERAL SOLUTION OF A HOMOGENEOUS EQUATION

Let $\{y_1, y_2, \dots, y_n\}$ be a linearly independent set of solutions of the homogeneous linear equation

$$b_0(x)y^{(n)} + b_1(x)y^{(n-1)} + \dots + b_{n-1}(x)y' + b_n(x)y = 0 \rightarrow (1)$$

for x on the interval $a < x < b$. Suppose further that $b_0(x), \dots, b_n(x)$ are continuous functions and that $b_0(x) \neq 0$ on $a < x < b$.

Then the general solution of equation (1) is

$$y = C_1 y_1 + C_2 y_2 + \dots + C_n y_n$$

where $C_1, C_2, \dots, C_n$ are arbitrary constants.

6.2 THE AUXILLIARY EQUATIONS

Any linear homogeneous differential equation with constant coefficients,

$$a_0 \frac{d^n y}{dx^n} + a_1 \frac{d^{n-1} y}{dx^{n-1}} + \dots + a_{n-1} \frac{dy}{dx} + a_n y = 0$$

May be written in the form

$$f(D)y = 0$$

where $f(D)$ is a linear differential operator.

If m is any root of the algebraic equation $f(m) = 0$, then

$$f(D)e^{mx} = 0$$

which means simply that $y = e^{mx}$ is a solution of the equation. The equation

$$f(m) = 0$$

is called the *auxiliary equation*.

HOMOGENEOUS LINEAR DIFFERENTIAL EQUATION WITH CONSTANTS COEFFICIENTS

Four Cases in the solution of $f(D)y = 0$

- CASE 1 : Real roots, distinct
- CASE 2 : Real roots, repeated
- CASE 3 : Imaginary roots, distinct
- CASE 4 : Imaginary roots, repeated

D-Operator: To facilitate writing equation

Let:

$$D = \frac{d}{dx}; \quad D^2 = \frac{d^2}{dx^2}; \quad D^k = \frac{d^k}{dx^k}$$

CASE 1: AUXILLIARY EQUATION HAS REAL ROOTS, DISTINCT

General Solution:

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_{n-1} e^{m_{n-1} x}$$

CASE 2: AUXILLIARY EQUATION HAS REAL ROOTS, REPEATED

General Solution:

$$y = (C_1 + C_2 x + C_3 x^2 + \dots + C_{n-2} x^{n-1} + C_{n-1} x^n) e^{mx}$$

CASE 3: AUXILLIARY EQUATION HAS IMAGINARY ROOTS, DISTINCT

General Solution:

$$y = e^{ax} (C_1 \cos bx + C_2 \sin bx)$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}; \quad b^2 < 4ac; \quad m = a \pm bi$$

CASE 4: AUXILLIARY EQUATION HAS IMAGINARY ROOTS, REPEATED

General Solution:

$$y = e^{ax} [(C_1 + C_2 x + \dots + C_{n-1} x^{n-1}) \cos bx + (C_3 + C_4 x + \dots + C_{n-1} x^{n-1}) \sin bx]$$

$$m = a \pm bi$$

CASE 1: AUXILLIARY EQUATION - REAL ROOTS, DISTINCT

CASE 1: AUXILLIARY EQUATION HAS REAL ROOTS, DISTINCT

General Solution:

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_{n-1} e^{m_{n-1} x}$$

1. In exercises 1 through 22, find the general solution. When the operator D is used, it is implied that the independent variable is x

1. $(D^2 - D - 2)y = 0$

$$m^2 - m - 2 = 0$$

$$(m+1)(m-2) = 0$$

$$m_1 = -1$$

$$m_2 = 2$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{-x} + C_2 e^{2x}$$

2. $(D^2 + 3D)y = 0$

$$m^2 + 3m = 0$$

$$m(m+3) = 0$$

$$m_1 = 0$$

$$m_2 = -3$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{0x} + C_2 e^{-3x}$$

$$y = C_1 + C_2 e^{-3x}$$

3. $(D^2 - D - 6)y = 0$

$$m^2 - m - 6 = 0$$

$$(m+2)(m-3) = 0$$

$$m_1 = -2$$

$$m_2 = 3$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{-2x} + C_2 e^{3x}$$

4. $(D^2 + 5D + 6)y = 0$

$$m^2 + 5m + 6 = 0$$

$$(m+2)(m+3) = 0$$

$$m_1 = -2$$

$$m_2 = -3$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{-2x} + C_2 e^{-3x}$$

5. $(D^3 + 2D^2 - 15D)y = 0$

$$m^3 + 2m^2 - 15m = 0$$

$$m(m-3)(m+5) = 0$$

$$m_1 = 0$$

$$m_2 = 3$$

$$m_3 = -5$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{0x} + C_2 e^{3x} + C_3 e^{-5x}$$

$$y = C_1 + C_2 e^{3x} + C_3 e^{-5x}$$

6. $(D^3 + 2D^2 - 8D)y = 0$

$$m^3 + 2m^2 - 8m = 0$$

$$m(m-2)(m+4) = 0$$

$$m_1 = 0$$

$$m_2 = 2$$

$$m_3 = -4$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{0x} + C_2 e^{2x} + C_3 e^{-4x}$$

$$y = C_1 + C_2 e^{2x} + C_3 e^{-4x}$$

7. $(D^3 - D^2 - 4D + 4)y = 0$

$$m^3 - 2m^2 - 4m + 4 = 0$$

$$(m+2)(m-1)(m-2) = 0$$

$$m_1 = -2$$

$$m_2 = 1$$

$$m_3 = 2$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{-2x} + C_2 e^x + C_3 e^{2x}$$

CASE 1: AUXILLIARY EQUATION - REAL ROOTS, DISTINCT

$$(D^3 - 3D^2 - D + 3)y = 0$$

$$m^3 - 3m^2 - m + 3 = 0$$

$$(m-3)(m-1)(m+1) = 0$$

$$m_1 = 3$$

$$m_2 = 1$$

$$m_3 = -1$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{3x} + C_2 e^x + C_3 e^{-x}$$

$$(4D^3 - 13D + 6)y = 0$$

$$4m^3 - 13m + 6 = 0$$

$$(2m-1)(2m-3)(m+2) = 0$$

$$m_1 = 1/2$$

$$m_2 = 3/2$$

$$m_3 = -2$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{x/2} + C_2 e^{3x/2} + C_3 e^{-2x}$$

10. $(4D^3 - 49D - 60)y = 0$

$$4m^3 - 49m - 60 = 0$$

$$(m-4)(2m+5)(2m+3) = 0$$

$$m_1 = 4$$

$$m_2 = -5/2$$

$$m_3 = -3/2$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{4x} + C_2 e^{-5x/2} + C_3 e^{-3x/2}$$

11. $\frac{d^3 x}{dt^3} - 2 \frac{d^2 x}{dt^2} - 3 \frac{dx}{dt} = 0$

$$m^3 - 2m^2 - 3m = 0$$

$$(m)(m+1)(m-3) = 0$$

$$m_1 = 0$$

$$m_2 = -1$$

$$m_3 = 3$$

$$x = C_1 e^{m_1 t} + C_2 e^{m_2 t} + C_3 e^{m_3 t}$$

$$x = C_1 e^{0t} + C_2 e^{-t} + C_3 e^{3t}$$

$$x = C_1 + C_2 e^{-t} + C_3 e^{3t}$$

12. $\frac{d^3 x}{dt^3} - 7 \frac{dx}{dt} + 6x = 0$

$$m^3 - 7m + 6 = 0$$

$$(m-1)(m-2)(m+3) = 0$$

$$m_1 = 1$$

$$m_2 = 2$$

$$m_3 = -3$$

$$x = C_1 e^{m_1 t} + C_2 e^{m_2 t} + C_3 e^{m_3 t}$$

$$x = C_1 e^t + C_2 e^{2t} + C_3 e^{-3t}$$

13. $(10D^3 + D^2 - 7D + 2)y = 0$

$$10m^3 + m^2 - 7m + 2 = 0$$

$$(m+1)(2m-1)(5m-2) = 0$$

$$m_1 = -1$$

$$m_2 = 1/2$$

$$m_3 = 2/5$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{-x} + C_2 e^{x/2} + C_3 e^{2x/5}$$

14. $(4D^3 - 13D - 6)y = 0$

$$4m^3 - 13m - 6 = 0$$

$$(m-2)(2m+3)(2m+1) = 0$$

$$m_1 = 2$$

$$m_2 = -3/2$$

$$m_3 = -1/2$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{2x} + C_2 e^{-3x/2} + C_3 e^{-x/2}$$

15. $(D^3 - 5D - 2)y = 0$

$$m^3 - 5m - 2 = 0$$

$$(m+2)(m-(1+\sqrt{2}))(m-(1-\sqrt{2})) = 0$$

$$m_1 = -2$$

$$m_2 = 1 + \sqrt{2}$$

$$m_3 = 1 - \sqrt{2}$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{-2x} + C_2 e^{(1+\sqrt{2})x} + C_3 e^{(1-\sqrt{2})x}$$

CASE I: AUXILIARY EQUATION - REAL ROOTS, DISTINCT

$$16. \quad (D^3 - 3D^2 - 3D + 1)y = 0$$

$$m^3 - 3m^2 - 3m + 1 = 0$$

$$(m+1)(m-(2+\sqrt{3}))(m-(2-\sqrt{3})) = 0$$

$$m_1 = -1$$

$$m_2 = 2 + \sqrt{3}$$

$$m_3 = 2 - \sqrt{3}$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{-x} + C_2 e^{(2+\sqrt{3})x} + C_3 e^{(2-\sqrt{3})x}$$

$$17. \quad (4D^4 - 15D^2 + 5D + 6)y = 0$$

$$4m^4 - 15m^2 + 5m + 6 = 0$$

$$(m+2)(2m+1)(2m-3)(m-1) = 0$$

$$m_1 = -2$$

$$m_2 = -1/2$$

$$m_3 = 3/2$$

$$m_4 = 1$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + C_4 e^{m_4 x}$$

$$y = C_1 e^{-2x} + C_2 e^{-x/2} + C_3 e^{3x/2} + C_4 e^x$$

$$18. \quad (D^4 - 2D^3 - 13D^2 + 38D - 24)y = 0$$

$$m^4 - 2m^3 - 13m^2 + 38m - 24 = 0$$

$$(m-1)(m-2)(m-3)(m+4) = 0$$

$$m_1 = 1$$

$$m_2 = 2$$

$$m_3 = 3$$

$$m_4 = -4$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + C_4 e^{m_4 x}$$

$$y = C_1 e^x + C_2 e^{2x} + C_3 e^{3x} + C_4 e^{-4x}$$

$$19. \quad (6D^4 + 23D^3 + 28D^2 + 13D + 2)y = 0$$

$$6m^4 + 23m^3 + 28m^2 + 13m + 2 = 0$$

$$(3m+1)(2m+1)(m+1)(m+2) = 0$$

$$m_1 = -1/3$$

$$m_2 = -1/2$$

$$m_3 = -1$$

$$m_4 = -2$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + C_4 e^{m_4 x}$$

$$y = C_1 e^{-x/3} + C_2 e^{-x/2} + C_3 e^{-x} + C_4 e^{-2x}$$

$$20. \quad (4D^4 - 45D^2 - 70D - 24)y = 0$$

$$4m^4 - 45m^2 - 70m - 24 = 0$$

$$(m-4)(m+2)(2m+1)(2m+3) = 0$$

$$m_1 = 4$$

$$m_2 = -2$$

$$m_3 = -1/2$$

$$m_4 = -3/2$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x} + C_4 e^{m_4 x}$$

$$y = C_1 e^{4x} + C_2 e^{-2x} + C_3 e^{-x/2} + C_4 e^{-3x/2}$$

$$21. \quad (D^2 - 4aD + 3a^2)y = 0; a \text{ real} \neq 0$$

$$m^2 - 4am + 3a^2 = 0$$

$$(m-a)(m-3a) = 0$$

$$m_1 = a$$

$$m_2 = 3a$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{ax} + C_2 e^{3ax}$$

$$22. \quad (D^2 - (a+b)D + ab)y = 0;$$

$$a \text{ and } b \text{ real and unequal} \neq 0$$

$$m^2 - (a+b)m + ab = 0$$

$$(m-a)(m-b) = 0$$

$$m_1 = a$$

$$m_2 = b$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{ax} + C_2 e^{bx}$$

II. In exercises 23 and 24, find the particular solution indicated.

$$23. \quad (D^2 - 2D - 3)y = 0$$

$$\text{when } x = 0, y = 0, y' = -4$$

$$m^2 - 2m - 3 = 0$$

$$(m+1)(m-3) = 0$$

$$m_1 = -1$$

$$m_2 = 3$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

CASE I: AUXILIARY EQUATION - REAL ROOTS, DISTINCT

$$y = C_1 e^{-x} + C_2 e^{3x}$$

$$y' = -C_1 e^{-x} + 3C_2 e^{3x}$$

$$\text{when } x = 0, y = 0,$$

$$0 = C_1 e^{-0} + C_2 e^{3(0)}$$

$$0 = C_1 + C_2$$

$$C_1 = -C_2$$

$$\text{when } x = 0, y' = -4$$

$$-4 = -C_1 e^{-0} + 3C_2 e^{3(0)}$$

$$-4 = -C_1 + 3C_2$$

$$-4 = -(-C_2) + 3C_2$$

$$-4 = C_2 + 3C_2$$

$$C_2 = -1$$

$$C_1 = 1$$

Thus,

$$y = e^{-x} - e^{3x} \rightarrow \text{Particular Solution}$$

$$24. \quad (D^2 - D - 6)y = 0$$

$$\text{when } x = 0, y = 0, \text{ and}$$

$$\text{when } x = 1, y = e^3$$

$$m^2 - m - 6 = 0$$

$$(m+2)(m-3) = 0$$

$$m_1 = -2$$

$$m_2 = 3$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{-2x} + C_2 e^{3x}$$

$$\text{when } x = 0, y = 0,$$

$$0 = C_1 e^{-2(0)} + C_2 e^{3(0)}$$

$$0 = C_1 + C_2$$

$$C_1 = -C_2$$

$$\text{when } x = 1, y = e^3$$

$$e^3 = C_1 e^{-2(1)} + C_2 e^{3(1)}$$

$$e^3 = C_1 e^{-2} + C_2 e^3$$

$$\text{But } C_1 = -C_2$$

$$e^3 = -C_2 e^{-2} + C_2 e^3$$

$$e^3 = C_2 (e^3 - e^{-2})$$

$$C_2 = \frac{e^3}{e^3 - e^{-2}}$$

$$= \frac{e^3}{e^3 - \frac{1}{e^2}}$$

$$= \frac{e^3}{\frac{e^5 - 1}{e^2}}$$

$$= \frac{e^5}{e^5 - 1}$$

$$= \frac{e^5 (e^5 - 1)^{-1}}{1 - e^{-5}}$$

$$C_1 = -C_2 = \frac{-1}{1 - e^{-5}}$$

Thus,

$$y = \frac{-1}{1 - e^{-5}} e^{-2x} + \frac{1}{1 - e^{-5}} e^{3x}$$

$$y = \frac{e^{3x} - e^{-2x}}{1 - e^{-5}} \rightarrow \text{Particular Solution}$$

III. In exercises 25 and 29, find for $x = 1$ the value for the particular solution required

$$25. \quad (D^2 - 2D - 3)y = 0$$

$$\text{when } x = 0, y = 4, y' = 0$$

$$m^2 - 2m - 3 = 0$$

$$(m+1)(m-3) = 0$$

$$m_1 = -1$$

$$m_2 = 3$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{-x} + C_2 e^{3x}$$

$$y' = -C_1 e^{-x} + 3C_2 e^{3x}$$

CASE 1: AUXILLIARY EQUATION - REAL ROOTS, DISTINCT

$$\rightarrow \text{when } x = 0, y = 4,$$

$$4 = C_1 e^{-0} + C_2 e^{3(0)}$$

$$4 = C_1 + C_2$$

$$C_1 = 4 - C_2$$

$$\rightarrow \text{when } x = 0, y' = 0$$

$$0 = -C_1 e^{-0} + 3C_2 e^{3(0)}$$

$$0 = -C_1 + 3C_2$$

$$\text{But } C_1 = 4 - C_2$$

$$0 = -(4 - C_2) + 3C_2$$

$$0 = -4 + C_2 + 3C_2$$

$$C_2 = 1$$

Then,

$$C_1 = 4 - 1 = 3$$

Thus,

$$y = 3e^{-x} + e^{3x}$$

When $x = 1$;

$$y = 3e^{-1} + e^{3(1)} = 21.2$$

26.

$$(D^3 - 4D)y = 0$$

$$\text{when } x = 0, y = 0, y' = 0, y'' = 2$$

$$m^3 - 4m = 0$$

$$(m)(m-2)(m+2) = 0$$

$$m_1 = 0$$

$$m_2 = 2$$

$$m_3 = -2$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 + C_2 e^{2x} + C_3 e^{-2x}$$

$$y' = 2C_2 e^{2x} - 2C_3 e^{-2x}$$

$$y'' = 4C_2 e^{2x} + 4C_3 e^{-2x}$$

$$\rightarrow \text{when } x = 0, y = 0,$$

$$0 = C_1 + C_2 e^{2(0)} + C_3 e^{-2(0)}$$

$$0 = C_1 + C_2 + C_3 \rightarrow (\text{Eq. 1})$$

$$\rightarrow \text{when } x = 0, y' = 0,$$

$$0 = 2C_2 e^{2(0)} - 2C_3 e^{-2(0)}$$

$$0 = 2C_2 - 2C_3$$

$$0 = C_2 - C_3 \rightarrow (\text{Eq. 2})$$

$$\rightarrow \text{when } x = 0, y'' = 2,$$

$$2 = 4C_2 e^{2(0)} + 4C_3 e^{-2(0)}$$

$$2 = 4C_2 + 4C_3$$

$$1 = 2C_2 + 2C_3 \rightarrow (\text{Eq. 3})$$

Solving 3 equations with 3 unknowns:

$$C_1 = -1/2$$

$$C_2 = 1/4$$

$$C_3 = 1/4$$

Then,

$$y = -\frac{1}{2} + \frac{1}{4}e^{2x} + \frac{1}{4}e^{-2x}$$

$$y = \frac{1}{4}(e^{2x} + e^{-2x} - 2)$$

$$y = \sinh^2 x$$

When $x = 1$;

$$y = \sinh^2 1$$

27.

$$(D^2 - D - 6)y = 0$$

$$\text{when } x = 0, y = 3, y' = -1$$

$$m^2 - m - 6 = 0$$

$$(m+2)(m-3) = 0$$

$$m_1 = -2$$

$$m_2 = 3$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{-2x} + C_2 e^{3x}$$

$$y' = -2C_1 e^{-2x} + 3C_2 e^{3x}$$

$$\rightarrow \text{when } x = 0, y = 3,$$

$$3 = C_1 e^{-2(0)} + C_2 e^{3(0)}$$

$$3 = C_1 + C_2$$

$$C_1 = 3 - C_2$$

$$\rightarrow \text{when } x = 0, y' = -1$$

$$-1 = -2C_1 e^{-2(0)} + 3C_2 e^{3(0)}$$

$$-1 = -2C_1 + 3C_2$$

CASE 1: AUXILLIARY EQUATION - REAL ROOTS, DISTINCT

$$\text{But } C_1 = 3 - C_2$$

$$-1 = -2(3 - C_2) + 3C_2$$

$$-1 = -6 + 2C_2 + 3C_2$$

$$C_2 = 1$$

Then,

$$C_1 = 3 - 1 = 2$$

Thus,

$$y = 2e^{-2x} + e^{3x}$$

When $x = 1$;

$$y = 2e^{-2(1)} + e^{3(1)} = 20.4$$

28.

$$(D^2 + 3D - 10)y = 0$$

$$\text{when } x = 0, y = 0, \text{ and when } x = 2, y = 1$$

$$m^2 + 3m - 10 = 0$$

$$(m-2)(m+5) = 0$$

$$m_1 = 2$$

$$m_2 = -5$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^{2x} + C_2 e^{-5x}$$

$$\rightarrow \text{when } x = 0, y = 0,$$

$$0 = C_1 e^{2(0)} + C_2 e^{-5(0)}$$

$$0 = C_1 + C_2$$

$$C_1 = -C_2$$

$$\rightarrow \text{when } x = 2, y = 1$$

$$1 = C_1 e^{2(2)} + C_2 e^{-5(2)}$$

$$1 = C_1 e^4 + C_2 e^{-10}$$

$$\text{But } C_1 = -C_2$$

$$1 = -C_2 e^4 + C_2 e^{-10}$$

$$C_2 = -0.01832$$

Then,

$$C_1 = 0.01832$$

Thus,

$$y = 0.01832e^{2x} - 0.01832e^{-5x}$$

When $x = 1$;

$$y = 0.01832e^{2(1)} - 0.01832e^{-5(1)} = 0.135$$

29.

$$(D^3 - 2D^2 - 5D + 6)y = 0$$

$$\text{when } x = 0, y = 1, y' = -7, y'' = -1$$

$$m^3 - 2m^2 - 5m + 6 = 0$$

$$(m-1)(m+2)(m-3) = 0$$

$$m_1 = 1$$

$$m_2 = -2$$

$$m_3 = 3$$

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^x + C_2 e^{-2x} + C_3 e^{3x}$$

$$y' = C_1 e^x - 2C_2 e^{-2x} + 3C_3 e^{3x}$$

$$y'' = C_1 e^x + 4C_2 e^{-2x} + 9C_3 e^{3x}$$

$$\rightarrow \text{when } x = 0, y = 1,$$

$$1 = C_1 e^{(0)} + C_2 e^{-2(0)} + C_3 e^{3(0)}$$

$$1 = C_1 + C_2 + C_3 \rightarrow (\text{Eq. 1})$$

$$\rightarrow \text{when } x = 0, y' = -7,$$

$$-7 = C_1 e^{(0)} - 2C_2 e^{-2(0)} + 3C_3 e^{3(0)}$$

$$-7 = C_1 - 2C_2 + 3C_3 \rightarrow (\text{Eq. 2})$$

$$\rightarrow \text{when } x = 0, y'' = -1,$$

$$-1 = C_1 e^{(0)} + 4C_2 e^{-2(0)} + 9C_3 e^{3(0)}$$

$$-1 = C_1 + 4C_2 + 9C_3 \rightarrow (\text{Eq. 3})$$

Solving 3 equations with 3 unknowns:

$$C_1 = 0$$

$$C_2 = 2$$

$$C_3 = -1$$

Thus,

$$y = 2e^{-2x} - e^{3x}$$

When $x = 1$;

$$y = 2e^{-2(1)} - e^{3(1)} = -19.8$$

CASE 2: AUXILLIARY EQUATION - REAL ROOTS, REPEATED

CASE 2: AUXILLIARY EQUATION HAS REAL ROOTS, REPEATED

General Solution:

$$y = (C_1 + C_2x + C_3x^2 + \dots + C_{n-2}x^{n-2} + C_{n-1}x^{n-1})e^{mx}$$

I. In exercises 1 through 20, find the general solution.

1. $(4D^2 - 4D + 1)y = 0$

$$4m^2 - 4m + 1 = 0$$

$$(2m - 1)(2m - 1) = 0$$

$$m_1 = 1/2$$

$$m_2 = 1/2$$

$$y = C_1e^{m_1x} + C_2xe^{m_1x}$$

$$y = C_1e^{x/2} + C_2xe^{x/2}$$

$$y = (C_1 + C_2x)e^{x/2}$$

2. $(D^2 + 6D + 9)y = 0$

$$m^2 + 6m + 9 = 0$$

$$(m + 3)(m + 3) = 0$$

$$m_1 = -3$$

$$m_2 = -3$$

$$y = C_1e^{m_1x} + C_2xe^{m_1x}$$

$$y = C_1e^{-3x} + C_2xe^{-3x}$$

$$y = (C_1 + C_2x)e^{-3x}$$

3. $(D^3 - 4D^2 + 4D)y = 0$

$$m^3 - 4m^2 + 4m = 0$$

$$(m)(m - 2)(m - 2) = 0$$

$$m_1 = 0$$

$$m_2 = 2$$

$$m_3 = 2$$

$$y = (C_1e^{m_1x}) + (C_2e^{m_2x} + C_3xe^{m_2x})$$

$$y = C_1e^{(0)x} + C_2e^{2x} + C_3xe^{2x}$$

$$y = C_1 + (C_2 + C_3x)e^{2x}$$

4. $(9D^3 + 6D^2 + D)y = 0$

$$m^3 - 4m^2 + 4m = 0$$

$$(m)(3m + 1)(3m + 1) = 0$$

$$m_1 = 0$$

$$m_2 = -1/3$$

$$m_3 = -1/3$$

$$y = (C_1e^{m_1x}) + (C_2e^{m_2x} + C_3xe^{m_2x})$$

$$y = C_1e^{(0)x} + C_2e^{-x/3} + C_3xe^{-x/3}$$

$$y = C_1 + (C_2 + C_3x)e^{-x/3}$$

5. $(2D^4 - 3D^3 - 2D^2)y = 0$

$$2m^4 - 3m^3 - 2m^2 = 0$$

$$(m)(m)(2m + 1)(m - 2) = 0$$

$$m_1 = 0$$

$$m_2 = 0$$

$$m_3 = 2$$

$$m_4 = -1/2$$

$$y = (C_1e^{m_1x} + C_2xe^{m_1x}) + (C_3e^{m_3x} + C_4e^{m_4x})$$

$$y = (C_1e^{0x} + C_2xe^{0x}) + (C_3e^{2x} + C_4e^{-x/2})$$

$$y = C_1 + C_2x + C_3e^{2x} + C_4e^{-x/2}$$

6. $(2D^4 - 5D^3 - 3D^2)y = 0$

$$2m^4 - 5m^3 - 3m^2 = 0$$

$$(m)(m)(2m + 1)(m - 3) = 0$$

$$m_1 = 0$$

$$m_2 = 0$$

$$m_3 = 3$$

$$m_4 = -1/2$$

$$y = (C_1e^{m_1x} + C_2xe^{m_1x}) + (C_3e^{m_3x} + C_4e^{m_4x})$$

$$y = (C_1e^{0x} + C_2xe^{0x}) + (C_3e^{3x} + C_4e^{-x/2})$$

$$y = C_1 + C_2x + C_3e^{3x} + C_4e^{-x/2}$$

CASE 2: AUXILLIARY EQUATION - REAL ROOTS, REPEATED

7. $(D^3 + 3D^2 - 4)y = 0$

$$m^3 + 3m^2 - 4 = 0$$

$$(m - 1)(m + 2)(m + 2) = 0$$

$$m_1 = 1$$

$$m_2 = -2$$

$$m_3 = -2$$

$$y = (C_1e^{m_1x}) + (C_2e^{m_2x} + C_3xe^{m_2x})$$

$$y = C_1e^x + C_2e^{-2x} + C_3xe^{-2x}$$

$$y = C_1e^x + (C_2 + C_3x)e^{-2x}$$

8. $(4D^3 - 27D + 27)y = 0$

$$4m^3 - 27m + 27 = 0$$

$$(m + 3)(2m - 3)(2m - 3) = 0$$

$$m_1 = -3$$

$$m_2 = 3/2$$

$$m_3 = 3/2$$

$$y = (C_1e^{m_1x}) + (C_2e^{m_2x} + C_3xe^{m_2x})$$

$$y = C_1e^{-3x} + C_2e^{3x/2} + C_3xe^{3x/2}$$

$$y = C_1e^{-3x} + (C_2 + C_3x)e^{3x/2}$$

9. $(D^3 + 3D^2 + 3D + 1)y = 0$

$$m^3 + 3m^2 + 3D + 1 = 0$$

$$(m + 1)(m + 1)(m + 1) = 0$$

$$m_1 = -1$$

$$m_2 = -1$$

$$m_3 = -1$$

$$y = (C_1e^{m_1x} + C_2xe^{m_1x} + C_3x^2e^{m_1x})$$

$$y = (C_1e^{-x} + C_2xe^{-x} + C_3x^2e^{-x})$$

$$y = (C_1 + C_2x + C_3x^2)e^{-x}$$

10. $(D^3 + 6D^2 + 12D + 8)y = 0$

$$m^3 + 6m^2 + 12D + 8 = 0$$

$$(m + 2)(m + 2)(m + 2) = 0$$

$$m_1 = -2$$

$$m_2 = -2$$

$$m_3 = -2$$

$$y = (C_1e^{m_1x} + C_2xe^{m_1x} + C_3x^2e^{m_1x})$$

$$y = (C_1e^{-2x} + C_2xe^{-2x} + C_3x^2e^{-2x})$$

$$y = (C_1 + C_2x + C_3x^2)e^{-2x}$$

11. $(D^5 - D^3)y = 0$

$$m^5 - m^3 = 0$$

$$(m)(m)(m)(m - 1)(m + 1) = 0$$

$$m_1 = 0$$

$$m_2 = 0$$

$$m_3 = 0$$

$$m_4 = 1$$

$$m_5 = -1$$

$$y = (C_1e^{m_1x} + C_2xe^{m_1x} + C_3x^2e^{m_1x}) + (C_4e^{m_4x} + C_5e^{m_5x})$$

$$y = (C_1e^{0x} + C_2xe^{0x} + C_3x^2e^{0x}) + (C_4e^x + C_5e^{-x})$$

$$y = C_1 + C_2x + C_3x^2 + C_4e^x + C_5e^{-x}$$

12. $(D^5 - 16D^3)y = 0$

$$m^5 - m^3 = 0$$

$$(m)(m)(m)(m - 4)(m + 4) = 0$$

$$m_1 = 0$$

$$m_2 = 0$$

$$m_3 = 0$$

$$m_4 = 4$$

$$m_5 = -4$$

$$y = (C_1e^{m_1x} + C_2xe^{m_1x} + C_3x^2e^{m_1x}) + (C_4e^{m_4x} + C_5e^{m_5x})$$

$$y = (C_1e^{0x} + C_2xe^{0x} + C_3x^2e^{0x}) + (C_4e^{4x} + C_5e^{-4x})$$

$$y = C_1 + C_2x + C_3x^2 + C_4e^{4x} + C_5e^{-4x}$$

13. $(4D^4 + 4D^3 - 3D^2 - 2D + 1)y = 0$

$$4m^4 + 4m^3 - 3m^2 - 2m + 1 = 0$$

$$(2m - 1)(2m - 1)(m + 1)(m + 1) = 0$$

$$m_1 = 1/2$$

$$m_2 = 1/2$$

$$m_3 = -1$$

$$m_4 = -1$$

CASE 2: AUXILIARY EQUATION - REAL ROOTS, REPEATED

$$y = (C_1 e^{m_1 x} + C_2 x e^{m_1 x} + C_3 e^{m_2 x} + C_4 x^2 e^{m_2 x})$$

$$y = (C_1 e^{x/2} + C_2 x e^{x/2} + C_3 e^{-x} + C_4 x e^{-x})$$

$$y = (C_1 + C_2 x) e^{x/2} + (C_3 + C_4 x) e^{-x}$$

$$14. \quad (4D^4 - 4D^3 - 23D^2 + 12D + 36)y = 0$$

$$4m^4 - 4m^3 - 23m^2 + 12m + 36 = 0$$

$$(2m + 3)(2m + 3)(m - 2)(m - 2) = 0$$

$$m_1 = -3/2$$

$$m_2 = -3/2$$

$$m_3 = 2$$

$$m_4 = 2$$

$$y = (C_1 e^{m_1 x} + C_2 x e^{m_1 x} + C_3 e^{m_2 x} + C_4 x e^{m_2 x})$$

$$y = (C_1 e^{-3x/2} + C_2 x e^{-3x/2} + C_3 e^{2x} + C_4 x e^{2x})$$

$$y = (C_1 + C_2 x) e^{-3x/2} + (C_3 + C_4 x) e^{2x}$$

$$15. \quad (D^4 + 3D^3 - 6D^2 - 28D - 24)y = 0$$

$$m^4 + 3m^3 - 6m^2 - 28m - 24 = 0$$

$$(m - 3)(m + 2)(m + 2)(m + 2) = 0$$

$$m_1 = 3$$

$$m_2 = -2$$

$$m_3 = -2$$

$$m_4 = -2$$

$$y = C_1 e^{m_1 x} + (C_2 e^{m_2 x} + C_3 x e^{m_2 x} + C_4 x^2 e^{m_2 x})$$

$$y = C_1 e^{3x} + (C_2 e^{-2x} + C_3 x e^{-2x} + C_4 x^2 e^{-2x})$$

$$y = C_1 e^{3x} + (C_2 + C_3 x + C_4 x^2) e^{-2x}$$

$$16. \quad (27D^4 - 18D^3 + 8D - 1)y = 0$$

$$27m^4 - 18m^3 + 8m - 1 = 0$$

$$(m + 1)(3m - 1)(3m - 1)(3m - 1) = 0$$

$$m_1 = -1$$

$$m_2 = 1/3$$

$$m_3 = 1/3$$

$$m_4 = 1/3$$

$$y = C_1 e^{m_1 x} + (C_2 e^{m_2 x} + C_3 x e^{m_2 x} + C_4 x^2 e^{m_2 x})$$

$$y = C_1 e^{-x} + (C_2 e^{x/3} + C_3 x e^{x/3} + C_4 x^2 e^{x/3})$$

$$y = C_1 e^{-x} + (C_2 + C_3 x + C_4 x^2) e^{x/3}$$

$$17. \quad (4D^5 - 23D^3 - 33D^2 - 17D - 3)y = 0$$

$$4m^5 - 23m^3 - 33m^2 - 17m - 3 = 0$$

$$(m - 3)(m + 1)(m + 1)(2m + 1)^2 = 0$$

$$m_1 = 3$$

$$m_2 = -1$$

$$m_3 = -1$$

$$m_4 = -1/2$$

$$m_5 = -1/2$$

$$y = C_1 e^{m_1 x} + (C_2 e^{m_2 x} + C_3 x e^{m_2 x} + C_4 e^{m_4 x} + C_5 x e^{m_4 x})$$

$$y = C_1 e^{3x} + (C_2 e^{-x} + C_3 x e^{-x} + C_4 e^{-x/2} + C_5 x e^{-x/2})$$

$$y = C_1 e^{3x} + (C_2 + C_3 x) e^{-x} + (C_4 + C_5 x) e^{-x/2}$$

$$18. \quad (4D^5 - 15D^3 - 5D^2 + 15D + 9)y = 0$$

$$4m^5 - 15m^3 - 5m^2 + 15m + 9 = 0$$

$$(m + 1)(m + 1)(m + 1)(2m - 3)^2 = 0$$

$$m_1 = -1$$

$$m_2 = -1$$

$$m_3 = -1$$

$$m_4 = 3/2$$

$$m_5 = 3/2$$

$$y = (C_1 e^{m_1 x} + C_2 x e^{m_1 x} + C_3 x^2 e^{m_1 x} + C_4 e^{m_4 x} + C_5 x e^{m_4 x})$$

$$y = (C_1 e^{-x} + C_2 x e^{-x} + C_3 x^2 e^{-x} + C_4 e^{3x/2} + C_5 x e^{3x/2})$$

$$y = (C_1 + C_2 x + C_3 x^2) e^{-x} + (C_4 + C_5 x) e^{3x/2}$$

CASE 2: AUXILIARY EQUATION - REAL ROOTS, REPEATED

$$19. \quad (D^4 - 5D^2 - 6D - 2)y = 0$$

$$m^4 - 5m^2 - 6m - 2 = 0$$

$$(m + 1)^2 (m - (1 + \sqrt{3})) (m - (1 - \sqrt{3})) = 0$$

$$m_1 = -1$$

$$m_2 = -1$$

$$m_3 = 1 + \sqrt{3}$$

$$m_4 = 1 - \sqrt{3}$$

$$y = (C_1 e^{m_1 x} + C_2 x e^{m_1 x} + C_3 e^{m_3 x} + C_4 e^{m_4 x})$$

$$y = (C_1 e^{-x} + C_2 x e^{-x} + C_3 e^{(1+\sqrt{3})x} + C_4 e^{(1-\sqrt{3})x})$$

$$y = (C_1 + C_2 x) e^{-x} + C_3 e^{(1+\sqrt{3})x} + C_4 e^{(1-\sqrt{3})x}$$

$$20. \quad (D^5 - 5D^4 + 7D^3 + D^2 - 8D + 4)y = 0$$

$$m^5 - 5m^4 + 7m^3 + m^2 - 8m + 4 = 0$$

$$(m - 1)(m - 1)(m + 1)(m - 2)(m - 2) = 0$$

$$m_1 = 1$$

$$m_2 = 1$$

$$m_3 = -1$$

$$m_4 = 2$$

$$m_5 = 2$$

$$y = (C_1 e^{m_1 x} + C_2 x e^{m_1 x} + C_3 e^{m_3 x} + C_4 e^{m_4 x} + C_5 x e^{m_4 x})$$

$$y = (C_1 e^x + C_2 x e^x + C_3 e^{-x} + C_4 e^{2x} + C_5 x e^{2x})$$

$$y = (C_1 + C_2 x) e^x + C_3 e^{-x} + (C_4 + C_5 x) e^{2x}$$

II. In exercises 21 through 26, find the particular solution indicated.

$$21. \quad (D^2 + 4D + 4)y = 0$$

when $x = 0, y = 1, y' = -1$

$$m^2 + 4m + 4 = 0$$

$$(m + 2)(m + 2) = 0$$

$$m_1 = -2$$

$$m_2 = -2$$

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x}$$

$$y = C_1 e^{-2x} + C_2 x e^{-2x}$$

$$y' = -2C_1 e^{-2x} + C_2 (-2x e^{-2x} + e^{-2x})$$

$$y' = -2C_1 e^{-2x} - 2C_2 x e^{-2x} + C_2 e^{-2x}$$

→ When $x = 0, y = 1,$

$$1 = C_1 e^{-2(0)} + C_2 (0) e^{-2(0)}$$

$$C_1 = 1$$

→ When $x = 0, y' = -1,$

$$-1 = -2C_1 e^{-2(0)} - 2C_2 (0) e^{-2(0)} + C_2 e^{-2(0)}$$

$$-1 = -2C_1 + C_2$$

But $C_1 = 1;$

$$-1 = -2(1) + C_2$$

$$C_2 = 1$$

Thus,

$$y = (1) e^{-2x} + (1) x e^{-2x}$$

$$y = (1 + x) e^{-2x}$$

22. The equation of exercise #21 with the condition that the graph of the solution pass through the points $(0, 2)$ and $(2, 0)$

$$y = C_1 e^{-2x} + C_2 x e^{-2x}$$

→ When $x = 0, y = 2,$

$$2 = C_1 e^{-2(0)} + C_2 (0) e^{-2(0)}$$

$$C_1 = 2$$

→ When $x = 2, y = 0,$

$$0 = C_1 e^{-2(2)} + C_2 (2) e^{-2(2)}$$

$$0 = C_1 e^{-4} + 2C_2 e^{-4}$$

But $C_1 = 2;$

CASE 2: AUXILIARY EQUATION – REAL ROOTS, REPEATED

$$0 = (2)e^{-4} + 2C_2e^{-4}$$

$$C_2 = -1$$

Thus,

$$y = 2e^{-2x} - xe^{-2x}$$

$$y = (2 - x)e^{-2x}$$

23. $(D^3 - 3D - 2)y = 0$
when $x = 0, y = 0, y' = 9, y'' = 0$

$$m^3 - 3m - 2 = 0$$

$$(m - 2)(m + 1)(m + 1) = 0$$

$$m_1 = 2$$

$$m_2 = -1$$

$$m_3 = -1$$

$$y = C_1e^{m_1x} + (C_2e^{m_2x} + C_3xe^{m_2x})$$

$$y = C_1e^{2x} + (C_2e^{-x} + C_3xe^{-x})$$

$$y' = 2C_1e^{2x} - C_2e^{-x} + C_3(-xe^{-x} + e^{-x})$$

$$y' = 2C_1e^{2x} - C_2e^{-x} - C_3xe^{-x} + C_3e^{-x}$$

$$y'' = 4C_1e^{2x} + C_2e^{-x} - C_3(-xe^{-x} + e^{-x})$$

$$- C_3e^{-x}$$

$$y'' = 4C_1e^{2x} + C_2e^{-x} + C_3xe^{-x} - C_3e^{-x}$$

$$- C_3e^{-x}$$

$$y'' = 4C_1e^{2x} + C_2e^{-x} + C_3xe^{-x} - 2C_3e^{-x}$$

→When $x = 0, y = 0,$

$$0 = C_1e^{2(0)} + (C_2e^{-(0)} + C_3(0)e^{-(0)})$$

$$0 = C_1 + C_2 \rightarrow (\text{Eq. 1})$$

→When $x = 0, y' = 9,$

$$9 = 2C_1e^{2(0)} - C_2e^{-(0)} - C_3(0)e^{-(0)}$$

$$+ C_3e^{-(0)}$$

$$9 = 2C_1 - C_2 + C_3 \rightarrow (\text{Eq. 2})$$

→When $x = 0, y'' = 0,$

$$0 = 4C_1e^{2(0)} + C_2e^{-(0)} + C_3(0)e^{-(0)}$$

$$- 2C_3e^{-(0)}$$

$$0 = 4C_1 + C_2 - 2C_3 \rightarrow (\text{Eq. 3})$$

Solving 3 equations with 3 unknowns,

$$C_1 = 2$$

$$C_2 = -2$$

$$C_3 = 3$$

Thus,

$$y = 2e^{2x} + (-2e^{-x} + 3xe^{-x})$$

$$y = 2e^{2x} + (3x - 2)e^{-x}$$

24. $(D^4 + 3D^3 + 2D^2)y = 0$
when $x = 0, y = 0, y' = 4, y'' = -6, y''' = 14$

$$m^4 + 3m^3 + 2m^2 = 0$$

$$(m)(m)(m + 1)(m + 2) = 0$$

$$m_1 = 0$$

$$m_2 = 0$$

$$m_3 = -1$$

$$m_4 = -2$$

$$y = (C_1e^{m_1x} + C_2xe^{m_1x}) + C_3e^{m_3x}$$

$$+ C_4e^{m_4x}$$

$$y = (C_1e^{0x} + C_2xe^{0x}) + C_3e^{-x} + C_4e^{-2x}$$

$$y = C_1 + C_2x + C_3e^{-x} + C_4e^{-2x}$$

$$y' = C_2 - C_3e^{-x} - 2C_4e^{-2x}$$

$$y'' = C_3e^{-x} + 4C_4e^{-2x}$$

$$y''' = -C_3e^{-x} - 8C_4e^{-2x}$$

→When $x = 0, y = 0,$

$$0 = C_1 + C_2(0) + C_3e^{-(0)} + C_4e^{-2(0)}$$

$$0 = C_1 + C_3 + C_4 \rightarrow (\text{Eq. 1})$$

→When $x = 0, y' = 4,$

$$4 = C_2 - C_3e^{-(0)} - 2C_4e^{-2(0)}$$

$$4 = C_2 - C_3 - 2C_4 \rightarrow (\text{Eq. 2})$$

→When $x = 0, y'' = -6,$

$$-6 = C_3e^{-(0)} + 4C_4e^{-2(0)}$$

$$-6 = C_3 + 4C_4 \rightarrow (\text{Eq. 3})$$

CASE 2: AUXILIARY EQUATION – REAL ROOTS, REPEATED

→When $x = 0, y''' = 14,$

$$14 = -C_3e^{-(0)} - 8C_4e^{-2(0)}$$

$$14 = -C_3 - 8C_4 \rightarrow (\text{Eq. 4})$$

Solving 4 equations with 4 unknowns,

$$C_1 = 0$$

$$C_2 = 2$$

$$C_3 = 2$$

$$C_4 = -2$$

Thus,

$$y = 0 + 2x + 2e^{-x} - 2e^{-2x}$$

$$y = 2(x + e^{-x} - e^{-2x})$$

25. The equation of exercise #24 with the conditions that
when $x = 0, y = 0, y' = 3, y'' = -5, y''' = 9$

$$(D^4 + 3D^3 + 2D^2)y = 0$$

$$m^4 + 3m^3 + 2m^2 = 0$$

$$(m)(m)(m + 1)(m + 2) = 0$$

$$m_1 = 0$$

$$m_2 = 0$$

$$m_3 = -1$$

$$m_4 = -2$$

$$y = (C_1e^{m_1x} + C_2xe^{m_1x}) + C_3e^{m_3x}$$

$$+ C_4e^{m_4x}$$

$$y = (C_1e^{0x} + C_2xe^{0x}) + C_3e^{-x} + C_4e^{-2x}$$

$$y = C_1 + C_2x + C_3e^{-x} + C_4e^{-2x}$$

$$y' = C_2 - C_3e^{-x} - 2C_4e^{-2x}$$

$$y'' = C_3e^{-x} + 4C_4e^{-2x}$$

$$y''' = -C_3e^{-x} - 8C_4e^{-2x}$$

→When $x = 0, y = 0,$

$$0 = C_1 + C_2(0) + C_3e^{-(0)} + C_4e^{-2(0)}$$

$$0 = C_1 + C_3 + C_4 \rightarrow (\text{Eq. 1})$$

→When $x = 0, y' = 3,$

$$3 = C_2 - C_3e^{-(0)} - 2C_4e^{-2(0)}$$

$$3 = C_2 - C_3 - 2C_4 \rightarrow (\text{Eq. 2})$$

→When $x = 0, y'' = -5,$

$$-5 = C_3e^{-(0)} + 4C_4e^{-2(0)}$$

$$-5 = C_3 + 4C_4 \rightarrow (\text{Eq. 3})$$

→When $x = 0, y''' = 9,$

$$9 = -C_3e^{-(0)} - 8C_4e^{-2(0)}$$

$$9 = -C_3 - 8C_4 \rightarrow (\text{Eq. 4})$$

Solving 4 equations with 4 unknowns,

$$C_1 = 2$$

$$C_2 = 0$$

$$C_3 = -1$$

$$C_4 = -1$$

Thus,

$$y = 2 - e^{-x} - e^{-2x}$$

26. $(D^3 + D^2 - D - 1)y = 0$
when $x = 0, y = 1,$
when $x = 2, y = 0,$
and also as $x \rightarrow \infty, y \rightarrow 0$

$$m^3 + D^2 - D - 1 = 0$$

$$(m - 1)(m + 1)(m + 1) = 0$$

$$m_1 = 1$$

$$m_2 = -1$$

$$m_3 = -1$$

$$y = C_1e^{m_1x} + (C_2e^{m_2x} + C_3xe^{m_2x})$$

$$y = C_1e^x + C_2e^{-x} + C_3xe^{-x}$$

→When $x = 0, y = 1,$

$$1 = C_1e^{(0)} + C_2e^{-(0)} + C_3(0)e^{-(0)}$$

$$1 = C_1 + C_2 \rightarrow (\text{Eq. 1})$$

→When $x = 2, y = 0,$

$$0 = C_1e^2 + C_2e^{-2} + C_3(2)e^{-2}$$

$$0 = C_1e^2 + C_2e^{-2} + 2C_3e^{-2} \rightarrow (\text{Eq. 2})$$

CASE 2: AUXILIARY EQUATION - REAL ROOTS, REPEATED

→When $x \rightarrow \infty$, $y \rightarrow 0$,

$$0 = C_1 e^{-x} + C_2 x e^{-x} + C_3 x^2 e^{-x}$$

$$0 = C_1 \rightarrow (\text{Eq. 3})$$

Solving three equations with three unknowns

$$C_1 = 0$$

$$C_2 = 1$$

$$C_3 = -1/2$$

Thus,

$$y = (0)e^x + (1)e^{-x} - \frac{1}{2}x e^{-x}$$

$$y = e^{-x} - \frac{1}{2}x e^{-x}$$

$$y = \frac{1}{2}(2-x)e^{-x}$$

III. In exercises 27 through 29, find for $x = 2$ the y value for the particular solution required.

27. $(4D^2 - 4D + 1)y = 0$
when $x = 0$, $y = -2$, $y' = 2$

$$4m^2 - 4m + 1 = 0$$

$$(2m-1)(2m-1) = 0$$

$$m_1 = 1/2$$

$$m_2 = 1/2$$

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x}$$

$$y = C_1 e^{x/2} + C_2 x e^{x/2}$$

$$y' = \frac{1}{2}C_1 e^{x/2} + C_2 \left(\frac{1}{2}x e^{x/2} + e^{x/2} \right)$$

$$y' = \frac{1}{2}C_1 e^{x/2} + \frac{1}{2}C_2 x e^{x/2} + C_2 e^{x/2}$$

→When $x = 0$, $y = -2$,

$$-2 = C_1 e^{0/2} + C_2(0)e^{0/2}$$

$$-2 = C_1$$

→When $x = 0$, $y' = 2$,

$$2 = \frac{1}{2}C_1 e^{0/2} + \frac{1}{2}C_2(0)e^{0/2} + C_2 e^{0/2}$$

$$2 = \frac{1}{2}C_1 + C_2$$

$$\text{But } C_1 = -2$$

$$2 = \frac{1}{2}(-2) + C_2$$

$$C_2 = 3$$

Thus,

$$y = -2e^{x/2} + 3xe^{x/2}$$

When $x = 2$;

$$y = -2e^2 + 3(2)e^2 = 4e$$

28. $(D^3 + 2D^2)y = 0$
when $x = 0$, $y = -3$, $y' = 0$, $y'' = 12$

$$m^3 + 2m^2 = 0$$

$$(m)(m)(m+2) = 0$$

$$m_1 = 0$$

$$m_2 = 0$$

$$m_3 = -2$$

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{0x} + C_2 x e^{0x} + C_3 e^{-2x}$$

$$y = C_1 + C_2 x + C_3 e^{-2x}$$

$$y' = C_2 - 2C_3 e^{-2x}$$

$$y'' = -4C_3 e^{-2x}$$

→When $x = 0$, $y = -3$,

$$-3 = C_1 + C_2(0) + C_3 e^{-2(0)}$$

$$-3 = C_1 + C_3 \rightarrow (\text{Eq. 1})$$

→When $x = 0$, $y' = 0$,

$$0 = C_2 - 2C_3 e^{-2(0)}$$

$$0 = C_2 - 2C_3 \rightarrow (\text{Eq. 2})$$

→When $x = 0$, $y'' = 12$,

$$12 = -4C_3 e^{-2(0)}$$

$$12 = -4C_3 \rightarrow (\text{Eq. 3})$$

Solving 3 equations with 3 unknowns.

CASE 2: AUXILIARY EQUATION - REAL ROOTS, REPEATED

$$C_1 = -6$$

$$C_2 = 6$$

$$C_3 = 3$$

Thus,

$$y = -6 + 6x + 3e^{-2x}$$

When $x = 2$;

$$y = -6 + 6(2) + 3e^{-2(2)} = 6 + 3e^{-4}$$

29.

$$(D^3 + 5D^2 + 3D - 9)y = 0$$

when $x = 0$, $y = -1$,
when $x = 1$, $y = 0$,
and also as $x \rightarrow \infty$, $y \rightarrow 0$

$$m^3 + 5m^2 + 3m - 9 = 0$$

$$(m+3)(m+3)(m-1) = 0$$

$$m_1 = -3$$

$$m_2 = -3$$

$$m_3 = 1$$

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x} + C_3 e^{m_3 x}$$

$$y = C_1 e^{-3x} + C_2 x e^{-3x} + C_3 e^x$$

→When $x = 0$, $y = -1$,

$$-1 = C_1 e^{-3(0)} + C_2(0)e^{-3(0)} + C_3 e^{(0)}$$

$$-1 = C_1 + C_3 \rightarrow (\text{Eq. 1})$$

→When $x = 1$, $y = 0$,

$$0 = C_1 e^{-3(1)} + C_2(1)e^{-3(1)} + C_3 e^1$$

$$0 = C_1 e^{-3} + C_2 e^{-3} + C_3 e^1 \rightarrow (\text{Eq. 2})$$

→When $x \rightarrow \infty$, $y \rightarrow 0$,

$$0 = C_1 e^{-3\infty} + C_2 x e^{-3\infty} + C_3 e^\infty$$

$$0 = C_3 \rightarrow (\text{Eq. 3})$$

Solving three equations with three unknowns

$$C_1 = -1$$

$$C_2 = 1$$

$$C_3 = 0$$

Thus,

$$y = (-1)e^{-3x} + (1)xe^{-3x} + (0)e^x$$

$$y = -e^{-3x} + xe^{-3x}$$

When $x = 2$;

$$y = -e^{-3(2)} + (2)e^{-3(2)} = e^{-4}$$

CASE 3 & 4: AUXILIARY EQUATION - IMAGINARY ROOTS

CASE 3: AUXILIARY EQUATION HAS IMAGINARY ROOTS, DISTINCT

General Solution:

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx)$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}; \quad b^2 < 4ac$$

$$m = a \pm bi$$

CASE 4: AUXILIARY EQUATION HAS IMAGINARY ROOTS, REPEATED

General Solution:

$$y = e^{ax} \left[(C_1 + C_2 x + \dots + C_{n-1} x^{n-1}) \cos bx + (C_3 + C_4 x + \dots + C_{n-1} x^{n-1}) \sin bx \right]$$

$$m = a \pm bi$$

A NOTE ON HYPERBOLIC FUNCTIONS:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\sinh^2 x = \frac{1}{4}(e^{2x} - 2 + e^{-2x})$$

$$\cosh^2 x = \frac{1}{4}(e^{2x} + 2 + e^{-2x})$$

$$\frac{d}{dx} \sinh u = \cosh u \frac{du}{dx}$$

$$\frac{d}{dx} \cosh u = \sinh u \frac{du}{dx}$$

Since,

$$D^2 \cosh ax = a^2 \cosh ax \text{ and } D^2 \sinh ax = a^2 \sinh ax$$

It follows that both $\cosh ax$ and $\sinh ax$ are solutions of

$$(D^2 - a^2)y = 0, \quad a \neq 0$$

Then the general solution may be written as

$$y = C_1 \cosh ax + C_2 \sinh ax$$

Instead of using the form

$$y = C_3 e^{ax} + C_4 e^{-ax}$$

1. Find the general solution except when the exercise stipulates otherwise.

1. Verify directly that the relation

$$y = C_3 e^{ax} \cos bx + C_4 e^{ax} \sin bx$$

satisfies the equation

$$[(D - a)^2 + b^2]y = 0$$

Solution:

$$(m - a)^2 + b^2 = 0$$

$$m^2 - 2am + a^2 + b^2 = 0$$

$$m = \frac{-(-2a) \pm \sqrt{(-2a)^2 - 4(1)(a^2 + b^2)}}{2(1)}$$

$$m = \frac{2a \pm \sqrt{-4b^2}}{2}$$

$$m = \frac{2a \pm 2b\sqrt{-1}}{2}$$

$$m = a \pm bi$$

Thus,

$$y = e^{ax}(C_3 \cos bx + C_4 \sin bx)$$

$$y = C_3 e^{ax} \cos bx + C_4 e^{ax} \sin bx$$

2. $(D^2 - 2D + 5)y = 0$

Verify your answer.

$$m^2 - 2m + 5 = 0$$

Using Quadratic Formula:

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(5)}}{2(1)}$$

$$m = \frac{2 \pm \sqrt{-16}}{2}$$

$$m = \frac{2 \pm 4\sqrt{-1}}{2}$$

$$m = 1 \pm 2i = a \pm bi$$

CASE 3 & 4: AUXILIARY EQUATION - IMAGINARY ROOTS

Thus,

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx)$$

$$y = e^x(C_1 \cos 2x + C_2 \sin 2x)$$

$$y = C_1 e^x \cos 2x + C_2 e^x \sin 2x$$

$$(D^2 - 2D + 2)y = 0$$

$$m^2 - 2m + 2 = 0$$

Using Quadratic Formula:

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(2)}}{2(1)}$$

$$m = \frac{2 \pm \sqrt{-4}}{2}$$

$$m = \frac{2 \pm 2\sqrt{-1}}{2}$$

$$m = 1 \pm i = a \pm bi$$

Thus,

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx)$$

$$y = e^x(C_1 \cos x + C_2 \sin x)$$

$$y = C_1 e^x \cos x + C_2 e^x \sin x$$

$$(D^2 + 9)y = 0$$

Verify your answer.

$$m^2 + 9 = 0$$

Using Quadratic Formula:

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-(0) \pm \sqrt{(0)^2 - 4(1)(9)}}{2(1)}$$

$$m = \frac{0 \pm \sqrt{-36}}{2}$$

$$m = \frac{0 \pm 6\sqrt{-1}}{2}$$

$$m = 0 \pm 3i = a \pm bi$$

Thus,

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx)$$

$$y = e^{2x}(C_1 \cos 3x + C_2 \sin 3x)$$

$$y = C_1 e^{2x} \cos 3x + C_2 e^{2x} \sin 3x$$

$$5. (D^2 - 9)y = 0$$

$$m^2 - 9 = 0$$

$$m = \pm 3$$

Thus,

$$y = C_1 e^{3x} + C_2 e^{-3x}$$

$$y = C_1 e^{3x} + C_2 e^{-3x}$$

$$y = C_1 \cosh 3x + C_2 \sinh 3x$$

$$6. (D^2 + 6D + 13)y = 0$$

Verify your answer.

$$m^2 + 6m + 13 = 0$$

Using Quadratic Formula:

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-(6) \pm \sqrt{(6)^2 - 4(1)(13)}}{2(1)}$$

$$m = \frac{-6 \pm \sqrt{-16}}{2}$$

$$m = \frac{-6 \pm 4\sqrt{-1}}{2}$$

$$m = -3 \pm 2i = a \pm bi$$

Thus,

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx)$$

$$y = e^{-3x}(C_1 \cos 2x + C_2 \sin 2x)$$

$$y = C_1 e^{-3x} \cos 2x + C_2 e^{-3x} \sin 2x$$

$$7. (D^2 - 4D + 7)y = 0$$

$$m^2 - 4m + 7 = 0$$

Using Quadratic Formula:

CASE 3 & 4: AUXILIARY EQUATION - IMAGINARY ROOTS

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-(-4) \pm \sqrt{(-4)^2 - 4(1)(7)}}{2(1)}$$

$$m = \frac{4 \pm \sqrt{12}}{2}$$

$$m = \frac{4 \pm (2\sqrt{3})\sqrt{-1}}{2}$$

$$m = 2 \pm \sqrt{3}i = a \pm bi$$

Thus,

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx)$$

$$y = e^{2x}(C_1 \cos \sqrt{3}x + C_2 \sin \sqrt{3}x)$$

$$y = C_1 e^{2x} \cos \sqrt{3}x + C_2 e^{2x} \sin \sqrt{3}x$$

8. $(D^3 + 2D^2 + D + 2)y = 0$
Verify your answer.

$$m^3 + 2m^2 + m + 2 = 0$$

$$(m+2)(m^2 + 1) = 0$$

$$m_1 = -2$$

$$m_2 = \pm i$$

Thus,

$$y = C_1 e^{m_1 x} + e^{ax}(C_2 \cos bx + C_3 \sin bx)$$

$$y = C_1 e^{-2x} + e^{0x}(C_2 \cos x + C_3 \sin x)$$

$$y = C_1 e^{-2x} + C_2 \cos x + C_3 \sin x$$

9. $(D^2 - 1)y = 0$
when $x = 0, y = y_0, y' = 0$

$$m^2 - 1 = 0$$

$$(m-1)(m+1) = 0$$

$$m_1 = 1$$

$$m_2 = -1$$

Thus,

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x}$$

$$y = C_1 e^x + C_2 e^{-x}$$

$$y = C_1 \cosh x + C_2 \sinh x$$

$$y' = C_1 \sinh x + C_2 \cosh x$$

$$\rightarrow \text{when } x = 0, y = y_0,$$

$$y_0 = C_1 \cosh 0 + C_2 \sinh 0$$

$$y_0 = C_1$$

$$\rightarrow \text{when } x = 0, y' = 0,$$

$$0 = C_1 \sinh 0 + C_2 \cosh 0$$

$$0 = C_2$$

Thus,

$$y = y_0 \cosh x + 0 \sinh x$$

$$y = y_0 \cosh x$$

10. $(D^2 + 1)y = 0$
when $x = 0, y = y_0, y' = 0$

$$m^2 + 1 = 0$$

$$m = \pm i$$

Thus,

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx)$$

$$y = e^{0x}(C_1 \cos x + C_2 \sin x)$$

$$y = C_1 \cos x + C_2 \sin x$$

$$y' = -C_1 \sin x + C_2 \cos x$$

$$\rightarrow \text{when } x = 0, y = y_0,$$

$$y_0 = C_1 \cos 0 + C_2 \sin 0$$

$$y_0 = C_1$$

$$\rightarrow \text{when } x = 0, y' = 0,$$

$$0 = -C_1 \sin 0 + C_2 \cos 0$$

$$0 = C_2$$

Thus,

$$y = y_0 \cos x + 0 \sin x$$

$$y = y_0 \cos x$$

CASE 3 & 4: AUXILIARY EQUATION - IMAGINARY ROOTS

$$(D^4 + 2D^3 + 10D^2)y = 0$$

11. $m^4 + 2m^3 + 10m^2 = 0$

$$m_1 = 0$$

$$m_2 = 0$$

$$m_{3,4} = -1 \pm 3i$$

Thus,

$$y = C_1 e^{m_1 x} + C_2 x e^{m_1 x}$$

$$+ e^{ax}(C_3 \cos bx + C_4 \sin bx)$$

$$y = C_1 e^{0x} + C_2 x e^{0x}$$

$$+ e^{-x}(C_3 \cos 3x + C_4 \sin 3x)$$

$$y = C_1 + C_2 x + C_3 e^{-x} \cos 3x$$

$$+ C_4 e^{-x} \sin 3x$$

12. $(D^3 + 7D^2 + 19D + 13)y = 0$
when $x = 0, y = 0, y' = 2, y'' = -12$

$$m^3 + 7m^2 + 19m + 13 = 0$$

$$m_1 = -1$$

$$m_{2,3} = -3 \pm 2i$$

Then,

$$y = C_1 e^{m_1 x} + e^{ax}(C_2 \cos bx + C_3 \sin bx)$$

$$y = C_1 e^{-x} + e^{-3x}(C_2 \cos 2x + C_3 \sin 2x)$$

$$y = C_1 e^{-x} + C_2 e^{-3x} \cos 2x + C_3 e^{-3x} \sin 2x$$

$$y' = -C_1 e^{-x}$$

$$+ C_2(-2e^{-3x} \sin 2x - 3 \cos 2x e^{-3x})$$

$$+ C_3(2e^{-3x} \cos 2x - 3 \sin 2x e^{-3x})$$

$$y' = -C_1 e^{-x} - 2C_2 e^{-3x} \sin 2x$$

$$- 3C_2 e^{-3x} \cos 2x + 2C_3 e^{-3x} \cos 2x$$

$$- 3C_3 e^{-3x} \sin 2x$$

$$y'' = C_1 e^{-x}$$

$$- 2C_2(2e^{-3x} \cos 2x - 3 \sin 2x e^{-3x})$$

$$- 3C_2(-2e^{-3x} \sin 2x - 3 \cos 2x e^{-3x})$$

$$+ 2C_3(-2e^{-3x} \sin 2x - 3 \cos 2x e^{-3x})$$

$$- 3C_3(2e^{-3x} \cos 2x - 3 \sin 2x e^{-3x})$$

$$\rightarrow \text{when } x = 0, y = 0,$$

$$0 = C_1 e^{-0} + C_2 e^{-3(0)} \cos 2(0)$$

$$+ C_3 e^{-3(0)} \sin 2(0)$$

$$0 = C_1 + C_2 \rightarrow (\text{Eq. 1})$$

$$\rightarrow \text{when } x = 0, y' = 2,$$

$$2 = -C_1 e^{-0} - 2C_2 e^{-3(0)} \sin 2(0)$$

$$- 3C_2 \cos 2(0) e^{-3(0)}$$

$$+ 2C_3 e^{-3(0)} \cos 2(0)$$

$$- 3C_3 \sin 2(0) e^{-3(0)}$$

$$2 = -C_1 - 3C_2 + 2C_3 \rightarrow (\text{Eq. 2})$$

$$\rightarrow \text{when } x = 0, y'' = -12,$$

$$-12 = C_1 e^{-0}$$

$$- 2C_2[2e^{-3(0)} \cos 2(0) - 3 \sin 2(0) e^{-3(0)}]$$

$$- 3C_2[-2e^{-3(0)} \sin 2(0) - 3 \cos 2(0) e^{-3(0)}]$$

$$+ 2C_3[-2e^{-3(0)} \sin 2(0) - 3 \cos 2(0) e^{-3(0)}]$$

$$- 3C_3(2e^{-3(0)} \cos 2(0) - 3 \sin 2(0) e^{-3(0)})$$

$$-12 = C_1 - 2C_2(2) - 3C_2(-3) + 2C_3(-3)$$

$$- 3C_3(2)$$

$$-12 = C_1 + 5C_2 - 12C_3 \rightarrow (\text{Eq. 3})$$

Solving three equations with three unknowns:

$$C_1 = 0$$

$$C_2 = 0$$

$$C_3 = 1$$

Thus,

$$y = (0)e^{-x} + (0)e^{-3x} \cos 2x$$

$$+ (1)e^{-3x} \sin 2x$$

$$y = e^{-3x} \sin 2x$$

13.

$$(D^5 + D^4 - 7D^3 - 11D^2 - 8D - 12)y = 0$$

$$m^5 + m^4 - 7m^3 - 11m^2 - 8m - 12 = 0$$

$$(m^2 + 1)(m + 2)(m + 3) = 0$$

$$m_{1,2} = \pm i$$

$$m_3 = -2$$

$$m_4 = -2$$

$$m_5 = 3$$

CASE 3 & 4: AUXILIARY EQUATION – IMAGINARY ROOTS

Thus,

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx) + (C_3 e^{m_3 x} + C_4 x e^{m_4 x}) + C_5 e^{m_5 x}$$

$$y = e^{0x}(C_1 \cos x + C_2 \sin x) + (C_3 e^{-2x} + C_4 x e^{-2x}) + C_5 e^{3x}$$

$$y = C_1 \cos x + C_2 \sin x + C_3 e^{-2x} + C_4 x e^{-2x} + C_5 e^{3x}$$

14. $(D^4 - 2D^3 + 2D^2 - 2D + 1)y = 0$
Verify your answer.

$$m^4 - 2m^3 + 2m^2 - 2m + 1 = 0$$

$$(m^2 + 1)(m - 1)(m - 1) = 0$$

$$m_{1\&2} = \pm i$$

$$m_3 = 1$$

$$m_4 = 1$$

Thus,

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx) + (C_3 e^{m_3 x} + C_4 x e^{m_4 x})$$

$$y = e^{0x}(C_1 \cos x + C_2 \sin x) + (C_3 e^x + C_4 x e^x)$$

$$y = C_1 \cos x + C_2 \sin x + C_3 e^x + C_4 x e^x$$

15. $(D^4 + 18D^2 + 81)y = 0$

$$m^4 + 18m^2 + 81 = 0$$

$$(m^2 + 9)(m^2 + 9) = 0$$

$$m_{1\&2} = \pm 3i$$

$$m_{3\&4} = \pm 3i$$

Thus,

$$y = e^{ax}(C_1 + C_2 x) \cos bx + e^{ax}(C_3 + C_4 x) \sin bx$$

$$y = e^{0x}(C_1 + C_2 x) \cos 3x + e^{0x}(C_3 + C_4 x) \sin 3x$$

$$y = (C_1 + C_2 x) \cos 3x + (C_3 + C_4 x) \sin 3x$$

16.

$$(2D^4 + 11D^3 - 4D^2 - 69D + 34)y = 0$$

Verify your answer.

$$2m^4 + 11m^3 - 4m^2 - 69m + 34 = 0$$

$$(m - 2)(2m - 1)(m^2 + 8m + 17) = 0$$

$$m_1 = 2$$

$$m_2 = 1/2$$

$$m_{3\&4} = -4 \pm i$$

Thus,

$$y = C_1 e^{m_1 x} + C_2 e^{m_2 x} + e^{ax}(C_3 \cos bx + C_4 \sin bx)$$

$$y = C_1 e^{2x} + C_2 e^{x/2} + e^{-4x}(C_3 \cos x + C_4 \sin x)$$

$$y = C_1 e^{2x} + C_2 e^{x/2} + C_3 e^{-4x} \cos x + C_4 e^{-4x} \sin x$$

17.

$$(D^6 + 9D^4 + 24D^2 + 16)y = 0$$

$$m^6 + 9m^4 + 24m^2 + 16 = 0$$

$$(m^2 + 1)(m^2 + 4)(m^2 + 4) = 0$$

$$m_{1\&2} = \pm i$$

$$m_{3\&4} = \pm 2i$$

$$m_{5\&6} = \pm 2i$$

Thus,

$$y = e^{ax}(C_1 \cos bx + C_2 \sin bx) + e^{ax}(C_3 + C_4 x) \cos bx + e^{ax}(C_5 + C_6 x) \sin bx$$

$$y = e^{0x}(C_1 \cos x + C_2 \sin x) + e^{0x}(C_3 + C_4 x) \cos 2x + e^{0x}(C_5 + C_6 x) \sin 2x$$

$$y = C_1 \cos x + C_2 \sin x + (C_3 + C_4 x) \cos 2x + (C_5 + C_6 x) \sin 2x$$

18.

$$(2D^3 - D^2 + 36D - 18)y = 0$$

$$2m^3 - m^2 + 36m - 18 = 0$$

$$(2m - 1)(m^2 + 18) = 0$$

$$m_1 = 1/2$$

$$m_{2\&3} = \pm 3\sqrt{2}i$$

CASE 3 & 4: AUXILIARY EQUATION - IMAGINARY ROOTS

21.

$$\frac{d^2x}{dt^2} + 2b\left(\frac{dx}{dt}\right) + k^2x = 0,$$

$$k > b > 0;$$

when $t = 0, x = 0, dx/dt = v_0$

$$(D^2 + 2bD + k^2)x = 0$$

$$m^2 + 2bm + k^2 = 0$$

Using Quadratic Formula:

$$m = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$m = \frac{-(2b) \pm \sqrt{(2b)^2 - 4(1)(k^2)}}{2(1)}$$

$$m = \frac{-2b \pm \sqrt{4b^2 - 4k^2}}{2}$$

$$m = \frac{-2b \pm 2\sqrt{b^2 - k^2}}{2}$$

$$m = -b \pm \sqrt{b^2 - k^2}$$

Let:

$$a = \sqrt{-b^2 + k^2}$$

$$a = \sqrt{k^2 - b^2}$$

$$m = -b \pm ai$$

Then,

$$x = e^{at}(C_1 \cos bt + C_2 \sin bt)$$

$$x = e^{-bt}(C_1 \cos at + C_2 \sin at)$$

$$x = C_1 e^{-bt} \cos at + C_2 e^{-bt} \sin at$$

$$\frac{dx}{dt} = C_1(-ae^{-bt} \sin at - be^{-bt} \cos at)$$

$$+ C_2(ae^{-bt} \cos at - be^{-bt} \sin at)$$

$$\rightarrow \text{when } t = 0, x = 0,$$

$$0 = C_1 e^{-b(0)} \cos a(0) + C_2 e^{-b(0)} \sin a(0)$$

$$0 = C_1$$

$$\rightarrow \text{when } t = 0, dx/dt = v_0,$$

$$v_0 = C_1(-ae^{-b(0)} \sin a(0) - be^{-b(0)} \cos a(0))$$

$$+ C_2(ae^{-b(0)} \cos a(0) - be^{-b(0)} \sin a(0))$$

$$v_0 = C_1(-b) + C_2(a)$$

$$\text{But } C_1 = 0$$

$$v_0 = C_2(a)$$

$$C_2 = \frac{v_0}{a}$$

Thus,

$$x = 0e^{-bt} \cos at + \frac{v_0}{a} e^{-bt} \sin at$$

$$x = \frac{v_0}{a} e^{-bt} \sin at; \text{ where } a = \sqrt{k^2 - b^2}$$

CHAPTER 7: NONHOMOGENEOUS DIFFERENTIAL EQUATIONS WITH CONSTANT COEFFICIENTS

7.1 GENERAL SOLUTION OF A NONHOMOGENEOUS EQUATION

Let y_p be any particular solution (not necessarily involving any arbitrary constants) of the equation

$$b_0 y^{(n)} + b_1 y^{(n-1)} + \dots + b_{n-1} y' + b_n y = R(x) \rightarrow (1)$$

and let y_c be a solution of the corresponding homogeneous equation

$$b_0 y^{(n)} + b_1 y^{(n-1)} + \dots + b_{n-1} y' + b_n y = 0 \rightarrow (2)$$

Then,

$$y = y_c + y_p \rightarrow (3)$$

is a solution of equation (1). For, using the y of equation (3) we see that

$$b_0 y^{(n)} + \dots + b_n y = (b_0 y_c^{(n)} + \dots + b_n y_c) + (b_0 y_p^{(n)} + \dots + b_n y_p) = 0 + R(x)$$

If $\{y_1, y_2, \dots, y_n\}$ are linearly independent solutions of equation (2), then

$$y_c = C_1 y_1 + C_2 y_2 + \dots + C_n y_n \rightarrow (4)$$

in which the C 's are arbitrary constants, is the general solution of equation (2). The right member of equation (4) is called the **complementary function** for equation (1).

The general solution of the nonhomogeneous equation (1) is the sum of the complementary function and any particular solution.

$$y = y_c + y_p$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

7.2. SOLUTION BY METHOD OF UNDETERMINED COEFFICIENTS

Consider the equation

$$f(D)y = R(x), \quad \text{where } R(x) \neq 0$$

General solution:

$$y = y_c + y_p$$

Where:

y_c = the complementary function

y_p = the particular function

Steps in finding y_p

1. Write the variable of $R(x)$ together with the variable part as unity
2. Take the variable parts and the derivative multiplied by the constant A, B, C , etc. Which are to be determined.
3. If y_p is similar to certain term in y_c , multiply y_p by the smallest degree of x which will make it different from y_c .

Note: The method of Undetermined Coefficients fails when the derivative of the functions in right neither terminate nor repeat.

1. In exercises 1 through 35, obtain the general solution.

$$1. \quad (D^2 - 3D + 2)y = 2x^3 - 9x^2 + 6x$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 - 3m + 2 &= 0 \\ (m-1)(m-2) &= 0 \\ m &= 1, 2 \end{aligned}$$

Then,

$$y_c = C_1 e^x + C_2 e^{2x}$$

For y_p :

Let:

$$(D^2 - 3D + 2)y_p = 2x^3 - 9x^2 + 6x$$

$$y_p = Ax^3 + Bx^2 + Cx + E$$

$$\begin{aligned} Dy_p &= 3Ax^2 + 2Bx + C \\ D^2 y_p &= 6Ax + 2B \end{aligned}$$

Substitute,

$$\begin{aligned} (6Ax + 2B) - 3(3Ax^2 + 2Bx + C) \\ + 2(Ax^3 + Bx^2 + Cx + E) \\ = 2x^3 - 9x^2 + 6x \end{aligned}$$

$$\begin{aligned} \text{Collecting } x^3 \text{ terms:} \\ 2A &= 2 \\ A &= 1 \end{aligned}$$

$$\begin{aligned} \text{Collecting } x^2 \text{ terms:} \\ -9A + 2B &= -9 \\ -9(1) + 2B &= -9 \\ B &= 0 \end{aligned}$$

$$\begin{aligned} \text{Collecting } x \text{ terms:} \\ 6A - 6B + 2C &= 6 \\ 6(1) - 6(0) + 2C &= 6 \\ C &= 0 \end{aligned}$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

$$\begin{aligned} \text{Collecting constant terms:} \\ -3C + 2E &= 0 \\ -3(0) + 2E &= 0 \\ E &= 0 \end{aligned}$$

Then,

$$\begin{aligned} y_p &= (1)x^3 + (0)x^2 + (0)x + (0) \\ y_p &= x^3 \end{aligned}$$

Therefore,

$$y = C_1 e^x + C_2 e^{2x} + x^3$$

$$(D^2 + 4)y = 5e^x - 4x$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 + 4 &= 0 \\ m &= \pm 2i \end{aligned}$$

Then,

$$y_c = C_1 \cos 2x + C_2 \sin 2x$$

For y_p :

Let:

$$(D^2 + 4)y_p = 5e^x - 4x$$

$$\begin{aligned} y_p &= Ae^x + Bx + C \\ Dy_p &= Ae^x + B \\ D^2 y_p &= Ae^x \end{aligned}$$

Substitute,

$$(Ae^x) + 4(Ae^x + Bx + C) = 5e^x - 4x$$

$$\begin{aligned} \text{Collecting } e^x \text{ terms:} \\ A + 4A &= 5 \\ A &= 1 \end{aligned}$$

$$\begin{aligned} \text{Collecting } x \text{ terms:} \\ 4B &= -4 \\ B &= -1 \end{aligned}$$

$$\begin{aligned} \text{Collecting constant terms:} \\ 4C &= 0 \\ C &= 0 \end{aligned}$$

Then,

$$\begin{aligned} y_p &= (1)e^x + (-1)x + (0) \\ y_p &= e^x - x \end{aligned}$$

Therefore,

$$y = C_1 \cos 2x + C_2 \sin 2x + e^x - x$$

$$3. \quad (D^2 + 4)y = 5e^x - 4x^2$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 + 4 &= 0 \\ m &= \pm 2i \end{aligned}$$

Then,

$$y_c = C_1 \cos 2x + C_2 \sin 2x$$

For y_p :

Let:

$$(D^2 + 4)y_p = 5e^x - 4x^2$$

$$\begin{aligned} y_p &= Ae^x + Bx^2 + Cx + E \\ Dy_p &= Ae^x + 2Bx + C \\ D^2 y_p &= Ae^x + 2B \end{aligned}$$

Substitute,

$$(Ae^x + 2B) + 4(Ae^x + Bx^2 + Cx + E) = 5e^x - 4x^2$$

$$\begin{aligned} \text{Collecting } e^x \text{ terms:} \\ A + 4A &= 5 \\ A &= 1 \end{aligned}$$

$$\begin{aligned} \text{Collecting } x^2 \text{ terms:} \\ 4B &= -4 \\ B &= -1 \end{aligned}$$

$$\begin{aligned} \text{Collecting } x \text{ terms:} \\ 4C &= 0 \\ C &= 0 \end{aligned}$$

$$\begin{aligned} \text{Collecting constant terms:} \\ 2B + 4E &= 0 \\ 2(-1) + 4E &= 0 \\ E &= 1/2 \end{aligned}$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

Then,

$$y_p = (1)e^x + (-1)x^2 + (0)x + \frac{1}{2}$$

$$y_p = e^x - x^2 + \frac{1}{2}$$

Therefore,

$$y = C_1 \cos 2x + C_2 \sin 2x + e^x - x^2 + \frac{1}{2}$$

4.

$$(D^2 + D)y = \sin x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 + m = 0$$

$$m(m+1) = 0$$

$$m = 0, -1$$

Then,

$$y_c = C_1 + C_2 e^{-x}$$

For y_p :

Let:

$$(D^2 + D)y_p = \sin x$$

$$y_p = A \sin x + B \cos x$$

$$Dy_p = A \cos x - B \sin x$$

$$D^2 y_p = -A \sin x - B \cos x$$

Substitute,

$$(-A \sin x - B \cos x) + (A \cos x - B \sin x) = \sin x$$

Collecting $\cos x$ terms:

$$-B + A = 0$$

$$A = B$$

Collecting $\sin x$ terms:

$$-A - B = 1$$

$$A + B = -1$$

$$A + A = -1$$

$$A = -1/2$$

$$B = -1/2$$

Then,

$$y_p = -\frac{1}{2} \sin x - \frac{1}{2} \cos x$$

Therefore,

$$y = C_1 + C_2 e^{-x} - \frac{1}{2} \sin x - \frac{1}{2} \cos x$$

5.

$$(D^2 - 4D + 4)y = e^x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - 4m + 4 = 0$$

$$(m-2)(m-2) = 0$$

$$m = 2, 2$$

Then,

$$y_c = (C_1 + C_2 x)e^{2x}$$

For y_p :

Let:

$$(D^2 - 4D + 4)y_p = e^x$$

$$y_p = Ae^x$$

$$Dy_p = Ae^x$$

$$D^2 y_p = Ae^x$$

Substitute,

$$(Ae^x) - 4(Ae^x) + 4Ae^x = e^x$$

Collecting e^x terms:

$$A - 4A + 4A = 1$$

$$A = 1$$

Then,

$$y_p = e^x$$

Therefore,

$$y = (C_1 + C_2 x)e^{2x} + e^x$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

$$(D^2 - 3D + 2)y = 2x^2 + 1$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - 3m + 2 = 0$$

$$(m-1)(m-2) = 0$$

$$m = 1, 2$$

Then,

$$y_c = C_1 e^x + C_2 e^{2x}$$

For y_p :

Let:

$$(D^2 - 3D + 2)y_p = 2x^2 + 1$$

$$y_p = Ax^2 + Bx + C$$

$$Dy_p = 2Ax + B$$

$$D^2 y_p = 2A$$

Substitute,

$$(2A) - 3(2Ax + B) + 2(Ax^2 + Bx + C) = 2x^2 + 1$$

Collecting x^2 terms:

$$2A = 2$$

$$A = 1$$

Collecting x terms:

$$-6A + 2B = 0$$

$$-6(1) + 2B = 0$$

$$B = 3$$

Collecting constant terms:

$$2A - 3B + 2C = 1$$

$$2(1) - 3(3) + 2C = 1$$

$$C = 4$$

Then,

$$y_p = x^2 + 3x + 4$$

Therefore,

$$y = C_1 e^x + C_2 e^{2x} + x^2 + 3x + 4$$

7.

$$y'' - 3y' - 4y = 6e^x$$

$$(D^2 - 3D - 4)y = 6e^x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - 3m - 4 = 0$$

$$(m-4)(m+1) = 0$$

$$m = 4, -1$$

Then,

$$y_c = C_1 e^{4x} + C_2 e^{-x}$$

For y_p :

Let:

$$(D^2 - 3D - 4)y_p = 6e^x$$

$$y_p = Ae^x$$

$$Dy_p = Ae^x$$

$$D^2 y_p = Ae^x$$

Substitute,

$$(Ae^x) - 3(Ae^x) - 4(Ae^x) = 6e^x$$

Collecting e^x terms:

$$A - 3A - 4A = 6$$

$$A = -1$$

Then,

$$y_p = -e^x$$

Therefore,

$$y = C_1 e^{4x} + C_2 e^{-x} - e^x$$

8.

$$y'' - 3y' - 4y = 5e^{4x}$$

$$(D^2 - 3D - 4)y = 5e^{4x}$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - 3m - 4 = 0$$

$$(m-4)(m+1) = 0$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

$$m = 4, -1$$

Then,

$$y_c = C_1 e^{4x} + C_2 e^{-x}$$

For y_p :

Let:

$$(D^2 - 3D - 4)y_p = 5e^{4x}$$

$$y_p = Axe^{4x}$$

$$Dy_p = A(4xe^{4x} + e^{4x}) = 4Axe^{4x} + Ae^{4x}$$

$$D^2 y_p = 4A(4xe^{4x} + e^{4x}) + 4Ae^{4x} = 16Axe^{4x} + 8Ae^{4x}$$

Substitute,

$$(16Axe^{4x} + 8Ae^{4x}) - 3(4Axe^{4x} + Ae^{4x}) - 4(Axe^{4x}) = 5e^{4x}$$

Collecting e^{4x} terms:

$$8A - 3A = 5$$

$$A = 1$$

Then,

$$y_p = xe^{4x}$$

Therefore,

$$y = C_1 e^{4x} + C_2 e^{-x} + xe^{4x}$$

$$y = (C_1 + x)e^{4x} + C_2 e^{-x}$$

$$9. (D^2 - 4)y = 8e^{2x} - 12$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - 4 = 0$$

$$(m - 2)(m + 2) = 0$$

$$m = \pm 2$$

Then,

$$y_c = C_1 e^{2x} + C_2 e^{-2x}$$

For y_p :

Let:

$$(D^2 - 4)y_p = 8e^{2x} - 12$$

$$y_p = Axe^{2x} + B$$

$$Dy_p = A(2xe^{2x} + e^{2x}) = 2Axe^{2x} + Ae^{2x}$$

$$D^2 y_p = 2A(xe^{2x} + e^{2x}) + 2Ae^{2x} = 2Axe^{2x} + 4Ae^{2x}$$

Substitute,

$$(2Axe^{2x} + 4Ae^{2x}) - 4(Axe^{2x} + B) = 8e^{2x} - 12$$

Collecting e^{2x} terms:

$$4A = 8$$

$$A = 2$$

Collecting constant terms:

$$-4B = -12$$

$$B = 3$$

Then,

$$y_p = 2xe^{2x} + 3$$

Therefore,

$$y = C_1 e^{2x} + C_2 e^{-2x} + 2xe^{2x} + 3$$

$$y = (C_1 + 2x)e^{2x} + C_2 e^{-2x} + 3$$

$$10. (D^2 - D - 2)y = 1 - 2x - 9e^{-x}$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - m - 2 = 0$$

$$(m + 1)(m - 2) = 0$$

$$m = -1, 2$$

Then,

$$y_c = C_1 e^{-x} + C_2 e^{2x}$$

For y_p :

Let:

$$(D^2 - D - 2)y_p = 1 - 2x - 9e^{-x}$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

$$y_p = A + Bx + Cxe^{-x}$$

$$Dy_p = B + C(-xe^{-x} + e^{-x})$$

$$= B - Cxe^{-x} + Ce^{-x}$$

$$D^2 y_p = -C(-xe^{-x} + e^{-x}) - Ce^{-x} = Cxe^{-x} - 2Ce^{-x}$$

Substitute,

$$(Cxe^{-x} - 2Ce^{-x}) - (B - Cxe^{-x} + Ce^{-x}) - 2(A + Bx + Cxe^{-x}) = 1 - 2x - 9e^{-x}$$

Collecting e^{-x} terms:

$$-2C - C = -9$$

$$C = 3$$

Collecting x terms:

$$-2B = -2$$

$$B = 1$$

Collecting constant terms:

$$-B - 2A = 1$$

$$-1 - 2A = 1$$

$$A = -1$$

Then,

$$y_p = -1 + x + 3xe^{-x}$$

Therefore,

$$y = C_1 e^{-x} + C_2 e^{2x} - 1 + x + 3xe^{-x}$$

$$y = (C_1 + 3x)e^{-x} + C_2 e^{2x} + x - 1$$

$$11. y'' - 4y' + 3y = 20 \cos x$$

$$(D^2 - 4D + 3)y = 20 \cos x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - 4m + 3 = 0$$

$$(m - 1)(m - 3) = 0$$

$$m = 1, 3$$

Then,

$$y_c = C_1 e^x + C_2 e^{3x}$$

For y_p :

Let:

$$(D^2 - 4D + 3)y_p = 20 \cos x$$

$$y_p = A \cos x + B \sin x$$

$$Dy_p = -A \sin x + B \cos x$$

$$D^2 y_p = -A \cos x - B \sin x$$

Substitute,

$$(-A \cos x - B \sin x)$$

$$-4(-A \sin x + B \cos x)$$

$$+ 3(A \cos x + B \sin x) = 20 \cos x$$

Collecting $\cos x$ terms:

$$-A - 4B + 3A = 20$$

$$2A - 4B = 20 \rightarrow (\text{Eq. 1})$$

Collecting $\sin x$ terms:

$$-B + 4A + 3B = 0$$

$$4A + 2B = 0 \rightarrow (\text{Eq. 2})$$

Solving two equations with two unknowns:

$$A = 2$$

$$B = -4$$

Then,

$$y_p = 2 \cos x - 4 \sin x$$

Therefore,

$$y = C_1 e^x + C_2 e^{3x} + 2 \cos x - 4 \sin x$$

$$12. y'' - 4y' + 3y = 2 \cos x + 4 \sin x$$

$$(D^2 - 4D + 3)y = 2 \cos x + 4 \sin x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - 4m + 3 = 0$$

$$(m - 1)(m - 3) = 0$$

$$m = 1, 3$$

Then,

$$y_c = C_1 e^x + C_2 e^{3x}$$

For y_p :

Let:

$$(D^2 - 4D + 3)y_p = 2 \cos x + 4 \sin x$$

$$\begin{aligned} y_p &= A \cos x + B \sin x \\ Dy_p &= -A \sin x + B \cos x \\ D^2 y_p &= -A \cos x - B \sin x \end{aligned}$$

Substitute,

$$\begin{aligned} (-A \cos x - B \sin x) \\ -4(-A \sin x + B \cos x) \\ +3(A \cos x + B \sin x) &= 2 \cos x + 4 \sin x \end{aligned}$$

Collecting $\cos x$ terms:

$$\begin{aligned} -A - 4B + 3A &= 2 \\ 2A - 4B &= 20 \rightarrow (\text{Eq. 1}) \end{aligned}$$

Collecting $\sin x$ terms:

$$\begin{aligned} -B + 4A + 3B &= 4 \\ 4A + 2B &= 0 \rightarrow (\text{Eq. 2}) \end{aligned}$$

Solving two equations with two unknowns:

$$\begin{aligned} A &= 1 \\ B &= 0 \end{aligned}$$

Then,

$$y_p = \cos x$$

Therefore,

$$y = C_1 e^x + C_2 e^{2x} + \cos x$$

$$13. \quad y'' + 2y' + y = 7 + 75 \sin 2x$$

$$(D^2 + 2D + 1)y = 7 + 75 \sin 2x$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 + 2m + 1 &= 0 \\ (m + 1)(m + 1) &= 0 \\ m &= -1, -1 \end{aligned}$$

Then,

$$\begin{aligned} y_c &= C_1 e^{-x} + C_2 x e^{-x} \\ y_c &= (C_1 + C_2 x) e^{-x} \end{aligned}$$

For y_p :

Let:

$$(D^2 + 2D + 1)y_p = 7 + 75 \sin 2x$$

$$\begin{aligned} y_p &= A + B \sin 2x + C \cos 2x \\ Dy_p &= 2B \cos 2x - 2C \sin 2x \\ D^2 y_p &= -4B \sin 2x - 4C \cos 2x \end{aligned}$$

Substitute,

$$\begin{aligned} (-4B \sin 2x - 4C \cos 2x) \\ +2(2B \cos 2x - 2C \sin 2x) \\ +(A + B \sin 2x + C \cos 2x) \\ &= 7 + 75 \sin 2x \end{aligned}$$

Collecting $\sin 2x$ terms:

$$\begin{aligned} -4B - 4C + B &= 75 \\ -3B - 4C &= 75 \rightarrow (\text{Eq. 1}) \end{aligned}$$

Collecting $\cos 2x$ terms:

$$\begin{aligned} -4C + 4B + C &= 0 \\ -3C + 4B &= 0 \rightarrow (\text{Eq. 2}) \end{aligned}$$

Collecting constant terms:

$$A = 7 \rightarrow (\text{Eq. 3})$$

Solving three equations with three unknowns:

$$\begin{aligned} A &= 7 \\ B &= -9 \\ C &= -12 \end{aligned}$$

Then,

$$y_p = 7 - 9 \sin 2x - 12 \cos 2x$$

Therefore,

$$y = (C_1 + C_2 x) e^{-x} + 7 - 9 \sin 2x - 12 \cos 2x$$

$$(D^2 + 4D + 5)y = 50x + 13e^{3x}$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 + 4m + 5 &= 0 \\ m &= -2 \pm i \end{aligned}$$

Then,

$$y = e^{-2x}(C_1 \cos x + C_2 \sin x)$$

For y_p :

Let:

$$(D^2 + 4D + 5)y_p = 50x + 13e^{3x}$$

$$\begin{aligned} y_p &= Ax + B + Ce^{3x} \\ Dy_p &= A + 3Ce^{3x} \\ D^2 y_p &= 9Ce^{3x} \end{aligned}$$

Substitute,

$$\begin{aligned} (9Ce^{3x}) + 4(A + 3Ce^{3x}) \\ +5(Ax + B + Ce^{3x}) &= 50x + 13e^{3x} \end{aligned}$$

Collecting e^{3x} terms:

$$\begin{aligned} 9C + 12C + 5C &= 13 \\ C &= 1/2 \end{aligned}$$

Collecting x terms:

$$\begin{aligned} 5A &= 50 \\ A &= 10 \end{aligned}$$

Collecting constant terms:

$$\begin{aligned} 4A + 5B &= 0 \\ 4(10) + 5B &= 0 \\ B &= -8 \end{aligned}$$

Then,

$$y_p = 10x - 8 + \frac{1}{2}e^{3x}$$

Therefore,

$$y = e^{-2x}(C_1 \cos x + C_2 \sin x) + 10x - 8 + \frac{1}{2}e^{3x}$$

15.

$$(D^2 + 1)y = \cos x$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 + 1 &= 0 \\ m &= \pm i \end{aligned}$$

Then,

$$y = C_1 \cos x + C_2 \sin x$$

For y_p :

Let:

$$(D^2 + 1)y_p = \cos x$$

$$\begin{aligned} y_p &= Ax \cos x + Bx \sin x \\ Dy_p &= A(-x \sin x + \cos x) \\ &\quad + B(x \cos x + \sin x) \\ &= -Ax \sin x + A \cos x + Bx \cos x \\ &\quad + B \sin x \end{aligned}$$

$$\begin{aligned} D^2 y_p &= -A(x \cos x + \sin x) - A \sin x \\ &\quad + B(-x \sin x + \cos x) + B \cos x \end{aligned}$$

Substitute,

$$\begin{aligned} -A(x \cos x + \sin x) - A \sin x \\ +B(-x \sin x + \cos x) + B \cos x \\ +(Ax \cos x + Bx \sin x) &= \cos x \end{aligned}$$

Collecting $\cos x$ terms:

$$\begin{aligned} B + B &= 1 \\ B &= 1/2 \end{aligned}$$

Collecting $\sin x$ terms:

$$\begin{aligned} -A - A &= 0 \\ A &= 0 \end{aligned}$$

Then,

$$y_p = \frac{1}{2}x \sin x$$

Therefore,

$$y = C_1 \cos x + C_2 \sin x + \frac{1}{2}x \sin x$$

16. $(D^2 - 4D + 4)y = e^{2x}$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 - 4m + 4 &= 0 \\ (m-2)(m-2) &= 0 \\ m &= 2, 2 \end{aligned}$$

Then,

$$y_c = C_1 e^{2x} + C_2 x e^{2x}$$

For y_p :

Let:

$$(D^2 - 4D + 4)y_p = e^{2x}$$

$$\begin{aligned} y_p &= Ax^2 e^{2x} \\ Dy_p &= A(2x^2 e^{2x} + 2x e^{2x}) \\ &= 2Ax^2 e^{2x} + 2Ax e^{2x} \\ D^2 y_p &= 2A(2x^2 e^{2x} + 2x e^{2x}) \\ &\quad + 2A(2x e^{2x} + e^{2x}) \\ &= 4Ax^2 e^{2x} + 8Ax e^{2x} + 2Ae^{2x} \end{aligned}$$

Substitute,

$$(4Ax^2 e^{2x} + 8Ax e^{2x} + 2Ae^{2x}) - 4(2Ax^2 e^{2x} + 2Ax e^{2x}) + 4(Ax^2 e^{2x}) = e^{2x}$$

Collecting e^{2x} terms:

$$\begin{aligned} 2A &= 1 \\ A &= 1/2 \end{aligned}$$

Then,

$$y_p = \frac{1}{2} x^2 e^{2x}$$

Therefore,

$$y = C_1 e^{2x} + C_2 x e^{2x} + \frac{1}{2} x^2 e^{2x}$$

$$y = \left(C_1 + C_2 x + \frac{1}{2} x^2 \right) e^{2x}$$

18. $(D^2 - 1)y = 8xe^x$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 - 1 &= 0 \\ (m+1)(m-1) &= 0 \\ m &= -1, 1 \end{aligned}$$

Then,

$$y_c = C_1 e^{-x} + C_2 e^x$$

For y_p :

Let:

$$(D^2 - 1)y_p = 8xe^x$$

$$\begin{aligned} y_p &= Axe^x + Bx^2 e^x \\ Dy_p &= A(xe^x + e^x) + B(x^2 e^x + 2xe^x) \\ &= Axe^x + Ae^x + Bx^2 e^x + 2Bxe^x \\ D^2 y_p &= A(xe^x + e^x) + Ae^x \\ &\quad + B(x^2 e^x + 2xe^x) + 2B(xe^x + e^x) \end{aligned}$$

Substitute,

$$A(xe^x + e^x) + Ae^x + B(x^2 e^x + 2xe^x) + 2B(xe^x + e^x) - A(xe^x + e^x) - B(x^2 e^x + 2xe^x) = 8xe^x$$

Collecting xe^x terms:

$$\begin{aligned} A + 2B + 2B - A &= 8 \\ B &= 2 \end{aligned}$$

Collecting e^x terms:

$$\begin{aligned} A + A + 2B &= 0 \\ 2A + 2(2) &= 0 \\ A &= -2 \end{aligned}$$

Then,

$$y_p = -2xe^x + 2x^2 e^x$$

Therefore,

$$\begin{aligned} y &= C_1 e^{-x} + C_2 e^x - 2xe^x + 2x^2 e^x \\ y &= C_1 e^{-x} + e^x(C_2 - 2x + 2x^2) \end{aligned}$$

19. $(D^3 - D)y = x$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^3 - m &= 0 \\ (m)(m-1)(m+1) &= 0 \\ m &= 0, 1, -1 \end{aligned}$$

Then,

$$y_c = C_1 + C_2 e^x + C_3 e^{-x}$$

For y_p :

Let:

$$\begin{aligned} (D^3 - D)y_p &= x \\ y_p &= Ax + Bx^2 \\ Dy_p &= A + 2Bx \\ D^2 y_p &= 2B \\ D^3 y_p &= 0 \end{aligned}$$

Substitute,

$$0 - (A + 2Bx) = x$$

Collecting x terms:

$$\begin{aligned} -2B &= 1 \\ B &= -1/2 \end{aligned}$$

Collecting constant terms:

$$\begin{aligned} 0 - A &= 0 \\ A &= 0 \end{aligned}$$

Then,

$$y_p = -\frac{1}{2} x^2$$

Therefore,

$$y = C_1 + C_2 e^x + C_3 e^{-x} - \frac{1}{2} x^2$$

20. $(D^3 - D^2 + D - 1)y = 4 \sin x$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^3 - m^2 + m - 1 &= 0 \\ (m-1)(m^2 + 1) &= 0 \\ m &= 1, \pm i \end{aligned}$$

Then,

$$y_c = C_1 e^x + C_2 \cos x + C_3 \sin x$$

For y_p :

Let:

$$(D^3 - D^2 + D - 1)y_p = 4 \sin x$$

$$\begin{aligned} y_p &= Ax \sin x + Bx \cos x \\ Dy_p &= A(x \cos x + \sin x) \\ &\quad + B(-x \sin x + \cos x) \end{aligned}$$

$$\begin{aligned} D^2 y_p &= A(-x \sin x + \cos x) + A \cos x \\ &\quad - B(x \cos x + \sin x) - B \sin x \\ D^2 y_p &= -Ax \sin x + 2A \cos x - Bx \cos x \\ &\quad - 2B \sin x \\ D^3 y_p &= -A(x \cos x + \sin x) - 2A \sin x \\ &\quad - B(-x \sin x + \cos x) - 2B \cos x \end{aligned}$$

Substitute,

$$\begin{aligned} -A(x \cos x + \sin x) - 2A \sin x \\ -B(-x \sin x + \cos x) - 2B \cos x \\ + Ax \sin x - 2A \cos x + Bx \cos x + 2B \sin x \\ + A(x \cos x + \sin x) + B(-x \sin x + \cos x) \\ - Ax \sin x - Bx \cos x = 4 \sin x \end{aligned}$$

Collecting $\sin x$ terms:

$$\begin{aligned} -A - 2A + 2B + A &= 4 \\ -2A + 2B &= 4 \\ -A + B &= 2 \rightarrow (\text{Eq 1}) \end{aligned}$$

Collecting $\cos x$ terms:

$$\begin{aligned} -B - 2B - 2A + B &= 0 \\ -2B - 2A &= 0 \rightarrow (\text{Eq 2}) \end{aligned}$$

Solving two equations with two unknowns:

$$\begin{aligned} A &= -1 \\ B &= 1 \end{aligned}$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

Then,

$$y_p = -x \sin x + x \cos x$$

Therefore,

$$y = C_1 e^x + C_2 \cos x + C_3 \sin x - x \sin x + x \cos x$$

$$y = C_1 e^x + (C_2 + x) \cos x + (C_3 - x) \sin x$$

21. $(D^3 + D^2 - 4D - 4)y = 3e^{-x} - 4x - 6$

$$y = y_c + y_p$$

For y_c :

$$m^3 + m^2 - 4m - 4 = 0$$

$$(m-2)(m+2)(m+1) = 0$$

$$m = 2, -2, -1$$

Then,

$$y_c = C_1 e^{2x} + C_2 e^{-2x} + C_3 e^{-x}$$

For y_p :

Let:

$$(D^3 + D^2 - 4D - 4)y_p = 3e^{-x} - 4x - 6$$

$$y_p = Axe^{-x} + Bx + C$$

$$Dy_p = A(-xe^{-x} + e^{-x}) + B$$

$$D^2 y_p = -A(-xe^{-x} + e^{-x}) - Ae^{-x}$$

$$D^3 y_p = A(-xe^{-x} + e^{-x}) + 2Ae^{-x}$$

Substitute,

$$A(-xe^{-x} + e^{-x}) + 2Ae^{-x}$$

$$-A(-xe^{-x} + e^{-x}) - Ae^{-x}$$

$$-4(A(-xe^{-x} + e^{-x}) + B)$$

$$-4(Axe^{-x} + Bx + C) = 3e^{-x} - 4x - 6$$

Collecting e^{-x} terms:

$$A + 2A - A - A - 4A = 3$$

$$A = -1$$

Collecting x terms:

$$-4B = -4$$

$$B = 1$$

Collecting constant terms:

$$-4B - 4C = -6$$

$$-4(1) - 4C = -6$$

$$C = 1/2$$

Then,

$$y_p = -xe^{-x} + x + \frac{1}{2}$$

Therefore,

$$y = C_1 e^{2x} + C_2 e^{-2x} + C_3 e^{-x} - xe^{-x} + x + \frac{1}{2}$$

$$y = C_1 e^{2x} + C_2 e^{-2x} + (C_3 - x)e^{-x} + x + \frac{1}{2}$$

23. $(D^4 - 1)y = e^{-x}$

$$y = y_c + y_p$$

For y_c :

$$m^4 - 1 = 0$$

$$(m-1)(m+1)(m^2+1) = 0$$

$$m = 1, -1, \pm i$$

Then,

$$y_c = C_1 e^x + C_2 e^{-x} + C_3 \cos x + C_4 \sin x$$

For y_p :

Let:

$$(D^4 - 1)y_p = e^{-x}$$

$$y_p = Axe^{-x}$$

$$Dy_p = A(-xe^{-x} + e^{-x})$$

$$D^2 y_p = -A(-xe^{-x} + e^{-x}) - Ae^{-x}$$

$$D^3 y_p = A(-xe^{-x} + e^{-x}) + 2Ae^{-x}$$

$$D^4 y_p = -A(-xe^{-x} + e^{-x}) - 3Ae^{-x}$$

Substitute,

$$-A(-xe^{-x} + e^{-x}) - 3Ae^{-x} - Ae^{-x}$$

$$= e^{-x}$$

Collecting e^{-x} terms:

$$-A - 3A = 1$$

$$A = -1/4$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

Then,

$$y_p = -\frac{1}{4}xe^{-x}$$

Therefore,

$$y = C_1 e^x + C_2 e^{-x} + C_3 \cos x + C_4 \sin x - \frac{1}{4}xe^{-x}$$

$$y = C_1 e^x + \left(C_2 - \frac{1}{4}x\right)e^{-x} + C_3 \cos x + C_4 \sin x$$

24. $(D^2 - 1)y = 10 \sin^2 x$

Use the identity $\sin^2 x = \frac{1}{2}(1 - \cos 2x)$

$$y = y_c + y_p$$

For y_c :

$$m^2 - 1 = 0$$

$$(m-1)(m+1) = 0$$

$$m = 1, -1$$

Then,

$$y_c = C_1 e^x + C_2 e^{-x}$$

For y_p :

Let:

$$(D^2 - 1)y_p = 10 \sin^2 x$$

$$(D^2 - 1)y = 10 \left(\frac{1 - \cos 2x}{2}\right)$$

$$(D^2 - 1)y = 5 - 5 \cos 2x$$

$$y_p = A + B \cos 2x + C \sin 2x$$

$$Dy_p = -2B \sin 2x + 2C \cos 2x$$

$$D^2 y_p = -4B \cos 2x - 4C \sin 2x$$

Substitute,

$$-4B \cos 2x - 4C \sin 2x$$

$$-(A + B \cos 2x + C \sin 2x) = 5 - 5 \cos 2x$$

Collecting $\cos 2x$ terms:

$$-4B - B = -5$$

$$B = 1$$

Collecting $\sin 2x$ terms:

$$-4C - C = 0$$

$$C = 0$$

Collecting constant terms:

$$-A = 5$$

$$A = -5$$

Then,

$$y_p = -5 + \cos 2x$$

Therefore,

$$y = C_1 e^x + C_2 e^{-x} - 5 + \cos 2x$$

25. $(D^2 + 1)y = 12 \cos^2 x$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 1 = 0$$

$$m = \pm i$$

Then,

$$y_c = C_1 \cos x + C_2 \sin x$$

For y_p :

Let:

$$(D^2 + 1)y_p = 12 \cos^2 x$$

$$(D^2 + 1)y = 12 \left(\frac{1 + \cos 2x}{2}\right)$$

$$(D^2 + 1)y = 6 + 6 \cos 2x$$

$$y_p = A + B \cos 2x + C \sin 2x$$

$$Dy_p = -2B \sin 2x + 2C \cos 2x$$

$$D^2 y_p = -4B \cos 2x - 4C \sin 2x$$

Substitute,

$$-4B \cos 2x - 4C \sin 2x$$

$$+(A + B \cos 2x + C \sin 2x) = 6 + 6 \cos 2x$$

Collecting $\cos 2x$ terms:

$$-4B + B = 6$$

$$C = -2$$

Collecting $\sin 2x$ terms:

$$-4C + C = 0$$

$$C = 0$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

Collecting constant terms:

$$A = 6$$

Then,

$$y_p = 6 - 2 \cos 2x$$

Therefore,

$$y = C_1 \cos x + C_2 \sin x + 6 - 2 \cos 2x$$

$$26. (D^2 + 4)y = 4 \sin^2 x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 4 = 0$$

$$m = \pm 2i$$

Then,

$$y_c = C_1 \cos 2x + C_2 \sin 2x$$

For y_p :

Let:

$$(D^2 + 4)y_p = 4 \sin^2 x$$

$$(D^2 + 4)y = 4 \left(\frac{1 - \cos 2x}{2} \right)$$

$$(D^2 + 4)y = 2 - 2 \cos 2x$$

$$y_p = A + Bx \cos 2x + Cx \sin 2x$$

$$Dy_p = B(-2x \sin 2x + \cos 2x) + C(2x \cos 2x + \sin 2x)$$

$$D^2 y_p = -2B(2x \cos 2x + \sin 2x) - 2B \sin 2x + 2C(-2x \sin 2x + \cos 2x) + 2C \cos 2x$$

Substitute,

$$-2B(2x \cos 2x + \sin 2x) - 2B \sin 2x + 2C(-2x \sin 2x + \cos 2x) + 2C \cos 2x = 2 - 2 \cos 2x$$

$$\text{Collecting } \cos 2x \text{ terms:}$$

$$2C + 2C = -2$$

$$C = -1/2$$

Collecting $\sin 2x$ terms:

$$-2B - 2B = 0$$

$$B = 0$$

Collecting constant terms:

$$4A = 2$$

$$A = 1/2$$

Then,

$$y_p = \frac{1}{2} - \frac{1}{2}x \sin 2x$$

Therefore,

$$y = C_1 \cos 2x + C_2 \sin 2x + \frac{1}{2} - \frac{1}{2}x \sin 2x$$

$$y = C_1 \cos 2x + C_2 \sin 2x + \frac{1}{2}(1 - x \sin 2x)$$

$$27. y'' - 3y' - 4y = 16x - 50 \cos 2x$$

$$(D^2 - 3D - 4)y = 16x - 50 \cos 2x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - 3m - 4 = 0$$

$$(m - 4)(m + 1) = 0$$

$$m = 4, -1$$

Then,

$$y_c = C_1 e^{4x} + C_2 e^{-x}$$

For y_p :

Let:

$$(D^2 - 3D - 4)y_p = 16x - 50 \cos 2x$$

$$y_p = Ax + B + C \cos 2x + D \sin 2x$$

$$Dy_p = A - 2C \sin 2x + 2D \cos 2x$$

$$D^2 y_p = -4C \cos 2x - 4D \sin 2x$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

Substitute,

$$(-4C \cos 2x - 4D \sin 2x) - 3(A - 2C \sin 2x + 2D \cos 2x) - 4(Ax + B + C \cos 2x + D \sin 2x) = 16x - 50 \cos 2x$$

Collecting $\cos 2x$ terms:

$$-4C - 6D - 4C = -50$$

$$-8C - 6D = -50 \rightarrow (\text{Eq. 1})$$

Collecting $\sin 2x$ terms:

$$-4D + 6C - 4C = -50$$

$$-8D + 6C = 0 \rightarrow (\text{Eq. 2})$$

Collecting x terms:

$$-4A = 16$$

$$A = -4$$

Collecting constant terms:

$$-3A - 4B = 0$$

$$-3(-4) - 4B = 0$$

$$B = 3$$

Solving two equations with two unknowns:

$$C = 4$$

$$D = 3$$

Then,

$$y_p = -4x + 3 + 4 \cos 2x + 3 \sin 2x$$

Therefore,

$$y = C_1 e^{4x} + C_2 e^{-x} - 4x + 3 + 4 \cos 2x + 3 \sin 2x$$

$$28. y'' + 4y' + 3y = 15e^{2x} + e^{-x}$$

$$(D^2 + 4D + 3)y = 15e^{2x} + e^{-x}$$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 4m + 3 = 0$$

$$(m + 1)(m + 3) = 0$$

$$m = -1, -3$$

Then,

$$y_c = C_1 e^{-x} + C_2 e^{-3x}$$

For y_p :

Let:

$$(D^2 + 4D + 3)y_p = 15e^{2x} + e^{-x}$$

$$y_p = Ae^{2x} + Bxe^{-x}$$

$$Dy_p = 2Ae^{2x} + B(-xe^{-x} + e^{-x})$$

$$D^2 y_p = 4Ae^{2x} - B(-xe^{-x} + e^{-x}) - Be^{-x}$$

Substitute,

$$[4Ae^{2x} - B(-xe^{-x} + e^{-x}) - Be^{-x}] + 4[2Ae^{2x} + B(-xe^{-x} + e^{-x})] + 3[Ae^{2x} + Bxe^{-x}] = 15e^{2x} + e^{-x}$$

Collecting e^{2x} terms:

$$4A + 8A + 3A = 15$$

$$A = 1$$

Collecting e^{-x} terms:

$$-B - B + 4B = 1$$

$$B = 1/2$$

Then,

$$y_p = e^{2x} + \frac{1}{2}xe^{-x}$$

Therefore,

$$y = C_1 e^{-x} + C_2 e^{-3x} + e^{2x} + \frac{1}{2}xe^{-x}$$

$$31. y'' - y' - 2y = 6x + 6e^{-x}$$

$$(D^2 - D - 2)y = 6x + 6e^{-x}$$

$$y = y_c + y_p$$

For y_c :

$$m^2 - m - 2 = 0$$

$$(m - 2)(m + 1) = 0$$

$$m = 2, -1$$

Then,

$$y_c = C_1 e^{2x} + C_2 e^{-x}$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

For y_p :

Let:

$$(D^2 - D - 2)y_p = 6x + 6e^{-x}$$

$$\begin{aligned} y_p &= Ax + B + Cxe^{-x} \\ Dy_p &= A + C(-xe^{-x} + e^{-x}) \\ D^2 y_p &= -C(-xe^{-x} + e^{-x}) - Ce^{-x} \end{aligned}$$

Substitute,

$$\begin{aligned} [-C(-xe^{-x} + e^{-x}) - Ce^{-x}] \\ -[A + C(-xe^{-x} + e^{-x})] \\ -2(Ax + B + Cxe^{-x}) = 6x + 6e^{-x} \end{aligned}$$

$$\begin{aligned} \text{Collecting } e^{-x} \text{ terms:} \\ -C - C - C = 6 \\ C = -2 \end{aligned}$$

$$\begin{aligned} \text{Collecting } x \text{ terms:} \\ -2A = 6 \\ A = -3 \end{aligned}$$

$$\begin{aligned} \text{Collecting constant terms:} \\ -A - 2B = 0 \\ -(-3) - 2B = 0 \\ B = 3/2 \end{aligned}$$

Then,

$$y_p = -3x + \frac{3}{2} - 2xe^{-x}$$

Therefore,

$$y = C_1 e^{2x} + C_2 e^{-x} - 3x + \frac{3}{2} - 2xe^{-x}$$

$$33. (D^3 - 3D^2 + 4)y = 6 + 80 \cos 2x$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^3 - 3m^2 + 4 &= 0 \\ (m+1)(m-2)(m-2) &= 0 \\ m &= -1, 2, 2 \end{aligned}$$

Then,

$$y_c = C_1 e^{-x} + e^{2x}(C_2 + C_3 x)$$

For y_p :

Let:

$$(D^3 - 3D^2 + 4)y_p = 6 + 80 \cos 2x$$

$$\begin{aligned} y_p &= A + B \cos 2x + C \sin 2x \\ Dy_p &= -2B \sin 2x + 2C \cos 2x \\ D^2 y_p &= -4B \cos 2x - 4C \sin 2x \\ D^3 y_p &= 8B \sin 2x - 8C \cos 2x \end{aligned}$$

Substitute,

$$\begin{aligned} (8B \sin 2x - 8C \cos 2x) \\ -3(-4B \cos 2x - 4C \sin 2x) \\ +4(A + B \cos 2x + C \sin 2x) \\ = 6 + 80 \cos 2x \end{aligned}$$

$$\begin{aligned} \text{Collecting } \cos 2x \text{ terms:} \\ -8C + 12B + 4B = 80 \\ -8C + 16B = 80 \rightarrow (\text{Eq. 1}) \end{aligned}$$

$$\begin{aligned} \text{Collecting } \sin 2x \text{ terms:} \\ 8B + 12C + 4C = 0 \\ 8B + 16C = 0 \rightarrow (\text{Eq. 2}) \end{aligned}$$

$$\begin{aligned} \text{Collecting constant terms:} \\ 4A = 6 \\ A = 3/2 \end{aligned}$$

Solving two equations with two unknowns:

$$\begin{aligned} B &= 4 \\ C &= -2 \end{aligned}$$

Then,

$$y_p = \frac{3}{2} + 4 \cos 2x - 2 \sin 2x$$

Therefore,

$$\begin{aligned} y = C_1 e^{-x} + e^{2x}(C_2 + C_3 x) + \frac{3}{2} \\ + 4 \cos 2x - 2 \sin 2x \end{aligned}$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

$$(D^3 + D^2 - 4D - 4)y = 8x + 8 + 6e^{-x}$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^3 + m^2 - 4m - 4 &= 0 \\ (m-2)(m+2)(m+1) &= 0 \\ m &= 2, -2, -1 \end{aligned}$$

Then,

$$y_c = C_1 e^{2x} + C_2 e^{-2x} + C_3 e^{-x}$$

For y_p :

Let:

$$(D^3 + D^2 - 4D - 4)y_p = 8x + 8 + 6e^{-x}$$

$$\begin{aligned} y_p &= Ax + B + Cxe^{-x} \\ Dy_p &= A + C(-xe^{-x} + e^{-x}) \\ D^2 y_p &= -C(-xe^{-x} + e^{-x}) - Ce^{-x} \\ D^3 y_p &= C(-xe^{-x} + e^{-x}) + 2Ce^{-x} \end{aligned}$$

Substitute,

$$\begin{aligned} [C(-xe^{-x} + e^{-x}) + 2Ce^{-x}] \\ + [-C(-xe^{-x} + e^{-x}) - Ce^{-x}] \\ -4[A + C(-xe^{-x} + e^{-x})] \\ -4(Ax + B + Cxe^{-x}) = 8x + 8 + 6e^{-x} \end{aligned}$$

$$\begin{aligned} \text{Collecting } e^{-x} \text{ terms:} \\ C + 2C - C - C - 4C = 6 \\ C = -2 \end{aligned}$$

$$\begin{aligned} \text{Collecting } x \text{ terms:} \\ -4A = 8 \\ A = -2 \end{aligned}$$

$$\begin{aligned} \text{Collecting constant terms:} \\ -4A - 4B = 8 \\ -4(-2) - 4B = 8 \\ B = 0 \end{aligned}$$

Then,

$$y_p = -2x - 2xe^{-x}$$

Therefore,

$$\begin{aligned} y &= C_1 e^{2x} + C_2 e^{-2x} + C_3 e^{-x} - 2x \\ &\quad - 2xe^{-x} \\ y &= C_1 \cosh 2x + C_2 \sinh 2x + C_3 e^{-x} \\ &\quad - 2x - 2xe^{-x} \end{aligned}$$

II. In exercises 36 through 44, find the particular solution indicated.

$$36. (D^2 + 1)y = 10e^{2x}, \text{ when } x=0, y=0, y'=0$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 + 1 &= 0 \\ m &= \pm i \end{aligned}$$

Then,

$$y_c = C_1 \cos x + C_2 \sin x$$

For y_p :

Let:

$$(D^2 + 1)y_p = 10e^{2x}$$

$$\begin{aligned} y_p &= Ae^{2x} \\ Dy_p &= 2Ae^{2x} \\ D^2 y_p &= 4Ae^{2x} \end{aligned}$$

Substitute,

$$(4Ae^{2x}) + (Ae^{2x}) = 10e^{2x}$$

$$\begin{aligned} \text{Collecting } e^{2x} \text{ terms:} \\ 4A + A &= 10 \\ A &= 2 \end{aligned}$$

$$\begin{aligned} \text{Then,} \\ y_p &= 2e^{2x} \end{aligned}$$

Therefore,

$$\begin{aligned} y &= C_1 \cos x + C_2 \sin x + 2e^{2x} \\ y' &= -C_1 \sin x + C_2 \cos x + 4e^{2x} \end{aligned}$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

→when $x = 0, y = 0$,

$$0 = C_1 \cos(0) + C_2 \sin(0) + 2e^{2(0)}$$

$$C_1 = -2$$

→when $x = 0, y' = 0$,

$$0 = -C_1 \sin(0) + C_2 \cos(0) + 4e^{2(0)}$$

$$C_2 = -4$$

Thus,

$$y = -2 \cos x - 4 \sin x + 2e^{2x}$$

$$y = 2(e^{2x} - \cos x - 2 \sin x)$$

37. $(D^2 - 4)y = 2 - 8x$
when $x = 0, y = 0, y' = 5$

$$y = y_c + y_p$$

◇ For y_c :

$$m^2 - 4 = 0$$

$$m = 2, -2$$

Then,

$$y_c = C_1 e^{2x} + C_2 e^{-2x}$$

◇ For y_p :

Let:

$$(D^2 - 4)y_p = 2 - 8x$$

$$y_p = A + Bx$$

$$Dy_p = B$$

$$D^2 y_p = 0$$

Substitute,

$$(0) - 4(A + Bx) = 2 - 8x$$

Collecting x terms:

$$-4B = -8$$

$$B = 2$$

Collecting constant terms:

$$-4A = 2$$

$$A = -1/2$$

Then,

$$y_p = -\frac{1}{2} + 2x$$

Therefore,

$$y = C_1 e^{2x} + C_2 e^{-2x} - \frac{1}{2} + 2x$$

and,

$$y' = 2C_1 e^{2x} - 2C_2 e^{-2x} + 2$$

→when $x = 0, y = 0$,

$$0 = C_1 e^{2(0)} + C_2 e^{-2(0)} - \frac{1}{2} + 2(0)$$

$$\frac{1}{2} = C_1 + C_2 \rightarrow (\text{Eq. 1})$$

→when $x = 0, y' = 5$,

$$5 = 2C_1 e^{2(0)} - 2C_2 e^{-2(0)} + 2$$

$$3 = 2C_1 - 2C_2 \rightarrow (\text{Eq. 2})$$

Solving two equations with two unknowns:

$$C_1 = 1$$

$$C_2 = -1/2$$

Thus,

$$y = e^{2x} - \frac{1}{2} e^{-2x} + 2x - \frac{1}{2}$$

38. $(D^2 + 3D)y = -18x$
when $x = 0, y = 0, y' = 5$

$$y = y_c + y_p$$

◇ For y_c :

$$m^2 + 3m = 0$$

$$m(m + 3) = 0$$

$$m = 0, -3$$

Then,

$$y_c = C_1 + C_2 e^{-3x}$$

◇ For y_p :

Let:

$$(D^2 + 3D)y_p = -18x$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

$$y_p = Ax + Bx^2$$

$$Dy_p = A + 2Bx$$

$$D^2 y_p = 2B$$

Substitute,

$$(2B) + 3(A + 2Bx) = -18x$$

Collecting x terms:

$$6B = -18$$

$$B = -3$$

Collecting constant terms:

$$2B + 3A = 0$$

$$2(-3) + 3A = 0$$

$$A = 2$$

Then,

$$y_p = 2x - 3x^2$$

Therefore,

$$y = C_1 + C_2 e^{-3x} + 2x - 3x^2$$

and,

$$y' = -3C_2 e^{-3x} + 2 - 6x$$

→when $x = 0, y = 0$,

$$0 = C_1 + C_2 e^{-3(0)} + 2(0) - 3(0)^2$$

$$0 = C_1 + C_2 \rightarrow (\text{Eq. 1})$$

→when $x = 0, y' = 5$,

$$5 = -3C_2 e^{-3(0)} + 2 - 6(0)$$

$$5 = -3C_2 + 2 \rightarrow (\text{Eq. 2})$$

$$C_2 = -1$$

$$C_1 = 1$$

Thus,

$$y = 1 - e^{-3x} + 2x - 3x^2$$

$$y = 1 + 2x - 3x^2 - e^{-3x}$$

40. $d^2x/dt^2 + 4(dx/dt) + 5x = 10$
when $t = 0, x = 0, dx/dt = 0$

$$(D^2 + 4D + 5)x = 10$$

$$x = x_c + x_p$$

◇ For x_c :

$$m^2 + 4m + 5 = 0$$

$$m = -2 \pm i$$

Then,

$$x_c = C_1 e^{-2i} \cos t + C_2 e^{-2i} \sin t$$

◇ For x_p :

Let:

$$(D^2 + 4D + 5)x_p = 10$$

$$x_p = A$$

$$Dx_p = 0$$

$$D^2 x_p = 0$$

Substitute,

$$(0) + 4(0) + 5(A) = 10$$

$$A = 2$$

Then,

$$x_p = 2$$

Therefore,

$$x = C_1 e^{-2i} \cos t + C_2 e^{-2i} \sin t + 2$$

and,

$$\frac{dx}{dt} = -2C_1 e^{-2i} \sin t + C_2(e^{-2i} \cos t - 2e^{-2i} \sin t)$$

→when $t = 0, x = 0$,

$$0 = C_1 e^{-2(0)} \cos(0) + C_2 e^{-2(0)} \sin(0) + 2$$

$$C_1 = -2$$

→when $t = 0, dx/dt = 0$,

$$0 = -2C_1 e^{-2(0)} \sin(0) + C_2(e^{-2(0)} \cos(0) - 2e^{-2(0)} \sin(0))$$

$$0 = -2C_1 + C_2$$

$$0 = -2(-2) + C_2$$

$$C_2 = -4$$

Thus,

$$x = -2e^{-2i} \cos t - 4e^{-2i} \sin t + 2$$

$$x = 2(1 - e^{-2i} \cos t - 2e^{-2i} \sin t)$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

→when $x = 0, y = 0$,

$$0 = C_1 \cos(0) + C_2 \sin(0) + 2e^{2(0)}$$

$$C_1 = -2$$

→when $x = 0, y' = 0$,

$$0 = -C_1 \sin(0) + C_2 \cos(0) + 4e^{2(0)}$$

$$C_2 = -4$$

Thus,

$$y = -2 \cos x - 4 \sin x + 2e^{2x}$$

$$y = 2(e^{2x} - \cos x - 2 \sin x)$$

37. $(D^2 - 4)y = 2 - 8x$;
when $x = 0, y = 0, y' = 5$

$$y = y_c + y_p$$

◇ For y_c :

$$m^2 - 4 = 0$$

$$m = 2, -2$$

Then,

$$y_c = C_1 e^{2x} + C_2 e^{-2x}$$

◇ For y_p :

Let:

$$(D^2 - 4)y_p = 2 - 8x$$

$$y_p = A + Bx$$

$$Dy_p = B$$

$$D^2 y_p = 0$$

Substitute,

$$(0) - 4(A + Bx) = 2 - 8x$$

Collecting x terms:

$$-4B = -8$$

$$B = 2$$

Collecting constant terms:

$$-4A = 2$$

$$A = -1/2$$

Then,

$$y_p = -\frac{1}{2} + 2x$$

Therefore,

$$y = C_1 e^{2x} + C_2 e^{-2x} - \frac{1}{2} + 2x$$

and,

$$y' = 2C_1 e^{2x} - 2C_2 e^{-2x} + 2$$

→when $x = 0, y = 0$,

$$0 = C_1 e^{2(0)} + C_2 e^{-2(0)} - \frac{1}{2} + 2(0)$$

$$\frac{1}{2} = C_1 + C_2 \rightarrow (\text{Eq. 1})$$

→when $x = 0, y' = 5$,

$$5 = 2C_1 e^{2(0)} - 2C_2 e^{-2(0)} + 2$$

$$3 = 2C_1 - 2C_2 \rightarrow (\text{Eq. 2})$$

Solving two equations with two unknowns:

$$C_1 = 1$$

$$C_2 = -1/2$$

Thus,

$$y = e^{2x} - \frac{1}{2} e^{-2x} + 2x - \frac{1}{2}$$

38. $(D^2 + 3D)y = -18x$;
when $x = 0, y = 0, y' = 5$

$$y = y_c + y_p$$

◇ For y_c :

$$m^2 + 3m = 0$$

$$m(m + 3) = 0$$

$$m = 0, -3$$

Then,

$$y_c = C_1 + C_2 e^{-3x}$$

◇ For y_p :

Let:

$$(D^2 + 3D)y_p = -18x$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

$$y_p = Ax + Bx^2$$

$$Dy_p = A + 2Bx$$

$$D^2 y_p = 2B$$

Substitute,

$$(2B) + 3(A + 2Bx) = -18x$$

Collecting x terms:

$$6B = -18$$

$$B = -3$$

Collecting constant terms:

$$2B + 3A = 0$$

$$2(-3) + 3A = 0$$

$$A = 2$$

Then,

$$y_p = 2x - 3x^2$$

Therefore,

$$y = C_1 + C_2 e^{-3x} + 2x - 3x^2$$

and,

$$y' = -3C_2 e^{-3x} + 2 - 6x$$

→when $x = 0, y = 0$,

$$0 = C_1 + C_2 e^{-3(0)} + 2(0) - 3(0)^2$$

$$0 = C_1 + C_2 \rightarrow (\text{Eq. 1})$$

→when $x = 0, y' = 5$,

$$5 = -3C_2 e^{-3(0)} + 2 - 6(0)$$

$$5 = -3C_2 + 2 \rightarrow (\text{Eq. 2})$$

$$C_2 = -1$$

$$C_1 = 1$$

Thus,

$$y = 1 - e^{-3x} + 2x - 3x^2$$

$$y = 1 + 2x - 3x^2 - e^{-3x}$$

40.

$$d^2 x/dt^2 + 4(dx/dt) + 5x = 10$$

$$\text{when } t = 0, x = 0, dx/dt = 0$$

$$(D^2 + 4D + 5)x = 10$$

$$x = x_c + x_p$$

◇ For x_c :

$$m^2 + 4m + 5 = 0$$

$$m = -2 \pm i$$

Then,

$$x_c = C_1 e^{-2i} \cos t + C_2 e^{-2i} \sin t$$

◇ For x_p :

Let:

$$(D^2 + 4D + 5)x_p = 10$$

$$x_p = A$$

$$Dx_p = 0$$

$$D^2 x_p = 0$$

Substitute,

$$(0) + 4(0) + 5(A) = 10$$

$$A = 2$$

Then,

$$x_p = 2$$

Therefore,

$$x = C_1 e^{-2i} \cos t + C_2 e^{-2i} \sin t + 2$$

and,

$$\frac{dx}{dt} = -2C_1 e^{-2i} \sin t + C_2 (e^{-2i} \cos t - 2e^{-2i} \sin t)$$

→when $t = 0, x = 0$,

$$0 = C_1 e^{-2i(0)} \cos(0) + C_2 e^{-2i(0)} \sin(0) + 2$$

$$C_1 = -2$$

→when $t = 0, dx/dt = 0$,

$$0 = -2C_1 e^{-2i(0)} \sin(0) + C_2 (e^{-2i(0)} \cos(0) - 2e^{-2i(0)} \sin(0))$$

$$0 = -2C_1 + C_2$$

$$0 = -2(-2) + C_2$$

$$C_2 = -4$$

Thus,

$$x = -2e^{-2i} \cos t - 4e^{-2i} \sin t + 2$$

$$x = 2(1 - e^{-2i} \cos t - 2e^{-2i} \sin t)$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

41. $\ddot{x} + 4\dot{x} + 5x = 8 \sin t$,
when $t = 0, x = 0, \dot{x} = 0$

$$(D^2 + 4D + 5)x = 8 \sin t$$

$$x = x_c + x_p$$

◆ For x_c :

$$m^2 + 4m + 5 = 0$$

$$m = -2 \pm i$$

Then,

$$x_c = C_1 e^{-2t} \cos t + C_2 e^{-2t} \sin t$$

◆ For x_p :

Let:

$$(D^2 + 4D + 5)x_p = 8 \sin t$$

$$x_p = A \sin t + B \cos t$$

$$Dx_p = A \cos t - B \sin t$$

$$D^2 x_p = -A \sin t - B \cos t$$

Substitute,

$$(-A \sin t - B \cos t) + 4(A \cos t - B \sin t) + 5(A \sin t + B \cos t) = 8 \sin t$$

Collecting $\sin t$ terms:

$$-A - 4B + 5A = 8$$

$$4A - 4B = 8 \rightarrow (\text{Eq. 1})$$

Collecting $\cos t$ terms:

$$-B + 4A + 5B = 0$$

$$4A + 4B = 0 \rightarrow (\text{Eq. 2})$$

Solving two equations with two unknowns:

$$A = 1$$

$$B = -1$$

Then,

$$x_p = \sin t - \cos t$$

Therefore,

$$x = C_1 e^{-2t} \cos t + C_2 e^{-2t} \sin t + \sin t - \cos t$$

and,

$$\dot{x} = -2C_1 e^{-2t} + C_2(e^{-2t} \cos t - 2e^{-2t} \sin t) + \cos t + \sin t$$

→when $t = 0, x = 0$,

$$0 = C_1 e^{-2(0)} \cos(0) + C_2 e^{-2(0)} \sin(0) + \sin(0) - \cos(0)$$

$$0 = C_1 - 1$$

$$C_1 = 1$$

→when $t = 0, \dot{x} = 0$,

$$0 = -2C_1 e^{-2(0)} + C_2(e^{-2(0)} \cos(0) - 2e^{-2(0)} \sin(0)) + \cos(0) + \sin(0)$$

$$0 = -2C_1 + C_2 + 1$$

$$0 = -2(1) + C_2 + 1$$

$$C_2 = 1$$

Thus,

$$x = e^{-2t} \cos t + e^{-2t} \sin t + \sin t - \cos t$$

$$x = \sin t + e^{-2t} \sin t - \cos t + e^{-2t} \cos t$$

$$x = (1 + e^{-2t}) \sin t - (1 - e^{-2t}) \cos t$$

44. $y'' + 2y' + 5y = 8e^{-x}$
when $x = 0, y = 0, y' = 8$

$$(D^2 + 2D + 5)y = 8e^{-x}$$

$$y = y_c + y_p$$

◆ For y_c :

$$m^2 + 2m + 5 = 0$$

$$m = -1 \pm 2i$$

Then,

$$y_c = C_1 e^{-x} \cos 2x + C_2 e^{-x} \sin 2x$$

◆ For y_p :

Let:

$$(D^2 + 2D + 5)y_p = 8e^{-x}$$

$$y_p = Ae^{-x}$$

$$Dy_p = -Ae^{-x}$$

$$D^2 y_p = Ae^{-x}$$

NONHOMOGENEOUS EQUATIONS: UNDETERMINED COEFFICIENTS

Substitute,

$$Ae^{-x} + 2(-Ae^{-x}) + 5(Ae^{-x}) = 8e^{-x}$$

Collecting e^{-x} terms:

$$A - 2A + 5A = 8$$

$$A = 2$$

Then,

$$y_p = 2e^{-x}$$

Therefore,

$$y = C_1 e^{-x} \cos 2x + C_2 e^{-x} \sin 2x + 2e^{-x}$$

And

$$y' = C_1(-2e^{-x} \sin 2x - e^{-x} \cos 2x) + C_2(2e^{-x} \cos 2x - e^{-x} \sin 2x) - 2e^{-x}$$

→when $x = 0, y = 0$,

$$0 = C_1 e^{-0} \cos 2(0) + C_2 e^{-0} \sin 2(0) + 2e^{-0}$$

$$0 = C_1 + 2$$

$$C_1 = -2$$

→when $x = 0, y' = 8$,

$$8 = C_1(-2e^{-0} \sin 2(0) - e^{-0} \cos 2(0)) + C_2(2e^{-0} \cos 2(0) - e^{-0} \sin 2(0)) - 2e^{-0}$$

$$8 = (-2)(-1) + C_2(2) - 2$$

$$C_2 = 4$$

Thus,

$$y = -2e^{-x} \cos 2x + 4e^{-x} \sin 2x + 2e^{-x}$$

$$y = e^{-x}(2 + 4 \sin 2x - \cos 2x)$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

7.3 SOLUTION BY VARIATION OF PARAMETERS

Consider the differential equation,

$$y'' + q(x)y' + r(x)y = f(x)$$

Assume that $y_1(x)$ and $y_2(x)$ are a fundamental set of solutions for

$$y'' + q(x)y' + r(x)y = 0$$

Then a complimentary solution is,

$$y_c = C_1 y_1(x) + C_2 y_2(x)$$

Replacing both of the constants C_1 and C_2 by A and B respectively with functions of x ,

$$y_p = A y_1(x) + B y_2(x)$$

And then a particular solution to the nonhomogeneous differential equation is,

$$y_p = -y_1 \int \frac{y_2 f(x)}{W(y_1, y_2)} dx + y_2 \int \frac{y_1 f(x)}{W(y_1, y_2)} dx$$

Where,

$$A = - \int \frac{y_2 f(x)}{W(y_1, y_2)} dx$$

$$B = \int \frac{y_1 f(x)}{W(y_1, y_2)} dx$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

Exercise 1 through 9 use variation of parameters.

$$(D^2 - 1)y = e^x + 1$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 - 1 &= 0 \\ (m-1)(m+1) &= 0 \\ m &= 1, -1 \end{aligned}$$

Then,

$$y_c = C_1 e^x + C_2 e^{-x}$$

For y_p :

$$(D^2 - 1)y_p = e^x + 1 \rightarrow (1)$$

Replace C_1 and C_2 by A and B , respectively:
where A and B are the functions of x

$$y_p = A e^x + B e^{-x} \rightarrow (2)$$

Differentiate (2):

$$\begin{aligned} D y_p &= (A e^x + e^x A') + (-B e^{-x} + e^{-x} B') \\ D y_p &= A e^x - B e^{-x} + A' e^x + B' e^{-x} \end{aligned}$$

Set Condition:

$$A' e^x + B' e^{-x} = 0 \rightarrow (3)$$

Then,

$$D y_p = A e^x - B e^{-x} \rightarrow (4)$$

Differentiate (4):

$$\begin{aligned} D^2 y_p &= (A e^x + e^x A') - (-B e^{-x} + e^{-x} B') \\ D^2 y_p &= A e^x + A' e^x + B e^{-x} - B' e^{-x} \end{aligned}$$

Substitute (2) and (5) in (1):

$$\begin{aligned} (A e^x + A' e^x + B e^{-x} - B' e^{-x}) - (A e^x - B e^{-x}) &= e^x + 1 \\ A' e^x - B' e^{-x} &= e^x + 1 \rightarrow (6) \end{aligned}$$

Solving for A' from (3):

$$A' = -\frac{B' e^{-x}}{e^x} \rightarrow (7)$$

Substitute (7) in (6):

$$\begin{aligned} -\frac{B' e^{-x}}{e^x} e^x - B' e^{-x} &= e^x + 1 \\ -B' e^{-x} - B' e^{-x} &= e^x + 1 \\ -2B' e^{-x} &= e^x + 1 \\ B' &= \frac{e^x + 1}{-2e^{-x}} \\ B' &= -\frac{1}{2} e^{2x} - \frac{1}{2} e^x \rightarrow (8) \end{aligned}$$

Then,

$$\begin{aligned} A' &= -\frac{\left(-\frac{1}{2} e^{2x} - \frac{1}{2} e^x\right) e^{-x}}{e^x} \\ A' &= -\frac{\left(-\frac{1}{2} e^{2x} - \frac{1}{2} e^x\right)}{e^{2x}} \\ A' &= \frac{1}{2} + \frac{1}{2} e^{-x} \rightarrow (9) \end{aligned}$$

Integrate (8) and (9):

$$\begin{aligned} \int B' &= -\frac{1}{2} \int e^{2x} - \frac{1}{2} \int e^x \\ B &= -\frac{1}{4} e^{2x} - \frac{1}{2} e^x \end{aligned}$$

and

$$\begin{aligned} \int A' &= \int \frac{1}{2} + \int \frac{1}{2} e^{-x} \\ A &= \frac{1}{2} x - \frac{1}{2} e^{-x} \end{aligned}$$

Substitute value of A and B in (2):

$$\begin{aligned} y_p &= \left(\frac{1}{2} x - \frac{1}{2} e^{-x}\right) e^x + \left(-\frac{1}{4} e^{2x} - \frac{1}{2} e^x\right) e^{-x} \\ y_p &= \left(\frac{1}{2} x e^x - \frac{1}{2}\right) + \left(-\frac{1}{4} e^x - \frac{1}{2}\right) \\ y_p &= \frac{1}{2} x e^x - \frac{1}{4} e^x - 1 \end{aligned}$$

Therefore,

$$y = C_1 e^x + C_2 e^{-x} + \frac{1}{2} x e^x - \frac{1}{4} e^x - 1$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

7.3 SOLUTION BY VARIATION OF PARAMETERS

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Replacing both of the constants C_1 and C_2 by A and B respectively with functions of x ,

$$y_p = A y_1(x) + B y_2(x)$$

And then a particular solution to the nonhomogeneous differential equation is,

$$y_p = -y_1 \int \frac{y_2 f(x)}{W(y_1, y_2)} dx + y_2 \int \frac{y_1 f(x)}{W(y_1, y_2)} dx$$

Where,

$$A = - \int \frac{y_2 f(x)}{W(y_1, y_2)} dx$$

$$B = \int \frac{y_1 f(x)}{W(y_1, y_2)} dx$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

Exercise 1 through 9 use variation of parameters.

$$(D^2 - 1)y = e^x + 1$$

$$y = y_c + y_p$$

For y_c :

$$\begin{aligned} m^2 - 1 &= 0 \\ (m-1)(m+1) &= 0 \\ m &= 1, -1 \end{aligned}$$

Then,

$$y_c = C_1 e^x + C_2 e^{-x}$$

For y_p :

$$(D^2 - 1)y_p = e^x + 1 \rightarrow (1)$$

Replace C_1 and C_2 by A and B , respectively; where A and B are the functions of x

$$y_p = A e^x + B e^{-x} \rightarrow (2)$$

Differentiate (2):

$$\begin{aligned} D y_p &= (A e^x + e^x A') + (-B e^{-x} + e^{-x} B') \\ D y_p &= A e^x - B e^{-x} + A' e^x + B' e^{-x} \end{aligned}$$

Set Condition:

$$A' e^x + B' e^{-x} = 0 \rightarrow (3)$$

Then,

$$D y_p = A e^x - B e^{-x} \rightarrow (4)$$

Differentiate (4):

$$\begin{aligned} D^2 y_p &= (A e^x + e^x A') - (-B e^{-x} + e^{-x} B') \\ D^2 y_p &= A e^x + A' e^x + B e^{-x} - B' e^{-x} \\ &\rightarrow (5) \end{aligned}$$

Substitute (2) and (5) in (1):

$$\begin{aligned} (A e^x + A' e^x + B e^{-x} - B' e^{-x}) - (A e^x - B e^{-x}) &= e^x + 1 \\ A' e^x - B' e^{-x} &= e^x + 1 \rightarrow (6) \end{aligned}$$

Solving for A' from (3):

$$A' = -\frac{B' e^{-x}}{e^x} \rightarrow (7)$$

Substitute (7) in (6):

$$\begin{aligned} -\frac{B' e^{-x}}{e^x} e^x - B' e^{-x} &= e^x + 1 \\ -B' e^{-x} - B' e^{-x} &= e^x + 1 \\ -2B' e^{-x} &= e^x + 1 \\ B' &= \frac{e^x + 1}{-2e^{-x}} \\ B' &= -\frac{1}{2} e^{2x} - \frac{1}{2} e^x \rightarrow (8) \end{aligned}$$

Then,

$$\begin{aligned} A' &= -\left(-\frac{1}{2} e^{2x} - \frac{1}{2} e^x\right) e^{-x} \\ A' &= \left(\frac{1}{2} e^{2x} + \frac{1}{2} e^x\right) e^{-x} \\ A' &= \frac{1}{2} + \frac{1}{2} e^{-x} \rightarrow (9) \end{aligned}$$

Integrate (8) and (9):

$$\begin{aligned} \int B' &= -\frac{1}{2} \int e^{2x} - \frac{1}{2} \int e^x \\ B &= -\frac{1}{4} e^{2x} - \frac{1}{2} e^x \end{aligned}$$

and

$$\begin{aligned} \int A' &= \int \frac{1}{2} + \frac{1}{2} e^{-x} \\ A &= \frac{1}{2} x - \frac{1}{2} e^{-x} \end{aligned}$$

Substitute value of A and B in (2):

$$\begin{aligned} y_p &= \left(\frac{1}{2} x - \frac{1}{2} e^{-x}\right) e^x + \left(-\frac{1}{4} e^{2x} - \frac{1}{2} e^x\right) e^{-x} \\ y_p &= \left(\frac{1}{2} x e^x - \frac{1}{2}\right) + \left(-\frac{1}{4} e^x - \frac{1}{2}\right) \\ y_p &= \frac{1}{2} x e^x - \frac{1}{4} e^x - 1 \end{aligned}$$

Therefore,

$$y = C_1 e^x + C_2 e^{-x} + \frac{1}{2} x e^x - \frac{1}{4} e^x - 1$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

$$2. (D^2 + 1)y = \csc x \cot x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 1 = 0$$

$$m = \pm i$$

Then,

$$y_c = C_1 \cos x + C_2 \sin x$$

For y_p :

$$(D^2 + 1)y_p = \csc x \cot x \rightarrow (1)$$

Replace C_1 and C_2 by A and B , respectively:
where A and B are the functions of x

$$y_p = A \cos x + B \sin x \rightarrow (2)$$

Differentiate (2):

$$Dy_p = (-A \sin x + A' \cos x) + (B \cos x + B' \sin x)$$

$$Dy_p = -A \sin x + B \cos x + A' \cos x + B' \sin x$$

Set Condition:

$$A' \cos x + B' \sin x = 0 \rightarrow (3)$$

Then,

$$Dy_p = -A \sin x + B \cos x \rightarrow (4)$$

Differentiate (4):

$$D^2 y_p = -(A \cos x + A' \sin x) + (-B \sin x + B' \cos x)$$

$$D^2 y_p = -A \cos x - A' \sin x - B \sin x + B' \cos x \rightarrow (5)$$

Substitute (2) and (5) in (1):

$$(-A \cos x - A' \sin x - B \sin x + B' \cos x) + (A \cos x + B \sin x) = \csc x \cot x$$

$$-A' \sin x + B' \cos x = \csc x \cot x \rightarrow (6)$$

Solving for A' from (3):

$$A' = -\frac{B' \sin x}{\cos x} \rightarrow (7)$$

Substitute (7) in (6):

$$\frac{B' \sin x}{\cos x} \sin x + B' \cos x = \csc x \cot x$$

$$B' \sin^2 x + B' \cos^2 x = \frac{\cos^2 x}{\sin^2 x}$$

$$B'(\sin^2 x + \cos^2 x) = \frac{\cos^2 x}{\sin^2 x}$$

$$B' = \frac{\cos^2 x}{\sin^2 x} \rightarrow (8)$$

Then,

$$A' = -\frac{\frac{\cos^2 x}{\sin^2 x} \sin x}{\cos x}$$

$$A' = -\frac{\cos x}{\sin x} \rightarrow (9)$$

Integrate (8) and (9):

$$\int B' = \int \frac{\cos^2 x}{\sin^2 x} dx$$

$$B = \int \cot^2 x dx$$

$$B = \int \cot^2 x dx$$

$$B = \int (\csc^2 x - 1) dx$$

$$B = -\cot x - x$$

and

$$\int A' = -\int \frac{\cos x}{\sin x} dx$$

$$A = -\ln(\sin x)$$

Substitute value of A and B in (2):

$$y_p = -\ln(\sin x) \cos x + (-\cot x - x) \sin x$$

$$y_p = -\ln(\sin x) \cos x - \cos x - x \sin x$$

$$y_p = -\cos x [\ln(\sin x) + 1] - x \sin x$$

Therefore,

$$y = C_1 \cos x + C_2 \sin x - \cos x [\ln(\sin x) + 1] - x \sin x$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

$$(D^2 + 1)y = \csc x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 1 = 0$$

$$m = \pm i$$

Then,

$$y_c = C_1 \cos x + C_2 \sin x$$

For y_p :

$$(D^2 + 1)y_p = \csc x \rightarrow (1)$$

Replace C_1 and C_2 by A and B , respectively:
where A and B are the functions of x

$$y_p = A \cos x + B \sin x \rightarrow (2)$$

Differentiate (2):

$$Dy_p = (-A \sin x + A' \cos x) + (B \cos x + B' \sin x)$$

$$Dy_p = -A \sin x + B \cos x + A' \cos x + B' \sin x$$

Set Condition:

$$A' \cos x + B' \sin x = 0 \rightarrow (3)$$

Then,

$$Dy_p = -A \sin x + B \cos x \rightarrow (4)$$

Differentiate (4):

$$D^2 y_p = -(A \cos x + A' \sin x) + (-B \sin x + B' \cos x)$$

$$D^2 y_p = -A \cos x - A' \sin x - B \sin x + B' \cos x \rightarrow (5)$$

Substitute (2) and (5) in (1):

$$(-A \cos x - A' \sin x - B \sin x + B' \cos x) + (A \cos x + B \sin x) = \csc x$$

$$-A' \sin x + B' \cos x = \csc x \rightarrow (6)$$

Solving for A' from (3):

$$A' = -\frac{B' \sin x}{\cos x} \rightarrow (7)$$

Substitute (7) in (6):

$$\frac{B' \sin x}{\cos x} \sin x + B' \cos x = \csc x$$

$$B' \sin^2 x + B' \cos^2 x = \frac{\cos x}{\sin x}$$

$$B'(\sin^2 x + \cos^2 x) = \frac{\cos x}{\sin x}$$

$$B' = \frac{\cos x}{\sin x} \rightarrow (8)$$

Then,

$$A' = -\frac{\frac{\cos x}{\sin x} \sin x}{\cos x}$$

$$A' = -1 \rightarrow (9)$$

Integrate (8) and (9):

$$\int B' = \int \frac{\cos x}{\sin x} dx$$

$$B = \ln(\sin x)$$

and

$$\int A' = -\int dx$$

$$A = -x$$

Substitute value of A and B in (2):

$$y_p = -x \cos x + \ln(\sin x) \sin x$$

$$y_p = -x \cos x + \sin x \ln(\sin x)$$

Therefore,

$$y = C_1 \cos x + C_2 \sin x - x \cos x + \sin x \ln(\sin x)$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

$$4. \quad (D^2 + 2D + 2)y = e^{-x} \csc x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 2m + 2 = 0$$

$$m = -1 \pm i$$

Then,

$$y_c = C_1 e^{-x} \cos x + C_2 e^{-x} \sin x$$

For y_p :

$$(D^2 + 2D + 2)y_p = e^{-x} \csc x \rightarrow (1)$$

Replace C_1 and C_2 by A and B , respectively, where A and B are the functions of x

$$y_p = Ae^{-x} \cos x + Be^{-x} \sin x \rightarrow (2)$$

Differentiate (2):

$$Dy_p = [-Ae^{-x} \sin x + \cos x (-Ae^{-x} + A'e^{-x})] + [Be^{-x} \cos x + \sin x (-Be^{-x} + B'e^{-x})]$$

$$Dy_p = -Ae^{-x} \sin x - Ae^{-x} \cos x + A'e^{-x} \cos x + Be^{-x} \cos x - Be^{-x} \sin x + B'e^{-x} \sin x$$

$$Dy_p = -Ae^{-x} \sin x - Ae^{-x} \cos x + Be^{-x} \cos x - Be^{-x} \sin x + A'e^{-x} \cos x + B'e^{-x} \sin x$$

Set Condition:

$$A'e^{-x} \cos x + B'e^{-x} \sin x = 0 \rightarrow (3)$$

Then,

$$Dy_p = -Ae^{-x} \sin x - Ae^{-x} \cos x + Be^{-x} \cos x - Be^{-x} \sin x \rightarrow (4)$$

Differentiate (4):

$$\begin{aligned} D^2 y_p &= -[Ae^{-x} \cos x + \sin x (-Ae^{-x} + A'e^{-x})] \\ &\quad -[-Ae^{-x} \sin x + \cos x (-Ae^{-x} + A'e^{-x})] \\ &\quad +[-Be^{-x} \sin x + \cos x (-Be^{-x} + B'e^{-x})] \\ &\quad -[Be^{-x} \cos x + \sin x (-Be^{-x} + B'e^{-x})] \end{aligned}$$

$$\begin{aligned} D^2 y_p &= -Ae^{-x} \cos x + Ae^{-x} \sin x - \\ &\quad A'e^{-x} \sin x + Ae^{-x} \sin x + Ae^{-x} \cos x - \\ &\quad A'e^{-x} \cos x - Be^{-x} \sin x - Be^{-x} \cos x + \\ &\quad B'e^{-x} \cos x - Be^{-x} \cos x + Be^{-x} \sin x - \\ &\quad B'e^{-x} \sin x \end{aligned}$$

$$\begin{aligned} D^2 y_p &= 2Ae^{-x} \sin x - A'e^{-x} \sin x - \\ &\quad A'e^{-x} \cos x - 2Be^{-x} \cos x + B'e^{-x} \cos x - \\ &\quad B'e^{-x} \sin x \rightarrow (5) \end{aligned}$$

Substitute (2) and (5) in (1):

$$\begin{aligned} (2Ae^{-x} \sin x - A'e^{-x} \sin x - A'e^{-x} \cos x - \\ 2Be^{-x} \cos x + B'e^{-x} \cos x - B'e^{-x} \sin x) + 2(-Ae^{-x} \sin x - \\ Ae^{-x} \cos x + Be^{-x} \cos x - Be^{-x} \sin x) + \\ 2(Ae^{-x} \cos x + Be^{-x} \sin x) = e^{-x} \csc x \end{aligned}$$

$$\begin{aligned} (-A'e^{-x} \sin x - A'e^{-x} \cos x + \\ B'e^{-x} \cos x - B'e^{-x} \sin x) = e^{-x} \csc x \end{aligned}$$

$$\begin{aligned} -A'e^{-x}(\sin x + \cos x) \\ + B'e^{-x}(\cos x - \sin x) = e^{-x} \csc x \\ -A'(\sin x + \cos x) + B'(\cos x - \sin x) \\ = \csc x \rightarrow (6) \end{aligned}$$

Solving for A' from (3):

$$A' = -\frac{B' \sin x}{\cos x} \rightarrow (7)$$

Substitute (7) in (6):

$$\begin{aligned} \frac{B' \sin x}{\cos x}(\sin x + \cos x) + B'(\cos x - \sin x) \\ = \csc x \\ B' \sin x(\sin x + \cos x) \\ + B'(\cos x - \sin x) \cos x = \frac{\cos x}{\sin x} \end{aligned}$$

$$\begin{aligned} B'[\sin^2 x + \sin x \cos x + \cos^2 x \\ - \sin x \cos x] = \frac{\cos x}{\sin x} \end{aligned}$$

$$B' = \frac{\cos x}{\sin x} \rightarrow (8)$$

Then,

$$\begin{aligned} A' &= -\frac{\cos x}{\sin x} \sin x \\ A' &= -1 \rightarrow (9) \end{aligned}$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

Integrate (8) and (9):

$$\begin{aligned} B &= \int \frac{\cos x}{\sin x} dx \\ B &= \ln(\sin x) \end{aligned}$$

and

$$\begin{aligned} A' &= -1 \\ A &= -x \end{aligned}$$

Substitute value of A and B in (2):

$$y_p = -xe^{-x} \cos x + \ln(\sin x) e^{-x} \sin x$$

$$(D^2 + 1)y = \sec^3 x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 1 = 0$$

$$m = \pm i$$

Then,

$$y_c = C_1 \cos x + C_2 \sin x$$

For y_p :

$$(D^2 + 1)y_p = \sec^3 x \rightarrow (1)$$

Replace C_1 and C_2 by A and B , respectively, where A and B are the functions of x

$$y_p = A \cos x + B \sin x \rightarrow (2)$$

Differentiate (2):

$$\begin{aligned} Dy_p &= (-A \sin x + A' \cos x) \\ &\quad + (B \cos x + B' \sin x) \\ Dy_p &= -A \sin x + B \cos x + A' \cos x \\ &\quad + B' \sin x \end{aligned}$$

Set Condition:

$$A' \cos x + B' \sin x = 0 \rightarrow (3)$$

Then,

$$Dy_p = -A \sin x + B \cos x \rightarrow (4)$$

Differentiate (4):

$$\begin{aligned} D^2 y_p &= -(A \cos x + A' \sin x) \\ &\quad + (-B \sin x + B' \cos x) \\ D^2 y_p &= -A \cos x - A' \sin x \\ &\quad - B \sin x + B' \cos x \rightarrow (5) \end{aligned}$$

Substitute (2) and (5) in (1):

$$\begin{aligned} (-A \cos x - A' \sin x - B \sin x + B' \cos x) \\ + (A \cos x + B \sin x) = \sec^3 x \\ -A' \sin x + B' \cos x = \sec^3 x \rightarrow (6) \end{aligned}$$

Solving for A' from (3):

$$A' = -\frac{B' \sin x}{\cos x} \rightarrow (7)$$

Substitute (7) in (6):

$$\begin{aligned} \frac{B' \sin x}{\cos x} \sin x + B' \cos x &= \sec^3 x \\ \frac{B' \sin^2 x}{\cos x} + B' \cos x &= \sec^2 x \\ B'(\sin^2 x + \cos^2 x) &= \sec^2 x \\ B' &= \sec^2 x \rightarrow (8) \end{aligned}$$

Then,

$$\begin{aligned} A' &= -\frac{\sec^2 x \sin x}{\cos x} \\ A' &= -\frac{\sin x}{\cos^3 x} \rightarrow (9) \end{aligned}$$

Integrate (8) and (9):

$$\begin{aligned} B &= \int \sec^2 x dx \\ B &= \tan x \end{aligned}$$

and

$$\begin{aligned} A' &= -\frac{\sin x}{\cos^3 x} \\ A &= -\frac{1}{2 \cos^2 x} \end{aligned}$$

Substitute value of A and B in (2):

$$\begin{aligned} y_p &= -\frac{1}{2 \cos^2 x} \cos x + \tan x \sin x \\ y_p &= -\frac{1}{2 \cos x} + \tan x \sin x \end{aligned}$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

$$y_p = -\frac{1}{2} \sec x + \tan x \sin x$$

Therefore,

$$y = C_1 \cos x + C_2 \sin x - \frac{1}{2} \sec x + \tan x \sin x$$

6. $(D^2 + 1)y = \sec^4 x$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 1 = 0$$

$$m = \pm i$$

Then,

$$y_c = C_1 \cos x + C_2 \sin x$$

For y_p :

Replacing C_1 and C_2 by A and B , respectively, where A and B are a function of x

$$y_p = A \cos x + B \sin x \rightarrow (1)$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$W(\cos x, \sin x) = \cos x \cos x + \sin x \sin x$$

$$W(\cos x, \sin x) = \cos^2 x + \sin^2 x = 1$$

Solving for A and B :

$$A = - \int \frac{y_2 f(x)}{W(y_1, y_2)} dx$$

$$A = - \int \frac{\sin x \sec^4 x}{1} dx$$

$$A = - \int \sin x dx$$

$$A = - \frac{1}{\cos^3 x}$$

And,

$$B = \int \frac{y_1 f(x)}{W(y_1, y_2)} dx$$

$$B = \int \frac{\cos x \sec^4 x}{1} dx$$

$$B = \int \sec^3 x dx$$

$$B = \frac{\ln(\tan x + \sec x) + \sec x \tan x}{2}$$

Substitute value of A and B in (1):

$$y_p = -\frac{1}{3 \cos^3 x} \cos x + \frac{\ln(\tan x + \sec x) + \sec x \tan x}{2} \sin x$$

$$y_p = -\frac{1}{3 \cos^2 x} + \frac{1}{2} \sec x \tan x \sin x + \frac{1}{2} \sin x \ln(\tan x + \sec x)$$

$$y_p = -\frac{1}{3 \cos^2 x} + \frac{1}{2} \frac{\sin^2 x}{\cos^2 x} + \frac{1}{2} \sin x \ln(\tan x + \sec x)$$

$$y_p = -\frac{1}{3 \cos^2 x} + \frac{1}{2} \left(\frac{1 - \cos^2 x}{\cos^2 x} \right) + \frac{1}{2} \sin x \ln(\tan x + \sec x)$$

$$y_p = -\frac{1}{3 \cos^2 x} + \frac{1}{2 \cos^2 x} - \frac{1}{2} + \frac{1}{2} \sin x \ln(\tan x + \sec x)$$

$$y_p = \frac{1}{6 \cos^2 x} - \frac{1}{2} + \frac{1}{2} \sin x \ln(\tan x + \sec x)$$

$$y_p = -\frac{1}{2} + \frac{1}{6} \sec^2 x + \frac{1}{2} \sin x \ln(\tan x + \sec x)$$

Therefore,

$$y = y_c - \frac{1}{2} + \frac{1}{6} \sec^2 x + \frac{1}{2} \sin x \ln(\tan x + \sec x)$$

NONHOMOGENEOUS EQUATIONS: VARIATION OF PARAMETERS

$$(D^2 + 1)y = \tan x$$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 1 = 0$$

$$m = \pm i$$

Then,

$$y_c = C_1 \cos x + C_2 \sin x$$

For y_p :

Replacing C_1 and C_2 by A and B , respectively, where A and B are a function of x

$$y_p = A \cos x + B \sin x \rightarrow (1)$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$W(\cos x, \sin x) = \cos x \cos x + \sin x \sin x$$

$$W(\cos x, \sin x) = \cos^2 x + \sin^2 x = 1$$

Solving for A and B :

$$A = - \int \frac{y_2 f(x)}{W(y_1, y_2)} dx$$

$$A = - \int \frac{\sin x \tan x}{1} dx$$

$$A = - \int \sec x \sin^2 x dx$$

$$A = - \int \sec x (1 - \cos^2 x) dx$$

$$A = - \int (\sec x - \cos x) dx$$

$$A = - \ln(\tan x + \sec x) + \sin x$$

And,

$$B = \int \frac{y_1 f(x)}{W(y_1, y_2)} dx$$

$$B = \int \frac{\cos x \tan x}{1} dx$$

$$B = \int \sin x dx$$

$$B = -\cos x$$

Substitute value of A and B in (1):

$$y_p = [-\ln(\tan x + \sec x) + \sin x] \cos x - \cos x \sin x$$

$$y_p = [-\ln(\tan x + \sec x) + \sin x] \cos x - \cos x \sin x$$

$$y_p = -\cos x \ln(\tan x + \sec x)$$

Therefore,

$$y = y_c - \cos x \ln(\tan x + \sec x)$$

8. $(D^2 + 1)y = \tan^2 x$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 1 = 0$$

$$m = \pm i$$

Then,

$$y_c = C_1 \cos x + C_2 \sin x$$

For y_p :

Replacing C_1 and C_2 by A and B , respectively, where A and B are a function of x

$$y_p = A \cos x + B \sin x \rightarrow (1)$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$W(\cos x, \sin x) = \cos x \cos x + \sin x \sin x$$

$$W(\cos x, \sin x) = \cos^2 x + \sin^2 x = 1$$

Solving for A and B :

$$A = - \int \frac{y_2 f(x)}{W(y_1, y_2)} dx$$

$$A = - \int \frac{\sin x \tan^2 x}{1} dx$$

$$A = - \int \sin x \tan^2 x dx$$

$$A = - \int \sin x (\sec^2 x - 1) dx$$

$$A = - \int \sec x \tan x \, dx + \int \sin x \, dx$$

$$A = - \sec x - \cos x$$

And,

$$B = \int \frac{y_1 f(x)}{W(y_1, y_2)} \, dx$$

$$B = \int \frac{\cos x \tan^2 x}{1} \, dx$$

$$B = \int \cos x \tan^2 x \, dx$$

$$B = \int \cos x (\sec^2 x - 1) \, dx$$

$$B = \int \sec x \, dx - \int \cos x \, dx$$

$$B = \ln(\tan x + \sec x) - \sin x$$

Substitute value of A and B in (1):

$$y_p = (-\sec x - \cos x) \cos x$$

$$+ [\ln(\tan x + \sec x) - \sin x] \sin x$$

$$y_p = -1 - \cos^2 x - \sin^2 x$$

$$+ \sin x \ln(\tan x + \sec x)$$

$$y_p = -1 - (\cos^2 x + \sin^2 x)$$

$$+ \sin x \ln(\tan x + \sec x)$$

$$y_p = -1 - 1 + \sin x \ln(\tan x + \sec x)$$

$$y_p = -2 + \sin x \ln(\tan x + \sec x)$$

Therefore,

$$y = y_c - 2 + \sin x \ln(\tan x + \sec x)$$

9. $(D^2 + 1)y = \sec x \csc x$

$$y = y_c + y_p$$

For y_c :

$$m^2 + 1 = 0$$

$$m = \pm i$$

Then,

$$y_c = C_1 \cos x + C_2 \sin x$$

For y_p :

Replacing C_1 and C_2 by A and B, respectively, where A and B are a function of x

$$y_p = A \cos x + B \sin x \rightarrow (1)$$

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

$$W(\cos x, \sin x) = \begin{vmatrix} \cos x & \sin x \\ -\sin x & \cos x \end{vmatrix}$$

$$W(\cos x, \sin x) = \cos x \cos x + \sin x \sin x$$

$$W(\cos x, \sin x) = \cos^2 x + \sin^2 x = 1$$

Solving for A and B:

$$A = - \int \frac{y_2 f(x)}{W(y_1, y_2)} \, dx$$

$$A = - \int \frac{\sin x \sec x \csc x}{1} \, dx$$

$$A = - \int \sec x \, dx$$

$$A = - \ln(\tan x + \sec x)$$

And,

$$B = \int \frac{y_1 f(x)}{W(y_1, y_2)} \, dx$$

$$B = \int \frac{\cos x \sec x \csc x}{1} \, dx$$

$$B = \int \csc x \, dx$$

$$B = - \ln(\csc x + \cot x)$$

Substitute value of A and B in (1):

$$y_p = [- \ln(\tan x + \sec x)] \cos x$$

$$- \ln(\csc x + \cot x) \sin x$$

$$y_p = - \cos x \ln(\tan x + \sec x)$$

$$- \sin x \ln(\csc x + \cot x)$$

Therefore,

$$y = y_c - \cos x \ln(\tan x + \sec x)$$

$$- \sin x \ln(\csc x + \cot x)$$

CHAPTER 8: THE LAPLACE TRANSFORM

8.1 THE TRANSFORM CONCEPT

Let $F(t)$ be any function such that the integrations encountered may be legitimately performed on $F(t)$. The Laplace transform of $F(t)$ is denoted by $\mathcal{L}\{F(t)\}$ and is defined by

$$\mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) \, dt \rightarrow (8.1)$$

The integral in (Eq 8.1) is a function of the parameter s ; call that function $f(s)$. We may write

$$\mathcal{L}\{F(t)\} = \int_0^{\infty} e^{-st} F(t) \, dt = f(s) \rightarrow (8.2)$$

It is customary to refer to $f(s)$, as well as to the symbol $\mathcal{L}\{F(t)\}$, as the transform, or the Laplace transform, of $F(t)$. We may also look upon (Eq 8.2) as a definition of a Laplace operator $\mathcal{L}$, which transforms each function $F(t)$ of a certain set of functions into some function $f(s)$.

It is important that the operator $\mathcal{L}$, like the differential operator D , is a linear operator. If $F_1(t)$ and $F_2(t)$ have Laplace transforms and if c_1 and c_2 are any constants,

$$\mathcal{L}\{c_1 F_1(t) + c_2 F_2(t)\} = c_1 \mathcal{L}\{F_1(t)\} + c_2 \mathcal{L}\{F_2(t)\}$$

8.2 TRANSFORMS OF ELEMENTARY FUNCTIONS

The transforms of certain exponential and trigonometric functions and of polynomials will now be obtained. These results enter our work frequently.

Common Functions $F(t)$ and Their Laplace Transforms $\mathcal{L}\{F(t)\}$			
$F(t)$	$\mathcal{L}\{F(t)\}$	$F(t)$	$\mathcal{L}\{F(t)\}$
1	$\frac{1}{s}$	$\cos \omega t$	$\frac{s}{s^2 + \omega^2}$
t	$\frac{1}{s^2}$	$\sin \omega t$	$\frac{\omega}{s^2 + \omega^2}$
t^2	$\frac{2}{s^3}$	$\cosh at$	$\frac{s}{s^2 - a^2}$
t^n ($n = 0, 1, \dots$)	$\frac{n!}{s^{n+1}}$	$\sinh at$	$\frac{a}{s^2 - a^2}$
e^{at}	$\frac{1}{s - a}$	$e^{at} \cos \omega t$	$\frac{s - a}{(s - a)^2 + \omega^2}$
$t^n e^{at}$	$\frac{n!}{(s - a)^{n+1}}$	$e^{at} \sin \omega t$	$\frac{\omega}{(s - a)^2 + \omega^2}$

EXERCISES:

1. Solve the Laplace Transform of the following elementary functions:

1. Show that $\mathcal{L}\{1\} = 1/s$

$$\begin{aligned}\mathcal{L}\{F(t)\} &= \int_0^{\infty} e^{-st} F(t) dt \\ \mathcal{L}\{1\} &= \int_0^{\infty} e^{-st} (1) dt \\ \mathcal{L}\{1\} &= -\frac{1}{s} [e^{-st}]_0^{\infty} \\ \mathcal{L}\{1\} &= -\frac{1}{s} [e^{-s(\infty)} - e^{-s(0)}] \\ \mathcal{L}\{1\} &= \frac{1}{s}\end{aligned}$$

2. Show that $\mathcal{L}\{t\} = 1/s^2$

$$\begin{aligned}\mathcal{L}\{F(t)\} &= \int_0^{\infty} e^{-st} F(t) dt \\ \mathcal{L}\{t\} &= \int_0^{\infty} e^{-st} t dt\end{aligned}$$

Using Integration by Parts:

Let:

$$\begin{aligned}u &= t & dv &= e^{-st} dt \\ du &= dt & v &= -\frac{1}{s} e^{-st}\end{aligned}$$

$$\begin{aligned}\mathcal{L}\{t\} &= -\left[\frac{t}{s} e^{-st}\right]_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt \\ \mathcal{L}\{t\} &= -\left[\frac{t}{s} e^{-st}\right]_0^{\infty} - \left[\frac{1}{s^2} e^{-st}\right]_0^{\infty} \\ \mathcal{L}\{t\} &= -\left[\frac{0}{s} e^{-s(\infty)} - \frac{0}{s} e^{-s(0)}\right] \\ &\quad - \left[\frac{1}{s^2} e^{-s(\infty)} - \frac{1}{s^2} e^{-s(0)}\right] \\ \mathcal{L}\{t\} &= 0 - \left[0 - \frac{1}{s^2}\right] \\ \mathcal{L}\{t\} &= \frac{1}{s^2}\end{aligned}$$

3. Show that $\mathcal{L}\{t^2\} = 2/s^3$

$$\begin{aligned}\mathcal{L}\{F(t)\} &= \int_0^{\infty} e^{-st} F(t) dt \\ \mathcal{L}\{t^2\} &= \int_0^{\infty} e^{-st} t^2 dt\end{aligned}$$

Considering:

$$\int_0^{\infty} e^{-st} t^2 dt$$

Using Integration by Parts:

Let:

$$\begin{aligned}u &= t^2 & dv &= e^{-st} dt \\ du &= 2t dt & v &= -\frac{1}{s} e^{-st}\end{aligned}$$

Then,

$$\int_0^{\infty} e^{-st} t^2 dt = -\left[\frac{t^2}{2} e^{-st}\right]_0^{\infty} + \frac{2}{s} \int_0^{\infty} t e^{-st} dt$$

Considering:

$$\int_0^{\infty} t e^{-st} dt$$

Let:

$$\begin{aligned}u &= t & dv &= e^{-st} dt \\ du &= dt & v &= -\frac{1}{s} e^{-st}\end{aligned}$$

Then,

$$\begin{aligned}\int_0^{\infty} e^{-st} t^2 dt &= -\left[\frac{t^2}{2} e^{-st}\right]_0^{\infty} \\ &\quad + \frac{2}{s} \left[-\frac{t}{s} e^{-st}\right]_0^{\infty} + \frac{1}{s} \int_0^{\infty} e^{-st} dt\end{aligned}$$

$$\begin{aligned}\int_0^{\infty} e^{-st} t^2 dt &= -\left[\frac{t^2}{2} e^{-st}\right]_0^{\infty} - \frac{2}{s} \left[\frac{t}{s} e^{-st}\right]_0^{\infty} \\ &\quad - \left[\frac{2}{s^2} e^{-st}\right]_0^{\infty}\end{aligned}$$

$$\int_0^{\infty} e^{-st} t^2 dt = -[0] - \frac{2}{s} [0] - \left[0 - \frac{2}{s^3}\right]$$

$$\begin{aligned}\int_0^{\infty} e^{-st} t^2 dt &= -[0] - \frac{2}{s} [0] - \left[0 - \frac{2}{s^3}\right] \\ &= \frac{2}{s^3}\end{aligned}$$

Thus,

$$\mathcal{L}\{t^2\} = \frac{2}{s^3}$$

$\mathcal{L}\{e^{at}\}$

$$\begin{aligned}\mathcal{L}\{F(t)\} &= \int_0^{\infty} e^{-st} F(t) dt \\ \mathcal{L}\{e^{at}\} &= \int_0^{\infty} e^{-st} e^{at} dt \\ \mathcal{L}\{e^{at}\} &= \int_0^{\infty} e^{-(s-a)t} dt \\ \mathcal{L}\{e^{at}\} &= -\left[\frac{1}{s-a} e^{-(s-a)t}\right]_0^{\infty} \\ \mathcal{L}\{e^{at}\} &= -\left[0 - \frac{1}{s-a}\right] \\ \mathcal{L}\{e^{at}\} &= \frac{1}{s-a}\end{aligned}$$

5. Show that $\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}$

$$\begin{aligned}\mathcal{L}\{F(t)\} &= \int_0^{\infty} e^{-st} F(t) dt \\ \mathcal{L}\{\cos \omega t\} &= \int_0^{\infty} e^{-st} \cos \omega t dt\end{aligned}$$

Using Integration by Parts twice:

Considering:

$$\int_0^{\infty} e^{-st} \cos \omega t dt$$

Let:

$$\begin{aligned}u &= \cos \omega t & dv &= e^{-st} dt \\ du &= -\omega \sin \omega t dt & v &= -\frac{1}{s} e^{-st}\end{aligned}$$

Then,

$$\begin{aligned}\int_0^{\infty} e^{-st} \cos \omega t dt &= -\left[\frac{1}{s} e^{-st} \cos \omega t\right]_0^{\infty} \\ &\quad - \frac{\omega}{s} \int_0^{\infty} e^{-st} \sin \omega t dt\end{aligned}$$

Considering:

$$\int_0^{\infty} e^{-st} \sin \omega t dt$$

Let:

$$\begin{aligned}u &= \sin \omega t & dv &= e^{-st} dt \\ du &= \omega \cos \omega t dt & v &= -\frac{1}{s} e^{-st}\end{aligned}$$

Then,

$$\begin{aligned}\int_0^{\infty} e^{-st} \cos \omega t dt &= -\left[\frac{1}{s} e^{-st} \cos \omega t\right]_0^{\infty} \\ &\quad - \frac{\omega}{s} \left[-\frac{1}{s} e^{-st} \sin \omega t\right]_0^{\infty} \\ &\quad + \frac{\omega}{s} \int_0^{\infty} e^{-st} \sin \omega t dt \\ \left(1 + \frac{\omega^2}{s^2}\right) \int_0^{\infty} e^{-st} \cos \omega t dt &= -\left[\frac{1}{s} e^{-st} \cos \omega t\right]_0^{\infty} + \frac{\omega}{s} \left[\frac{1}{s} e^{-st} \sin \omega t\right]_0^{\infty}\end{aligned}$$

$$\begin{aligned}\left(1 + \frac{\omega^2}{s^2}\right) \int_0^{\infty} e^{-st} \cos \omega t dt &= -\left[-\frac{1}{s}\right] + \frac{\omega}{s} [0]\end{aligned}$$

$$\begin{aligned}\int_0^{\infty} e^{-st} \cos \omega t dt &= \frac{1/s}{\left(1 + \frac{\omega^2}{s^2}\right)} \\ \int_0^{\infty} e^{-st} \cos \omega t dt &= \frac{s}{s^2 + \omega^2}\end{aligned}$$

Thus,

$$\mathcal{L}\{\cos \omega t\} = \frac{s}{s^2 + \omega^2}$$

THE LAPLACE TRANSFORM

5. Show that $\mathcal{L}(\sin \omega t) = \frac{\omega}{s^2 + \omega^2}$

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st} F(t) dt$$

$$\mathcal{L}(\sin \omega t) = \int_0^\infty e^{-st} \sin \omega t dt$$

Using Integration by Parts twice:

Considering:

$$\int_0^\infty e^{-st} \sin \omega t dt$$

Let:

$$u = \sin \omega t \quad dv = e^{-st} dt$$

$$du = \omega \cos \omega t dt \quad v = -\frac{1}{s} e^{-st}$$

Then,

$$\int_0^\infty e^{-st} \sin \omega t dt$$

$$= -\left[\frac{1}{s} e^{-st} \sin \omega t\right]_0^\infty + \frac{\omega}{s} \int_0^\infty e^{-st} \cos \omega t dt$$

Considering:

$$\int_0^\infty e^{-st} \cos \omega t dt$$

Let:

$$u = \cos \omega t \quad dv = e^{-st} dt$$

$$du = -\omega \sin \omega t dt \quad v = -\frac{1}{s} e^{-st}$$

Then,

$$\int_0^\infty e^{-st} \sin \omega t dt$$

$$= -\left[\frac{1}{s} e^{-st} \sin \omega t\right]_0^\infty$$

$$+ \frac{\omega}{s} \left[-\frac{1}{s} e^{-st} \cos \omega t\right]_0^\infty$$

$$- \frac{\omega}{s} \int_0^\infty e^{-st} \sin \omega t dt$$

$$\left(1 + \frac{\omega^2}{s^2}\right) \int_0^\infty e^{-st} \sin \omega t dt$$

$$= -\left[\frac{1}{s} e^{-st} \sin \omega t\right]_0^\infty - \frac{\omega}{s} \left[\frac{1}{s} e^{-st} \cos \omega t\right]_0^\infty$$

$$\left(1 + \frac{\omega^2}{s^2}\right) \int_0^\infty e^{-st} \sin \omega t dt$$

$$= -[0] - \frac{\omega}{s} \left[0 - \frac{1}{s}\right]$$

$$\int_0^\infty e^{-st} \sin \omega t dt = \frac{\frac{\omega}{s^2}}{\left(1 + \frac{\omega^2}{s^2}\right)}$$

$$\int_0^\infty e^{-st} \sin \omega t dt = \frac{\omega}{s^2 + \omega^2}$$

Thus,

$$\mathcal{L}(\sin \omega t) = \frac{\omega}{s^2 + \omega^2}$$

6.

Show that $\mathcal{L}(\cosh at) = \frac{s}{s^2 - a^2}$

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st} F(t) dt$$

$$\mathcal{L}(\cosh at) = \int_0^\infty e^{-st} \cosh at dt$$

Using Integration by Parts twice:

Considering:

$$\int_0^\infty e^{-st} \cosh at dt$$

Let:

$$u = \cosh at \quad dv = e^{-st} dt$$

$$du = a \sinh at dt \quad v = -\frac{1}{s} e^{-st}$$

Then,

$$\int_0^\infty e^{-st} \cosh at dt$$

$$= -\left[\frac{1}{s} e^{-st} \cosh at\right]_0^\infty$$

$$+ \frac{a}{s} \int_0^\infty e^{-st} \sinh at dt$$

THE LAPLACE TRANSFORM

Considering:

$$\int_0^\infty e^{-st} \sinh at dt$$

Let:

$$u = \sinh at \quad dv = e^{-st} dt$$

$$du = a \cosh at dt \quad v = -\frac{1}{s} e^{-st}$$

Then,

$$\int_0^\infty e^{-st} \cosh at dt$$

$$= -\left[\frac{1}{s} e^{-st} \cosh at\right]_0^\infty$$

$$+ \frac{a}{s} \left[-\frac{1}{s} e^{-st} \sinh at\right]_0^\infty$$

$$+ \frac{a}{s} \int_0^\infty e^{-st} \cosh at dt$$

$$\left(1 - \frac{a^2}{s^2}\right) \int_0^\infty e^{-st} \cosh at dt$$

$$= -\left[\frac{1}{s} e^{-st} \cosh at\right]_0^\infty - \frac{a}{s} \left[\frac{1}{s} e^{-st} \sinh at\right]_0^\infty$$

$$\left(1 - \frac{a^2}{s^2}\right) \int_0^\infty e^{-st} \cosh at dt$$

$$= -\left[-\frac{1}{s}\right] - \frac{a}{s} [0]$$

$$\int_0^\infty e^{-st} \cosh at dt = \frac{1/s}{\left(1 - \frac{a^2}{s^2}\right)}$$

$$\int_0^\infty e^{-st} \cosh at dt = \frac{s}{s^2 - a^2}$$

Thus,

$$\mathcal{L}(\cosh at) = \frac{s}{s^2 - a^2}$$

7.

Show that $\mathcal{L}(\sinh at) = \frac{a}{s^2 - a^2}$

$$\mathcal{L}\{F(t)\} = \int_0^\infty e^{-st} F(t) dt$$

$$\mathcal{L}(\sinh at) = \int_0^\infty e^{-st} \sinh at dt$$

Using Integration by Parts twice:

Considering:

$$\int_0^\infty e^{-st} \sinh at dt$$

Let:

$$u = \sinh at \quad dv = e^{-st} dt$$

$$du = a \cosh at dt \quad v = -\frac{1}{s} e^{-st}$$

$$\int_0^\infty e^{-st} \sinh at dt$$

$$= -\left[\frac{1}{s} e^{-st} \sinh at\right]_0^\infty$$

$$+ \frac{a}{s} \int_0^\infty e^{-st} \cosh at dt$$

Considering:

$$\int_0^\infty e^{-st} \cosh at dt$$

Let:

$$u = \cosh at \quad dv = e^{-st} dt$$

$$du = a \sinh at dt \quad v = -\frac{1}{s} e^{-st}$$

Then,

$$\int_0^\infty e^{-st} \sinh at dt$$

$$= -\left[\frac{1}{s} e^{-st} \sinh at\right]_0^\infty$$

$$+ \frac{a}{s} \left[-\frac{1}{s} e^{-st} \cosh at\right]_0^\infty$$

$$+ \frac{a}{s} \int_0^\infty e^{-st} \sinh at dt$$

$$\left(1 - \frac{a^2}{s^2}\right) \int_0^\infty e^{-st} \sinh at dt$$

$$= -\left[\frac{1}{s} e^{-st} \sinh at\right]_0^\infty - \frac{a}{s} \left[\frac{1}{s} e^{-st} \cosh at\right]_0^\infty$$

$$\left(1 - \frac{a^2}{s^2}\right) \int_0^\infty e^{-st} \sinh at dt$$

$$= -[0] - \frac{a}{s} \left[-\frac{1}{s}\right]$$

THE LAPLACE TRANSFORM

$$\int_0^{\infty} e^{-st} \sinh at \, dt = \frac{a/s^2}{(1 - \frac{a^2}{s^2})}$$

$$\int_0^{\infty} e^{-st} \sinh at \, dt = \frac{a}{s^2 - a^2}$$

Thus,

$$\mathcal{L}(\sinh at) = \frac{a}{s^2 - a^2}$$

8. Show that $\mathcal{L}(e^{at} \cos \omega t) = \frac{s-a}{(s-a)^2 + \omega^2}$

$$\mathcal{L}(F(t)) = \int_0^{\infty} e^{-st} F(t) \, dt$$

$$\mathcal{L}(e^{at} \cos \omega t) = \int_0^{\infty} e^{-st} e^{at} \cos \omega t \, dt$$

$$\mathcal{L}(e^{at} \cos \omega t) = \int_0^{\infty} e^{-(s-a)t} \cos \omega t \, dt$$

Using Integration by Parts twice:

Considering:

$$\int_0^{\infty} e^{-(s-a)t} \cos \omega t \, dt$$

Let:

$$u = \cos \omega t \quad dv = e^{-(s-a)t} \, dt$$

$$du = -\omega \sin \omega t \, dt \quad v = -\frac{1}{s-a} e^{-(s-a)t}$$

Then,

$$\int_0^{\infty} e^{-(s-a)t} \cos \omega t \, dt$$

$$= -\left[\frac{1}{s-a} e^{-(s-a)t} \cos \omega t \right]_0^{\infty}$$

$$- \frac{\omega}{s-a} \int_0^{\infty} e^{-(s-a)t} \sin \omega t \, dt$$

Considering:

$$\int_0^{\infty} e^{-(s-a)t} \sin \omega t \, dt$$

Let:

$$u = \sin \omega t \quad dv = e^{-(s-a)t} \, dt$$

$$du = \omega \cos \omega t \, dt \quad v = -\frac{1}{s-a} e^{-(s-a)t}$$

Then,

$$\int_0^{\infty} e^{-(s-a)t} \cos \omega t \, dt$$

$$= -\left[\frac{1}{s-a} e^{-(s-a)t} \cos \omega t \right]_0^{\infty}$$

$$- \frac{\omega}{s-a} \left[-\left[\frac{1}{s-a} e^{-(s-a)t} \sin \omega t \right]_0^{\infty} \right]$$

$$+ \frac{\omega}{s-a} \int_0^{\infty} e^{-(s-a)t} \cos \omega t \, dt$$

$$\left(1 + \frac{\omega^2}{(s-a)^2} \right) \int_0^{\infty} e^{-(s-a)t} \cos \omega t \, dt$$

$$= -\left[\frac{1}{s-a} e^{-(s-a)t} \cos \omega t \right]_0^{\infty}$$

$$+ \frac{\omega}{s-a} \left[\frac{1}{s-a} e^{-(s-a)t} \sin \omega t \right]_0^{\infty}$$

$$\left(1 + \frac{\omega^2}{(s-a)^2} \right) \int_0^{\infty} e^{-(s-a)t} \cos \omega t \, dt$$

$$= -\left[-\frac{1}{s-a} \right] + \frac{\omega}{s-a} [0]$$

$$\int_0^{\infty} e^{-(s-a)t} \cos \omega t \, dt = \frac{\frac{1}{s-a}}{\left(1 + \frac{\omega^2}{(s-a)^2} \right)}$$

$$\int_0^{\infty} e^{-(s-a)t} \cos \omega t \, dt = \frac{s-a}{(s-a)^2 + \omega^2}$$

Thus,

$$\mathcal{L}(e^{at} \cos \omega t) = \frac{s-a}{s^2 - 2as + a^2 + \omega^2}$$

THE LAPLACE TRANSFORM

9. Find the Laplace transform of the following:

Obtain $\mathcal{L}(t^2 + 4t - 5)$

$$\mathcal{L}(t^2 + 4t - 5) = \mathcal{L}(t^2) + 4\mathcal{L}(t) - 5\mathcal{L}(1)$$

From the Table:

$$\mathcal{L}(t^2 + 4t - 5) = \frac{2!}{s^3} + 4 \frac{1!}{s^2} - 5 \frac{1}{s}$$

$$\mathcal{L}(t^2 + 4t - 5) = \frac{2}{s^3} + \frac{4}{s^2} - \frac{5}{s}$$

Obtain $\mathcal{L}(t^3 - t^2 + 4t)$

$$\mathcal{L}(t^3 - t^2 + 4t) = \mathcal{L}(t^3) - \mathcal{L}(t^2) + 4\mathcal{L}(t)$$

From the Table:

$$\mathcal{L}(t^3 - t^2 + 4t) = \frac{3!}{s^4} - \frac{2!}{s^3} + 4 \frac{1!}{s^2}$$

$$\mathcal{L}(t^3 - t^2 + 4t) = \frac{6}{s^4} - \frac{2}{s^3} + \frac{4}{s^2}$$

10.

Evaluate $\mathcal{L}(e^{-2t} + 4e^{-3t})$

$$\mathcal{L}(e^{-2t} + 4e^{-3t}) = \mathcal{L}(e^{-2t}) + 4\mathcal{L}(e^{-3t})$$

From the Table:

$$\mathcal{L}(e^{-2t} + 4e^{-3t}) = \frac{1}{s+2} + 4 \frac{1}{s+3}$$

$$\mathcal{L}(e^{-2t} + 4e^{-3t}) = \frac{(s+2) + 4(s+3)}{(s+2)(s+3)}$$

$$\mathcal{L}(e^{-2t} + 4e^{-3t}) = \frac{5s+11}{(s+2)(s+3)}$$

11.

Evaluate $\mathcal{L}(3e^{4t} - e^{-2t})$

$$\mathcal{L}(3e^{4t} - e^{-2t}) = 3\mathcal{L}(e^{4t}) - \mathcal{L}(e^{-2t})$$

From the Table:

$$\mathcal{L}(3e^{4t} - e^{-2t}) = 3 \frac{1}{s-4} - \frac{1}{s+2}$$

$$\mathcal{L}(3e^{4t} - e^{-2t}) = \frac{3(s+2) - (s-4)}{(s-4)(s+2)}$$

$$\mathcal{L}(3e^{4t} - e^{-2t}) = \frac{2s+10}{(s-4)(s+2)}$$

12.

Evaluate $\mathcal{L}(5 \sin 4t)$

From the Table:

$$\mathcal{L}(5 \sin 4t) = 5\mathcal{L}(\sin 4t) = 5 \left(\frac{4}{s^2 + 4^2} \right)$$

$$\mathcal{L}(5 \sin 4t) = \frac{20}{s^2 + 16}$$

13.

Evaluate $\mathcal{L}(4 \cos 3t - 10 \sin 3t + 2 \cos 10t)$

$$\mathcal{L}(4 \cos 3t - 10 \sin 3t + 2 \cos 10t) = 4\mathcal{L}(\cos 3t) - 10\mathcal{L}(\sin 3t) + 2\mathcal{L}(\cos 10t)$$

From the Table:

$$\mathcal{L}(4 \cos 3t - 10 \sin 3t + 2 \cos 10t) = 4 \left(\frac{s}{s^2 + 3^2} \right) - 10 \left(\frac{3}{s^2 + 3^2} \right) + 2 \left(\frac{s}{s^2 + 10^2} \right)$$

$$\mathcal{L}(4 \cos 3t - 10 \sin 3t + 2 \cos 10t) = \frac{4s}{s^2 + 9} - \frac{30}{s^2 + 9} + \frac{2s}{s^2 + 100}$$

14.

Evaluate $\mathcal{L}(6 \sinh 3t + 5 \sin 2t)$

$$\mathcal{L}(6 \sinh 3t + 5 \sin 2t) = 6\mathcal{L}(\sinh 3t) + 5\mathcal{L}(\sin 2t)$$

From the Table:

$$\mathcal{L}(6 \sinh 3t + 5 \sin 2t) = 6 \left(\frac{3}{s^2 - 3^2} \right) + 5 \left(\frac{2}{s^2 + 2^2} \right)$$

$$\mathcal{L}(6 \sinh 3t + 5 \sin 2t) = \frac{18}{s^2 - 9} + \frac{10}{s^2 + 4}$$

INVERSE LAPLACE TRANSFORM

8.3 INVERSE LAPLACE TRANSFORM

Suppose that the function $F(t)$ is to be determined from a differential equation with initial conditions. The Laplace operator $\mathcal{L}$ is used to transform the original problem into a new problem from which the transform $f(s)$ is to be found. If the Laplace transformation is to be effective, the new problem must be simpler than the original problem. We first find $f(s)$ and then must obtain $F(t)$ from $f(s)$. It is therefore desirable to develop methods for finding the object function $F(t)$ when its transform $f(s)$ is known.

If
$$\mathcal{L}\{F(t)\} = f(s)$$

We say that $F(t)$ is an *inverse Laplace Transform*, or an inverse transform, of $f(s)$ and we write

$$F(t) = \mathcal{L}^{-1}\{f(s)\}$$

EXERCISES:

1. In exercises 1 through 16, obtain $\mathcal{L}^{-1}\{f(s)\}$ from the given $f(s)$.

1.
$$f(s) = \frac{6}{s} - \frac{1}{s-8} + \frac{4}{s-3}$$

$$f(t) = 6\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-8}\right\} + 4\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\}$$

From the Table:

$$f(t) = 6(1) - e^{8t} + 4e^{3t}$$

2.
$$f(s) = \frac{19}{s+2} - \frac{1}{3s-5} + \frac{7}{s^3}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{19}{s+2}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{3s-5}\right\} + \mathcal{L}^{-1}\left\{\frac{7}{s^3}\right\}$$

$$f(t) = 19\mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} - \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s-5/3}\right\} + \frac{7}{4!}\mathcal{L}^{-1}\left\{\frac{4!}{s^3}\right\}$$

3.
$$f(s) = \frac{6s}{s^2+25} + \frac{3}{s^2+25}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{6s}{s^2+25}\right\} + \mathcal{L}^{-1}\left\{\frac{3}{s^2+25}\right\}$$

$$f(t) = 6\mathcal{L}^{-1}\left\{\frac{s}{s^2+5^2}\right\} + \frac{3}{5}\mathcal{L}^{-1}\left\{\frac{5}{s^2+5^2}\right\}$$

From the Table:

$$f(t) = 6 \cos 5t + \frac{3}{5} \sin 5t$$

4.
$$f(s) = \frac{8}{3s^2+12} + \frac{3}{s^2+49}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{8}{3s^2+12}\right\} + \mathcal{L}^{-1}\left\{\frac{3}{s^2+49}\right\}$$

$$f(t) = \frac{8}{6}\mathcal{L}^{-1}\left\{\frac{2}{s^2+2^2}\right\} + \frac{3}{7}\mathcal{L}^{-1}\left\{\frac{7}{s^2+7^2}\right\}$$

From the Table:

$$f(t) = \frac{4}{3} \sin 2t + \frac{3}{7} \sin 7t$$

INVERSE LAPLACE TRANSFORM

$$f(s) = \frac{6s-5}{s^2+7}$$

$$f(s) = \frac{6s-5}{s^2+7} = \frac{6s}{s^2+7} - \frac{5}{s^2+7}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{6s}{s^2+7}\right\} - \mathcal{L}^{-1}\left\{\frac{5}{s^2+7}\right\}$$

$$f(t) = 6\mathcal{L}^{-1}\left\{\frac{s}{s^2+(\sqrt{7})^2}\right\} - \frac{5}{\sqrt{7}}\mathcal{L}^{-1}\left\{\frac{\sqrt{7}}{s^2+(\sqrt{7})^2}\right\}$$

From the Table:

$$f(t) = 6 \cos \sqrt{7}t - \frac{5}{\sqrt{7}} \sin \sqrt{7}t$$

5. Find $\mathcal{L}^{-1}\left\{\frac{12}{s^3} + \frac{5}{s^2+25} - \frac{7s}{s^2-9}\right\}$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{12}{s^3}\right\} + \mathcal{L}^{-1}\left\{\frac{5}{s^2+25}\right\} - \mathcal{L}^{-1}\left\{\frac{7s}{s^2-9}\right\}$$

$$f(t) = \frac{12}{4!}\mathcal{L}^{-1}\left\{\frac{4!}{s^3}\right\} + \mathcal{L}^{-1}\left\{\frac{5}{s^2+5^2}\right\} - 7\mathcal{L}^{-1}\left\{\frac{s}{s^2-3^2}\right\}$$

From the Table:

$$f(t) = \frac{1}{2}t^4 + \sin 5t - 7 \cosh 3t$$

7.
$$f(s) = \frac{s+7}{s^2-3s-10}$$

$$f(s) = \frac{s-3+10}{s^2-3s+9-19} = \frac{(s-3)+10}{(s-3)^2-19}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{(s-3)}{(s-3)^2-19}\right\} + \mathcal{L}^{-1}\left\{\frac{10}{(s-3)^2-19}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{(s-3)}{(s-3)^2-19}\right\} + \frac{10}{\sqrt{19}}\mathcal{L}^{-1}\left\{\frac{\sqrt{19}}{(s-3)^2-(\sqrt{19})^2}\right\}$$

From the Table:

$$f(t) = e^{3t} \cosh \sqrt{19}t + \frac{10}{\sqrt{19}} e^{3t} \sinh \sqrt{19}t$$

8.
$$\frac{1}{s^2+2s+10}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2+9}\right\}$$

$$f(t) = \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{3}{(s+1)^2+3^2}\right\}$$

From the Table:

$$f(t) = \frac{1}{3}e^{-t} \sin 3t$$

9.
$$\frac{1}{s^2-4s+8}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2+4}\right\}$$

$$f(t) = \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{2}{(s-2)^2+2^2}\right\}$$

From the Table:

$$f(t) = \frac{1}{2}e^{2t} \sin 2t$$

10.
$$\frac{3s}{s^2+4s+13}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{(3s+6)-6}{(s+2)^2+9}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{3s+6}{(s+2)^2+3^2}\right\} - \mathcal{L}^{-1}\left\{\frac{6}{(s+2)^2+3^2}\right\}$$

INVERSE LAPLACE TRANSFORM

$$f(t) = 3\mathcal{L}^{-1}\left\{\frac{s+2}{(s+2)^2+3^2}\right\} - 2\mathcal{L}^{-1}\left\{\frac{3}{(s+2)^2+3^2}\right\}$$

From the Table:

$$f(t) = 3e^{-2t}\cos 3t - 2e^{-2t}\sin 3t$$

$$f(t) = e^{-2t}(3\cos 3t - 2\sin 3t)$$

$$11. \frac{s}{s^2+6s+13}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{s+3-3}{(s+3)^2+4}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{s+3}{(s+3)^2+2^2}\right\} - \mathcal{L}^{-1}\left\{\frac{3}{(s+3)^2+2^2}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{s+3}{(s+3)^2+2^2}\right\} - \frac{3}{2}\mathcal{L}^{-1}\left\{\frac{2}{(s+3)^2+2^2}\right\}$$

From the Table:

$$f(t) = e^{-3t}\cos 2t - \frac{3}{2}e^{-3t}\sin 2t$$

$$f(t) = e^{-3t}\left(\cos 2t - \frac{3}{2}\sin 2t\right)$$

$$12. \frac{1}{s^2+4s+4}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2}\right\}$$

From the Table:

$$f(t) = te^{-2t}$$

$$14. \frac{s}{s^2+4s+4}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{(s+2)-2}{(s+2)^2}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s+2)}\right\} - 2\mathcal{L}^{-1}\left\{\frac{1}{(s+2)^2}\right\}$$

From the Table:

$$f(t) = e^{-2t} - 2te^{-2t}$$

$$f(t) = e^{-2t}(1-2t)$$

$$14. \frac{2s-3}{s^2-4s+8}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{2s-4+1}{(s-2)^2+4}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{2s-4}{(s-2)^2+2^2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2+2^2}\right\}$$

$$f(t) = 2\mathcal{L}^{-1}\left\{\frac{s-2}{(s-2)^2+2^2}\right\} + \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{2}{(s-2)^2+2^2}\right\}$$

From the Table:

$$f(t) = 2e^{2t}\cos 2t + \frac{1}{2}e^{2t}\sin 2t$$

$$f(t) = e^{2t}\left(2\cos 2t + \frac{1}{2}\sin 2t\right)$$

$$15. \frac{3s+1}{s^2+6s+13}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{3s+9-8}{(s+3)^2+4}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{3s+9}{(s+3)^2+2^2}\right\} - \mathcal{L}^{-1}\left\{\frac{8}{(s+3)^2+2^2}\right\}$$

$$f(t) = 3\mathcal{L}^{-1}\left\{\frac{s+3}{(s+3)^2+2^2}\right\} - \frac{8}{2}\mathcal{L}^{-1}\left\{\frac{2}{(s+3)^2+2^2}\right\}$$

From the Table:

$$f(t) = 3e^{-3t}\cos 2t - 4e^{-3t}\sin 2t$$

$$f(t) = e^{-3t}(3\cos 2t - 4\sin 2t)$$

INVERSE LAPLACE TRANSFORM

$$16. \frac{2s+3}{(s+4)^3}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{2(s+4)-5}{(s+4)^3}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{2}{(s+4)^2}\right\} - \mathcal{L}^{-1}\left\{\frac{5}{(s+4)^3}\right\}$$

$$f(t) = 2\mathcal{L}^{-1}\left\{\frac{1}{(s+4)^2}\right\} - \frac{5}{2}\mathcal{L}^{-1}\left\{\frac{2}{(s+4)^{2+1}}\right\}$$

From the Table:

$$f(t) = 2te^{-4t} - \frac{5}{2}t^2e^{-4t}$$

$$f(t) = e^{-4t}\left(2t - \frac{5}{2}t^2\right)$$

$$17. \frac{s^2}{(s-1)^4}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{(s-1)^2+2(s-1)+1}{(s-1)^4}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{(s-1)^2}{(s-1)^4}\right\} + \mathcal{L}^{-1}\left\{\frac{2(s-1)}{(s-1)^4}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{(s-1)^4}\right\}$$

$$f(t) = \mathcal{L}^{-1}\left\{\frac{1}{(s-1)^{2+1}}\right\} + \mathcal{L}^{-1}\left\{\frac{2}{(s-1)^{2+1}}\right\} + \frac{1}{6}\mathcal{L}^{-1}\left\{\frac{6}{(s-1)^{3+1}}\right\}$$

From the Table:

$$f(t) = te^t + t^2e^t + \frac{1}{6}t^3e^t$$

$$f(t) = e^t\left(t + t^2 + \frac{1}{6}t^3\right)$$

INVERSE LAPLACE TRANSFORM: PARTIAL FRACTIONS

8.4 PARTIAL FRACTIONS

In using the Laplace transform to solve differential equations, we often need to obtain the inverse transform of a rational fraction

$$F(s) = \frac{N(s)}{D(s)}$$

The numerator $N(s)$ and denominator $D(s)$ are polynomials in s and the degree of $D(s)$ is larger than the degree of $N(s)$. The fraction has the partial fractions expansion used in calculus. Because of the linearity of the inverse operator $\mathcal{L}^{-1}$, the partial fractions expansion permits us to replace a complicated problem in obtaining an inverse transform with a set of simpler problems.

EXERCISES

In exercises 1 through 10, find an inverse transform of the given $f(s)$

1. $\frac{1}{s^2 + as}$

Using Partial Fraction:

$$\frac{1}{s(s+a)} = \frac{A}{s} + \frac{B}{s+a}$$

$$1 = A(s+a) + Bs$$

If

$$s=0 : A = 1/a$$

$$s=-a : B = -1/a$$

Then,

$$\mathcal{L}^{-1}\left\{\frac{1}{s(s+a)}\right\} = \frac{1}{a}\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \frac{1}{a}\mathcal{L}^{-1}\left\{\frac{1}{s+a}\right\}$$

From the Table:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+as}\right\} = \frac{1}{a}(1) - \frac{1}{a}e^{-at}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+as}\right\} = \frac{1}{a}(1 - e^{-at})$$

2. $\frac{s+2}{s^2-6s+8}$

Using Partial Fraction:

$$\frac{s+2}{(s-4)(s-2)} = \frac{A}{s-4} + \frac{B}{s-2}$$

$$s+2 = A(s-2) + B(s-4)$$

If

$$s=4 : A=3$$

$$s=2 : B=-2$$

Then,

$$\mathcal{L}^{-1}\left\{\frac{s+2}{s^2-6s+8}\right\} = 3\mathcal{L}^{-1}\left\{\frac{1}{s-4}\right\} - 2\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\}$$

From the Table:

$$\mathcal{L}^{-1}\left\{\frac{s+2}{s^2-6s+8}\right\} = 3e^{4t} - 2e^{2t}$$

3. $\frac{2s^2+5s-4}{s^3+s^2-2s}$

Using Partial Fraction:

$$\frac{2s^2+5s-4}{s(s+2)(s-1)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{s-1}$$

$$2s^2+5s-4 = A(s+2)(s-1) + Bs(s-1) + Cs(s+2)$$

If

$$s=0 : A=-2$$

$$s=-2 : B=-1$$

$$s=1 : C=1$$

INVERSE LAPLACE TRANSFORM - PARTIAL FRACTIONS

Then,

$$\mathcal{L}^{-1}\left\{\frac{2s^2+5s-4}{s^3+s^2-2s}\right\} = 2\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\}$$

From the Table

$$\mathcal{L}^{-1}\left\{\frac{2s^2+5s-4}{s^3+s^2-2s}\right\} = 2 - e^{-2t} + e^{-t}$$

4.

$$\frac{2s^2+1}{s(s+1)^2}$$

Using Partial Fraction:

$$\frac{2s^2+1}{s(s+1)^2} = \frac{A}{s} + \frac{B}{s+1} + \frac{C}{(s+1)^2}$$

$$2s^2+1 = A(s+1)^2 + Bs(s+1) + Cs$$

Collecting constant terms:

$$1 = A$$

Collecting s^2 terms:

$$2 = A + B$$

$$2 = 1 + B$$

$$B = 1$$

Collecting s terms:

$$0 = 2A + B + C$$

$$0 = 2(1) + 1 + C$$

$$C = -3$$

Then,

$$\mathcal{L}^{-1}\left\{\frac{2s^2+1}{s(s+1)^2}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} - 3\mathcal{L}^{-1}\left\{\frac{1}{(s+1)^2}\right\}$$

From the Table:

$$\mathcal{L}^{-1}\left\{\frac{2s^2+1}{s(s+1)^2}\right\} = 1 + e^{-t} - 3te^{-t}$$

5.

$$\frac{4s+4}{s^2(s-2)}$$

Using Partial Fraction:

$$\frac{4s+4}{s^2(s-2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-2}$$

$$4s+4 = As(s-2) + B(s-2) + Cs^2$$

Collecting constant terms:

$$4 = -2B$$

$$B = -2$$

Collecting s terms:

$$4 = -2A + B$$

$$4 = -2A - 2$$

$$A = -3$$

Collecting s^2 terms:

$$0 = A + C$$

$$0 = -3 + C$$

$$C = 3$$

Then,

$$\mathcal{L}^{-1}\left\{\frac{4s+4}{s^2(s-2)}\right\} = -3\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - 2\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + 3\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\}$$

From the Table:

$$\mathcal{L}^{-1}\left\{\frac{4s+4}{s^2(s-2)}\right\} = -3 - 2t + 3e^{2t}$$

6.

$$\frac{1}{s^3(s^2+1)}$$

Using Partial Fraction:

$$\frac{1}{s^3(s^2+1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s^2+1} + \frac{Es}{s^2+1}$$

$$1 = As^2(s^2+1) + Bs(s^2+1) + Cs(s^2+1) + Ds^3 + Es^4$$

Collecting constant terms:

$$1 = C$$

INVERSE LAPLACE TRANSFORM – PARTIAL FRACTIONS

Collecting s terms:
 $0 = B$

Collecting s^2 terms:
 $0 = A + C$
 $0 = A + 1$
 $A = -1$

Collecting s^3 terms:
 $0 = B + D$
 $0 = 0 + D$
 $D = 0$

Collecting s^4 terms:
 $0 = A + E$
 $0 = -1 + E$
 $E = 1$

Then,

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3(s^2+1)}\right\} = -\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} + \mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\}$$

From the Table:

$$\mathcal{L}^{-1}\left\{\frac{1}{s^3(s^2+1)}\right\} = -1 + \frac{1}{2}t^2 + \cos t$$

7.

$$\frac{5s-2}{s^2(s+2)(s-1)}$$

Using Partial Fraction:

$$\frac{5s-2}{s^2(s+2)(s-1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s+2} + \frac{D}{s-1}$$

$$\begin{aligned} 5s-2 &= As(s+2)(s-1) + B(s+2)(s-1) + Cs^2(s-1) + Ds^2(s+2) \\ 5s-2 &= As(s^2+s-2) + B(s^2-1) + Cs^3 - Cs^2 + Ds^3 + 2Ds^2 \end{aligned}$$

Collecting constant terms:
 $-2 = -2B$
 $B = 1$

Collecting s terms:
 $5 = -2A + B$
 $5 = -2A + 1$
 $A = -2$

Collecting s^2 terms:
 $0 = A + B - C + 2D$
 $0 = -2 + 1 - C + 2D$
 $C - 2D = -1 \rightarrow (1)$

Collecting s^3 terms:
 $0 = A + C + D$
 $0 = -2 + C + D$
 $C + D = 2 \rightarrow (2)$

Solving two equations with two unknowns:

$$\begin{aligned} C &= 1 \\ D &= 1 \end{aligned}$$

Then,

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{5s-2}{s^2(s+2)(s-1)}\right\} &= -2\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} \end{aligned}$$

From the Table:

$$\mathcal{L}^{-1}\left\{\frac{5s-2}{s^2(s+2)(s-1)}\right\} = -2 + t + e^{-2t} + e^t$$

8.

$$\frac{1}{(s^2+a^2)(s^2+b^2)}; a^2 \neq b^2, ab \neq 0$$

Using Partial Fraction:

$$\frac{1}{(s^2+a^2)(s^2+b^2)} = \frac{A}{s^2+a^2} + \frac{Bs}{s^2+a^2} + \frac{C}{s^2+b^2} + \frac{Ds}{s^2+b^2}$$

$$1 = A(s^2+b^2) + Bs(s^2+b^2) + C(s^2+a^2) + Ds(s^2+a^2)$$

Collecting constant terms:
 $1 = Ab^2 + Ca^2 \rightarrow (1)$

Collecting s terms:
 $0 = Bb^2 + Da^2 \rightarrow (2)$

Collecting s^2 terms:
 $0 = A + C$
 $A = -C \rightarrow (3)$

INVERSE LAPLACE TRANSFORM – PARTIAL FRACTIONS

Collecting s^3 terms:
 $0 = B + D$
 $B = -D \rightarrow (4)$

Substitute Equation (3) in Equation (1):

$$\begin{aligned} 1 &= -Cb^2 + Ca^2 \\ 1 &= -Cb^2 + Ca^2 \\ C &= \frac{1}{-b^2 + a^2} \\ A &= \frac{1}{-b^2 + a^2} \end{aligned}$$

Substitute Equation (4) in Equation (2):

$$\begin{aligned} 0 &= Bb^2 + Da^2 \\ 0 &= -Db^2 + Da^2 \\ D &= 0 \\ B &= 0 \end{aligned}$$

Then,

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\} &= \left(\frac{-1}{-b^2+a^2}\right)\mathcal{L}^{-1}\left\{\frac{1}{s^2+a^2}\right\} + \left(\frac{1}{-b^2+a^2}\right)\mathcal{L}^{-1}\left\{\frac{1}{s^2+b^2}\right\} \end{aligned}$$

From the Table:

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\} &= \left(\frac{-1}{-b^2+a^2}\right)\frac{1}{a}\sin at + \left(\frac{1}{-b^2+a^2}\right)\frac{1}{b}\sin bt \end{aligned}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\} = \left(\frac{-1}{-b^2+a^2}\right)\left(\frac{1}{a}\sin at - \frac{1}{b}\sin bt\right)$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\} = \left(\frac{1}{-b^2+a^2}\right)\left(\frac{1}{a}\sin at - \frac{1}{b}\sin bt\right)$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)(s^2+b^2)}\right\} = \frac{b \sin at - a \sin bt}{ab(b^2 - a^2)}$$

9.

$$\frac{s}{(s^2+a^2)(s^2+b^2)}; a^2 \neq b^2, ab \neq 0$$

Using Partial Fraction:

$$\frac{s}{(s^2+a^2)(s^2+b^2)} = \frac{A}{s^2+a^2} + \frac{Bs}{s^2+a^2} + \frac{C}{s^2+b^2} + \frac{Ds}{s^2+b^2}$$

$$s = A(s^2+b^2) + Bs(s^2+b^2) + C(s^2+a^2) + Ds(s^2+a^2)$$

Collecting constant terms:
 $0 = Ab^2 + Ca^2 \rightarrow (1)$

Collecting s terms:
 $1 = Bb^2 + Da^2 \rightarrow (2)$

Collecting s^2 terms:
 $0 = A + C$
 $A = -C \rightarrow (3)$

Collecting s^3 terms:
 $0 = B + D$
 $B = -D \rightarrow (4)$

Substitute Equation (3) in Equation (1):

$$\begin{aligned} 0 &= -Cb^2 + Ca^2 \\ 0 &= -Cb^2 + Ca^2 \\ C &= 0 \\ A &= 0 \end{aligned}$$

Substitute Equation (4) in Equation (2):

$$\begin{aligned} 1 &= Bb^2 + Da^2 \\ 1 &= -Db^2 + Da^2 \\ D &= \frac{1}{-b^2+a^2} \\ B &= \frac{-1}{-b^2+a^2} \end{aligned}$$

Then,

$$\begin{aligned} \mathcal{L}^{-1}\left\{\frac{s}{(s^2+a^2)(s^2+b^2)}\right\} &= \left(\frac{-1}{-b^2+a^2}\right)\mathcal{L}^{-1}\left\{\frac{s}{s^2+a^2}\right\} + \left(\frac{1}{-b^2+a^2}\right)\mathcal{L}^{-1}\left\{\frac{s}{s^2+b^2}\right\} \end{aligned}$$

INVERSE LAPLACE TRANSFORM - PARTIAL FRACTIONS

From the Table:

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2+a^2)(s^2+b^2)}\right\}$$

$$= \left(\frac{-1}{-b^2+a^2}\right)\cos at + \left(\frac{1}{-b^2+a^2}\right)\cos bt$$

$$\mathcal{L}^{-1}\left\{\frac{-1}{(s^2+a^2)(s^2+b^2)}\right\}$$

$$= \left(\frac{-1}{-b^2+a^2}\right)(\cos at - \cos bt)$$

$$\mathcal{L}^{-1}\left\{\frac{s}{(s^2+a^2)(s^2+b^2)}\right\}$$

$$= \left(\frac{1}{-b^2+a^2}\right)(\cos at - \cos bt)$$

10. $\frac{s^2}{(s^2+a^2)(s^2+b^2)}; a^2 \neq b^2, ab \neq 0$

Using Partial Fraction:

$$\frac{s^2}{(s^2+a^2)(s^2+b^2)} = \frac{A}{(s^2+a^2)} + \frac{Bs}{(s^2+b^2)} + \frac{Cs}{(s^2+b^2)}$$

$$s^2 = A(s^2+b^2) + Bs(s^2+a^2) + Cs(s^2+a^2)$$

Collecting constant terms:
 $0 = Ab^2 + Ca^2 \rightarrow (1)$

Collecting s terms:
 $0 = Bb^2 + Da^2 \rightarrow (2)$

Collecting s^2 terms:
 $1 = A + C$
 $A = 1 - C \rightarrow (3)$

Collecting s^3 terms:
 $0 = B + D$
 $B = -D \rightarrow (4)$

Substitute Equation (3) in Equation (1):

$$0 = -Cb^2 + Ca^2$$

$$0 = (1-C)b^2 + Ca^2$$

$$C = \frac{-1}{-b^2+a^2}$$

$$A = \frac{1}{-b^2+a^2}$$

Substitute Equation (4) in Equation (2):

$$0 = Bb^2 + Da^2$$

$$0 = -Db^2 + Da^2$$

$$D = 0$$

$$B = 0$$

Then,

$$\mathcal{L}^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$$

$$= \left(\frac{1}{-b^2+a^2}\right)\mathcal{L}^{-1}\left\{\frac{1}{(s^2+a^2)}\right\}$$

$$- \left(\frac{1}{-b^2+a^2}\right)\mathcal{L}^{-1}\left\{\frac{1}{(s^2+b^2)}\right\}$$

From the Table:

$$\mathcal{L}^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$$

$$= \left(\frac{1}{-b^2+a^2}\right)\frac{1}{a}\sin at$$

$$- \left(\frac{1}{-b^2+a^2}\right)\frac{1}{b}\sin bt$$

$$\mathcal{L}^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$$

$$= \left(\frac{1}{-b^2+a^2}\right)\left(\frac{1}{a}\sin at - \frac{1}{b}\sin bt\right)$$

$$\mathcal{L}^{-1}\left\{\frac{s^2}{(s^2+a^2)(s^2+b^2)}\right\}$$

$$= \frac{b \sin at - a \sin bt}{a^2 - b^2}$$

TRANSFORMS OF DERIVATIVES

15 TRANSFORMS OF DERIVATIVES

If $F(t), F'(t), \dots, F^{(n-1)}(t)$ are continuous for $t \geq 0$ and of exponential order as $t \rightarrow \infty$, then

$$\mathcal{L}\{F(t)\} = f(s)$$

It follows that

$$\mathcal{L}\{F^{(n)}(t)\} = s^n f(s) - \sum_{k=0}^{n-1} s^{n-1-k} F^{(k)}(0)$$

Thus;

$$\mathcal{L}\{F(t)\} = f(s)$$

$$\mathcal{L}\{F'(t)\} = sf(s) - f(0)$$

$$\mathcal{L}\{F''(t)\} = s^2 f(s) - sf(0) - f'(0)$$

$$\mathcal{L}\{F'''(t)\} = s^3 f(s) - s^2 f(0) - sf'(0) - f''(0)$$

EXERCISES:

In exercises 1 through 22, solve the problem by the Laplace transform method.

1. $y' = e^t; y(0) = 2$

$$\mathcal{L}\{y'\} = \mathcal{L}\{e^t\}$$

$$sy(s) - y(0) = \frac{1}{s-1}$$

Substitute $y(0) = 2$:

$$sy(s) - 2 = \frac{1}{s-1}$$

$$sy(s) = \frac{1}{s-1} + 2$$

$$sy(s) = \frac{1+2(s-1)}{s-1}$$

$$y(s) = \frac{2s-1}{s(s-1)}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{2s-1}{s(s-1)}\right\}$$

Using Partial Fraction:

$$\frac{2s-1}{s(s-1)} = \frac{A}{s} + \frac{B}{s-1}$$

$$2s-1 = A(s-1) + Bs$$

If

$$s = 0; A = 1$$

$$s = 1; B = 1$$

$$\mathcal{L}^{-1}\left\{\frac{2s-1}{s(s-1)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+as}\right\} = 1 + e^t$$

Thus,

$$y(t) = 1 + e^t$$

TRANSFORMS OF DERIVATIVES

$$2. \quad y' = 2e^t; y(0) = -1$$

$$\mathcal{L}\{y'\} = \mathcal{L}\{2e^t\}$$

$$sy(s) - y(0) = \frac{2}{s-1}$$

$$\text{Substitute } y(0) = -1;$$

$$sy(s) + 1 = \frac{2}{s-1}$$

$$sy(s) = \frac{2}{s-1} - 1$$

$$sy(s) = \frac{2 - (s-1)}{s-1}$$

$$y(s) = \frac{3-s}{s(s-1)}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{3-s}{s(s-1)}\right\}$$

Using Partial Fraction:

$$\frac{3-s}{s(s-1)} = \frac{A}{s} + \frac{B}{s-1}$$

$$3-s = A(s-1) + Bs$$

$$\text{If } s=0; A=-3$$

$$s=1; B=2$$

$$\mathcal{L}^{-1}\left\{\frac{3-s}{s(s-1)}\right\} = -3\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + 2\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{3-s}{s(s-1)}\right\} = -3 + 2e^t$$

Thus,

$$y(t) = -3 + 2e^t$$

$$3. \quad y' + y = e^{2t}; y(0) = 0$$

$$\mathcal{L}\{y'\} + \mathcal{L}\{y\} = \mathcal{L}\{e^{2t}\}$$

$$sy(s) - y(0) + y(s) = \frac{1}{s-2}$$

$$\text{Substitute } y(0) = 0;$$

$$sy(s) - 0 + y(s) = \frac{1}{s-2}$$

$$(s+1)y(s) = \frac{1}{s-2}$$

$$y(s) = \frac{1}{(s-2)(s+1)}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{(s-2)(s+1)}\right\}$$

Using Partial Fraction:

$$\frac{1}{(s-2)(s+1)} = \frac{A}{(s-2)} + \frac{B}{(s+1)}$$

$$1 = A(s+1) + B(s-2)$$

$$\text{If } s=2; A=1/3$$

$$s=-1; B=-1/3$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-2)(s+1)}\right\} = \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - \frac{1}{3}\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-2)(s+1)}\right\} = \frac{1}{3}e^{2t} - \frac{1}{3}e^{-t}$$

Thus,

$$y(t) = \frac{1}{3}e^{2t} - \frac{1}{3}e^{-t}$$

$$4. \quad y' - y = e^{-t}; y(0) = 1$$

$$\mathcal{L}\{y'\} - \mathcal{L}\{y\} = \mathcal{L}\{e^{-t}\}$$

$$sy(s) - y(0) - y(s) = \frac{1}{s+1}$$

$$\text{Substitute } y(0) = 1;$$

$$sy(s) - 1 - y(s) = \frac{1}{s+1}$$

$$(s-1)y(s) = \frac{1}{s+1} + 1$$

$$y(s) = \frac{1+s+1}{(s+1)(s-1)}$$

$$y(s) = \frac{s+2}{(s+1)(s-1)}$$

TRANSFORMS OF DERIVATIVES

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{s+2}{(s+1)(s-1)}\right\}$$

Using Partial Fraction:

$$\frac{s+2}{(s+1)(s-1)} = \frac{A}{(s+1)} + \frac{B}{(s-1)}$$

$$s+2 = A(s-1) + B(s+1)$$

$$\text{If } s=-1; A=-1/2$$

$$s=1; B=3/2$$

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s+1)(s-1)}\right\} = \frac{-1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} + \frac{3}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{s+2}{(s+1)(s-1)}\right\} = \frac{-1}{2}e^{-t} + \frac{3}{2}e^t$$

Thus,

$$y(t) = \frac{3}{2}e^t - \frac{1}{2}e^{-t}$$

$$5. \quad y'' + a^2y = 0; y(0) = 1, y'(0) = 0$$

$$\mathcal{L}\{y''\} + a^2\mathcal{L}\{y\} = \mathcal{L}\{0\}$$

$$s^2y(s) - sy(0) - y'(0) + a^2y(s) = 0$$

$$\text{Substitute } y(0) = 1, y'(0) = 0;$$

$$s^2y(s) - s(1) - 0 + a^2y(s) = 0$$

$$s^2y(s) - s + a^2y(s) = 0$$

$$(s^2 + a^2)y(s) = s$$

$$y(s) = \frac{s}{s^2 + a^2}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2 + a^2}\right\}$$

Using Partial Fraction:

$$\frac{s}{s^2 + a^2} = \frac{A}{s^2 + a^2} + \frac{Bs}{s^2 + a^2}$$

$$s = A + Bs$$

Collecting s terms:

$$B = 1$$

Collecting constant terms:
 $A = 0$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + a^2}\right\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2 + a^2}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2 + a^2}\right\} = \cos at$$

Thus,

$$y(t) = \cos at$$

$$6. \quad y'' + a^2y = 0; y(0) = 0, y'(0) = a$$

$$\mathcal{L}\{y''\} + a^2\mathcal{L}\{y\} = \mathcal{L}\{0\}$$

$$s^2y(s) - sy(0) - y'(0) + a^2y(s) = 0$$

$$\text{Substitute } y(0) = 0, y'(0) = a;$$

$$s^2y(s) - s(0) - a + a^2y(s) = 0$$

$$s^2y(s) - a + a^2y(s) = 0$$

$$(s^2 + a^2)y(s) = a$$

$$y(s) = \frac{a}{s^2 + a^2}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{a}{s^2 + a^2}\right\}$$

Using Partial Fraction:

$$\frac{a}{s^2 + a^2} = \frac{A}{s^2 + a^2} + \frac{Bs}{s^2 + a^2}$$

$$a = A + Bs$$

Collecting s terms:
 $B = 0$

Collecting constant terms:
 $A = a$

$$\mathcal{L}^{-1}\left\{\frac{a}{s^2 + a^2}\right\} = \mathcal{L}^{-1}\left\{\frac{a}{s^2 + a^2}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{a}{s^2 + a^2}\right\} = \sin at$$

Thus,

$$y(t) = \sin at$$

TRANSFORMS OF DERIVATIVES

$$7. \quad y'' - 3y' + 2y = e^{3t}; y(0) = 0, y'(0) = 0$$

$$\mathcal{L}(y'') - 3\mathcal{L}(y') + 2\mathcal{L}(y) = \mathcal{L}(e^{3t})$$

$$[s^2 y(s) - sy(0) - y'(0)] - 3[sy(s) - y(0)] + 2y(s) = \frac{1}{s-3}$$

$$-3[sy(s) - y(0)] + 2y(s) = \frac{1}{s-3}$$

Substitute $y(0) = 0, y'(0) = 0$:

$$[s^2 y(s) - s(0) - 0] - 3[sy(s) - 0] + 2y(s) = \frac{1}{s-3}$$

$$+ 2y(s) = \frac{1}{s-3}$$

$$s^2 y(s) - 3sy(s) + 2y(s) = \frac{1}{s-3}$$

$$(s^2 - 3s + 2)y(s) = \frac{1}{s-3}$$

$$y(s) = \frac{1}{(s-3)(s^2 - 3s + 2)}$$

$$y(s) = \frac{1}{(s-3)(s-2)(s-1)}$$

$$\mathcal{L}^{-1}(y(s)) = \mathcal{L}^{-1}\left\{\frac{1}{(s-3)(s-2)(s-1)}\right\}$$

Using Partial Fraction:

$$\frac{1}{(s-3)(s-2)(s-1)} = \frac{A}{s-3} + \frac{B}{s-2} + \frac{C}{s-1}$$

$$1 = A(s-2)(s-1) + B(s-3)(s-1) + C(s-3)(s-2)$$

$$1 = A(s^2 - 3s + 2) + B(s^2 - 4s + 3) + C(s^2 - 5s + 6)$$

Collecting s^2 terms:

$$0 = A + B + C \rightarrow (1)$$

Collecting s terms:

$$0 = -3A - 4B - 5C \rightarrow (2)$$

Collecting constant terms:

$$1 = 2A + 3B + 6C \rightarrow (3)$$

Solving three equations with three unknowns:

$$A = 1/2$$

$$B = -1$$

$$C = 1/2$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-3)(s-2)(s-1)}\right\}$$

$$= \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} + \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s-3)(s-2)(s-1)}\right\}$$

$$= \frac{1}{2}e^{3t} - e^{2t} + \frac{1}{2}e^t$$

Thus,

$$y(t) = \frac{1}{2}e^{3t} - e^{2t} + \frac{1}{2}e^t$$

$$8. \quad y'' + y = e^{-t}; y(0) = 0, y'(0) = 0$$

$$\mathcal{L}(y'') + \mathcal{L}(y) = \mathcal{L}(e^{-t})$$

$$[s^2 y(s) - sy(0) - y'(0)] + y(s) = \frac{1}{s+1}$$

Substitute $y(0) = 0, y'(0) = 0$:

$$[s^2 y(s) - s(0) - 0] + y(s) = \frac{1}{s+1}$$

$$s^2 y(s) + y(s) = \frac{1}{s+1}$$

$$(s^2 + 1)y(s) = \frac{1}{s+1}$$

$$y(s) = \frac{1}{(s+1)(s^2 + 1)}$$

$$\mathcal{L}^{-1}(y(s)) = \mathcal{L}^{-1}\left\{\frac{1}{(s+1)(s^2 + 1)}\right\}$$

Using Partial Fraction:

$$\frac{1}{(s+1)(s^2 + 1)} = \frac{A}{s+1} + \frac{B}{s^2 + 1} + \frac{Cs}{s^2 + 1}$$

$$1 = A(s^2 + 1) + B(s+1) + Cs(s+1)$$

Collecting s^2 terms:

$$0 = A + C \rightarrow (1)$$

Collecting s terms:

$$0 = B + C \rightarrow (2)$$

TRANSFORMS OF DERIVATIVES

Collecting constant terms:
 $1 = A + B \rightarrow (3)$

Solving three equations with three unknowns:

$$A = 1/2$$

$$B = 1/2$$

$$C = -1/2$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)(s^2 + 1)}\right\}$$

$$= \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s+1}\right\} + \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 1}\right\}$$

$$- \frac{1}{2}\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{1}{(s+1)(s^2 + 1)}\right\}$$

$$= \frac{1}{2}e^{-t} + \frac{1}{2}\sin t - \frac{1}{2}\cos t$$

Thus,

$$y(t) = \frac{1}{2}e^{-t} + \frac{1}{2}\sin t - \frac{1}{2}\cos t$$

$$y(t) = \frac{1}{2}(e^{-t} + \sin t - \cos t)$$

$$9. \quad y'' - 2y' = -4; y(0) = 0, y'(0) = 4$$

$$\mathcal{L}(y'') - 2\mathcal{L}(y') = \mathcal{L}(-4)$$

$$[s^2 y(s) - sy(0) - y'(0)] - 2[sy(s) - y(0)] = \frac{-4}{s}$$

$$-2[sy(s) - y(0)] = \frac{-4}{s}$$

Substitute $y(0) = 0, y'(0) = 4$:

$$[s^2 y(s) - s(0) - 4] - 2[sy(s) - 0] = \frac{-4}{s}$$

$$s^2 y(s) - 4 - 2sy(s) = \frac{-4}{s}$$

$$(s^2 - 2s)y(s) = \frac{-4}{s} + 4$$

$$(s^2 - 2s)y(s) = \frac{4s - 4}{s}$$

$$y(s) = \frac{4s - 4}{s(s^2 - 2s)}$$

$$y(s) = \frac{4s - 4}{s^2(s-2)}$$

$$\mathcal{L}^{-1}(y(s)) = \mathcal{L}^{-1}\left\{\frac{4s - 4}{s^2(s-2)}\right\}$$

Using Partial Fraction:

$$\frac{4s - 4}{s^2(s-2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s-2}$$

$$4s - 4 = As(s-2) + B(s-2) + Cs^2$$

Collecting constant terms:

$$-4 = -B$$

$$B = 4$$

Collecting s terms:

$$4 = -2A + B$$

$$4 = -2A + 4$$

$$A = -1$$

Collecting s^2 terms:

$$0 = A + C$$

$$0 = -1 + C$$

$$C = 1$$

$$\mathcal{L}^{-1}\left\{\frac{4s - 4}{s^2(s-2)}\right\} = -\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + 4\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}$$

$$+ \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{4s - 4}{s^2(s-2)}\right\} = -1 + 2t + e^{2t}$$

Thus,

$$y(t) = e^{2t} + 2t - 1$$

$$10. \quad y'' + y' - 2y = -4; y(0) = 2, y'(0) = 3$$

$$\mathcal{L}(y'') + \mathcal{L}(y') - 2\mathcal{L}(y) = \mathcal{L}(-4)$$

$$[s^2 y(s) - sy(0) - y'(0)] + [sy(s) - y(0)] - 2y(s) = \frac{-4}{s}$$

$$- 2y(s) = \frac{-4}{s}$$

Substitute $y(0) = 2, y'(0) = 3$:

$$[s^2 y(s) - s(2) - 3] + [sy(s) - 2] - 2y(s) = \frac{-4}{s}$$

$$= \frac{-4}{s}$$

TRANSFORMS OF DERIVATIVES

$$s^2 y(s) - 2s - 3 + sy(s) - 2 - 2y(s) = \frac{-4}{s}$$

$$(s^2 + s - 2)y(s) = \frac{-4}{s} + 2s + 5$$

$$(s^2 + s - 2)y(s) = \frac{2s^2 + 5s - 4}{s}$$

$$y(s) = \frac{2s^2 + 5s - 4}{s(s-1)(s+2)}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{2s^2 + 5s - 4}{s(s-1)(s+2)}\right\}$$

Using Partial Fraction:

$$\frac{2s^2 + 5s - 4}{s(s-1)(s+2)} = \frac{A}{s} + \frac{B}{s-1} + \frac{C}{s+2}$$

$$2s^2 + 5s - 4 = A(s-1)(s+2) + Bs(s+2) + Cs(s-1)$$

$$2s^2 + 5s - 4 = A(s^2 + s - 2) + Bs(s+2) + Cs(s-1)$$

Collecting constant terms:

$$-4 = -2A$$

$$A = 2 \rightarrow (1)$$

Collecting s terms:

$$5 = A + 2B - C \rightarrow (2)$$

Collecting s^2 terms:

$$2 = A + B + C \rightarrow (3)$$

Solving three equations with three unknowns:

$$A = 2$$

$$B = 1$$

$$C = -1$$

$$\mathcal{L}^{-1}\left\{\frac{2s^2 + 5s - 4}{s(s-1)(s+2)}\right\}$$

$$= 2\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s+2}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{2s^2 + 5s - 4}{s(s-1)(s+2)}\right\} = 2 + e^t - e^{-2t}$$

Thus,

$$y(t) = 2 + e^t - e^{-2t}$$

$$11. \quad x''(t) - 4x'(t) + 4x(t) = 4e^{2t};$$

$$x(0) = -1, x'(0) = -4$$

$$\mathcal{L}\{x''(t)\} - 4\mathcal{L}\{x'(t)\} + 4\mathcal{L}\{x(t)\} = \mathcal{L}\{4e^{2t}\}$$

$$[s^2 x(s) - sx(0) - x'(0)] - 4[sx(s) - x(0)] + 4x(s) = \frac{4}{s-2}$$

$$[s^2 x(s) - s(-1) - (-4)] - 4[sx(s) - (-1)] + 4x(s) = \frac{4}{s-2}$$

$$\text{Substituting } x(0) = -1, x'(0) = -4,$$

$$[s^2 x(s) - s(-1) - (-4)] - 4[sx(s) - (-1)] + 4x(s) = \frac{4}{s-2}$$

$$-4(-1) - (-1) + 4x(s) = \frac{4}{s-2}$$

$$s^2 x(s) + s + 1 - 4sx(s) - 4 + 4x(s) = \frac{4}{s-2}$$

$$(s^2 - 4s + 4)x(s) = \frac{4}{s-2} - s$$

$$(s^2 - 4s + 4)x(s) = \frac{4 - s^2 + 2s}{s-2}$$

$$x(s) = \frac{4 - s^2 + 2s}{(s-2)(s-2)^2}$$

$$x(s) = \frac{4 - s^2 + 2s}{(s-2)^3}$$

$$\mathcal{L}^{-1}\{x(s)\} = \mathcal{L}^{-1}\left\{\frac{4 - s^2 + 2s}{(s-2)^3}\right\}$$

Using Partial Fraction:

$$\frac{4 - s^2 + 2s}{(s-2)^3} = \frac{A}{s-2} + \frac{B}{(s-2)^2} + \frac{C}{(s-2)^3}$$

$$4 - s^2 + 2s = A(s-2)^2 + B(s-2) + C$$

$$4 - s^2 + 2s = A(s^2 - 4s + 4) + B(s-2) + C$$

Collecting s^2 terms:

$$-1 = A$$

Collecting s terms:

$$2 = -4A + B$$

$$2 = -4(-1) + B$$

$$B = -2$$

Collecting constant terms:

$$4 = 4A - 2B + C$$

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$$4 = 4(-1) - 2(-2) + C$$

$$C = 4$$

$$\mathcal{L}^{-1}\left\{\frac{4 - s^2 + 2s}{(s-2)^3}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s-2}\right\} - 2\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2}\right\} + 4\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^3}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{4 - s^2 + 2s}{(s-2)^3}\right\} = -e^{2t} - 2te^{2t} + 2t^2e^{2t}$$

Thus,

$$x(t) = e^{2t}(-1 - 2t + 2t^2)$$

$$x(t) = e^{2t}(2t^2 - 2t - 1)$$

12.

$$x''(t) + x(t) = 6 \sin 2t;$$

$$x(0) = 3, x'(0) = 1$$

$$\mathcal{L}\{x''(t)\} + \mathcal{L}\{x(t)\} = \mathcal{L}\{6 \sin 2t\}$$

$$[s^2 x(s) - sx(0) - x'(0)] + x(s) = \frac{6}{s^2 + 4}$$

$$\text{Substitute } x(0) = 3, x'(0) = 1;$$

$$[s^2 x(s) - s(3) - 1] + x(s) = \frac{6}{s^2 + 4}$$

$$s^2 x(s) - 3s - 1 + x(s) = \frac{12}{s^2 + 4}$$

$$(s^2 + 1)x(s) = \frac{12}{s^2 + 4} + 3s + 1$$

$$(s^2 + 1)x(s) = \frac{12 + (3s+1)(s^2+4)}{s^2+4}$$

$$x(s) = \frac{12 + (3s^3 + 12s + s^2 + 4)}{(s^2+4)(s^2+1)}$$

$$x(s) = \frac{3s^3 + s^2 + 12s + 16}{(s^2+4)(s^2+1)}$$

$$\mathcal{L}^{-1}\{x(s)\} = \mathcal{L}^{-1}\left\{\frac{3s^3 + s^2 + 12s + 16}{(s^2+4)(s^2+1)}\right\}$$

Using Partial Fraction:

$$\frac{3s^3 + s^2 + 12s + 16}{(s^2+4)(s^2+1)} = \frac{A}{s^2+4} + \frac{Bs}{s^2+4} + \frac{C}{s^2+1} + \frac{Ds}{s^2+1}$$

$$3s^3 + s^2 + 12s + 16 = A(s^2+1) + Bs(s^2+1) + C(s^2+4) + Ds(s^2+4)$$

Collecting s^3 terms:

$$3 = B + D \rightarrow (1)$$

Collecting s^2 terms:

$$1 = A + C \rightarrow (2)$$

Collecting s terms:

$$12 = B + 4D \rightarrow (3)$$

Collecting constant terms:

$$16 = A + 4C \rightarrow (4)$$

Solving four equations with four unknowns:

$$A = -4$$

$$B = 0$$

$$C = 5$$

$$D = 3$$

$$\mathcal{L}^{-1}\left\{\frac{3s^3 + s^2 + 12s + 16}{(s^2+4)(s^2+1)}\right\} = -4\mathcal{L}^{-1}\left\{\frac{1}{s^2+4}\right\} + 5\mathcal{L}^{-1}\left\{\frac{1}{s^2+1}\right\} + 3\mathcal{L}^{-1}\left\{\frac{s}{s^2+1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{3s^3 + s^2 + 12s + 16}{(s^2+4)(s^2+1)}\right\} = -2 \sin 2t + 5 \sin t + 3 \cos t$$

Thus,

$$x(t) = 5 \sin t + 3 \cos t - 2 \sin 2t$$

TRANSFORMS OF DERIVATIVES

$$13. \quad y''(t) - y'(t) = 4 \cos t; \\ y(0) = 0, y'(0) = 1 \\ \mathcal{L}\{y''(t) - y'(t)\} = \mathcal{L}\{4 \cos t\} \\ [s^2 y(s) - sy'(0) - y(0)] - [sy(s) - y(0)] = \frac{4s}{s^2 + 1}$$

Substitute $y(0) = 0, y'(0) = 1$:

$$[s^2 y(s) - s(0) - 1] - y(s) = 4 \left(\frac{s}{s^2 + 1} \right)$$

$$s^2 y(s) - 1 - y(s) = \frac{4s}{s^2 + 1}$$

$$(s^2 - 1)y(s) = \frac{4s}{s^2 + 1} + 1$$

$$(s^2 - 1)y(s) = \frac{s^2 + 4s + 1}{s^2 + 1}$$

$$y(s) = \frac{s^2 + 4s + 1}{(s^2 + 1)(s^2 - 1)}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{s^2 + 4s + 1}{(s^2 + 1)(s^2 - 1)}\right\}$$

Using Partial Fraction:

$$\frac{s^2 + 4s + 1}{(s^2 + 1)(s^2 - 1)} = \frac{A}{s^2 + 1} + \frac{Bs}{s^2 - 1} + \frac{Cs}{s^2 - 1} + \frac{Ds}{s^2 - 1}$$

$$s^2 + 4s + 1 = A(s^2 - 1) + Bs(s^2 - 1) + Cs(s^2 + 1) + Ds(s^2 + 1)$$

Collecting s^2 terms:

$$0 = B + D \rightarrow (1)$$

Collecting s terms:

$$1 = A + C \rightarrow (2)$$

Collecting s terms:

$$4 = -B + D \rightarrow (3)$$

Collecting constant terms:

$$1 = -A + C \rightarrow (4)$$

Solving four equations with four unknowns:

$$A = 0$$

$$B = -2$$

$$C = 1$$

$$D = 2$$

$$\mathcal{L}^{-1}\left\{\frac{s^2 + 4s + 1}{(s^2 + 1)(s^2 - 1)}\right\} \\ = -2\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 1}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2 - 1}\right\} \\ + 2\mathcal{L}^{-1}\left\{\frac{s}{s^2 - 1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{s^2 + 4s + 1}{(s^2 + 1)(s^2 - 1)}\right\} \\ = -2 \cos t + \sinh t + 2 \cosh t$$

Thus,

$$y(t) = 2 \cosh t + \sinh t - 2 \cos t$$

$$14. \quad y''(t) - 6y'(t) + 9y(t) = 6t^2 e^{3t}; \\ y(0) = 0, y'(0) = 0$$

$$\mathcal{L}\{y''(t) - 6y'(t) + 9y(t)\} = \mathcal{L}\{6t^2 e^{3t}\}$$

$$[s^2 y(s) - sy'(0) - y(0)] - 6[sy(s) - y(0)] + 9y(s) = 6 \left(\frac{2}{(s - 3)^3} \right)$$

Substitute $y(0) = 0, y'(0) = 0$:

$$[s^2 y(s) - s(0) - 0] - 6[sy(s) - 0] + 9y(s) = 6 \left(\frac{2}{(s - 3)^3} \right)$$

$$s^2 y(s) - 6sy(s) + 9y(s) = \frac{12}{(s - 3)^3}$$

$$(s^2 - 6s + 9)y(s) = \frac{12}{(s - 3)^3}$$

$$y(s) = \frac{12}{(s - 3)^2 (s - 3)^3}$$

$$y(s) = \frac{12}{(s - 3)^5}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{12}{(s - 3)^5}\right\}$$

Using Partial Fraction:

$$\frac{12}{(s - 3)^5} = \frac{A}{(s - 3)} + \frac{B}{(s - 3)^2} + \frac{C}{(s - 3)^3} + \frac{D}{(s - 3)^4} + \frac{E}{(s - 3)^5}$$

TRANSFORMS OF DERIVATIVES

Substitute $x(0) = 1, x'(0) = 0$:

$$[s^2 x(s) - sx(0) - x'(0)] + 4x(s) = \frac{1}{s^2} + \frac{4}{s}$$

$$s^2 x(s) - s + 4x(s) = \frac{1}{s^2} + \frac{4}{s}$$

$$(s^2 + 4)x(s) = \frac{1}{s^2} + \frac{4}{s} + s$$

$$(s^2 + 4)x(s) = \frac{s + 4s^2 + s^4}{s^3}$$

$$x(s) = \frac{1 + 4s + s^3}{s^3(s^2 + 4)}$$

$$\mathcal{L}^{-1}\{x(s)\} = \mathcal{L}^{-1}\left\{\frac{1 + 4s + s^3}{s^3(s^2 + 4)}\right\}$$

Using Partial Fraction:

$$\frac{1 + 4s + s^3}{s^3(s^2 + 4)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^2 + 4} + \frac{Ds}{s^2 + 4}$$

$$1 + 4s + s^3 = As(s^2 + 4) + B(s^2 + 4) + Cs^2 + Ds^3$$

Collecting constant terms:

$$1 = 4B$$

$$B = 1/4$$

Collecting s terms:

$$4 = 4A$$

$$A = 1$$

Collecting s^2 terms:

$$0 = B + C$$

$$0 = 1/4 + C$$

$$C = -1/4$$

Collecting s^3 terms:

$$1 = A + D$$

$$1 = 1 + D$$

$$D = 0$$

$$\mathcal{L}^{-1}\left\{\frac{1 + 4s + s^3}{s^3(s^2 + 4)}\right\} = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} - \frac{1}{4}\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 4}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{1 + 4s + s^3}{s^3(s^2 + 4)}\right\} = 1 + \frac{1}{4}t - \frac{1}{8}\sin 2t$$

$$12 = A(s - 3)^4 + B(s - 3)^3 + C(s - 3)^2 + D(s - 3) + E$$

$$12 = A(s^4 - 12s^3 + 54s^2 - 108s + 81) + B(s^3 - 9s^2 + 27s - 27) + C(s^2 - 6s + 9) + D(s - 3) + E$$

Collecting s^4 terms:

$$0 = A$$

Collecting s^3 terms:

$$0 = -12A + B$$

$$0 = -12(0) + B$$

$$B = 0$$

Collecting s^2 terms:

$$0 = 54A - 9B + 5C$$

$$0 = 54(0) - 9(0) + 5C$$

$$C = 0$$

Collecting s terms:

$$0 = -108A + 27B - 6C + D$$

$$0 = -108(0) + 27(0) - 6(0) + D$$

$$D = 0$$

Collecting constant terms:

$$12 = 81A - 27B + 9C - 3D + E$$

$$12 = 81(0) - 27(0) + 9(0) - 3(0) + E$$

$$E = 12$$

$$\mathcal{L}^{-1}\left\{\frac{12}{(s - 3)^5}\right\} = 12\mathcal{L}^{-1}\left\{\frac{1}{(s - 3)^5}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{12}{(s - 3)^5}\right\} = \frac{12}{4!}\mathcal{L}^{-1}\left\{\frac{4!}{(s - 3)^5}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{12}{(s - 3)^5}\right\} = \frac{1}{2}t^4 e^{3t}$$

Thus,

$$y(t) = \frac{1}{2}t^4 e^{3t}$$

$$15. \quad 0$$

$$x''(t) + 4x(t) = t + 4; x(0) = 1, x'(0) = 0$$

$$\mathcal{L}\{x''(t) + 4x(t)\} = \mathcal{L}\{t + 4\} \\ [s^2 x(s) - sx(0) - x'(0)] + 4x(s) = \frac{1}{s^2} + \frac{4}{s}$$

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$$16. \quad x''(t) - 2x'(t) = 6 - 4t; \\ x(0) = 2, x'(0) = 0 \\ \mathcal{L}\{x''(t) - 2x'(t)\} = \mathcal{L}\{6 - 4t\} \\ [s^2x(s) - sx(0) - x'(0)] \\ - 2[sx(s) - x(0)] = \frac{6}{s} - \frac{4}{s^2}$$

$$\text{Substitute } x(0) = 2, x'(0) = 0;$$

$$[s^2x(s) - s(2) - 0] - 2[sx(s) - 2] \\ = \frac{6}{s} - \frac{4}{s^2}$$

$$s^2x(s) - 2s - 2sx(s) + 4 = \frac{6}{s} - \frac{4}{s^2}$$

$$(s^2 - 2s)x(s) = \frac{6}{s} - \frac{4}{s^2} - 4 + 2s$$

$$x(s - 2)x(s) = \frac{6s - 4 - 4s^2 + 2s^3}{s^2}$$

$$x(s) = \frac{2s^3 - 4s^2 + 6s - 4}{s^2(s - 2)}$$

$$\mathcal{L}^{-1}\{x(s)\} = \mathcal{L}^{-1}\left\{\frac{2s^3 - 4s^2 + 6s - 4}{s^2(s - 2)}\right\}$$

Using Partial Fraction:

$$\frac{2s^3 - 4s^2 + 6s - 4}{s^2(s - 2)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s - 2}$$

$$2s^3 - 4s^2 + 6s - 4 = As^2(s - 2) \\ + Bs(s - 2) + C(s - 2) + Ds^3$$

Collecting constant terms:

$$-4 = -2C$$

$$C = 2$$

Collecting s terms:

$$6 = -2B + C$$

$$6 = -2B + 2$$

$$B = -2$$

Collecting s^2 terms:

$$-4 = -2A + B$$

$$-4 = -2A - 2$$

$$A = 1$$

Collecting s^3 terms:

$$2 = A + D$$

$$2 = 1 + D$$

$$D = 1$$

$$\mathcal{L}^{-1}\left\{\frac{2s^3 - 4s^2 + 6s - 4}{s^2(s - 2)}\right\} \\ = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - 2\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + 2\mathcal{L}^{-1}\left\{\frac{1}{s - 2}\right\} \\ + \mathcal{L}^{-1}\left\{\frac{1}{s - 2}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{2s^3 - 4s^2 + 6s - 4}{s^2(s - 2)}\right\} \\ = 1 - 2t + t^2 + e^{2t}$$

Thus,

$$x(t) = t^2 - 2t + 1 + e^{2t}$$

$$17. \quad x''(t) + x(t) = 4e^t; x(0) = 1, x'(0) = 3$$

$$\mathcal{L}\{x''(t) + x(t)\} = \mathcal{L}\{4e^t\}$$

$$[s^2x(s) - sx(0) - x'(0)] + x(s) = \frac{4}{s - 1}$$

$$\text{Substitute } x(0) = 1, x'(0) = 3;$$

$$[s^2x(s) - s(1) - 3] + x(s) = \frac{4}{s - 1}$$

$$s^2x(s) - s - 3 + x(s) = \frac{4}{s - 1}$$

$$(s^2 + 1)x(s) = \frac{4}{s - 1} + s + 3$$

$$(s^2 + 1)x(s) = \frac{4 + s^2 - s + 3s - 3}{s - 1}$$

$$x(s) = \frac{s^2 + 2s + 1}{(s - 1)(s^2 + 1)}$$

$$\mathcal{L}^{-1}\{x(s)\} = \mathcal{L}^{-1}\left\{\frac{s^2 + 2s + 1}{(s - 1)(s^2 + 1)}\right\}$$

Using Partial Fraction:

$$\frac{s^2 + 2s + 1}{(s - 1)(s^2 + 1)} = \frac{A}{s - 1} + \frac{B}{s^2 + 1} + \frac{Cs}{s^2 + 1}$$

$$s^2 + 2s + 1 = A(s^2 + 1) + B(s - 1) \\ + Cs(s - 1)$$

Collecting s^2 terms:

$$1 = A + C \rightarrow (1)$$

TRANSFORMS OF DERIVATIVES

Collecting s terms:

$$2 = B - C \rightarrow (2)$$

Collecting constant terms:

$$1 = A - B \rightarrow (3)$$

Solving three equations with three unknowns:

$$A = 2$$

$$B = 1$$

$$C = -1$$

$$\mathcal{L}^{-1}\left\{\frac{s^2 + 2s + 1}{(s - 1)(s^2 + 1)}\right\} \\ = 2\mathcal{L}^{-1}\left\{\frac{1}{s - 1}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 1}\right\} \\ - \mathcal{L}^{-1}\left\{\frac{s}{s^2 + 1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{s^2 + 2s + 1}{(s - 1)(s^2 + 1)}\right\} \\ = 2e^t + \sin t - \cos t$$

Thus,

$$x(t) = \sin t - \cos t + 2e^t$$

$$18. \quad x''(t) + x'(t) - 2x(t) = 6; \\ x(0) = 1, x'(0) = 1$$

$$\mathcal{L}\{x''(t) + x'(t) - 2x(t)\} = \mathcal{L}\{6\} \\ [s^2x(s) - sx(0) - x'(0)] + [sx(s) - x(0)] \\ - 2x(s) = \frac{6}{s}$$

$$\text{Substitute } x(0) = 1, x'(0) = 1;$$

$$[s^2x(s) - s(1) - 1] + [sx(s) - 1] - 2x(s) \\ = \frac{6}{s}$$

$$s^2x(s) - s - 1 + sx(s) - 1 - 2x(s) = \frac{6}{s}$$

$$(s^2 + s - 2)x(s) = \frac{6}{s} + 2 + s$$

$$(s + 2)(s - 1)x(s) = \frac{6 + 2s + s^2}{s}$$

$$x(s) = \frac{6 + 2s + s^2}{s(s + 2)(s - 1)}$$

$$\mathcal{L}^{-1}\{x(s)\} = \mathcal{L}^{-1}\left\{\frac{6 + 2s + s^2}{s(s + 2)(s - 1)}\right\}$$

Using Partial Fraction:

$$\frac{6 + 2s + s^2}{s(s + 2)(s - 1)} = \frac{A}{s} + \frac{B}{s + 2} + \frac{C}{s - 1}$$

$$6 + 2s + s^2 = A(s + 2)(s - 1) \\ + Bs(s - 1) + Cs(s + 2)$$

$$6 + 2s + s^2 = A(s^2 + s - 2) + Bs(s - 1) \\ + Cs(s + 2)$$

Collecting constant terms:

$$6 = -2A$$

$$A = -3 \rightarrow (1)$$

Collecting s terms:

$$2 = A - B + 2C \rightarrow (2)$$

Collecting s^2 terms:

$$1 = A + B + C \rightarrow (3)$$

Solving three equations with three unknowns:

$$A = -3$$

$$B = 1$$

$$C = 3$$

$$\mathcal{L}^{-1}\left\{\frac{6 + 2s + s^2}{s(s + 2)(s - 1)}\right\} \\ = -3\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s + 2}\right\} \\ + 3\mathcal{L}^{-1}\left\{\frac{1}{s - 1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{6 + 2s + s^2}{s(s + 2)(s - 1)}\right\} = -3 + e^{-2t} + 3e^t$$

Thus,

$$x(t) = e^{-2t} + 3e^t - 3$$

19. $y''(x) + 9y(x) = 40e^x$;
 $y(0) = 5, y'(0) = -2$
 $\mathcal{L}\{y''(x) + 9y(x)\} = \mathcal{L}\{40e^x\}$
 $[s^2y(s) - sy(0) - y'(0)] + 9y(s) = \frac{40}{s-1}$

Substitute $y(0) = 5, y'(0) = -2$;

$$[s^2y(s) - s(5) - (-2)] + 9y(s) = \frac{40}{s-1}$$

$$s^2y(s) - 5s + 2 + 9y(s) = \frac{40}{s-1}$$

$$(s^2 + 9)y(s) = \frac{40}{s-1} + 5s - 2$$

$$(s^2 + 9)y(s) = \frac{40 + 5s^2 - 5s - 2s + 2}{s-1}$$

$$y(s) = \frac{5s^2 - 7s + 42}{(s-1)(s^2 + 9)}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{5s^2 - 7s + 42}{(s-1)(s^2 + 9)}\right\}$$

Using Partial Fraction:

$$\frac{5s^2 - 7s + 42}{(s-1)(s^2 + 9)} = \frac{A}{s-1} + \frac{B}{s^2 + 9} + \frac{Cs}{s^2 + 9}$$

$$5s^2 - 7s + 42 = A(s^2 + 9) + B(s-1) + Cs(s-1)$$

Collecting s^2 terms:
 $5 = A + C \rightarrow (1)$

Collecting s terms:
 $-7 = B - C \rightarrow (2)$

Collecting constant terms:
 $42 = 9A - B \rightarrow (3)$

Solving three equations with three unknowns:
 $A = 4$
 $B = -6$
 $C = 1$

$$\mathcal{L}^{-1}\left\{\frac{5s^2 - 7s + 42}{(s-1)(s^2 + 9)}\right\} = 4\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - 6\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 9}\right\} + \mathcal{L}^{-1}\left\{\frac{s}{s^2 + 9}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{5s^2 - 7s + 42}{(s-1)(s^2 + 9)}\right\} = 4e^x - 2\sin 3x + \cos 3x$$

Thus,

$$y(x) = 4e^x + \cos 3x - 2\sin 3x$$

20.

$$y''(x) + y(x) = 4e^x; y(0) = 0, y'(0) = 0$$

$$\mathcal{L}\{y''(x) + y(x)\} = \mathcal{L}\{4e^x\}$$

$$[s^2y(s) - sy(0) - y'(0)] + y(s) = \frac{4}{s-1}$$

Substitute $y(0) = 0, y'(0) = 0$;

$$[s^2y(s) - s(0) - 0] + y(s) = \frac{4}{s-1}$$

$$s^2y(s) + y(s) = \frac{4}{s-1}$$

$$(s^2 + 1)y(s) = \frac{4}{s-1}$$

$$y(s) = \frac{4}{(s-1)(s^2 + 1)}$$

$$\mathcal{L}^{-1}\{y(s)\} = \mathcal{L}^{-1}\left\{\frac{4}{(s-1)(s^2 + 1)}\right\}$$

Using Partial Fraction:

$$\frac{4}{(s-1)(s^2 + 1)} = \frac{A}{s-1} + \frac{B}{s^2 + 1} + \frac{Cs}{s^2 + 1}$$

$$4 = A(s^2 + 1) + B(s-1) + Cs(s-1)$$

Collecting s^2 terms:
 $0 = A + C \rightarrow (1)$

Collecting s terms:
 $0 = B - C \rightarrow (2)$

Collecting constant terms:
 $4 = A - B \rightarrow (3)$

Solving three equations with three unknowns:

$$A = 2$$

$$B = -2$$

$$C = -2$$

$$\mathcal{L}^{-1}\left\{\frac{4}{(s-1)(s^2 + 1)}\right\} = 2\mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} - 2\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 1}\right\} - 2\mathcal{L}^{-1}\left\{\frac{s}{s^2 + 1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{4}{(s-1)(s^2 + 1)}\right\} = 2e^x - 2\sin x - 2\cos x$$

Thus,

$$y(x) = 2e^x - 2\cos x - 2\sin x$$

$$y(x) = 2(e^x - \cos x - \sin x)$$

21.

$$x''(t) + 3x'(t) + 2x(t) = 4t^2$$

$$x(0) = 0, x'(0) = 0$$

$$\mathcal{L}\{x''(t) + 3x'(t) + 2x(t)\} = \mathcal{L}\{4t^2\}$$

$$[s^2x(s) - sx(0) - x'(0)] + 3[sx(s) - x(0)] + 2x(s) = \frac{8}{s^3}$$

$$+ 3[sx(s) - x(0)] + 2x(s) = \frac{8}{s^3}$$

Substitute $x(0) = 0, x'(0) = 0$;

$$[s^2x(s) - s(0) - 0] + 3[sx(s) - 0] + 2x(s) = \frac{8}{s^3}$$

$$+ 2x(s) = \frac{8}{s^3}$$

$$s^2x(s) + 3sx(s) + 2x(s) = \frac{8}{s^3}$$

$$(s^2 + 3s + 2)x(s) = \frac{8}{s^3}$$

$$(s + 2)(s + 1)x(s) = \frac{8}{s^3}$$

$$x(s) = \frac{8}{s^3(s + 2)(s + 1)}$$

$$\mathcal{L}^{-1}\{x(s)\} = \mathcal{L}^{-1}\left\{\frac{8}{s^3(s + 2)(s + 1)}\right\}$$

Using Partial Fraction:

$$\frac{8}{s^3(s + 2)(s + 1)} = \frac{A}{s} + \frac{B}{s^2} + \frac{C}{s^3} + \frac{D}{s + 2} + \frac{E}{s + 1}$$

$$8 = As^2(s + 2)(s + 1) + Bs(s + 2)(s + 1) + C(s + 2)(s + 1) + Ds^3(s + 1) + Es^3(s + 2)$$

$$8 = As^2(s^2 + 3s + 2) + Bs(s^2 + 3s + 2) + C(s^2 + 3s + 2) + Ds^3(s + 1) + Es^3(s + 2)$$

Collecting constant terms:

$$8 = 2C$$

$$C = 4$$

Collecting s terms:

$$0 = 2B + 3C$$

$$0 = 2B + 3(4)$$

$$B = -6$$

Collecting s^2 terms:

$$0 = 2A + 3B + C$$

$$0 = 2A + 3(-6) + 4$$

$$A = 7$$

Collecting s^3 terms:

$$0 = 3A + B + D + 2E$$

$$0 = 3(7) - 6 + D + 2E$$

$$0 = 15 + D + 2E \rightarrow (1)$$

Collecting s^4 terms:

$$0 = A + D + E$$

$$0 = 7 + D + E \rightarrow (2)$$

Solving two equations with two unknowns:

$$D = 1$$

$$E = -8$$

$$\mathcal{L}^{-1}\left\{\frac{8}{s^3(s + 2)(s + 1)}\right\} = 7\mathcal{L}^{-1}\left\{\frac{1}{s}\right\} - 6\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} + 4\mathcal{L}^{-1}\left\{\frac{1}{s^3}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s + 2}\right\} - 8\mathcal{L}^{-1}\left\{\frac{1}{s + 1}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{8}{s^3(s + 2)(s + 1)}\right\} = 7 - 6t + 2t^2 + e^{-2t} - 8e^{-t}$$

Thus,

$$x(t) = 2t^2 - 6t + 7 - 8e^{-t} + e^{-2t}$$

$$22. \quad x''(t) - 4x'(t) + 4x(t) = 4 \cos 2t;$$

$$x(0) = 2, x'(0) = 5$$

$$\mathcal{L}\{x''(t)\} - 4\mathcal{L}\{x'(t)\} + 4\mathcal{L}\{x(t)\} = \mathcal{L}\{4 \cos 2t\}$$

$$[s^2x(s) - sx(0) - x'(0)] - 4[sx(s) - x(0)] + 4x(s) = \frac{4s}{s^2 + 4}$$

$$-4[sx(s) - x(0)] + 4x(s) = \frac{4s}{s^2 + 4}$$

$$\text{Substitute } x(0) = 2, x'(0) = 5:$$

$$[s^2x(s) - s(2) - 5] - 4[sx(s) - 2] + 4x(s) = \frac{4s}{s^2 + 4}$$

$$+ 4x(s) = \frac{4s}{s^2 + 4}$$

$$s^2x(s) - 2s - 5 - 4sx(s) + 8 + 4x(s) = \frac{4s}{s^2 + 4}$$

$$= \frac{4s}{s^2 + 4}$$

$$(s^2 - 4s + 4)x(s) = \frac{4s}{s^2 + 4} - 3 + 2s$$

$$(s-2)^2x(s) = \frac{4s - 3s^2 - 12 + 2s^3 + 8s}{s^2 + 4}$$

$$x(s) = \frac{2s^3 - 3s^2 + 12s - 12}{(s^2 + 4)(s-2)^2}$$

$$\mathcal{L}^{-1}\{x(s)\} = \mathcal{L}^{-1}\left\{\frac{2s^3 - 3s^2 + 12s - 12}{(s^2 + 4)(s-2)^2}\right\}$$

Using Partial Fraction:

$$\frac{2s^3 - 3s^2 + 12s - 12}{(s^2 + 4)(s-2)^2} = \frac{A}{s^2 + 4} + \frac{Bs}{s^2 + 4} + \frac{C}{(s-2)} + \frac{D}{(s-2)^2}$$

$$2s^3 - 3s^2 + 12s - 12 = A(s-2)^2 + Bs(s-2)^2 + C(s^2 + 4)(s-2) + D(s^2 + 4)$$

$$2s^3 - 3s^2 + 12s - 12 = A(s^2 - 4s + 4) + B(s^3 - 4s^2 + 4s) + C(s^3 - 2s^2 + 4s - 8) + D(s^2 + 4)$$

$$\text{Collecting constant terms:}$$

$$-12 = 4A - 8C + 4D \rightarrow (1)$$

$$C = 4$$

$$\text{Collecting } s \text{ terms:}$$

$$12 = -4A + 4B + 4C \rightarrow (2)$$

$$\text{Collecting } s^2 \text{ terms:}$$

$$-3 = A - 4B - 2C + D \rightarrow (3)$$

$$\text{Collecting } s^3 \text{ terms:}$$

$$2 = B + C \rightarrow (4)$$

Solving four equations with four unknowns:

$$A = -1$$

$$B = 0$$

$$C = 2$$

$$D = 2$$

$$\mathcal{L}^{-1}\left\{\frac{2s^3 - 3s^2 + 12s - 12}{(s^2 + 4)(s-2)^2}\right\}$$

$$= -1\mathcal{L}^{-1}\left\{\frac{1}{s^2 + 4}\right\} + 2\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2}\right\}$$

$$+ 2\mathcal{L}^{-1}\left\{\frac{1}{(s-2)^2}\right\}$$

$$\mathcal{L}^{-1}\left\{\frac{2s^3 - 3s^2 + 12s - 12}{(s^2 + 4)(s-2)^2}\right\}$$

$$= \frac{-1}{2} \sin 2t + 2e^{2t} + 2te^{2t}$$

Thus,

$$x(t) = 2e^{2t} + 2te^{2t} - \frac{1}{2} \sin 2t$$

$$x(t) = 2e^{2t}(1+t) - \frac{1}{2} \sin 2t$$

INTEGRATION FORMULAS

$$(1) \quad \int a \, du = a \int du = au + C$$

$$(2) \quad \int a^u \, du = \frac{a^u}{\ln a} + C, a > 1, a \neq 1$$

$$(3) \quad \int u^n \, du = \frac{1}{n+1} u^{n+1} + C, \text{ for } n \neq -1$$

$$(4) \quad \int e^u \, du = e^u + C$$

$$(5) \quad \int u^{-1} \, du = \int \frac{du}{u} = \ln|u| + C$$

$$(6) \quad \int \ln u \, du = u \ln|u| - u + C$$

$$(7) \quad \int \sin u \, du = -\cos u + C$$

$$(8) \quad \int \cos u \, du = \sin u + C$$

$$(9) \quad \int \tan u \, du = \ln|\sec u| + C$$

$$(10) \quad \int \cot u \, du = \ln|\sin u| + C$$

$$(11) \quad \int \sec u \, du = \ln|\sec u + \tan u| + C$$

$$(12) \quad \int \csc u \, du = \ln|\csc u - \cot u| + C = -\ln|\csc u + \cot u| + C$$

$$(13) \quad \int \sec u \tan u \, du = \sec u + C$$

$$(14) \quad \int \csc u \cot u \, du = -\csc u + C$$

$$(15) \quad \int \sec^2 u \, du = \tan u + C$$

$$(16) \quad \int \csc^2 u \, du = -\cot u + C$$

$$(17) \quad \int \frac{du}{\sqrt{a^2 - u^2}} = \text{Arcsin} \frac{u}{a} + C$$

$$(18) \quad \int \frac{du}{a^2 + u^2} = \frac{1}{a} \text{Arctan} \frac{u}{a} + C$$

$$(19) \quad \int \frac{du}{u\sqrt{u^2 - a^2}} = \frac{1}{a} \text{Arcsec} \frac{u}{a} + C$$

$$(20) \quad \int \text{Arcsin} u \, du = u \text{Arcsin} u + \sqrt{1 - u^2} + C$$

APPENDIX A: INTEGRATION FORMULAS

Hyperbolic Functions

$$(21) \int \operatorname{Arctan} u \, du = u \operatorname{Arctan} u - \ln \sqrt{1+u^2} + C$$

$$(22) \int \sinh u \, du = \cosh u + C$$

$$(23) \int \cosh u \, du = \sinh u + C$$

$$(24) \int \tanh u \, du = \ln \cosh u + C$$

$$(25) \int \coth u \, du = \ln |\sinh u| + C$$

$$(26) \int \operatorname{sech}^2 u \, du = \tanh u + C$$

$$(27) \int \operatorname{csch}^2 u \, du = -\coth u + C$$

$$(28) \int \operatorname{sech} u \tanh u \, du = -\operatorname{sech} u + C$$

$$(29) \int \operatorname{csch} u \coth u \, du = -\operatorname{csch} u + C$$

$$(30) \int \frac{du}{\sqrt{u^2+a^2}} = \sinh^{-1} \frac{u}{a} + C$$

$$(31) \int \frac{du}{\sqrt{u^2-a^2}} = \cosh^{-1} \frac{u}{a} + C, u > a > 0$$

$$(32) \int \frac{du}{a^2-u^2} = \frac{1}{a} \tanh^{-1} \frac{u}{a} + C, u^2 < a^2$$

$$(33) \int \frac{du}{a^2-u^2} = -\frac{1}{a} \coth^{-1} \frac{u}{a} + C, u^2 > a^2$$

$$(34) \int \frac{du}{\sqrt{u^2 \pm a^2}} = \ln \left| u + \sqrt{u^2 \pm a^2} \right| + C$$

$$(35) \int \frac{du}{a^2-u^2} = \frac{1}{2a} \ln \left| \frac{a+u}{a-u} \right| + C, u^2 < a^2$$

$$(36) \int \frac{du}{u^2-a^2} = \frac{1}{2a} \ln \left| \frac{u-a}{u+a} \right| + C, u^2 > a^2$$

$$(37) \int \sqrt{a^2-u^2} \, du = \frac{u}{2} \sqrt{a^2-u^2} + \frac{a^2}{2} \operatorname{Arcsin} \frac{u}{a} + C$$

$$(38) \int \sqrt{u^2 \pm a^2} \, du = \frac{u}{2} \sqrt{u^2 \pm a^2} + \frac{a^2}{2} \ln \left| u + \sqrt{u^2 \pm a^2} \right| + C$$

$$(39) \int u \, dv = uv - \int v \, du$$

Other Functions

Integration by Parts

APPENDIX B: DIFFERENTIATION FORMULAS

DIFFERENTIATION FORMULAS

Algebraic Functions

$$(1) \frac{d}{dx}(C) = 0$$

$$(2) \frac{d}{dx}(Cu) = C \frac{d(u)}{dx}$$

$$(3) \frac{d}{dx}(u+v) = \frac{d(u)}{dx} + \frac{d(v)}{dx}$$

$$(4) \frac{d}{dx}(uv) = u \frac{d(v)}{dx} + v \frac{d(u)}{dx}$$

$$(5) \frac{d}{dx}(u^n) = nu^{n-1} \frac{d(u)}{dx}$$

$$(6) \frac{d}{dx}(\sqrt{u}) = \frac{\frac{d(u)}{dx}}{2\sqrt{u}}; u \neq 0$$

$$(7) d\left(\frac{u}{v}\right) = \frac{v \frac{d(u)}{dx} - u \frac{d(v)}{dx}}{v^2}; v \neq 0$$

$$(8) d\left(\frac{C}{v}\right) = \frac{-C \frac{d(v)}{dx}}{v^2}$$

Logarithmic & Exponential Functions

$$(9) \frac{d}{dx}(\log_a u) = \frac{\frac{d(u)}{dx}}{u \ln a}$$

$$(10) \frac{d}{dx}(\log_{10} u) = \log_{10} e \cdot \frac{\frac{d(u)}{dx}}{u} = 0.43429 \frac{\frac{d(u)}{dx}}{u}$$

$$(11) \frac{d}{dx}(\ln u) = \frac{\frac{d(u)}{dx}}{u}; u \neq 0$$

$$(12) \frac{d}{dx}(a^u) = a^u \ln a \frac{d(u)}{dx}$$

$$(13) \frac{d}{dx}(e^u) = e^u \frac{d(u)}{dx}$$

$$(14) \frac{d}{dx}(u^v) = vu^{v-1} \frac{d(u)}{dx} + u^v \ln u \frac{d(v)}{dx}$$

$$(15) \frac{d}{dx}(\sin u) = \cos u \frac{d(u)}{dx}$$

$$(16) \frac{d}{dx}(\cos u) = -\sin u \frac{d(u)}{dx}$$

$$(17) \frac{d}{dx}(\tan u) = \sec^2 u \frac{d(u)}{dx}$$

$$(18) \frac{d}{dx}(\cot u) = -\csc^2 u \frac{d(u)}{dx}$$

$$(19) \frac{d}{dx}(\sec u) = \sec u \tan u \frac{d(u)}{dx}$$

Trigonometric Functions

APPENDIX B: DIFFERENTIATION FORMULAS

Inverse Trigonometric Functions

$$(20) \frac{d}{dx}(\csc u) = -\csc u \cot u \frac{d(u)}{dx}$$

$$(21) \frac{d}{dx}(\operatorname{Arcsin} u) = \frac{d(u)/dx}{\sqrt{1-u^2}}$$

$$(22) \frac{d}{dx}(\operatorname{Arcos} u) = -\frac{d(u)/dx}{\sqrt{1-u^2}}$$

$$(23) \frac{d}{dx}(\operatorname{Arctan} u) = \frac{d(u)/dx}{1+u^2}$$

$$(24) \frac{d}{dx}(\operatorname{Arccot} u) = -\frac{d(u)/dx}{1+u^2}$$

$$(25) \frac{d}{dx}(\operatorname{Arcsec} u) = \frac{d(u)/dx}{u\sqrt{u^2-1}}$$

$$(26) \frac{d}{dx}(\operatorname{Arccsc} u) = -\frac{d(u)/dx}{u\sqrt{u^2-1}}$$

Hyperbolic Functions

where:

$$\sinh x = \frac{e^x - e^{-x}}{2}$$

$$\cosh x = \frac{e^x + e^{-x}}{2}$$

$$\tanh x = \frac{\sinh x}{\cosh x}$$

$$\coth x = \frac{1}{\tanh x}$$

$$\operatorname{sech} x = \frac{1}{\cosh x}$$

$$\operatorname{csch} x = \frac{1}{\sinh x}$$

Inverse Hyperbolic Functions

where:

$$\operatorname{Arcsinh} x = \ln(x + \sqrt{x^2 + 1})$$

$$\operatorname{Arccosh} x = \ln(x + \sqrt{x^2 - 1})$$

$$\operatorname{Arctanh} x = \frac{1}{2} \ln \frac{1+x}{1-x}$$

$$\operatorname{Arcoth} x = \frac{1}{2} \ln \frac{x+1}{x-1}$$

$$\operatorname{Arcsech} x = \ln \frac{1+\sqrt{1-x^2}}{x}$$

$$\operatorname{Arccsch} x = \ln \frac{1+\sqrt{1+x^2}}{x}, x > 0$$

$$(27) \frac{d}{dx}(\sinh u) = \cosh u \frac{d(u)}{dx}$$

$$(28) \frac{d}{dx}(\cosh u) = \sinh u \frac{d(u)}{dx}$$

$$(29) \frac{d}{dx}(\tanh u) = \operatorname{sech}^2 u \frac{d(u)}{dx}$$

$$(30) \frac{d}{dx}(\coth u) = -\operatorname{csch}^2 u \frac{d(u)}{dx}$$

$$(31) \frac{d}{dx}(\operatorname{sech} u) = -\operatorname{sech} u \tanh u \frac{d(u)}{dx}$$

$$(32) \frac{d}{dx}(\operatorname{csch} u) = -\operatorname{csch} u \coth u \frac{d(u)}{dx}$$

$$(33) \frac{d}{dx}(\operatorname{Arcsinh} u) = \frac{d(u)/dx}{\sqrt{u^2 + 1}}$$

$$(34) \frac{d}{dx}(\operatorname{Arccosh} u) = \frac{d(u)/dx}{\sqrt{u^2 - 1}}$$

$$(35) \frac{d}{dx}(\operatorname{Arctanh} u) = \frac{d(u)/dx}{1-u^2}$$

$$(36) \frac{d}{dx}(\operatorname{Arcoth} u) = -\frac{d(u)/dx}{u^2-1} = \frac{d(u)/dx}{1-u^2}$$

$$(37) \frac{d}{dx}(\operatorname{Arcsech} u) = -\frac{d(u)/dx}{u\sqrt{1-u^2}}$$

$$(38) \frac{d}{dx}(\operatorname{Arccsch} u) = -\frac{d(u)/dx}{u\sqrt{1+u^2}}$$

APPENDIX C: TRIGONOMETRIC IDENTITIES

TRIGONOMETRIC IDENTITIES

Basic Identities

$$(1) \tan \theta = \frac{\sin \theta}{\cos \theta}$$

$$(2) \cot \theta = \frac{\cos \theta}{\sin \theta}$$

$$(3) \sec \theta = \frac{1}{\cos \theta}$$

$$(4) \csc \theta = \frac{1}{\sin \theta}$$

Pythagorean Identities

$$(5) \sin^2 \theta + \cos^2 \theta = 1$$

$$(6) \tan^2 \theta + 1 = \sec^2 \theta$$

$$(7) 1 + \cot^2 \theta = \csc^2 \theta$$

Sum and Difference of Two Angles

$$(8) \sin(x+y) = \sin x \cos y + \cos x \sin y$$

$$(9) \sin(x-y) = \sin x \cos y - \cos x \sin y$$

$$(10) \cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$(11) \cos(x-y) = \cos x \cos y + \sin x \sin y$$

$$(12) \tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$$

$$(13) \tan(x-y) = \frac{\tan x - \tan y}{1 + \tan x \tan y}$$

Double-Angle Formulas

$$(14) \sin(2x) = 2 \sin x \cos x$$

$$(15) \cos(2x) = 2 \cos^2 x - 1$$

$$(16) \tan(2x) = \frac{2 \tan x}{1 - \tan^2 x}$$

Half-Angle Formulas

$$(17) \sin\left(\frac{\theta}{2}\right) = \sqrt{\frac{1 - \cos \theta}{2}}$$

$$(18) \cos\left(\frac{\theta}{2}\right) = \sqrt{\frac{1 + \cos \theta}{2}}$$

$$(19) \tan\left(\frac{\theta}{2}\right) = \sqrt{\frac{1 - \cos \theta}{1 + \cos \theta}}$$

Power-Reduction Formulas

$$(20) \sin^2 x = \frac{1 - \cos 2x}{2}$$

$$(21) \cos^2 x = \frac{1 + \cos 2x}{2}$$

$$(22) \tan^2 x = \frac{1 - \cos 2x}{1 + \cos 2x}$$

Product-to-Sum Formulas

$$(23) \sin x \cos y = \frac{1}{2} [\sin(x+y) + \sin(x-y)]$$

APPENDIX C: TRIGONOMETRIC IDENTITIES

Sum and Difference of Functions (Factoring Formulas)

$$(24) \sin x \sin y = \frac{1}{2} [\cos(x - y) - \cos(x + y)]$$

$$(25) \cos x \cos y = \frac{1}{2} [\cos(x + y) + \cos(x - y)]$$

$$(26) \sin x + \sin y = 2 \sin\left(\frac{x + y}{2}\right) \cos\left(\frac{x - y}{2}\right)$$

$$(27) \sin x - \sin y = 2 \cos\left(\frac{x + y}{2}\right) \sin\left(\frac{x - y}{2}\right)$$

$$(28) \cos x + \cos y = 2 \cos\left(\frac{x + y}{2}\right) \cos\left(\frac{x - y}{2}\right)$$

$$(29) \cos x - \cos y = -2 \sin\left(\frac{x + y}{2}\right) \sin\left(\frac{x - y}{2}\right)$$

$$(30) \tan x + \tan y = \frac{\sin(x + y)}{\cos x \cos y}$$

$$(31) \tan x - \tan y = \frac{\sin(x - y)}{\cos x \cos y}$$

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